

HELM Autonomous Inspection Robot

Fabrication / Assembly / Wiring Drawing Set

Source: HELM_V13_LATEST.step

Revision V13-REISSUE-20260929 | Date 2026-09-29 | Units mm | 202 sheets
STEP SHA256: 6942914ab1ea14b2dece5bdde453c85a6a909255873ff5ab9afa32673fce11da

SERIES	CONTENTS	USE
A	General views, locations, exploded groups, frame	Assembly / orientation
F-101 onward	130 individual fabricated / modified components	Cut / drill / relief details
D	Fastening, camera, LiDAR, bumpers, battery	Assembly / adjustment
E	Power, signal, physical routing, connector access	Wiring / service reference
B	BOM and fastener quantities	Model-derived parts list
Q	Assembly order and pre-fabrication checklist	Build / verification

VERIFIED CAD: values measured from final model. ASSEMBLY ADJUST: position / relief may be fitted during assembly. VERIFY ON HARDWARE: purchased mating holes, connector cavities, polarity, voltage and load ratings are not inferred.

The final STEP controls shape and placement. Flat through-cut DXFs and individual part STEP files are included. Reference envelopes and schematic diagrams are not physical cutting parts. Purchased proxies are placement references, not manufacturer production drawings.

Detailed drawing index follows. PDF bookmarks and DRAWING_INDEX.csv use the same drawing numbers. Fabrication pages are dimensional source drawings; shaded CAD views support assembly understanding.

SHEET	DRAWING	TITLE
7	A-001	General assembly - major item balloons
8	A-002	Orthographic - FRONT / REAR
9	A-003	Orthographic - LEFT / RIGHT
10	A-004	Orthographic - TOP / BOTTOM
11	A-005-1	Major component positions / global coordinates
12	A-005-2	Major component positions / global coordinates
13	A-005-3	Major component positions / global coordinates
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16	A-007-2	Assembly groups 4 to 6
17	A-007-3	Assembly groups 7 to 9
18	A-008	Frame assembly / rods / brackets
19	F-101	Fabrication - BOTTOM_PLATE
20	F-101-H1	Coordinate schedule - BOTTOM_PLATE
21	F-101-H2	Coordinate schedule - BOTTOM_PLATE
22	F-101-H3	Coordinate schedule - BOTTOM_PLATE
23	F-101-H4	Coordinate schedule - BOTTOM_PLATE
24	F-101-H5	Coordinate schedule - BOTTOM_PLATE
25	F-102	Fabrication - BATTERY_PLATE
26	F-103	Fabrication - CHASSIS_PAD_0_0
27	F-104	Fabrication - CHASSIS_PAD_0_1
28	F-105	Fabrication - CHASSIS_PAD_1_0

SHEET	DRAWING	TITLE
29	F-106	Fabrication - CHASSIS_PAD_1_1
30	F-107	Fabrication - CHASSIS_PAD_2_0
31	F-108	Fabrication - CHASSIS_PAD_2_1
32	F-109	Fabrication - CHASSIS_PAD_3_0
33	F-110	Fabrication - CHASSIS_PAD_3_1
34	F-111	Fabrication - ELECTRONICS_PLATE
35	F-111-H1	Coordinate schedule - ELECTRONICS_PLATE
36	F-111-H2	Coordinate schedule - ELECTRONICS_PLATE
37	F-111-H3	Coordinate schedule - ELECTRONICS_PLATE
38	F-111-H4	Coordinate schedule - ELECTRONICS_PLATE
39	F-112	Fabrication - LEFT_PANEL
40	F-112-H1	Coordinate schedule - LEFT_PANEL
41	F-112-H2	Coordinate schedule - LEFT_PANEL
42	F-113	Fabrication - RIGHT_PANEL
43	F-113-H1	Coordinate schedule - RIGHT_PANEL
44	F-113-H2	Coordinate schedule - RIGHT_PANEL
45	F-114	Fabrication - REAR_PANEL
46	F-115	Fabrication - FRONT_PANEL
47	F-116	Fabrication - LIDAR_BRIDGE
48	F-117	Fabrication - LIDAR_PLATE
49	F-118	Fabrication - JETSON_CRADLE
50	F-119	Fabrication - BATTERY_STOP_FRONT

SHEET	DRAWING	TITLE
51	F-120	Fabrication - BATTERY_STOP_REAR
52	F-121	Fabrication - BATTERY_STOP_LEFT
53	F-122	Fabrication - BATTERY_STOP_RIGHT
54	F-123	Fabrication - HUB_TRAY
55	F-124	Fabrication - POWER_BACKPLANE
56	F-125	Fabrication - FUSE_SHELF
57	F-125-H1	Coordinate schedule - FUSE_SHELF
58	F-126	Fabrication - BOOST_BASE
59	F-127	Fabrication - FUSEBLOCK_TRAY
60	F-128	Fabrication - JETSON_BASE_RETAINER_LEFT
61	F-129	Fabrication - JETSON_BASE_RETAINER_RIGHT
62	F-130	Fabrication - RELAY_TRAY
63	F-131	Fabrication - CAMERA_BASE
64	F-132	Fabrication - PAN_SERVO_MOUNT
65	F-133	Fabrication - PAN_YOKE
66	F-134	Fabrication - PAN_ARM_LEFT
67	F-135	Fabrication - CAM_L_LEFT
68	F-136	Fabrication - PAN_ARM_RIGHT
69	F-137	Fabrication - CAM_L_RIGHT
70	F-138	Fabrication - TILT_SERVO_MOUNT
71	F-139	Fabrication - TILT_CHEEK_LEFT
72	F-140	Fabrication - TILT_CHEEK_RIGHT

SHEET	DRAWING	TITLE
73	F-141	Fabrication - C920_SADDLE
74	F-142	Fabrication - TILT_CORNER_LEFT0
75	F-143	Fabrication - TILT_CORNER_LEFT1
76	F-144	Fabrication - TILT_CORNER_RIGHT0
77	F-145	Fabrication - TILT_CORNER_RIGHT1
78	F-146	Fabrication - TILT_BUSH_LEFT
79	F-147	Fabrication - THERMAL_MOUNT
80	F-148	Fabrication - THERMAL_ANGLE_0
81	F-149	Fabrication - THERMAL_ANGLE_1
82	F-150	Fabrication - THERMAL_EDGE_0
83	F-151	Fabrication - THERMAL_EDGE_1
84	F-152	Fabrication - FRONT BUMPER
85	F-153	Fabrication - FRONT_STRIKER
86	F-154	Fabrication - FRONT_GUIDE_FOOT_0
87	F-155	Fabrication - FRONT_GUIDE_HANGER_0
88	F-156	Fabrication - FRONT_GUIDE_ANGLE_0
89	F-157	Fabrication - FRONT_GUIDE_BUSH_0
90	F-158	Fabrication - FRONT_STOPPER_0
91	F-159	Fabrication - FRONT_GUIDE_FOOT_1
92	F-160	Fabrication - FRONT_GUIDE_HANGER_1
93	F-161	Fabrication - FRONT_GUIDE_ANGLE_1
94	F-162	Fabrication - FRONT_GUIDE_BUSH_1

SHEET	DRAWING	TITLE
95	F-163	Fabrication - FRONT_STOPPER_1
96	F-164	Fabrication - FRONT_SWITCH_BRACKET
97	F-165	Fabrication - FRONT_SWITCH_FOOT
98	F-166	Fabrication - FRONT_SWITCH_ANGLE
99	F-167	Fabrication - FRONT_CORNER_NOSE_-1
100	F-168	Fabrication - FRONT_CORNER_NOSE_1
101	F-169	Fabrication - REAR BUMPER
102	F-170	Fabrication - REAR_STRIKER
103	F-171	Fabrication - REAR_GUIDE_FOOT_0
104	F-172	Fabrication - REAR_GUIDE_HANGER_0
105	F-173	Fabrication - REAR_GUIDE_ANGLE_0
106	F-174	Fabrication - REAR_GUIDE_BUSH_0
107	F-175	Fabrication - REAR_STOPPER_0
108	F-176	Fabrication - REAR_GUIDE_FOOT_1
109	F-177	Fabrication - REAR_GUIDE_HANGER_1
110	F-178	Fabrication - REAR_GUIDE_ANGLE_1
111	F-179	Fabrication - REAR_GUIDE_BUSH_1
112	F-180	Fabrication - REAR_STOPPER_1
113	F-181	Fabrication - REAR_SWITCH_BRACKET
114	F-182	Fabrication - REAR_SWITCH_FOOT
115	F-183	Fabrication - REAR_SWITCH_ANGLE
116	F-184	Fabrication - REAR_CORNER_NOSE_-1

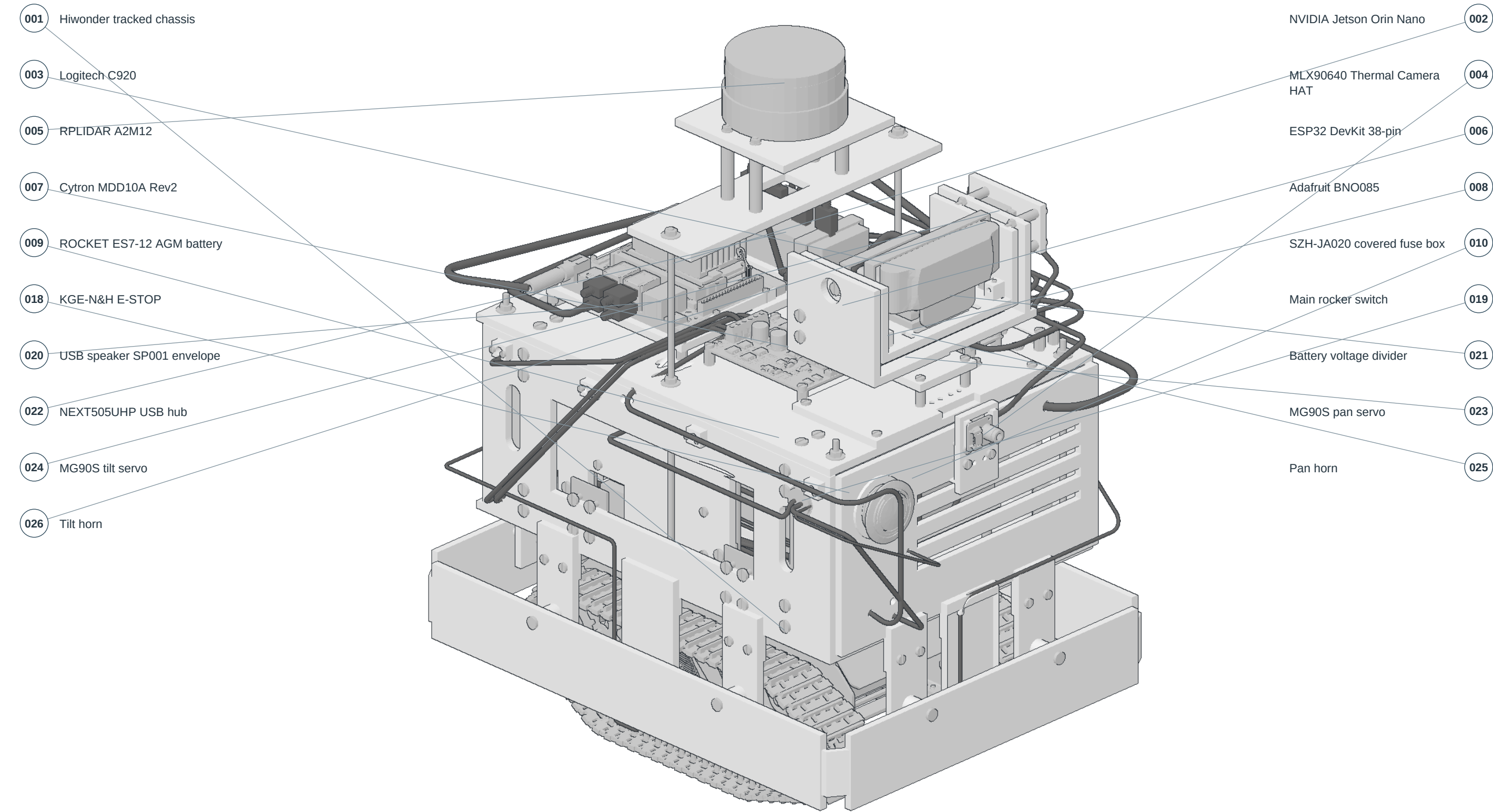
SHEET	DRAWING	TITLE
117	F-185	Fabrication - REAR_CORNER_NOSE_1
118	F-186	Fabrication - LEFT BUMPER
119	F-187	Fabrication - LEFT_STRIKER
120	F-188	Fabrication - LEFT_GUIDE_FOOT_0
121	F-189	Fabrication - LEFT_GUIDE_HANGER_0
122	F-190	Fabrication - LEFT_GUIDE_ANGLE_0
123	F-191	Fabrication - LEFT_GUIDE_BUSH_0
124	F-192	Fabrication - LEFT_STOPPER_0
125	F-193	Fabrication - LEFT_GUIDE_FOOT_1
126	F-194	Fabrication - LEFT_GUIDE_HANGER_1
127	F-195	Fabrication - LEFT_GUIDE_ANGLE_1
128	F-196	Fabrication - LEFT_GUIDE_BUSH_1
129	F-197	Fabrication - LEFT_STOPPER_1
130	F-198	Fabrication - LEFT_SWITCH_BRACKET
131	F-199	Fabrication - LEFT_SWITCH_FOOT
132	F-200	Fabrication - LEFT_SWITCH_ANGLE
133	F-201	Fabrication - RIGHT BUMPER
134	F-202	Fabrication - RIGHT_STRIKER
135	F-203	Fabrication - RIGHT_GUIDE_FOOT_0
136	F-204	Fabrication - RIGHT_GUIDE_HANGER_0
137	F-205	Fabrication - RIGHT_GUIDE_ANGLE_0
138	F-206	Fabrication - RIGHT_GUIDE_BUSH_0

SHEET	DRAWING	TITLE
139	F-207	Fabrication - RIGHT_STOPPER_0
140	F-208	Fabrication - RIGHT_GUIDE_FOOT_1
141	F-209	Fabrication - RIGHT_GUIDE_HANGER_1
142	F-210	Fabrication - RIGHT_GUIDE_ANGLE_1
143	F-211	Fabrication - RIGHT_GUIDE_BUSH_1
144	F-212	Fabrication - RIGHT_STOPPER_1
145	F-213	Fabrication - RIGHT_SWITCH_BRACKET
146	F-214	Fabrication - RIGHT_SWITCH_FOOT
147	F-215	Fabrication - RIGHT_SWITCH_ANGLE
148	F-216	Fabrication - ESP32_CARRIER
149	F-217	Fabrication - ESTOP_ADAPTER_RING
150	F-218	Fabrication - SPEAKER_MOUNT_0
151	F-219	Fabrication - SPEAKER_MOUNT_1
152	F-220	Fabrication - 5V_DISTRIBUTOR
153	F-221	Fabrication - ADC_CARRIER
154	F-222	Fabrication - HARNESS_CLAMP_0
155	F-223	Fabrication - HARNESS_CLAMP_1
156	F-224	Fabrication - HARNESS_CLAMP_2
157	F-225	Fabrication - HARNESS_CLAMP_3
158	F-226	Fabrication - HARNESS_CLAMP_4
159	F-227	Fabrication - HARNESS_CLAMP_5
160	F-228	Fabrication - HARNESS_CLAMP_6

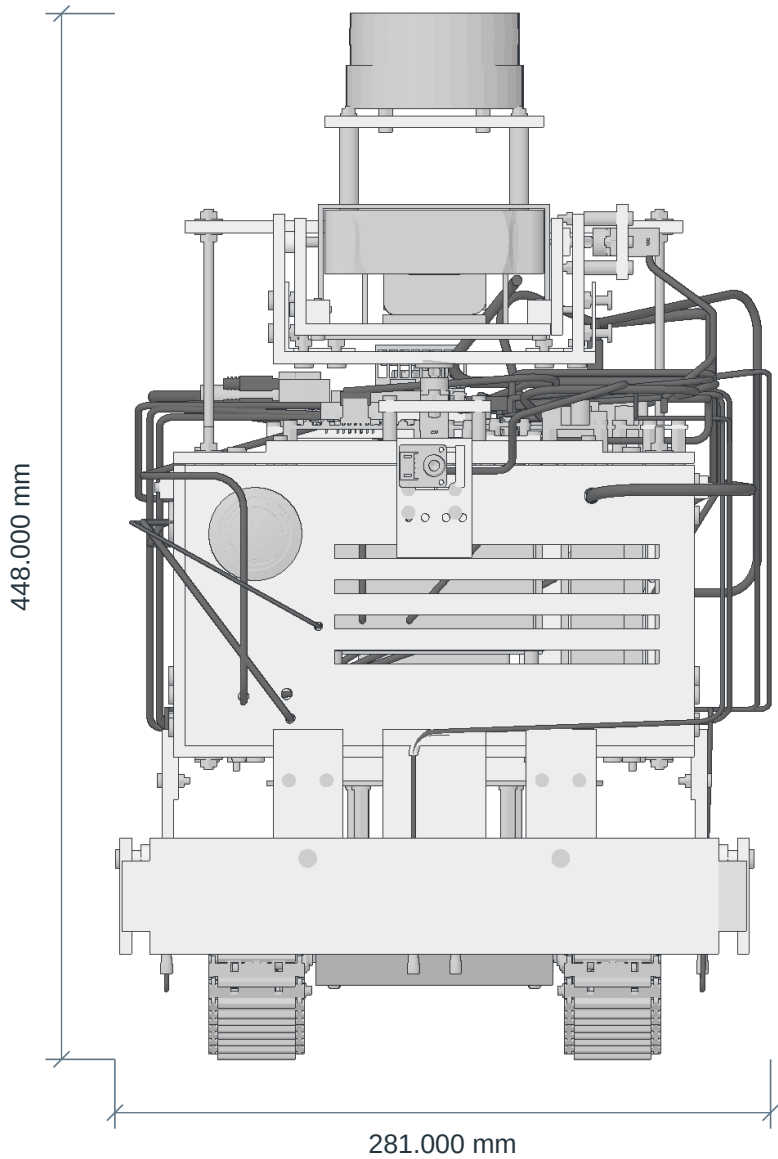
SHEET	DRAWING	TITLE
161	F-229	Fabrication - HARNESS_CLAMP_7
162	F-230	Fabrication - TILT_HORN_COUPLER
163	D-001	Detail A - upper deck / M4 posts
164	D-001-A	Detail A1 - deck clamp stack / rod cut lengths
165	D-002	Detail B - camera support fastening
166	D-003	Camera / thermal / Pan-Tilt - three views
167	D-004	Camera service loops / USB path
168	D-005	Detail C - LiDAR tower and scan reference
169	D-006	Detail D - FRONT limit switch / bumper
170	D-007	Detail D - REAR limit switch / bumper
171	D-008	Detail D - LEFT limit switch / bumper
172	D-009	Detail D - RIGHT limit switch / bumper
173	D-010	Detail E - battery mounting / removal
174	E-201	Power distribution component layout
175	E-202	Control / sensor component layout
176	E-203	Busbar / covered fuse box / rocker
177	E-204	Functional power diagram
178	E-205	Functional control / signal diagram
179	E-206	Physical cable routing - TOP
180	E-207	Physical cable routing - SIDE
181	E-208	Connector service access directions
182	E-209	Maintenance / service access

SHEET	DRAWING	TITLE
183	E-211	Electrical interface verification / open hardware items
184	E-210-1	Cable / connection schedule
185	E-210-2	Cable / connection schedule
186	E-210-3	Cable / connection schedule
187	B-001-1	BOM / drawing references
188	B-001-2	BOM / drawing references
189	B-001-3	BOM / drawing references
190	B-001-4	BOM / drawing references
191	B-001-5	BOM / drawing references
192	B-001-6	BOM / drawing references
193	B-001-7	BOM / drawing references
194	B-001-8	BOM / drawing references
195	B-001-9	BOM / drawing references
196	B-001-10	BOM / drawing references
197	B-001-11	BOM / drawing references
198	B-001-12	BOM / drawing references
199	B-002-1	Fastener list / model counts
200	B-002-2	Fastener list / model counts
201	Q-001	Fabrication and assembly sequence
202	Q-002	Pre-fabrication / commissioning checklist

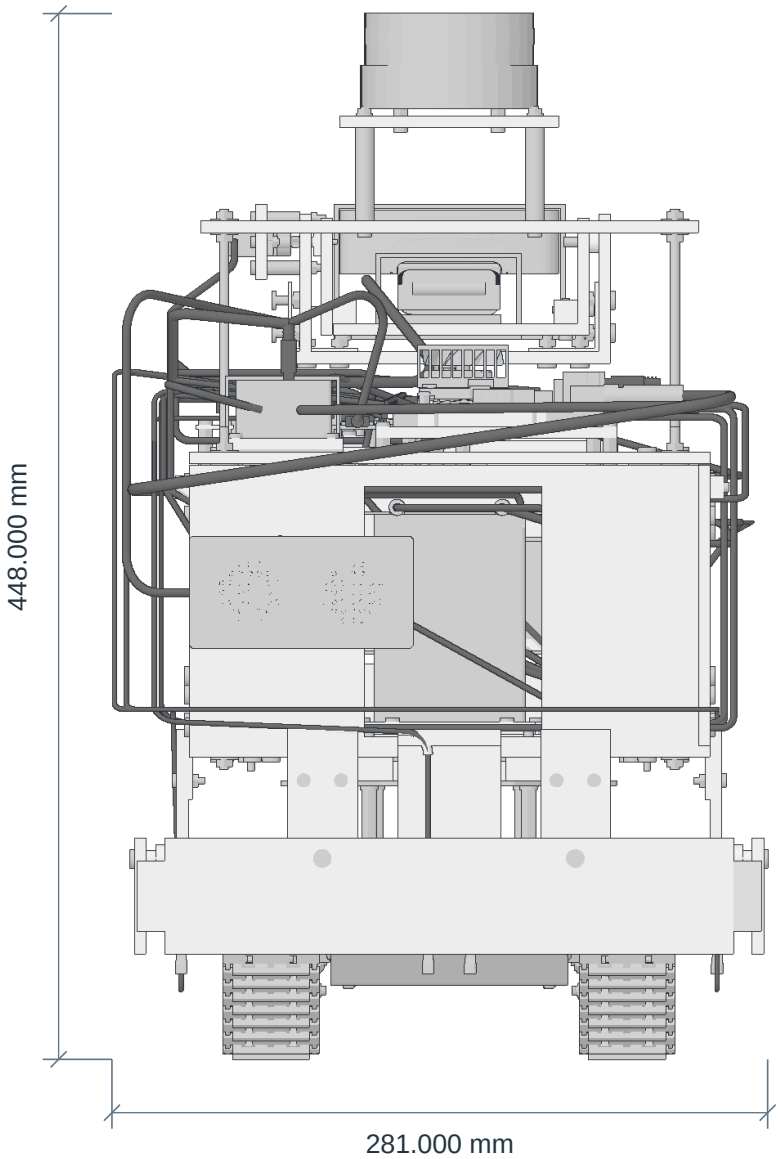
SHEET	DRAWING	TITLE
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Item numbers link to BOM and location tables. Purchased assemblies are gray; fabricated supports are light. CAD axes and physical component positions are unchanged.

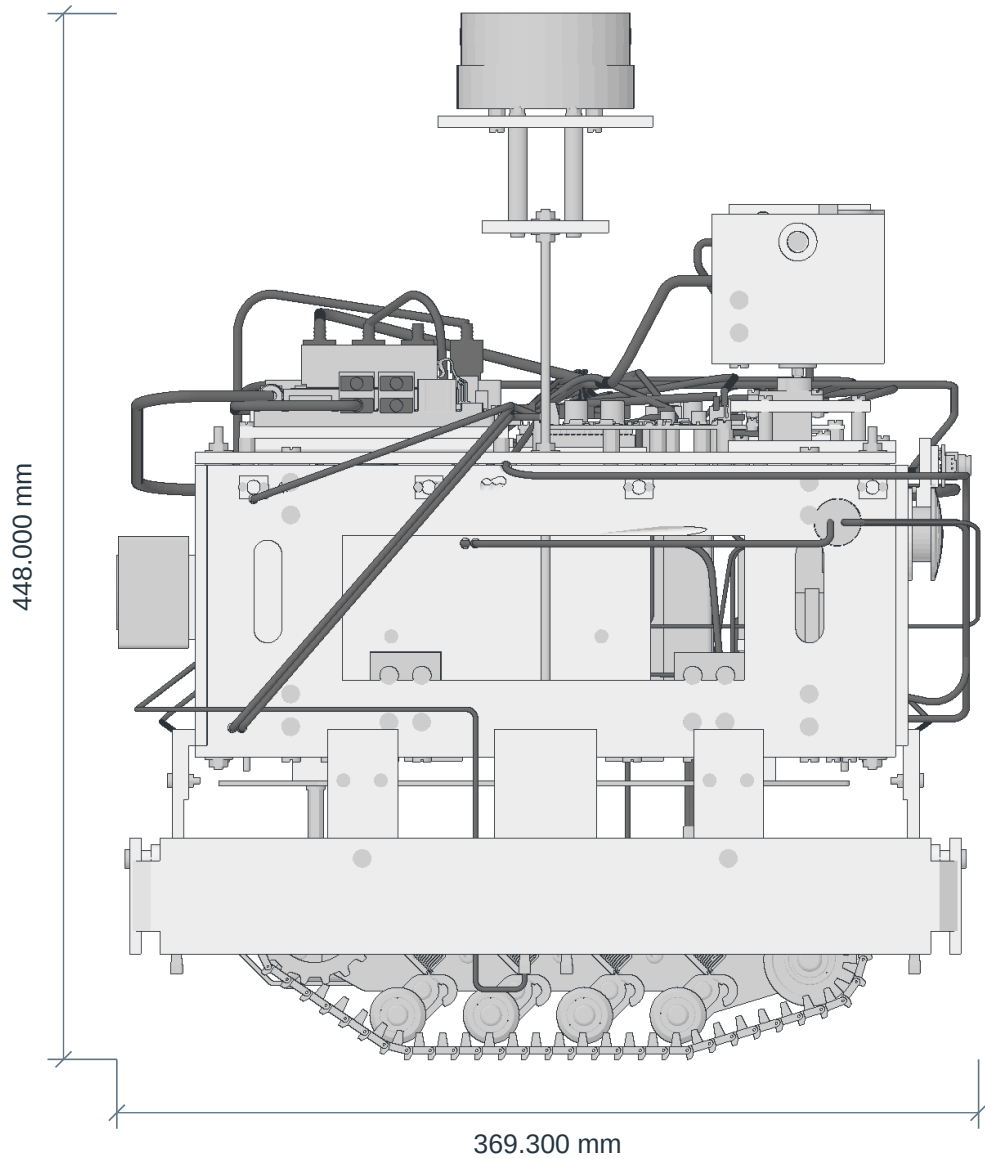


FRONT +X

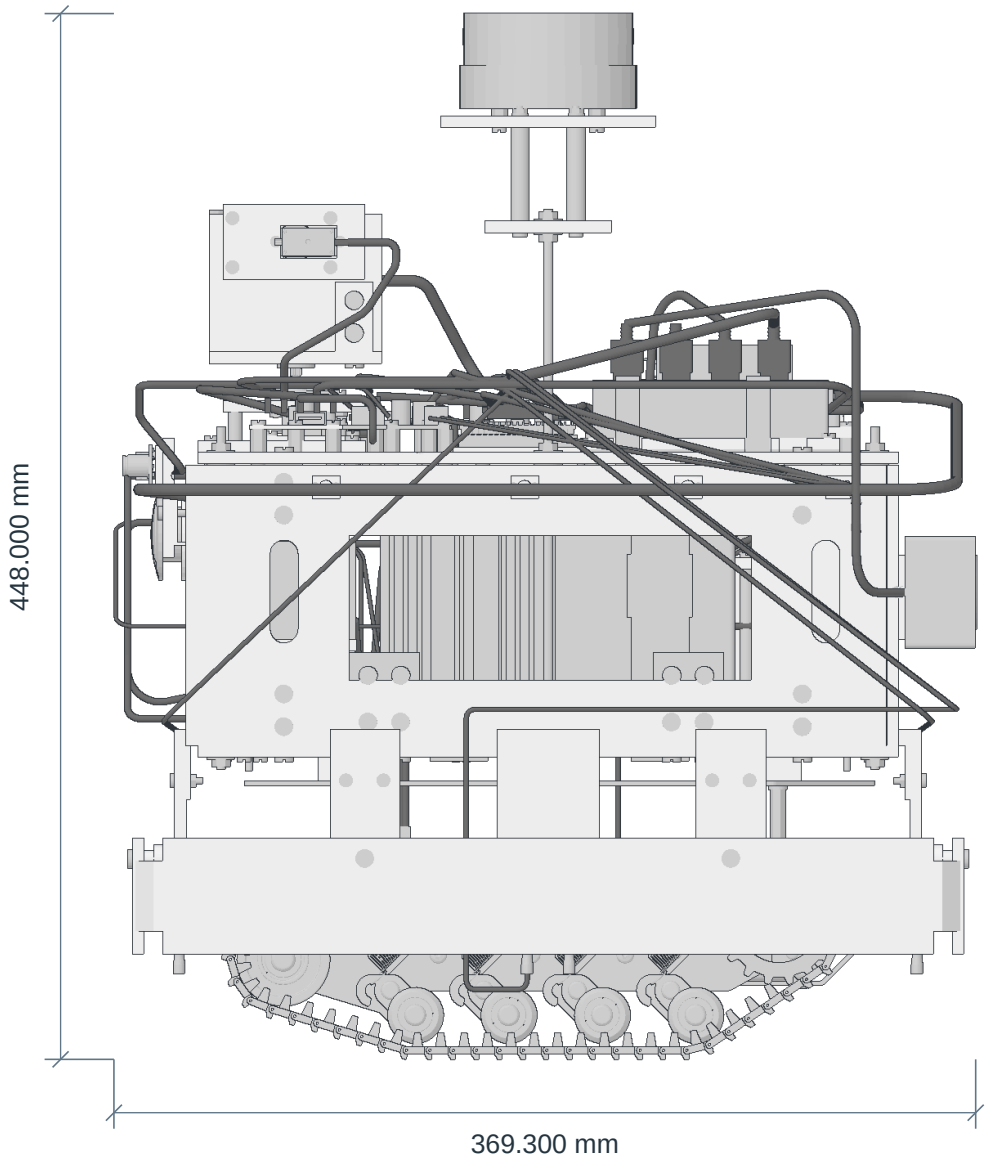


REAR -X

View directions preserve STEP axes: FRONT +X; REAR -X; LEFT -Y; RIGHT +Y; TOP +Z; BOTTOM -Z. Overall envelope includes cables and protrusions. Views fitted to sheet; use written dimensions.

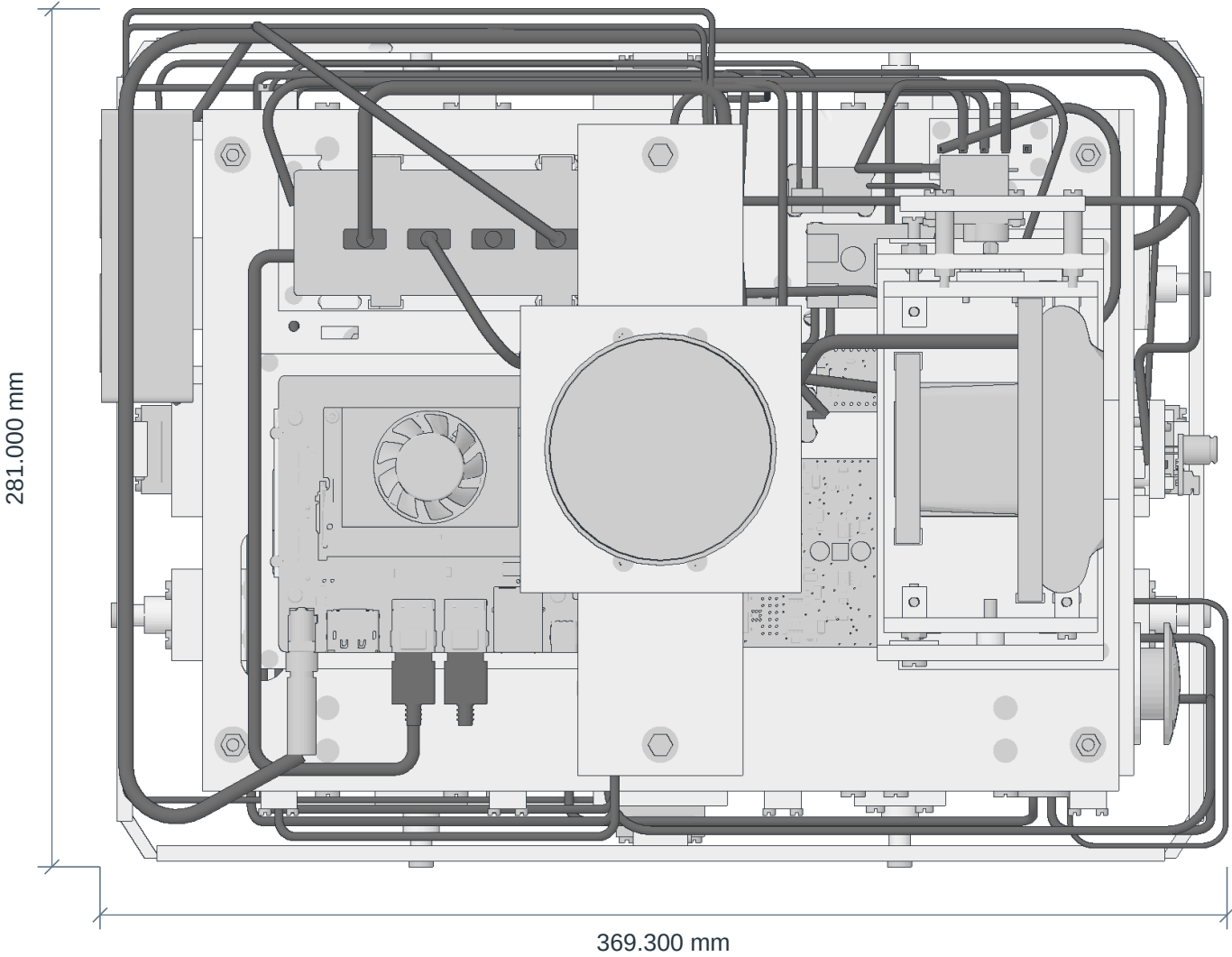


LEFT -Y

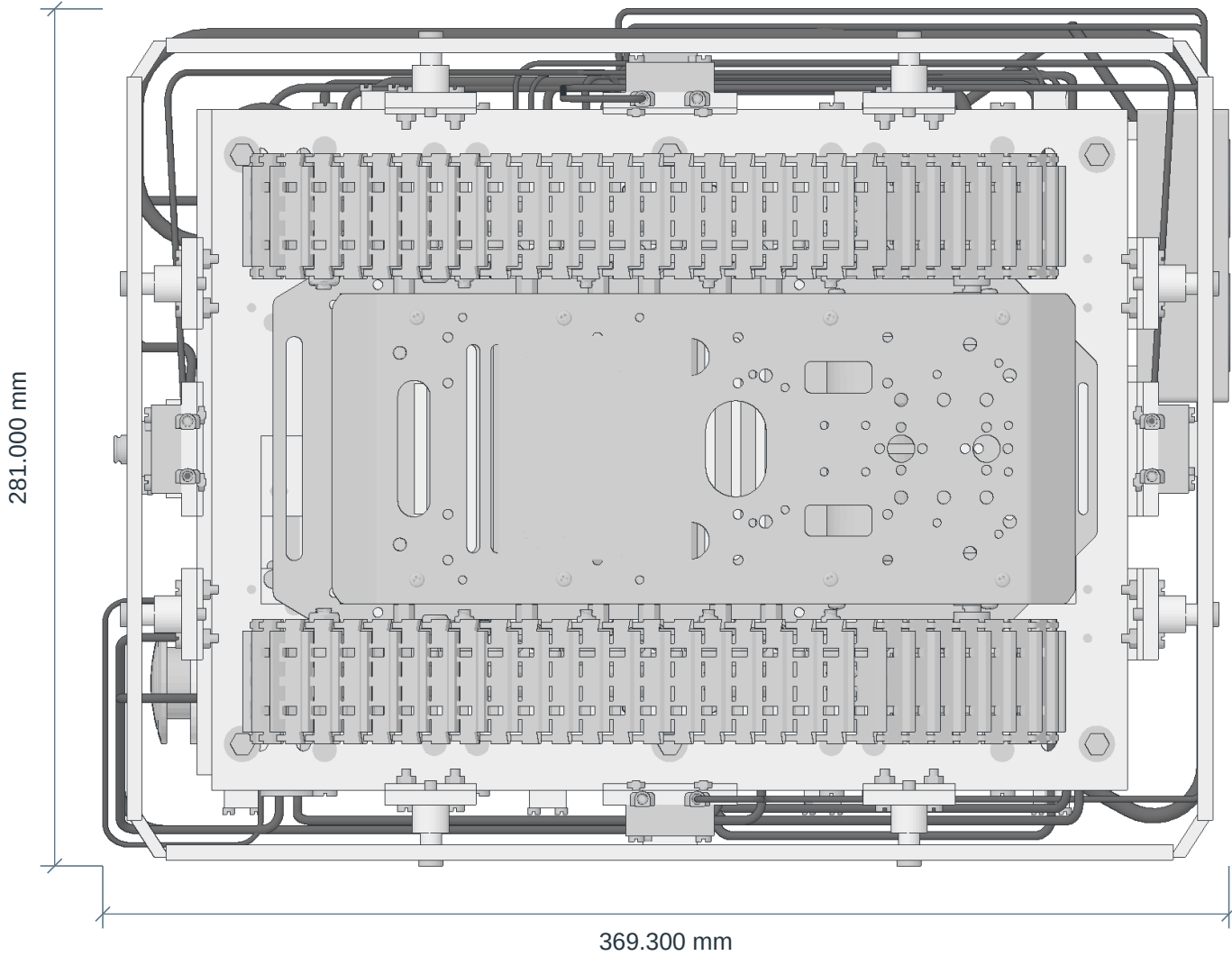


RIGHT +Y

View directions preserve STEP axes: FRONT +X; REAR -X; LEFT -Y; RIGHT +Y; TOP +Z; BOTTOM -Z. Overall envelope includes cables and protrusions. Views fitted to sheet; use written dimensions.



TOP +Z



BOTTOM -Z

View directions preserve STEP axes: FRONT +X; REAR -X; LEFT -Y; RIGHT +Y; TOP +Z; BOTTOM -Z. Overall envelope includes cables and protrusions. Views fitted to sheet; use written dimensions.

ITEM	PART	MIN XYZ	MAX XYZ	ORIENTATION / DATUM
001	Hiwonder tracked chassis	0.00, -0.00, 0.00	279.94, 193.43, 121.62	Global XYZ; orientation shown in A/D views
002	NVIDIA Jetson Orin Nano	15.00, 30.00, 271.00	118.00, 120.88, 305.77	Global XYZ; orientation shown in A/D views
003	Logitech C920	216.50, 49.72, 319.00	285.01, 143.71, 363.50	Global XYZ; orientation shown in A/D views
004	MLX90640 Thermal Camera HAT	305.00, 81.62, 245.00	322.50, 101.77, 263.17	Global XYZ; orientation shown in A/D views
005	RPLIDAR A2M12	101.97, 58.72, 407.00	177.97, 134.72, 448.00	Global XYZ; orientation shown in A/D views
006	ESP32 DevKit 38-pin	121.90, 122.66, 261.61	177.77, 149.34, 274.80	Global XYZ; orientation shown in A/D views
007	Cytron MDD10A Rev2	135.00, 32.00, 268.07	219.58, 93.72, 282.97	Global XYZ; orientation shown in A/D views
008	Adafruit BNO085	191.00, 109.00, 270.00	216.40, 131.86, 274.53	Global XYZ; orientation shown in A/D views
009	ROCKET ES7-12 AGM battery	60.00, 64.00, 139.39	211.00, 129.00, 237.39	Global XYZ; orientation shown in A/D views
010	SZH-JA020 covered fuse box	222.50, 53.00, 180.00	275.50, 139.00, 222.00	Global XYZ; orientation shown in A/D views
011	FHAC main fuse holder	53.00, 14.00, 215.00	103.00, 35.20, 230.00	Global XYZ; orientation shown in A/D views
012	FHAC motor fuse holder	119.00, 14.00, 215.00	169.00, 35.20, 230.00	Global XYZ; orientation shown in A/D views
013	12V relay / socket	228.00, 12.00, 164.00	262.00, 49.40, 200.00	Global XYZ; orientation shown in A/D views
014	Positive busbar	70.00, 10.00, 148.00	168.00, 21.00, 165.00	Global XYZ; orientation shown in A/D views
015	Negative busbar	70.00, 10.00, 177.00	168.00, 21.00, 194.00	Global XYZ; orientation shown in A/D views
016	SZH-EK366 5V converter	187.92, 143.00, 270.00	251.40, 170.50, 282.60	Global XYZ; orientation shown in A/D views

Coordinates are final assembly bounding positions, not mounting-hole locations. Use the matching fabrication detail and source mating geometry for installation. No part has been repositioned for these views.

Major component positions / global coordinates

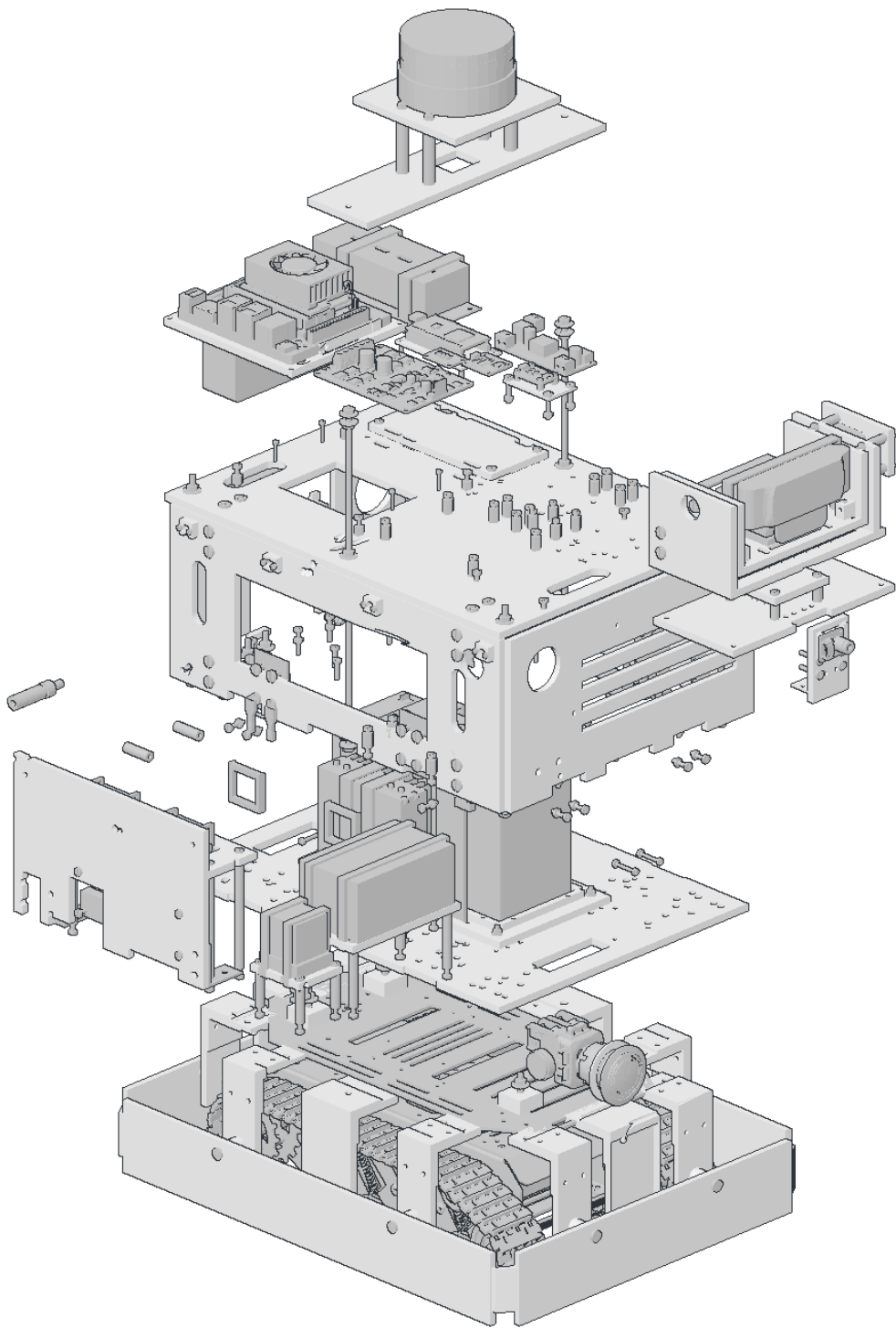
A-005-2

ITEM	PART	MIN XYZ	MAX XYZ	ORIENTATION / DATUM
017	12V to19V boost converter	137.00, 155.00, 151.00	212.00, 186.00, 229.00	Global XYZ; orientation shown in A/D views
018	KGE-N&H E-STOP	233.43, -0.00, 205.00	310.13, 40.00, 246.80	Global XYZ; orientation shown in A/D views
019	Main rocker switch	255.00, -17.00, 220.00	275.00, -10.00, 240.00	Global XYZ; orientation shown in A/D views
020	USB speaker SP001 envelope	-43.50, 112.00, 176.00	-13.00, 208.00, 224.00	Global XYZ; orientation shown in A/D views
021	Battery voltage divider	182.00, 174.00, 270.00	209.00, 189.00, 280.10	Global XYZ; orientation shown in A/D views
022	NEXT505UHP USB hub	20.00, 147.00, 266.00	121.00, 188.00, 291.00	Global XYZ; orientation shown in A/D views
023	MG90S pan servo	226.53, 90.46, 263.75	258.73, 102.79, 296.00	Global XYZ; orientation shown in A/D views
024	MG90S tilt servo	226.52, 160.72, 343.75	258.73, 192.97, 356.08	Global XYZ; orientation shown in A/D views
025	Pan horn	240.00, 88.72, 296.00	256.00, 104.72, 298.00	Global XYZ; orientation shown in A/D views
026	Tilt horn	240.00, 158.72, 342.00	256.00, 160.72, 358.00	Global XYZ; orientation shown in A/D views
027	LS-F FRONT V-J166	296.94, 82.22, 56.00	311.94, 111.22, 80.50	Global XYZ; orientation shown in A/D views
028	LS-R REAR V-J166	-32.00, 82.22, 56.00	-17.00, 111.22, 80.50	Global XYZ; orientation shown in A/D views
029	LS-L LEFT V-J166	125.47, -32.00, 56.00	154.47, -17.00, 80.50	Global XYZ; orientation shown in A/D views
030	LS-Rt RIGHT V-J166	125.47, 210.43, 56.00	154.47, 225.43, 80.50	Global XYZ; orientation shown in A/D views
031	WD SN530 2242 SSD	48.00, 63.00, 268.00	70.00, 105.00, 270.70	Global XYZ; orientation shown in A/D views
032	ADC carrier / hardware VERIFY	221.00, 117.00, 265.00	262.00, 142.00, 267.00	Global XYZ; orientation shown in A/D views

Coordinates are final assembly bounding positions, not mounting-hole locations. Use the matching fabrication detail and source mating geometry for installation. No part has been repositioned for these views.

ITEM	PART	MIN XYZ	MAX XYZ	ORIENTATION / DATUM
033	Encoder level interface	166.00, 100.00, 270.00	190.00, 116.00, 273.10	Global XYZ; orientation shown in A/D views

Coordinates are final assembly bounding positions, not mounting-hole locations. Use the matching fabrication detail and source mating geometry for installation. No part has been repositioned for these views.

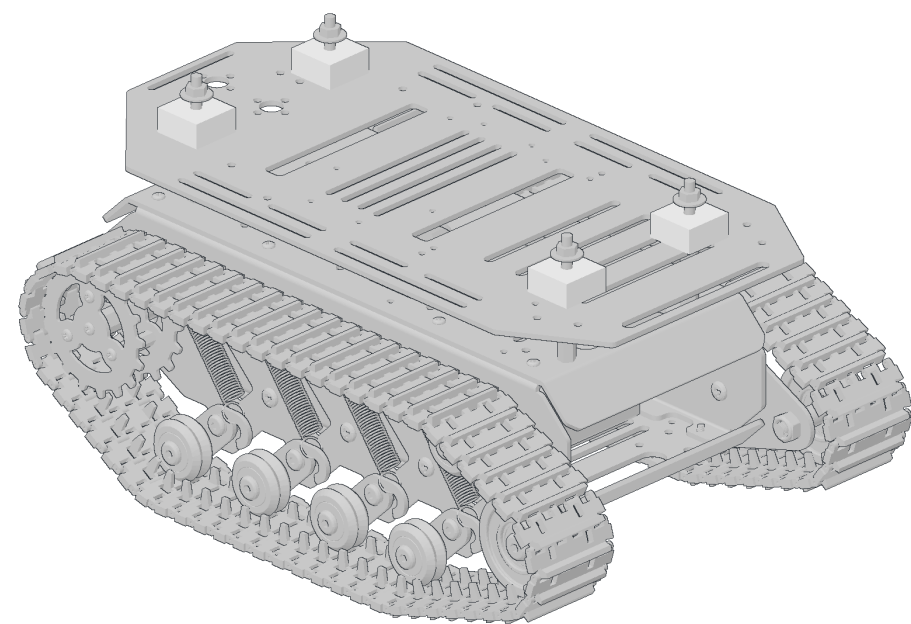


EXPLODED VIEW
Display offsets only. Final assembly coordinates remain unchanged.

- 1 Chassis
- 2 Lower frame / battery
- 3 Upper deck / posts
- 4 Compute / control
- 5 Power distribution
- 6 LiDAR tower
- 7 Camera / thermal / Pan-Tilt
- 8 E-STOP / rocker
- 9 Four bumper / switch modules



Remove electrical connections and retainers before separating modules. Assemble lower frame first; install power and battery before closing upper deck. Side panels and bumper modules attach laterally; top sensor modules lift off after their fasteners are removed.

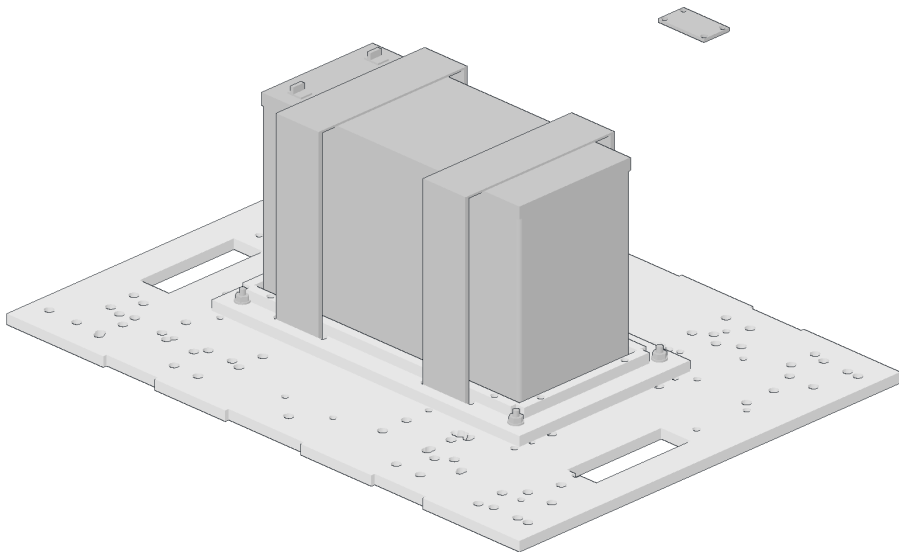


1 Chassis

Keep supplied chassis subassembly intact; mount pads and lower plate above the existing top plate.



Assembly order 1

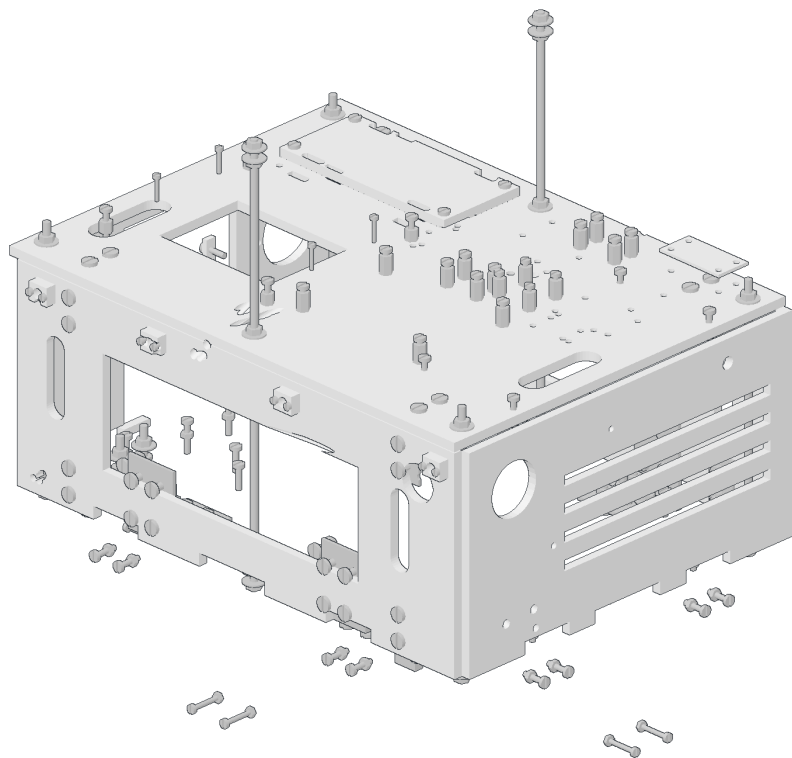


2 Lower frame / battery

Battery sits on its support plate between stops; straps restrain the case. Keep terminals accessible before upper deck closure.



Assembly order 2



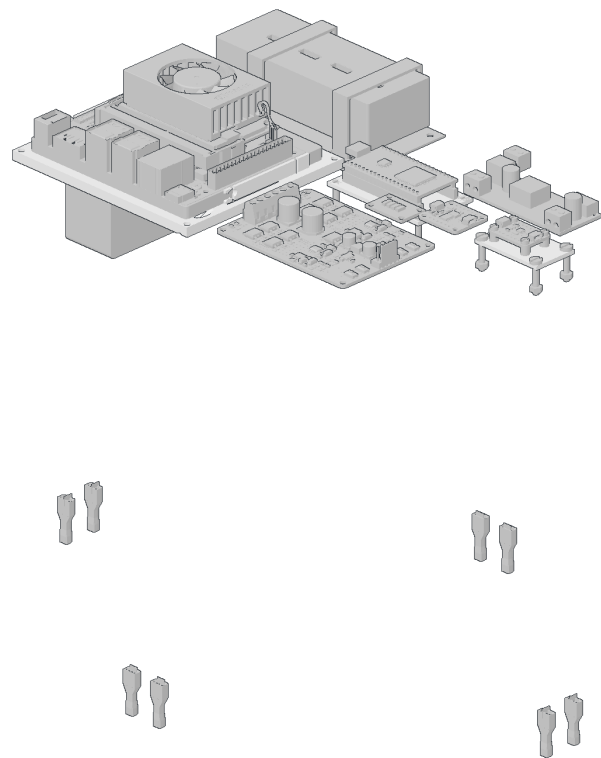
3 Upper deck / posts

M4 rods locate lower and upper plates. L brackets tie panels to plates; washers spread load on sheet.



Assembly order 3

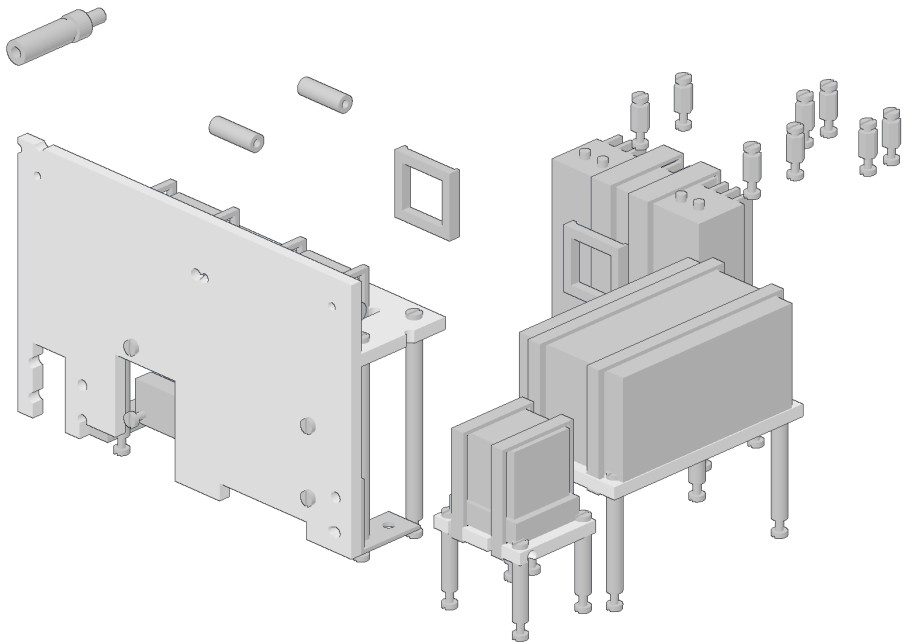
These views isolate groups for visibility; geometry and orientation within each group are unchanged. All actual fastener / plate references remain available in the part STEP files and BOM mapping.



4 Compute / control

Use the documented carrier / cradle and M3 standoffs. USB and terminal faces retain final STEP orientations.

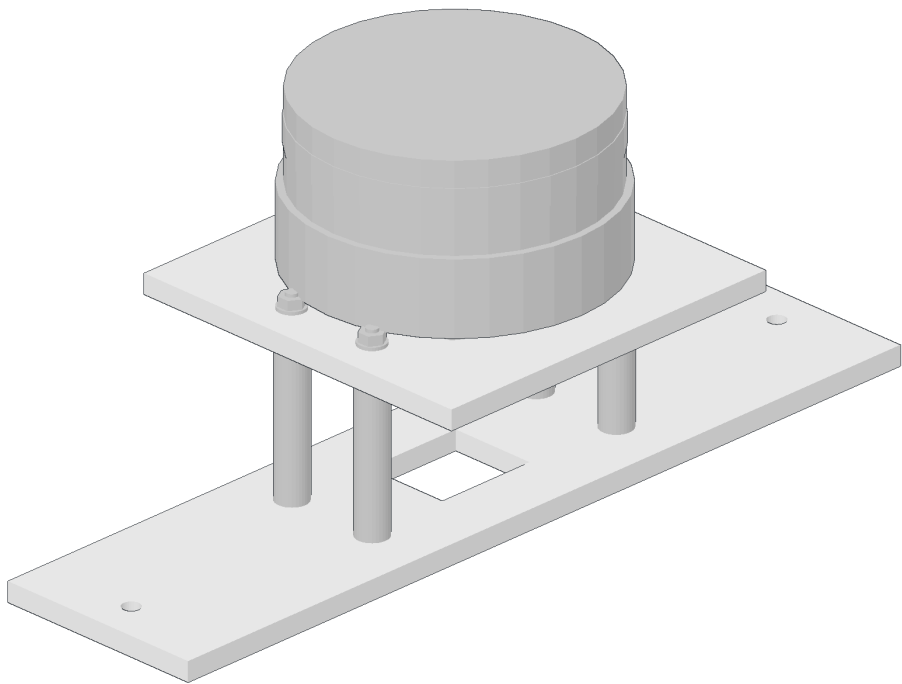
Assembly order 4



5 Power distribution

Install backplane, busbars, covered fuse box, relay and converters before closing side walls.

Assembly order 5

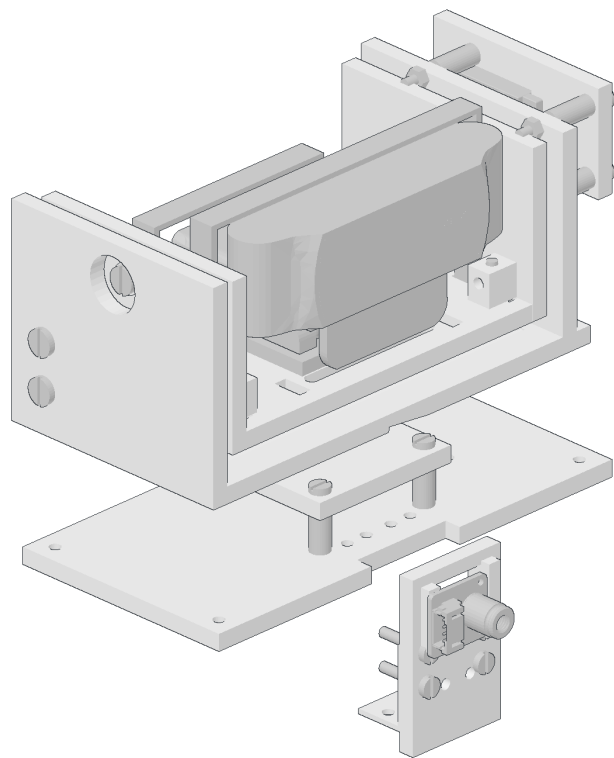


6 LiDAR tower

M3 through-bolts pass bridge and 40mm risers to the sensor plate. Top removal direction +Z.

Assembly order 6

These views isolate groups for visibility; geometry and orientation within each group are unchanged. All actual fastener / plate references remain available in the part STEP files and BOM mapping.

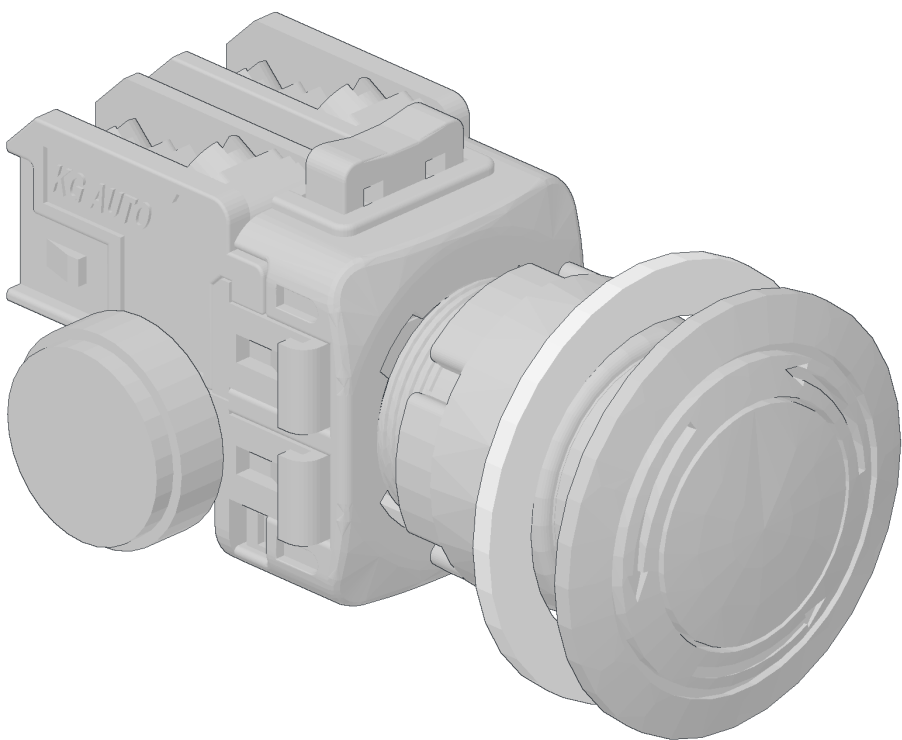


7 Camera / thermal / Pan-Tilt

Build the pan base, yoke, tilt frame and camera saddle as a module. Fit the thermal board to the front wall.



Assembly order 7

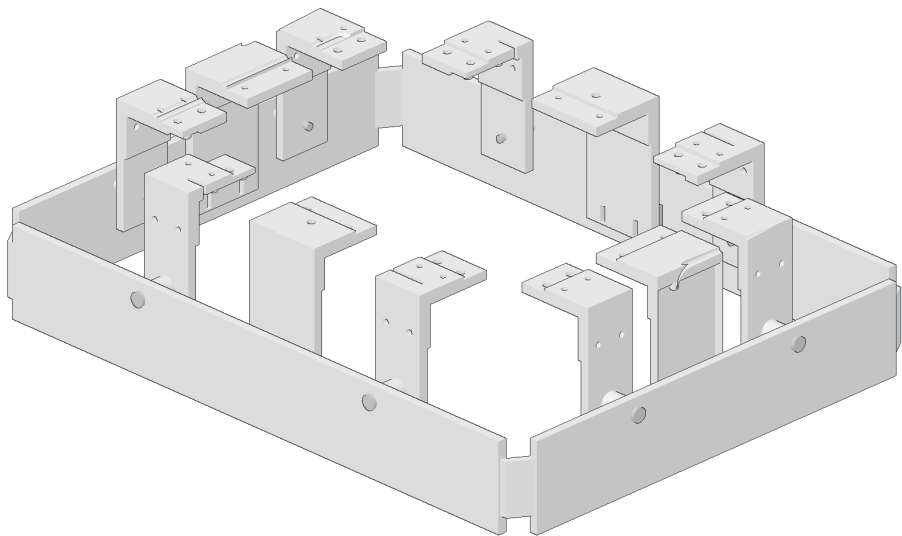


8 E-STOP / rocker

Rocker actuator is outward-facing; E-STOP head remains externally accessible. Terminal work requires panel access.



Assembly order 8



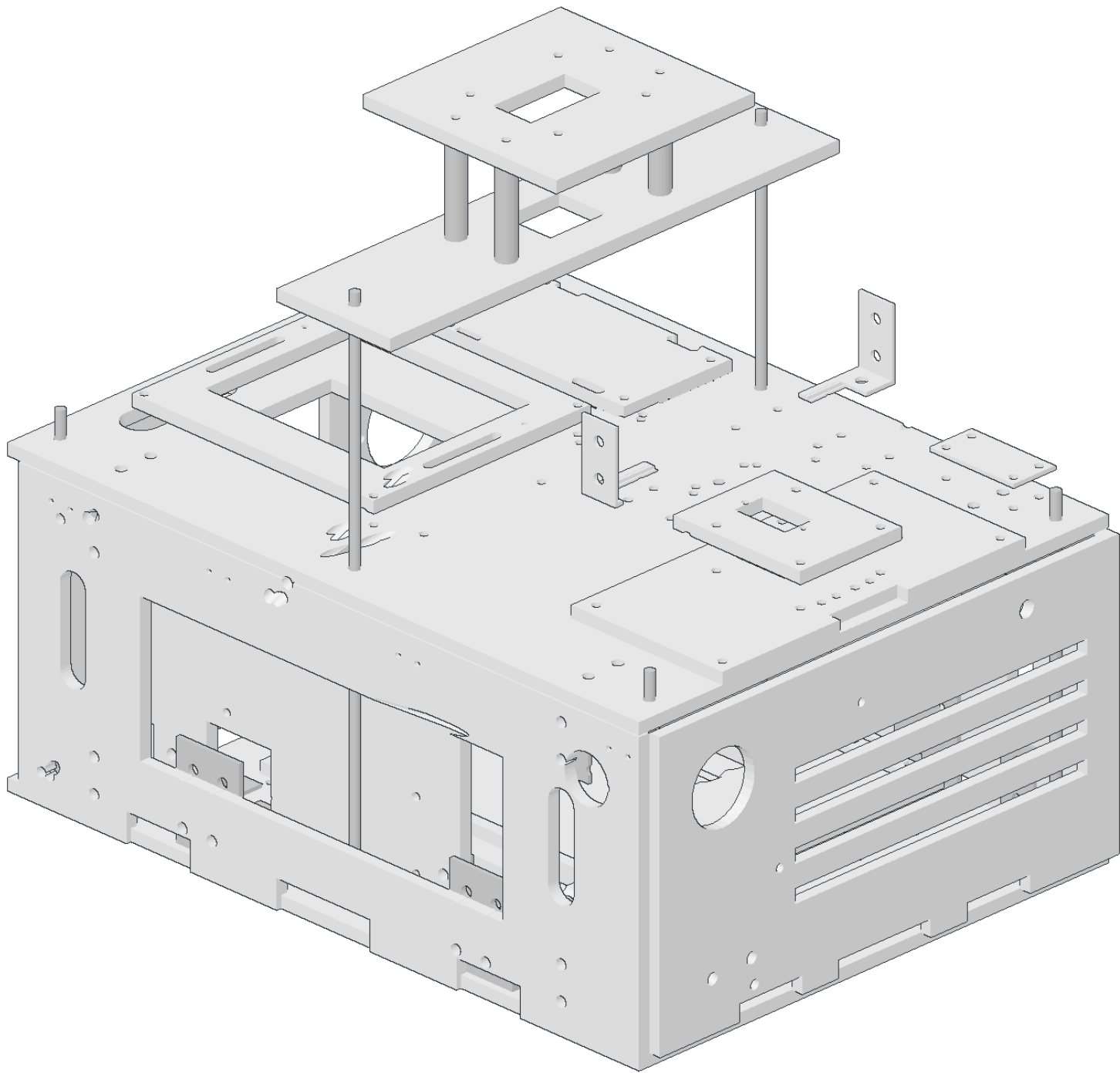
9 Four bumper / switch modules

Each bumper pushes its named lever through the modeled striker. Release must return under spring action.



Assembly order 9

These views isolate groups for visibility; geometry and orientation within each group are unchanged. All actual fastener / plate references remain available in the part STEP files and BOM mapping.



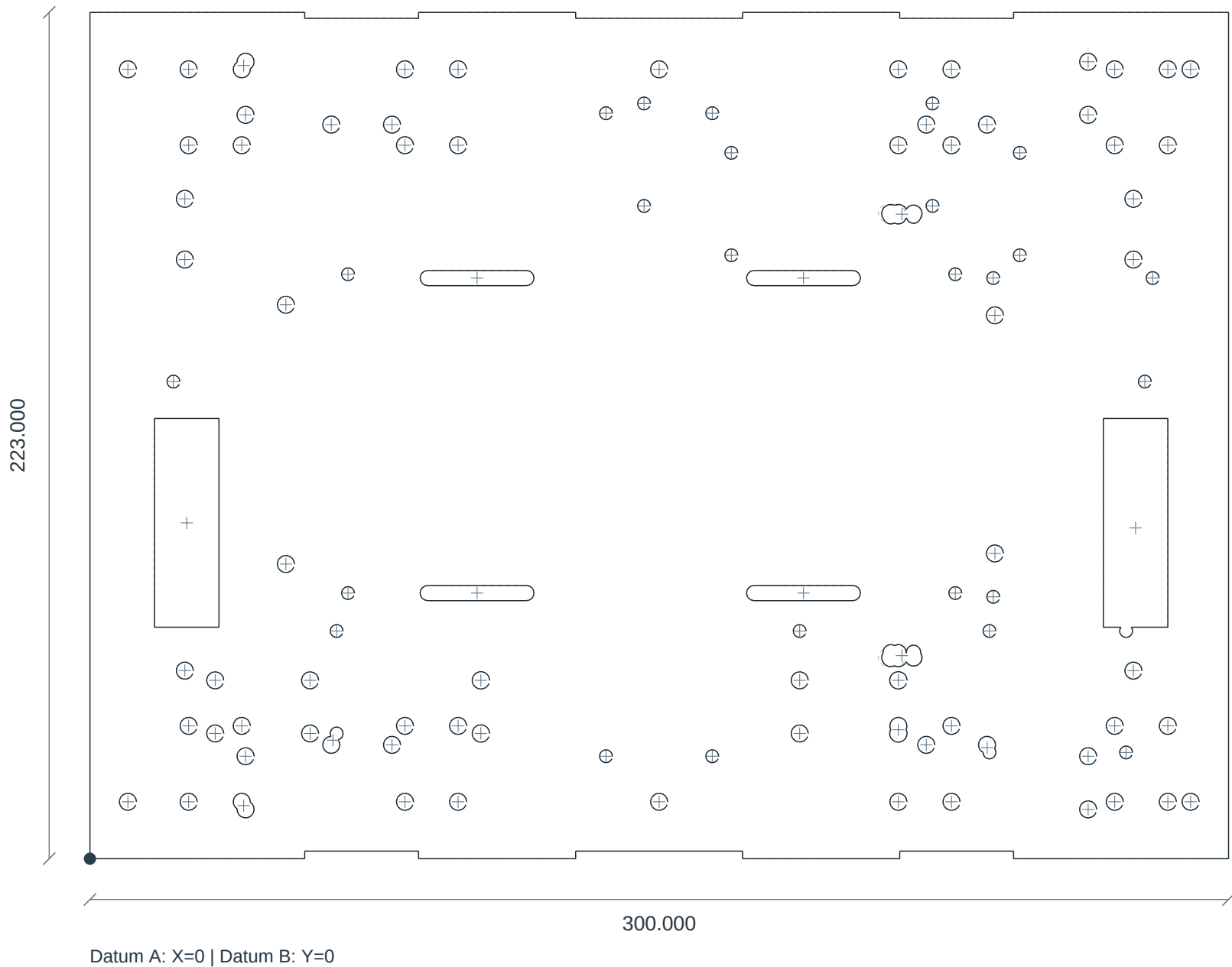
FRAME RELATIONSHIPS

- 01. Lower plate is located on the chassis pads; keep existing chassis mounting interfaces.
- 02. M4 rods pass lower and upper plate holes. Washers spread load on foam/PVC sheet; nuts lock plate height.
- 03. L brackets connect panels/plates; their measured stock hole pattern remains VERIFY ON HARDWARE.
- 04. M3 PCB standoffs support the electronic carriers. Straps retain battery and larger housings.

Use final STEP part relationships and the matching F drawing. Standard hardware heads / holes are nominal; verify purchased fasteners, clamp load and tool access on hardware.

CUSTOM_BOTTOM_PLATE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	263.000	13.000	DIA 4.500
H02	40.500	14.000	5.500 x 6.500
H03	10.000	15.000	DIA 4.500
H04	26.000	15.000	DIA 4.500
H05	83.000	15.000	DIA 4.500
H06	97.000	15.000	DIA 4.500
H07	149.970	15.000	DIA 4.500
H08	213.000	15.000	DIA 4.500
H09	227.000	15.000	DIA 4.500
H10	270.000	15.000	DIA 4.500
H11	284.000	15.000	DIA 4.500
H12	290.000	15.000	DIA 4.500
H13	41.000	27.000	DIA 4.500
H14	135.970	27.000	DIA 3.400
H15	163.970	27.000	DIA 3.400
H16	263.000	27.000	DIA 4.500
H17	273.000	28.000	DIA 3.400
H18	236.401	29.275	4.597 x 5.950

Full XY schedule follows; cross marks locate centers.

Assembly datum XYZ: -10.000, -15.000, 129.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H01	HOLE	263.000	13.000	4.500	4.500	4.500
H02	CUTOUT	40.500	14.000	-	5.500	6.500
H03	HOLE	10.000	15.000	4.500	4.500	4.500
H04	HOLE	26.000	15.000	4.500	4.500	4.500
H05	HOLE	83.000	15.000	4.500	4.500	4.500
H06	HOLE	97.000	15.000	4.500	4.500	4.500
H07	HOLE	149.970	15.000	4.500	4.500	4.500
H08	HOLE	213.000	15.000	4.500	4.500	4.500
H09	HOLE	227.000	15.000	4.500	4.500	4.500
H10	HOLE	270.000	15.000	4.500	4.500	4.500
H11	HOLE	284.000	15.000	4.500	4.500	4.500
H12	HOLE	290.000	15.000	4.500	4.500	4.500
H13	HOLE	41.000	27.000	4.500	4.500	4.500
H14	HOLE	135.970	27.000	3.400	3.400	3.400
H15	HOLE	163.970	27.000	3.400	3.400	3.400
H16	HOLE	263.000	27.000	4.500	4.500	4.500
H17	HOLE	273.000	28.000	3.400	3.400	3.400
H18	CUTOUT	236.401	29.275	-	4.597	5.950
H19	HOLE	79.587	30.000	4.500	4.500	4.500
H20	HOLE	220.353	30.000	4.500	4.500	4.500
H21	CUTOUT	64.018	31.225	-	5.363	6.950
H22	HOLE	33.000	33.000	4.500	4.500	4.500

Same datum as F-101. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H23	HOLE	58.000	33.000	4.500	4.500	4.500
H24	HOLE	103.000	33.000	4.600	4.600	4.600
H25	HOLE	187.000	33.000	4.500	4.500	4.500
H26	CUTOUT	213.000	33.975	-	4.600	6.550
H27	HOLE	26.000	35.000	4.500	4.500	4.500
H28	HOLE	40.000	35.000	4.500	4.500	4.500
H29	HOLE	83.000	35.000	4.500	4.500	4.500
H30	HOLE	97.000	35.000	4.500	4.500	4.500
H31	HOLE	227.000	35.000	4.500	4.500	4.500
H32	HOLE	270.000	35.000	4.500	4.500	4.500
H33	HOLE	284.000	35.000	4.500	4.500	4.500
H34	HOLE	33.000	47.000	4.500	4.500	4.500
H35	HOLE	58.000	47.000	4.500	4.500	4.500
H36	HOLE	103.000	47.000	4.600	4.600	4.600
H37	HOLE	187.000	47.000	4.500	4.500	4.500
H38	HOLE	213.000	47.000	4.600	4.600	4.600
H39	HOLE	25.000	49.555	4.500	4.500	4.500
H40	HOLE	274.939	49.555	4.500	4.500	4.500
H41	CUTOUT	213.925	53.511	-	10.650	5.821
H42	HOLE	65.000	60.000	3.400	3.400	3.400
H43	HOLE	187.000	60.000	3.400	3.400	3.400
H44	HOLE	237.000	60.000	3.400	3.400	3.400

Same datum as F-101. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H45	HOLE	238.000	69.000	3.400	3.400	3.400
H46	HOLE	68.000	70.000	3.400	3.400	3.400
H47	SLOT	102.000	70.000	-	30.000	4.000
H48	SLOT	188.000	70.000	-	30.000	4.000
H49	HOLE	228.000	70.000	3.400	3.400	3.400
H50	HOLE	51.625	77.648	4.500	4.500	4.500
H51	HOLE	238.438	80.433	4.500	4.500	4.500
H52	SLOT	275.500	87.150	-	17.000	57.700
H53	CUTOUT	25.500	88.500	-	17.000	55.000
H54	HOLE	22.000	125.715	3.400	3.400	3.400
H55	HOLE	277.939	125.715	3.400	3.400	3.400
H56	HOLE	238.438	143.180	4.500	4.500	4.500
H57	HOLE	51.625	145.965	4.500	4.500	4.500
H58	SLOT	102.000	153.000	-	30.000	4.000
H59	SLOT	188.000	153.000	-	30.000	4.000
H60	HOLE	238.000	153.000	3.400	3.400	3.400
H61	HOLE	280.000	153.000	3.400	3.400	3.400
H62	HOLE	68.000	154.000	3.400	3.400	3.400
H63	HOLE	228.000	154.000	3.400	3.400	3.400
H64	HOLE	25.000	157.875	4.500	4.500	4.500
H65	HOLE	274.939	157.875	4.500	4.500	4.500
H66	HOLE	169.000	159.000	3.400	3.400	3.400

Same datum as F-101. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H67	HOLE	245.000	159.000	3.400	3.400	3.400
H68	CUTOUT	213.925	169.825	-	10.650	5.150
H69	HOLE	146.000	172.000	3.400	3.400	3.400
H70	HOLE	222.000	172.000	3.400	3.400	3.400
H71	HOLE	25.000	173.875	4.500	4.500	4.500
H72	HOLE	274.939	173.875	4.500	4.500	4.500
H73	HOLE	169.000	186.000	3.400	3.400	3.400
H74	HOLE	245.000	186.000	3.400	3.400	3.400
H75	HOLE	26.000	188.000	4.500	4.500	4.500
H76	HOLE	40.000	188.000	4.500	4.500	4.500
H77	HOLE	83.000	188.000	4.500	4.500	4.500
H78	HOLE	97.000	188.000	4.500	4.500	4.500
H79	HOLE	213.000	188.000	4.500	4.500	4.500
H80	HOLE	227.000	188.000	4.500	4.500	4.500
H81	HOLE	270.000	188.000	4.500	4.500	4.500
H82	HOLE	284.000	188.000	4.500	4.500	4.500
H83	HOLE	63.587	193.430	4.500	4.500	4.500
H84	HOLE	79.587	193.430	4.500	4.500	4.500
H85	HOLE	220.353	193.430	4.500	4.500	4.500
H86	HOLE	236.353	193.430	4.500	4.500	4.500
H87	HOLE	41.000	196.000	4.500	4.500	4.500
H88	HOLE	263.000	196.000	4.500	4.500	4.500

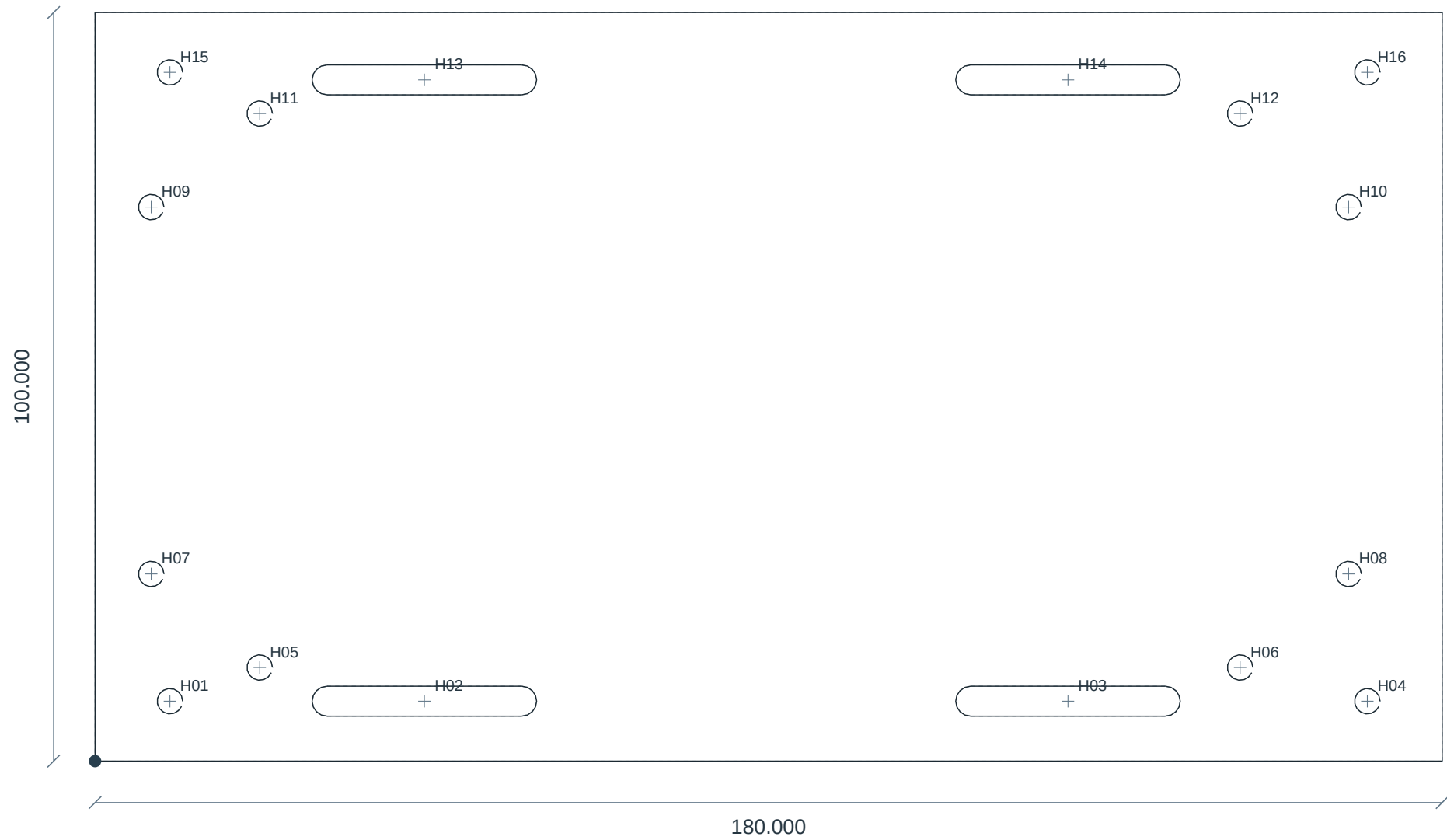
Same datum as F-101. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H89	HOLE	135.970	196.430	3.400	3.400	3.400
H90	HOLE	163.970	196.430	3.400	3.400	3.400
H91	HOLE	146.000	199.000	3.400	3.400	3.400
H92	HOLE	222.000	199.000	3.400	3.400	3.400
H93	HOLE	10.000	208.000	4.500	4.500	4.500
H94	HOLE	26.000	208.000	4.500	4.500	4.500
H95	HOLE	83.000	208.000	4.500	4.500	4.500
H96	HOLE	97.000	208.000	4.500	4.500	4.500
H97	HOLE	149.970	208.000	4.500	4.500	4.500
H98	HOLE	213.000	208.000	4.500	4.500	4.500
H99	HOLE	227.000	208.000	4.500	4.500	4.500
H100	HOLE	270.000	208.000	4.500	4.500	4.500
H101	HOLE	284.000	208.000	4.500	4.500	4.500
H102	HOLE	290.000	208.000	4.500	4.500	4.500
H103	CUTOUT	40.500	209.000	-	5.500	6.500
H104	HOLE	263.000	210.000	4.500	4.500	4.500

Same datum as F-101. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

CUSTOM_BATTERY_PLATE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	8.000	DIA 3.400
H02	44.000	8.000	30.000 x 4.000
H03	130.000	8.000	30.000 x 4.000
H04	170.000	8.000	DIA 3.400
H05	22.000	12.500	DIA 3.400
H06	153.000	12.500	DIA 3.400
H07	7.500	25.000	DIA 3.400
H08	167.500	25.000	DIA 3.400
H09	7.500	74.000	DIA 3.400
H10	167.500	74.000	DIA 3.400
H11	22.000	86.500	DIA 3.400
H12	153.000	86.500	DIA 3.400
H13	44.000	91.000	30.000 x 4.000
H14	130.000	91.000	30.000 x 4.000
H15	10.000	92.000	DIA 3.400
H16	170.000	92.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

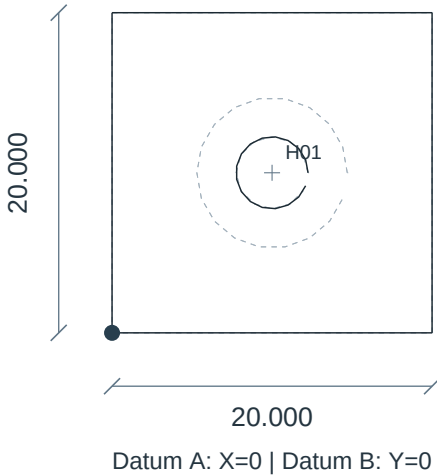
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 48.000, 47.000, 134.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_CHASSIS_PAD_0_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 31.625, 52.648, 119.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

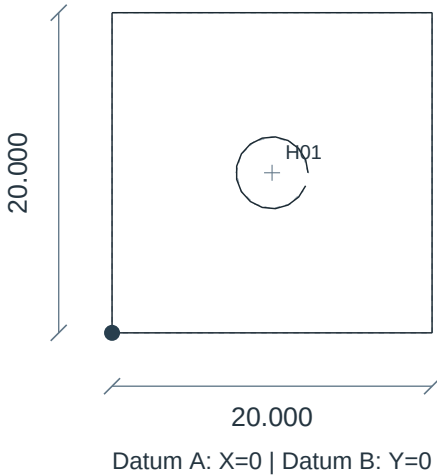
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_CHASSIS_PAD_0_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 31.625, 52.648, 124.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

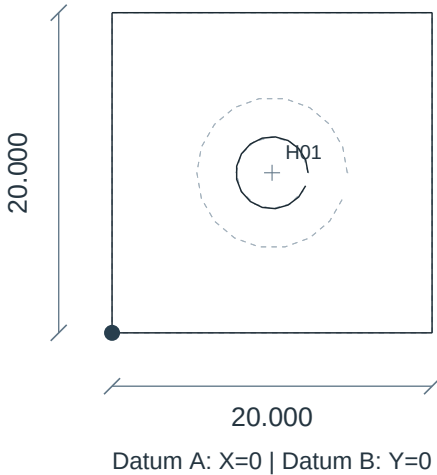
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_CHASSIS_PAD_1_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 31.625, 120.965, 119.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

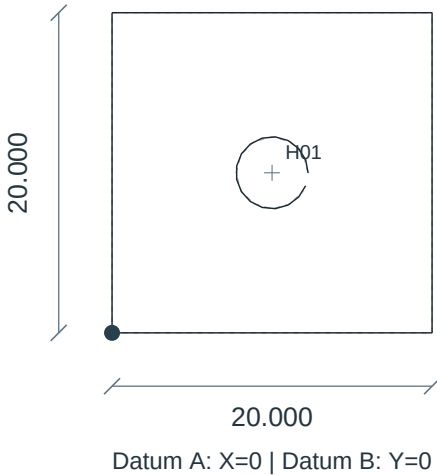
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_CHASSIS_PAD_1_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

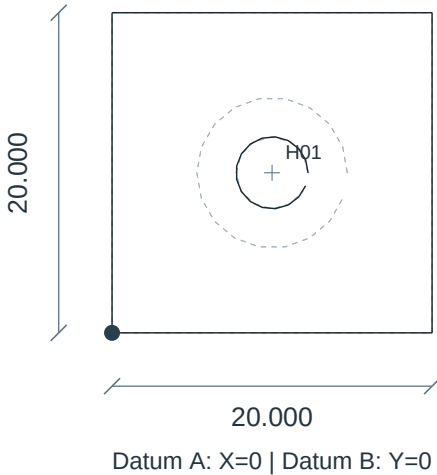
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 31.625, 120.965, 124.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_CHASSIS_PAD_2_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

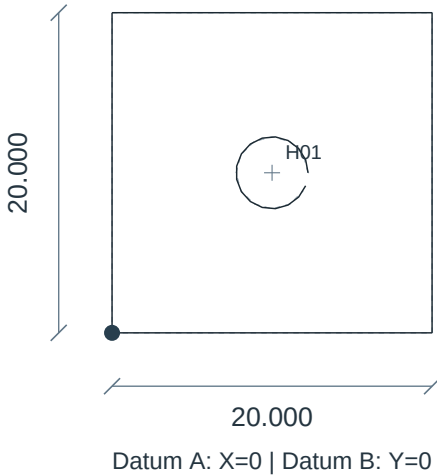
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 218.438, 55.433, 119.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_CHASSIS_PAD_2_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 218.438, 55.433, 124.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

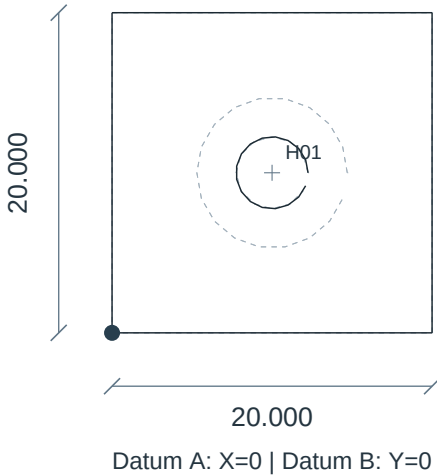
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_CHASSIS_PAD_3_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 218.438, 118.180, 119.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

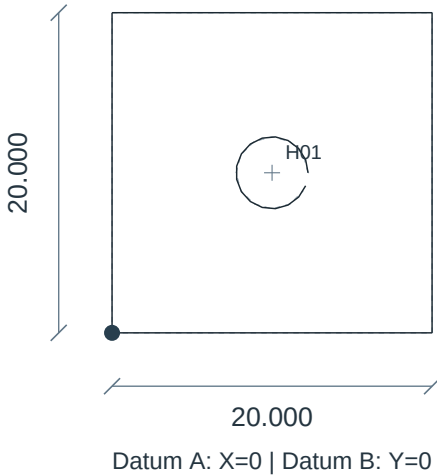
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_CHASSIS_PAD_3_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	10.000	DIA 4.500



SIDE EDGE PROJECTION - X / DEPTH



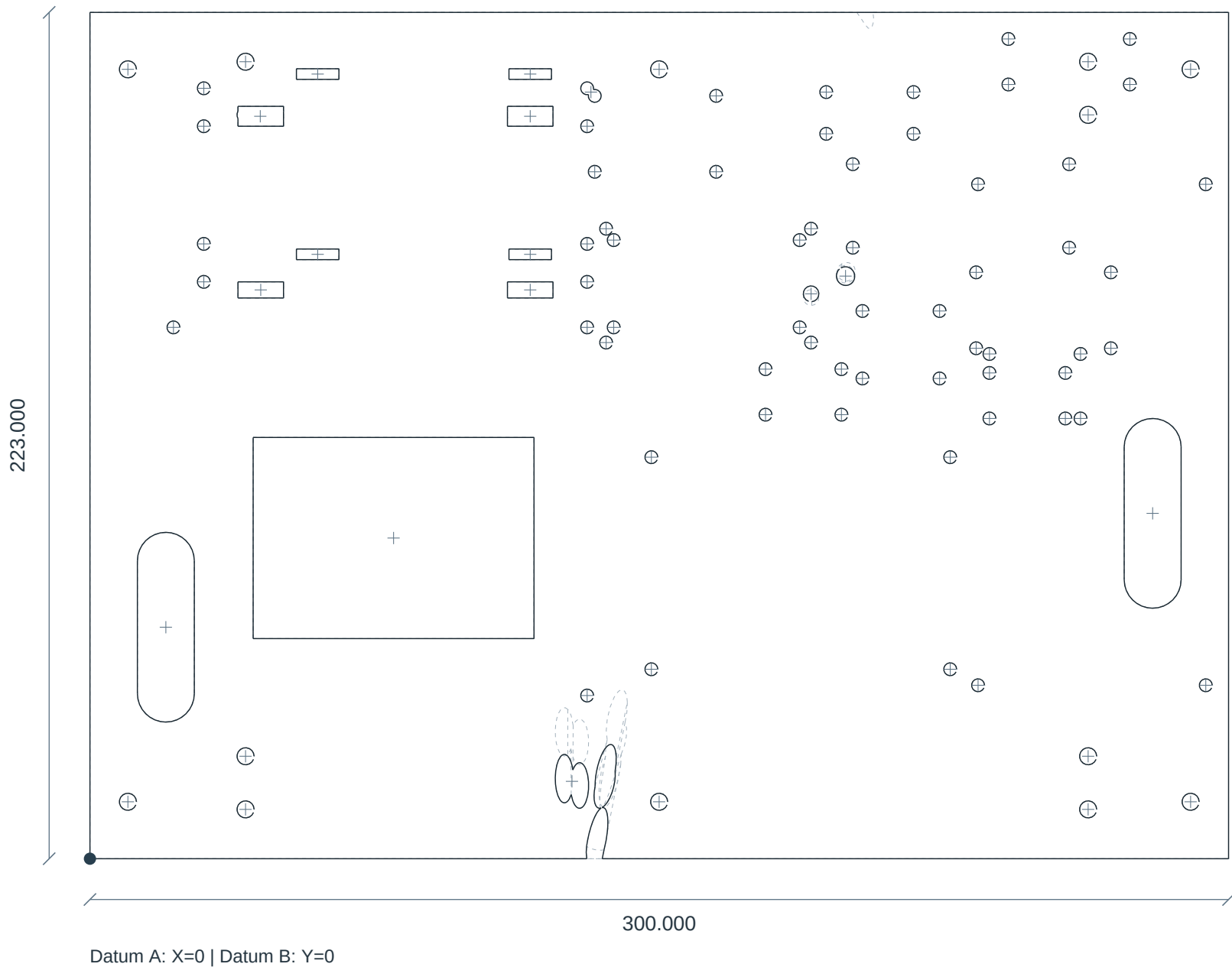
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 218.438, 118.180, 124.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_ELECTRONICS_PLATE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	41.000	13.000	DIA 4.500
H02	263.000	13.000	DIA 4.500
H03	10.000	15.000	DIA 4.500
H04	149.970	15.000	DIA 4.500
H05	290.000	15.000	DIA 4.500
H06	127.000	20.387	8.800 x 14.212
H07	41.000	27.000	DIA 4.500
H08	263.000	27.000	DIA 4.500
H09	131.000	43.000	DIA 3.400
H10	234.000	45.700	DIA 3.400
H11	294.000	45.700	DIA 3.400
H12	147.921	49.921	DIA 3.400
H13	226.661	49.921	DIA 3.400
H14	20.000	61.000	15.000 x 50.000
H15	80.000	84.500	74.000 x 53.000
H16	280.000	91.000	15.000 x 50.000
H17	147.921	105.801	DIA 3.400
H18	226.661	105.801	DIA 3.400

Full XY schedule follows; cross marks locate centers.

Assembly datum XYZ: -10.000, -15.000, 255.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H01	HOLE	41.000	13.000	4.500	4.500	4.500
H02	HOLE	263.000	13.000	4.500	4.500	4.500
H03	HOLE	10.000	15.000	4.500	4.500	4.500
H04	HOLE	149.970	15.000	4.500	4.500	4.500
H05	HOLE	290.000	15.000	4.500	4.500	4.500
H06	CUTOUT	127.000	20.387	-	8.800	14.212
H07	HOLE	41.000	27.000	4.500	4.500	4.500
H08	HOLE	263.000	27.000	4.500	4.500	4.500
H09	HOLE	131.000	43.000	3.400	3.400	3.400
H10	HOLE	234.000	45.700	3.400	3.400	3.400
H11	HOLE	294.000	45.700	3.400	3.400	3.400
H12	HOLE	147.921	49.921	3.400	3.400	3.400
H13	HOLE	226.661	49.921	3.400	3.400	3.400
H14	SLOT	20.000	61.000	-	15.000	50.000
H15	CUTOUT	80.000	84.500	-	74.000	53.000
H16	SLOT	280.000	91.000	-	15.000	50.000
H17	HOLE	147.921	105.801	3.400	3.400	3.400
H18	HOLE	226.661	105.801	3.400	3.400	3.400
H19	HOLE	237.000	116.000	3.400	3.400	3.400
H20	HOLE	257.000	116.000	3.400	3.400	3.400
H21	HOLE	261.000	116.000	3.400	3.400	3.400
H22	HOLE	178.000	117.000	3.400	3.400	3.400

Same datum as F-111. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H23	HOLE	198.000	117.000	3.400	3.400	3.400
H24	HOLE	203.540	126.540	3.400	3.400	3.400
H25	HOLE	223.860	126.540	3.400	3.400	3.400
H26	HOLE	237.000	128.000	3.400	3.400	3.400
H27	HOLE	257.000	128.000	3.400	3.400	3.400
H28	HOLE	178.000	129.000	3.400	3.400	3.400
H29	HOLE	198.000	129.000	3.400	3.400	3.400
H30	HOLE	237.000	133.000	3.400	3.400	3.400
H31	HOLE	261.000	133.000	3.400	3.400	3.400
H32	HOLE	233.500	134.500	3.400	3.400	3.400
H33	HOLE	269.000	134.500	3.400	3.400	3.400
H34	HOLE	136.000	136.000	3.400	3.400	3.400
H35	HOLE	190.000	136.000	3.400	3.400	3.400
H36	HOLE	22.000	140.000	3.400	3.400	3.400
H37	HOLE	131.000	140.000	3.400	3.400	3.400
H38	HOLE	138.000	140.000	3.400	3.400	3.400
H39	HOLE	187.000	140.000	3.400	3.400	3.400
H40	HOLE	203.540	144.320	3.400	3.400	3.400
H41	HOLE	223.860	144.320	3.400	3.400	3.400
H42	CUTOUT	190.000	148.843	-	4.100	4.178
H43	CUTOUT	45.000	149.875	-	12.000	4.250
H44	CUTOUT	116.000	149.875	-	12.000	4.250

Same datum as F-111. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H45	HOLE	30.000	152.000	3.400	3.400	3.400
H46	HOLE	131.000	152.000	3.400	3.400	3.400
H47	CUTOUT	199.083	153.516	-	4.826	4.924
H48	HOLE	233.500	154.500	3.400	3.400	3.400
H49	HOLE	269.000	154.500	3.400	3.400	3.400
H50	CUTOUT	60.000	159.250	-	11.200	2.800
H51	CUTOUT	116.000	159.250	-	11.200	2.800
H52	HOLE	201.000	161.000	3.400	3.400	3.400
H53	HOLE	258.000	161.000	3.400	3.400	3.400
H54	HOLE	30.000	162.000	3.400	3.400	3.400
H55	HOLE	131.000	162.000	3.400	3.400	3.400
H56	HOLE	138.000	163.000	3.400	3.400	3.400
H57	HOLE	187.000	163.000	3.400	3.400	3.400
H58	HOLE	136.000	166.000	3.400	3.400	3.400
H59	HOLE	190.000	166.000	3.400	3.400	3.400
H60	HOLE	234.000	177.700	3.400	3.400	3.400
H61	HOLE	294.000	177.700	3.400	3.400	3.400
H62	HOLE	133.000	181.000	3.400	3.400	3.400
H63	HOLE	165.000	181.000	3.400	3.400	3.400
H64	HOLE	201.000	183.000	3.400	3.400	3.400
H65	HOLE	258.000	183.000	3.400	3.400	3.400
H66	HOLE	194.000	191.000	3.400	3.400	3.400

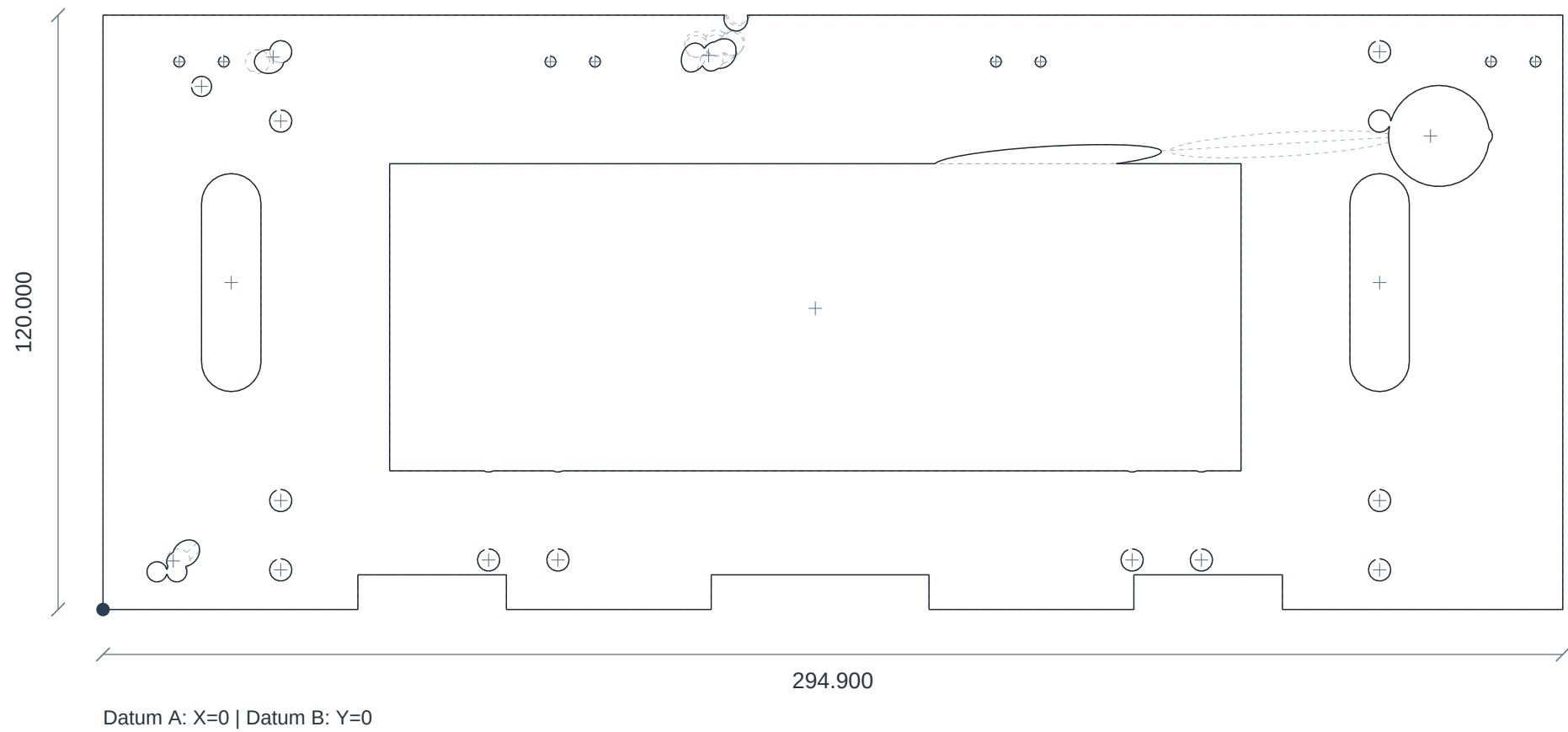
Same datum as F-111. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H67	HOLE	217.000	191.000	3.400	3.400	3.400
H68	HOLE	30.000	193.000	3.400	3.400	3.400
H69	HOLE	131.000	193.000	3.400	3.400	3.400
H70	CUTOUT	44.875	195.625	-	12.250	5.250
H71	CUTOUT	116.000	195.625	-	12.000	5.250
H72	HOLE	263.000	196.000	4.500	4.500	4.500
H73	HOLE	165.000	201.000	3.400	3.400	3.400
H74	CUTOUT	132.000	202.000	-	5.400	5.400
H75	HOLE	194.000	202.000	3.400	3.400	3.400
H76	HOLE	217.000	202.000	3.400	3.400	3.400
H77	HOLE	30.000	203.000	3.400	3.400	3.400
H78	HOLE	242.000	204.000	3.400	3.400	3.400
H79	HOLE	274.000	204.000	3.400	3.400	3.400
H80	CUTOUT	60.000	206.750	-	11.200	2.800
H81	CUTOUT	116.000	206.750	-	11.200	2.800
H82	HOLE	10.000	208.000	4.500	4.500	4.500
H83	HOLE	149.970	208.000	4.500	4.500	4.500
H84	HOLE	290.000	208.000	4.500	4.500	4.500
H85	HOLE	41.000	210.000	4.500	4.500	4.500
H86	HOLE	263.000	210.000	4.500	4.500	4.500
H87	HOLE	242.000	216.000	3.400	3.400	3.400
H88	HOLE	274.000	216.000	3.400	3.400	3.400

Same datum as F-111. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

CUSTOM_LEFT_PANEL

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

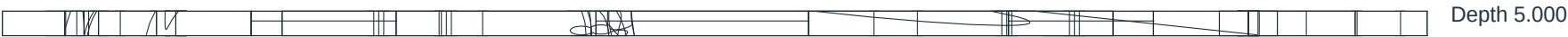


ID	X	Y	SIZE
H01	35.900	8.000	DIA 4.500
H02	257.900	8.000	DIA 4.500
H03	14.186	9.813	10.673 x 8.506
H04	77.900	10.000	DIA 4.500
H05	91.900	10.000	DIA 4.500
H06	207.900	10.000	DIA 4.500
H07	221.900	10.000	DIA 4.500
H08	35.900	22.000	DIA 4.500
H09	257.900	22.000	DIA 4.500
H10	143.900	60.807	172.000 x 66.114
H11	25.900	66.000	12.000 x 44.000
H12	257.900	66.000	12.000 x 44.000
H13	268.175	95.610	25.050 x 20.400
H14	35.900	98.610	DIA 4.500
H15	19.900	105.610	DIA 4.100
H16	15.400	110.610	DIA 2.200
H17	24.400	110.610	DIA 2.200
H18	90.400	110.610	DIA 2.200

Full XY schedule follows; cross marks locate centers.

Assembly datum XYZ: -4.900, -10.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H01	HOLE	35.900	8.000	4.500	4.500	4.500
H02	HOLE	257.900	8.000	4.500	4.500	4.500
H03	CUTOUT	14.186	9.813	-	10.673	8.506
H04	HOLE	77.900	10.000	4.500	4.500	4.500
H05	HOLE	91.900	10.000	4.500	4.500	4.500
H06	HOLE	207.900	10.000	4.500	4.500	4.500
H07	HOLE	221.900	10.000	4.500	4.500	4.500
H08	HOLE	35.900	22.000	4.500	4.500	4.500
H09	HOLE	257.900	22.000	4.500	4.500	4.500
H10	CUTOUT	143.900	60.807	-	172.000	66.114
H11	SLOT	25.900	66.000	-	12.000	44.000
H12	SLOT	257.900	66.000	-	12.000	44.000
H13	CUTOUT	268.175	95.610	-	25.050	20.400
H14	HOLE	35.900	98.610	4.500	4.500	4.500
H15	HOLE	19.900	105.610	4.100	4.100	4.100
H16	HOLE	15.400	110.610	2.200	2.200	2.200
H17	HOLE	24.400	110.610	2.200	2.200	2.200
H18	HOLE	90.400	110.610	2.200	2.200	2.200
H19	HOLE	99.400	110.610	2.200	2.200	2.200
H20	HOLE	180.400	110.610	2.200	2.200	2.200
H21	HOLE	189.400	110.610	2.200	2.200	2.200
H22	HOLE	280.400	110.610	2.200	2.200	2.200

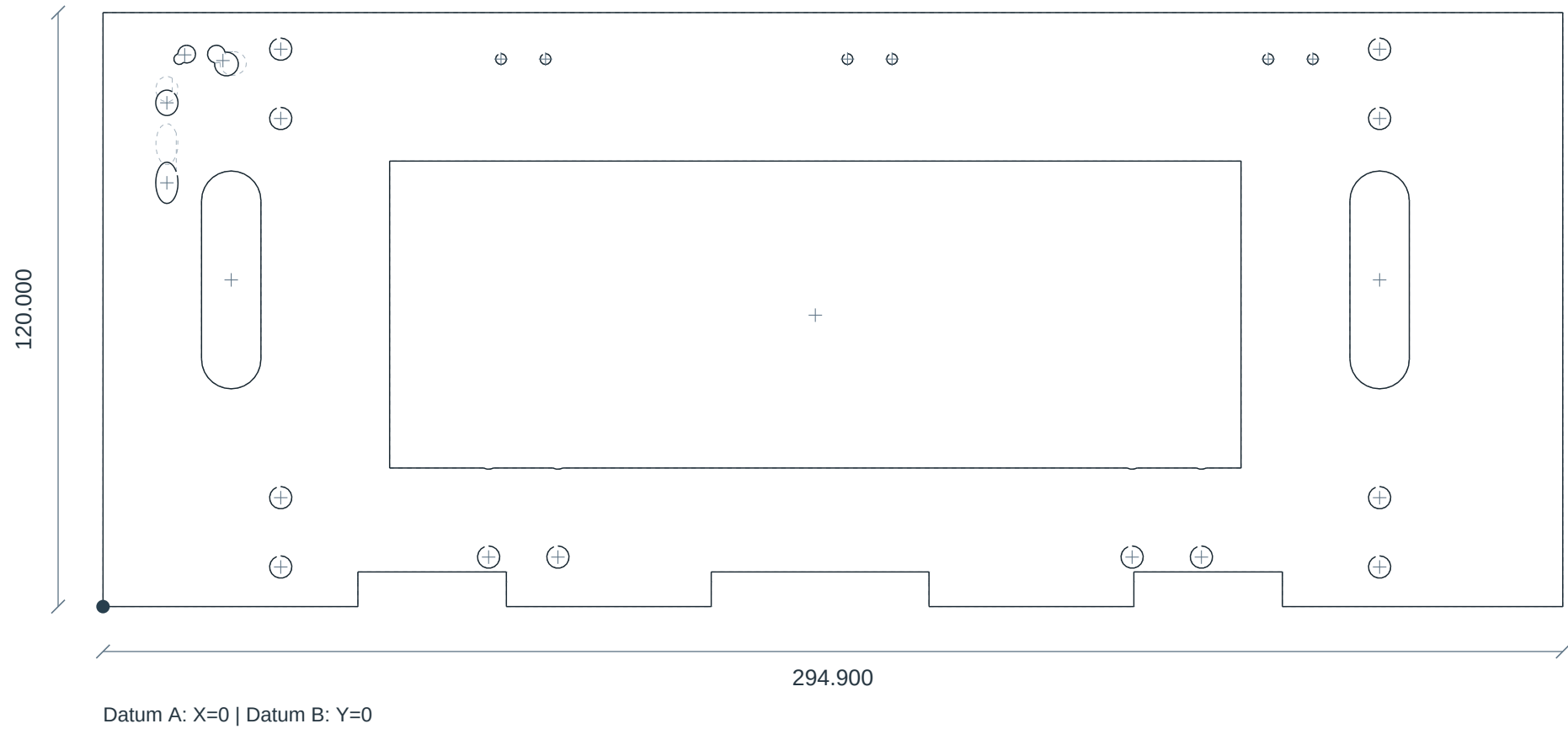
Same datum as F-112. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H23	HOLE	289.400	110.610	2.200	2.200	2.200
H24	CUTOUT	34.367	111.535	-	7.566	6.650
H25	CUTOUT	122.358	111.866	-	11.079	6.735
H26	HOLE	257.900	112.610	4.500	4.500	4.500

Same datum as F-112. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

CUSTOM_RIGHT_PANEL

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

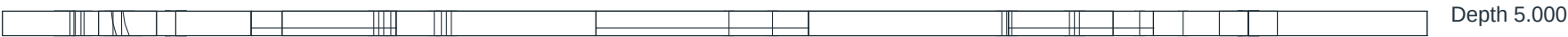


ID	X	Y	SIZE
H01	35.900	8.000	DIA 4.500
H02	257.900	8.000	DIA 4.500
H03	77.900	10.000	DIA 4.500
H04	91.900	10.000	DIA 4.500
H05	207.900	10.000	DIA 4.500
H06	221.900	10.000	DIA 4.500
H07	35.900	22.000	DIA 4.500
H08	257.900	22.000	DIA 4.500
H09	143.900	58.875	172.000 x 62.250
H10	25.900	66.000	12.000 x 44.000
H11	257.900	66.000	12.000 x 44.000
H12	12.900	85.610	4.500 x 8.348
H13	35.900	98.610	DIA 4.500
H14	257.900	98.610	DIA 4.500
H15	12.900	101.770	4.500 x 5.141
H16	24.221	110.312	6.243 x 6.197
H17	80.400	110.610	DIA 2.200
H18	89.400	110.610	DIA 2.200

Full XY schedule follows; cross marks locate centers.

Assembly datum XYZ: -4.900, 208.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H01	HOLE	35.900	8.000	4.500	4.500	4.500
H02	HOLE	257.900	8.000	4.500	4.500	4.500
H03	HOLE	77.900	10.000	4.500	4.500	4.500
H04	HOLE	91.900	10.000	4.500	4.500	4.500
H05	HOLE	207.900	10.000	4.500	4.500	4.500
H06	HOLE	221.900	10.000	4.500	4.500	4.500
H07	HOLE	35.900	22.000	4.500	4.500	4.500
H08	HOLE	257.900	22.000	4.500	4.500	4.500
H09	CUTOUT	143.900	58.875	-	172.000	62.250
H10	SLOT	25.900	66.000	-	12.000	44.000
H11	SLOT	257.900	66.000	-	12.000	44.000
H12	CUTOUT	12.900	85.610	-	4.500	8.348
H13	HOLE	35.900	98.610	4.500	4.500	4.500
H14	HOLE	257.900	98.610	4.500	4.500	4.500
H15	CUTOUT	12.900	101.770	-	4.500	5.141
H16	CUTOUT	24.221	110.312	-	6.243	6.197
H17	HOLE	80.400	110.610	2.200	2.200	2.200
H18	HOLE	89.400	110.610	2.200	2.200	2.200
H19	HOLE	150.400	110.610	2.200	2.200	2.200
H20	HOLE	159.400	110.610	2.200	2.200	2.200
H21	HOLE	235.400	110.610	2.200	2.200	2.200
H22	HOLE	244.400	110.610	2.200	2.200	2.200

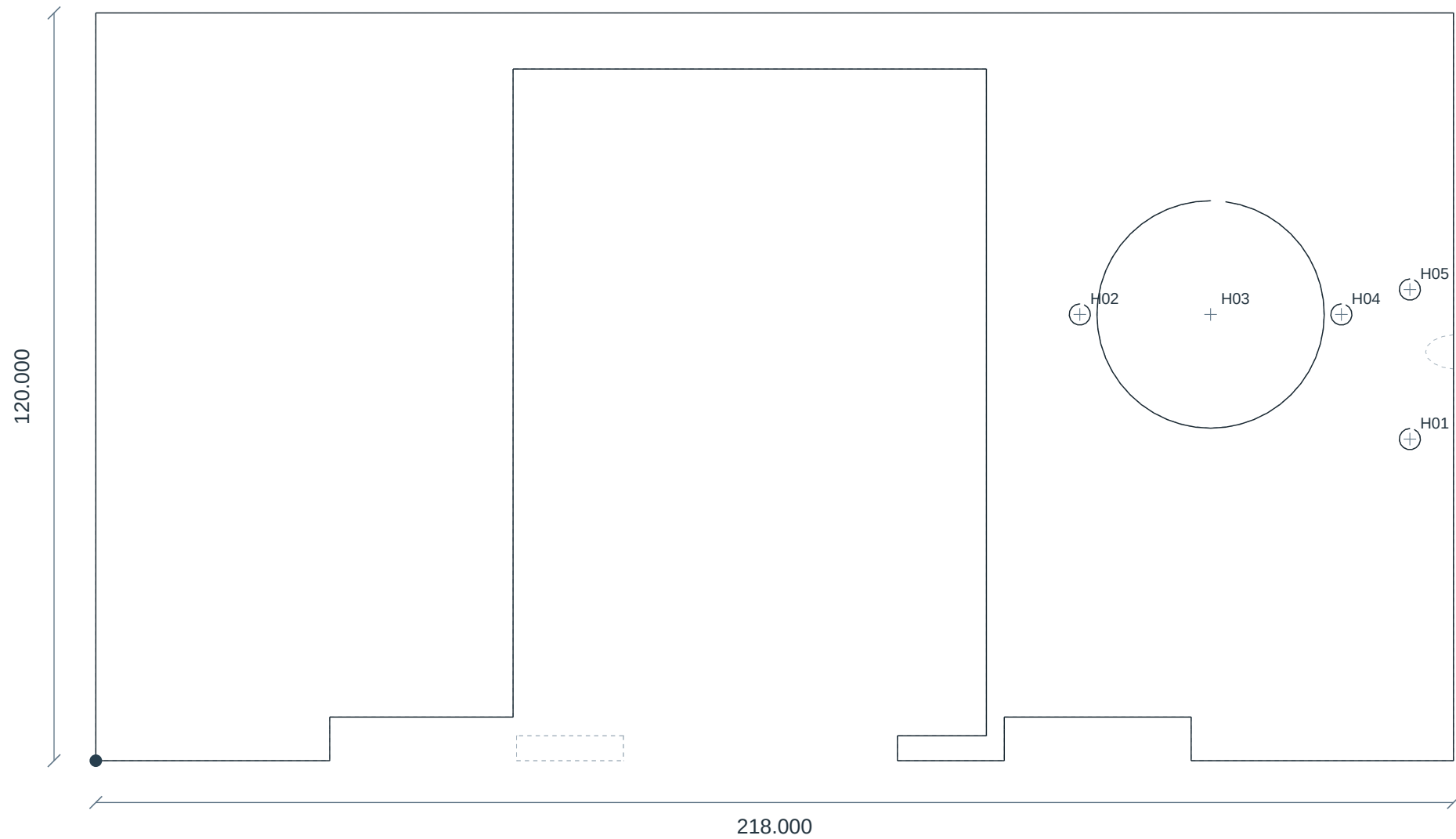
Same datum as F-113. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H23	CUTOUT	16.500	111.460	-	4.400	3.900
H24	HOLE	35.900	112.610	4.500	4.500	4.500
H25	HOLE	257.900	112.610	4.500	4.500	4.500

Same datum as F-113. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

CUSTOM_REAR_PANEL

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	211.000	51.610	DIA 3.400
H02	158.000	71.610	DIA 3.400
H03	179.000	71.610	DIA 36.500
H04	200.000	71.610	DIA 3.400
H05	211.000	75.610	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH

Assembly datum XYZ: -10.000, -10.000, 134.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, 1.0)

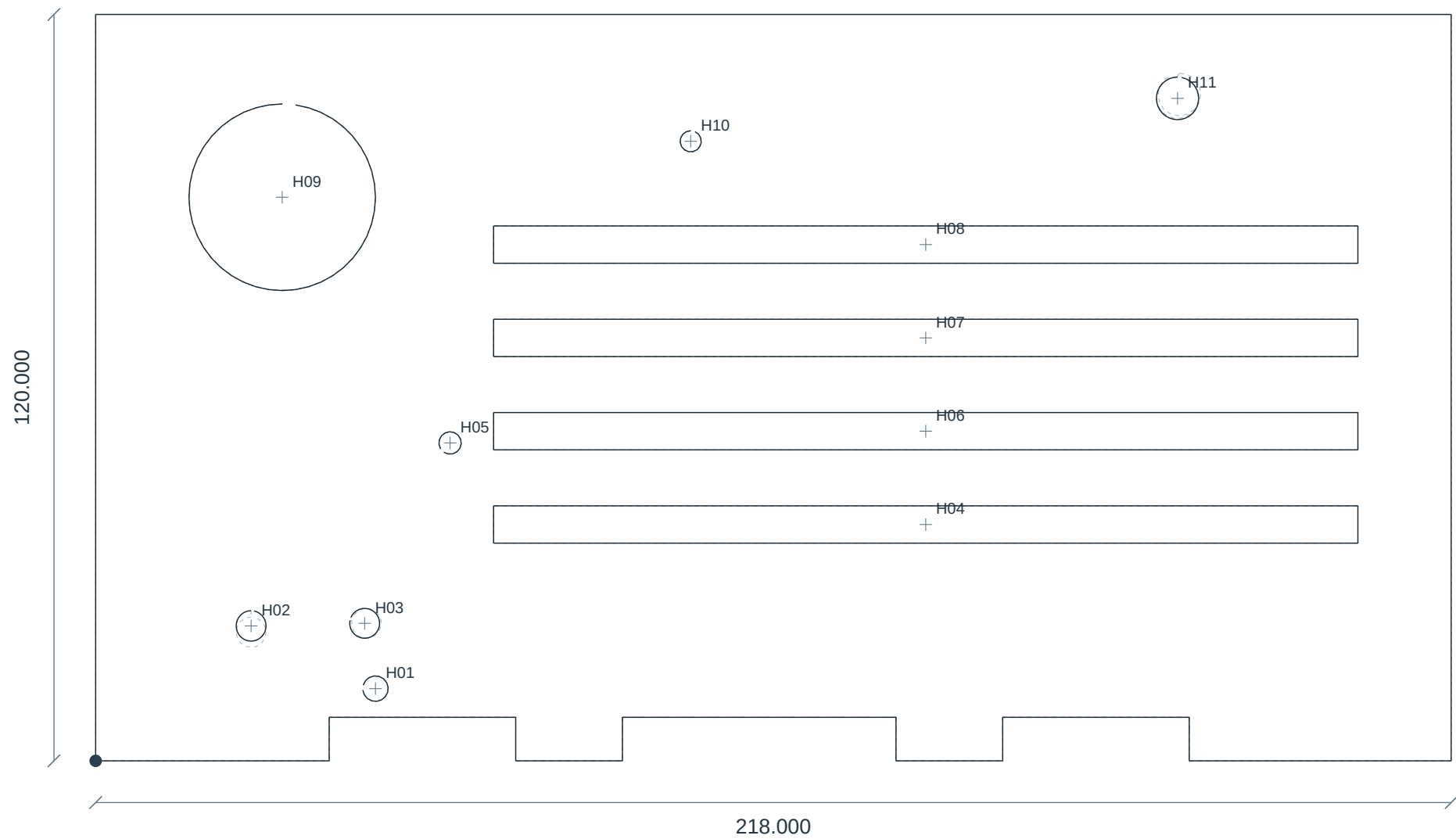


VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_PANEL

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	45.000	11.610	DIA 4.100
H02	25.000	21.697	4.800 x 4.898
H03	43.264	22.110	4.808 x 4.800
H04	133.500	38.000	139.000 x 6.000
H05	57.000	51.110	DIA 3.600
H06	133.500	53.000	139.000 x 6.000
H07	133.500	68.000	139.000 x 6.000
H08	133.500	83.000	139.000 x 6.000
H09	30.000	90.610	DIA 30.000
H10	95.700	99.610	DIA 3.400
H11	174.000	106.511	6.800 x 6.852

SIDE EDGE PROJECTION - X / DEPTH

Assembly datum XYZ: 290.000, -10.000, 134.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, 1.0)

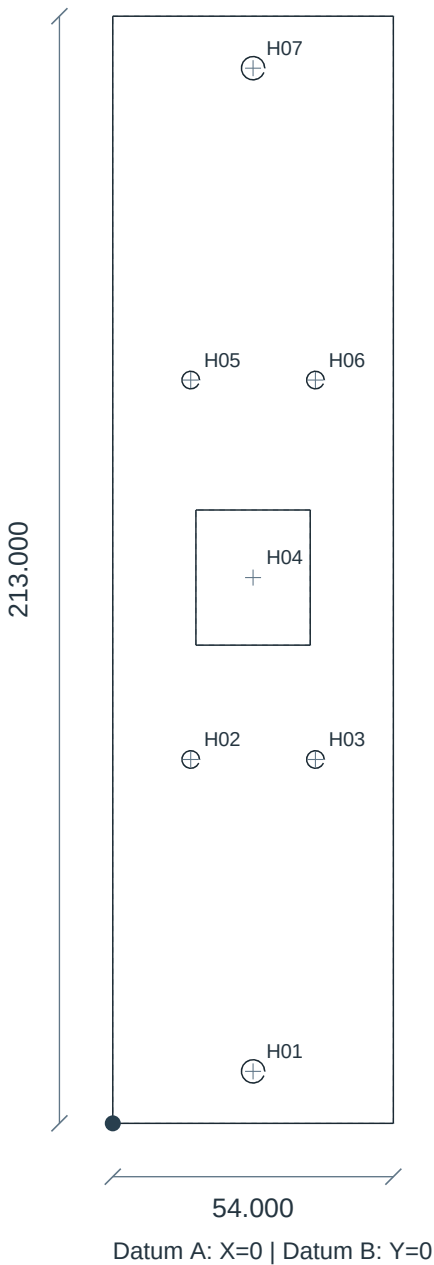


VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_LIDAR_BRIDGE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	27.000	10.000	DIA 4.500
H02	15.000	70.000	DIA 3.400
H03	39.000	70.000	DIA 3.400
H04	27.000	105.000	22.000 x 26.000
H05	15.000	143.000	DIA 3.400
H06	39.000	143.000	DIA 3.400
H07	27.000	203.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



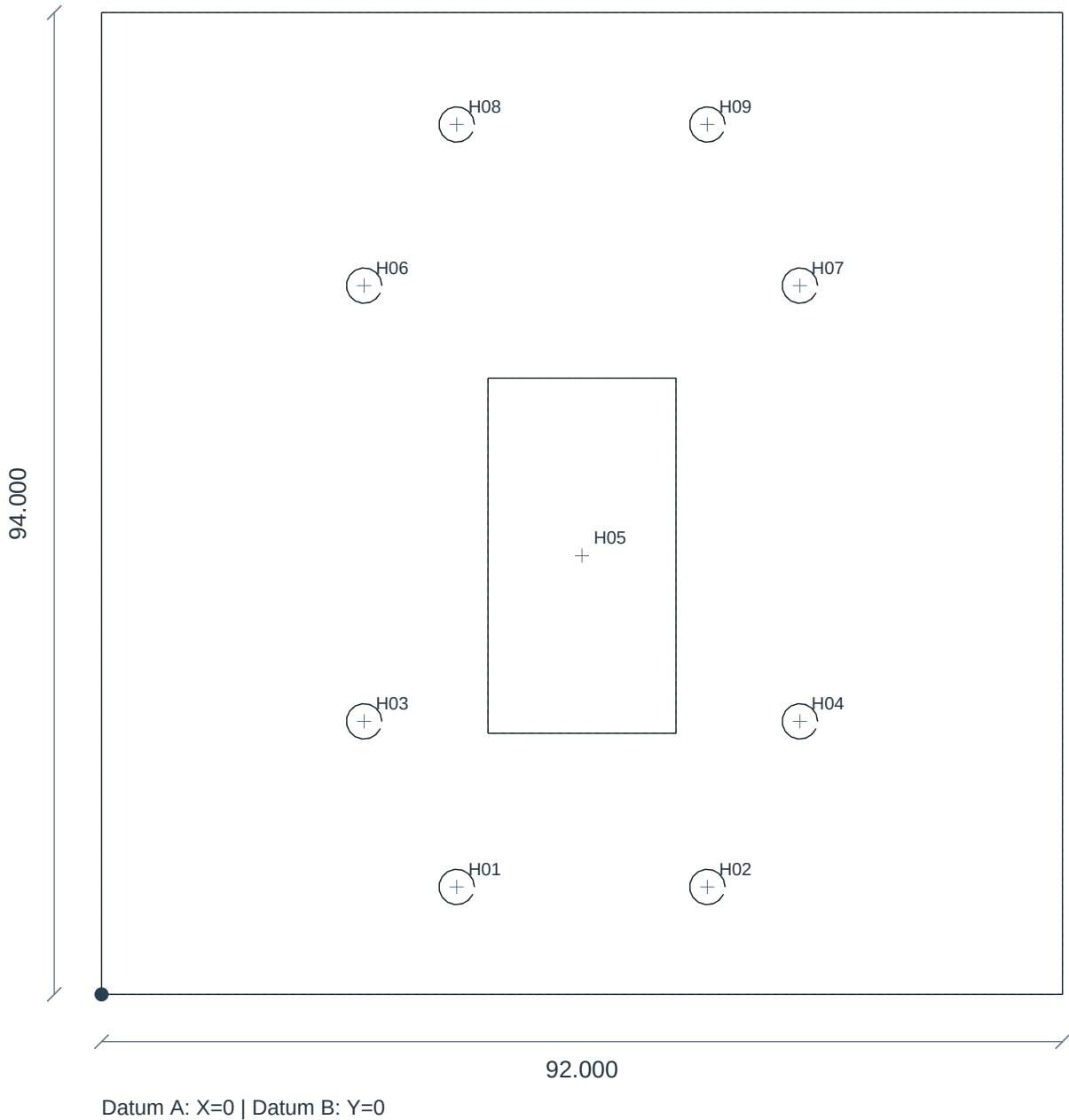
Assembly datum XYZ: 112.970, -10.000, 354.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_LIDAR_PLATE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

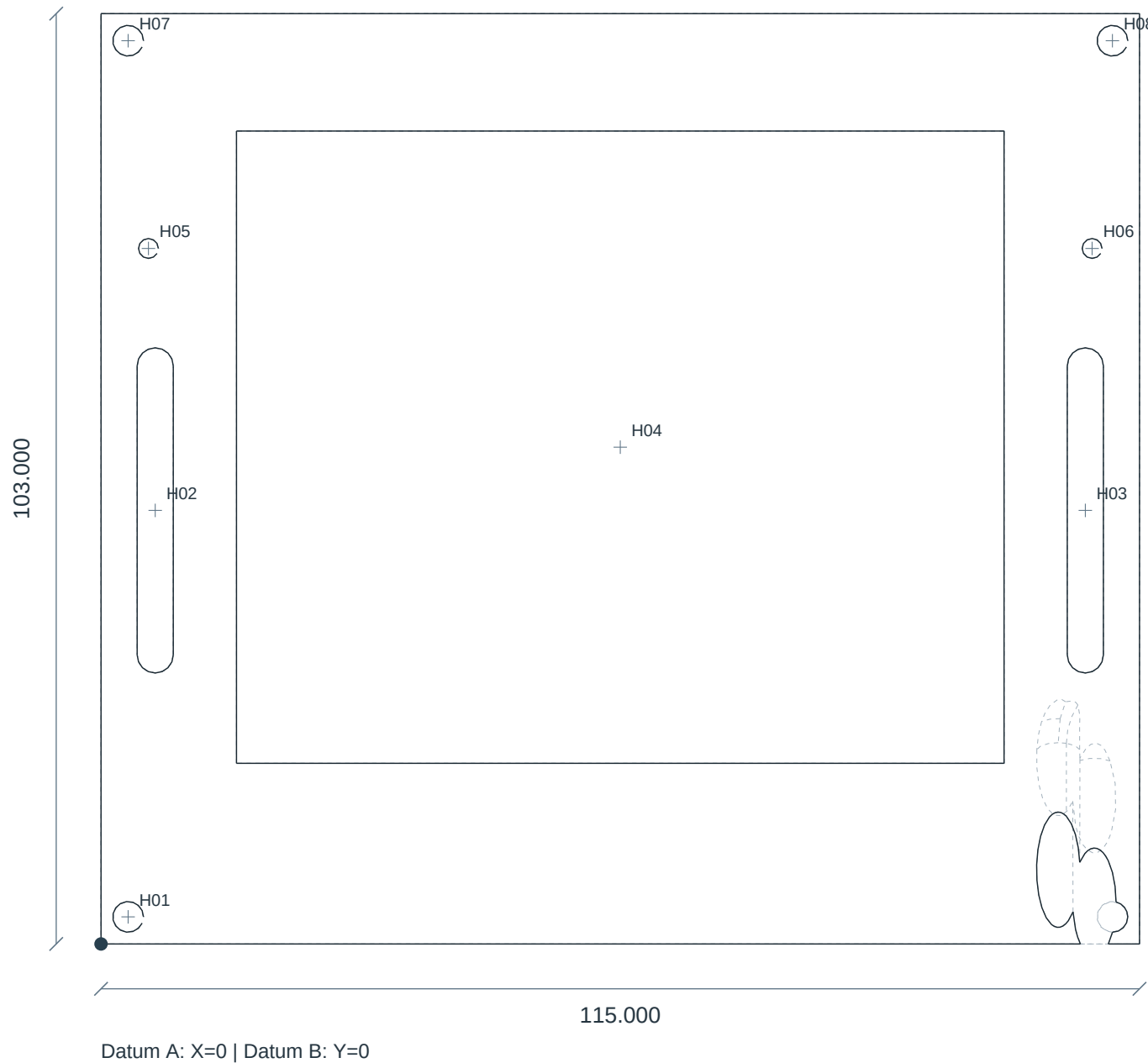
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	X	Y	SIZE
H01	34.000	10.285	DIA 3.400
H02	58.000	10.285	DIA 3.400
H03	25.140	26.140	DIA 3.400
H04	66.860	26.140	DIA 3.400
H05	46.000	42.000	18.000 x 34.000
H06	25.140	67.860	DIA 3.400
H07	66.860	67.860	DIA 3.400
H08	34.000	83.285	DIA 3.400
H09	58.000	83.285	DIA 3.400

Assembly datum XYZ: 93.970, 49.715, 399.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_JETSON_CRADLE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



Depth 5.000

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

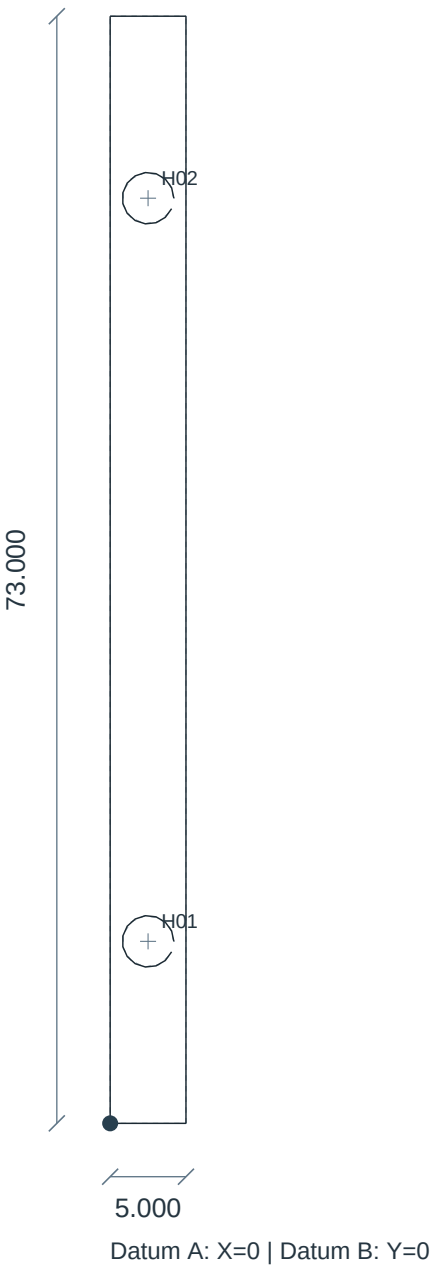
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

ID	X	Y	SIZE
H01	3.000	3.000	DIA 3.400
H02	6.000	48.000	4.000 x 36.000
H03	109.000	48.000	4.000 x 36.000
H04	57.500	55.000	85.000 x 70.000
H05	5.250	77.000	DIA 2.200
H06	109.750	77.000	DIA 2.200
H07	3.000	100.000	DIA 3.400
H08	112.000	100.000	DIA 3.400

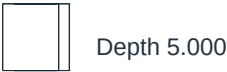
Assembly datum XYZ: 9.000, 25.000, 266.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_BATTERY_STOP_FRONT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

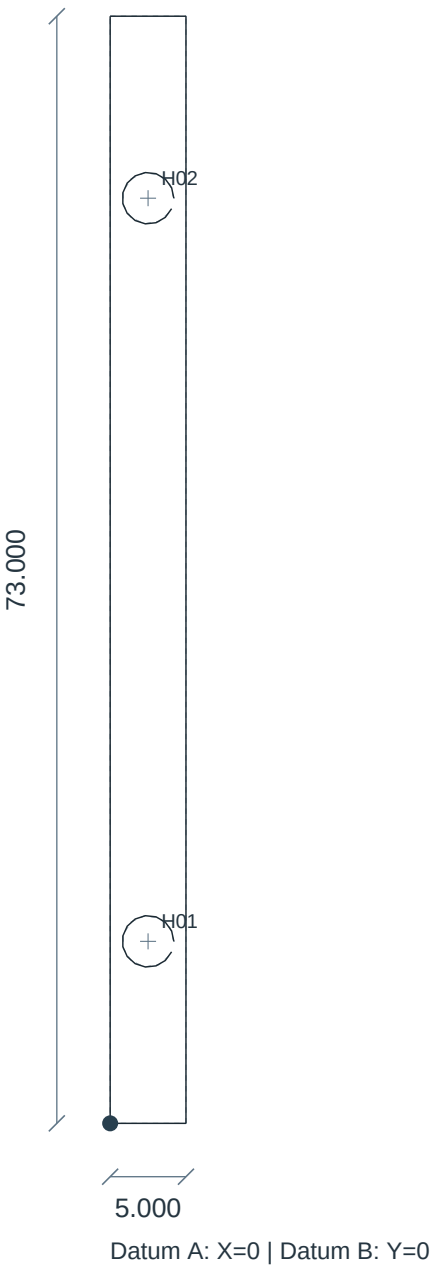
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

ID	X	Y	SIZE
H01	2.500	12.000	DIA 3.400
H02	2.500	61.000	DIA 3.400

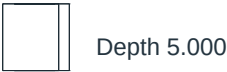
Assembly datum XYZ: 213.000, 60.000, 139.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_BATTERY_STOP_REAR

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



Depth 5.000

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

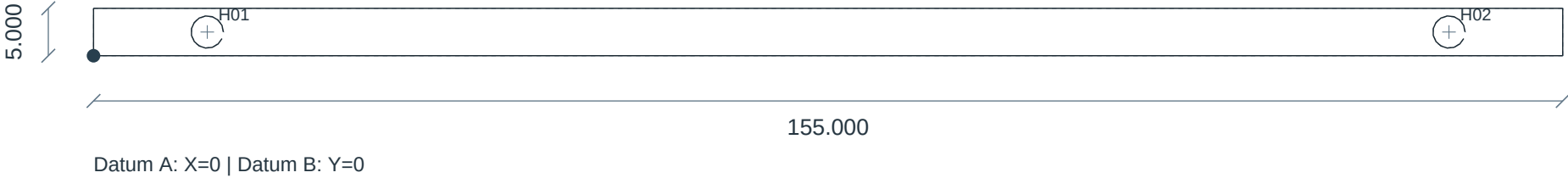
Assembly datum XYZ: 53.000, 60.000, 139.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

ID	X	Y	SIZE
H01	2.500	12.000	DIA 3.400
H02	2.500	61.000	DIA 3.400

CUSTOM_BATTERY_STOP_LEFT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	12.000	2.500	DIA 3.400
H02	143.000	2.500	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

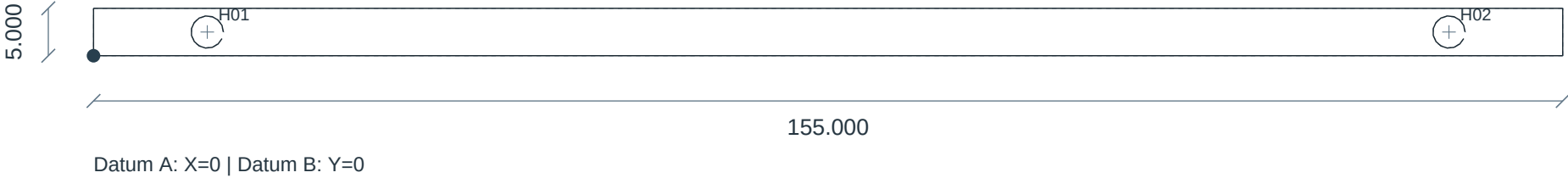
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 58.000, 57.000, 139.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_BATTERY_STOP_RIGHT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	12.000	2.500	DIA 3.400
H02	143.000	2.500	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



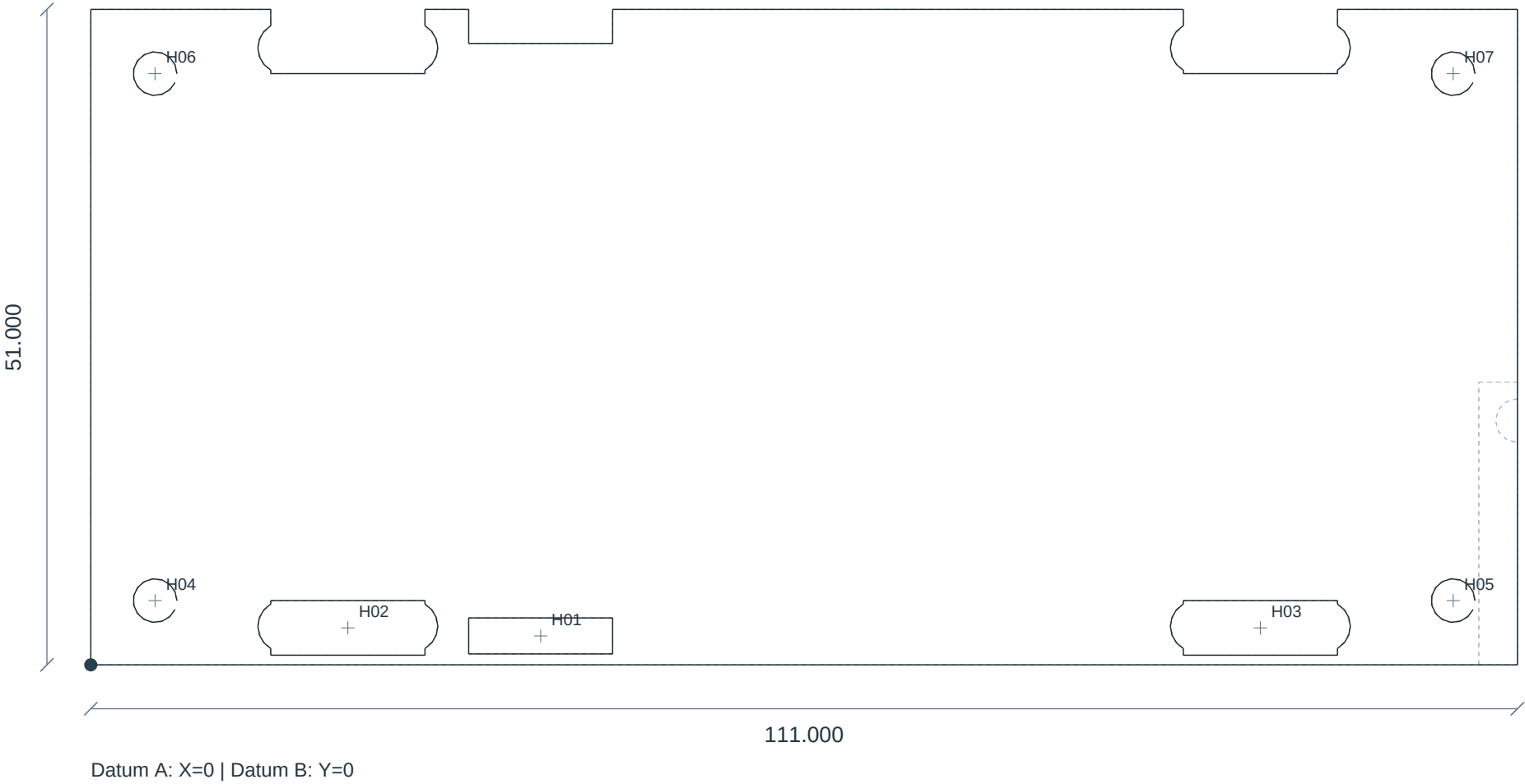
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 58.000, 131.000, 139.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_HUB_TRAY

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	35.000	2.250	11.200 x 2.800
H02	20.000	2.875	14.000 x 4.250
H03	91.000	2.875	14.000 x 4.250
H04	5.000	5.000	DIA 3.400
H05	106.000	5.000	DIA 3.400
H06	5.000	46.000	DIA 3.400
H07	106.000	46.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



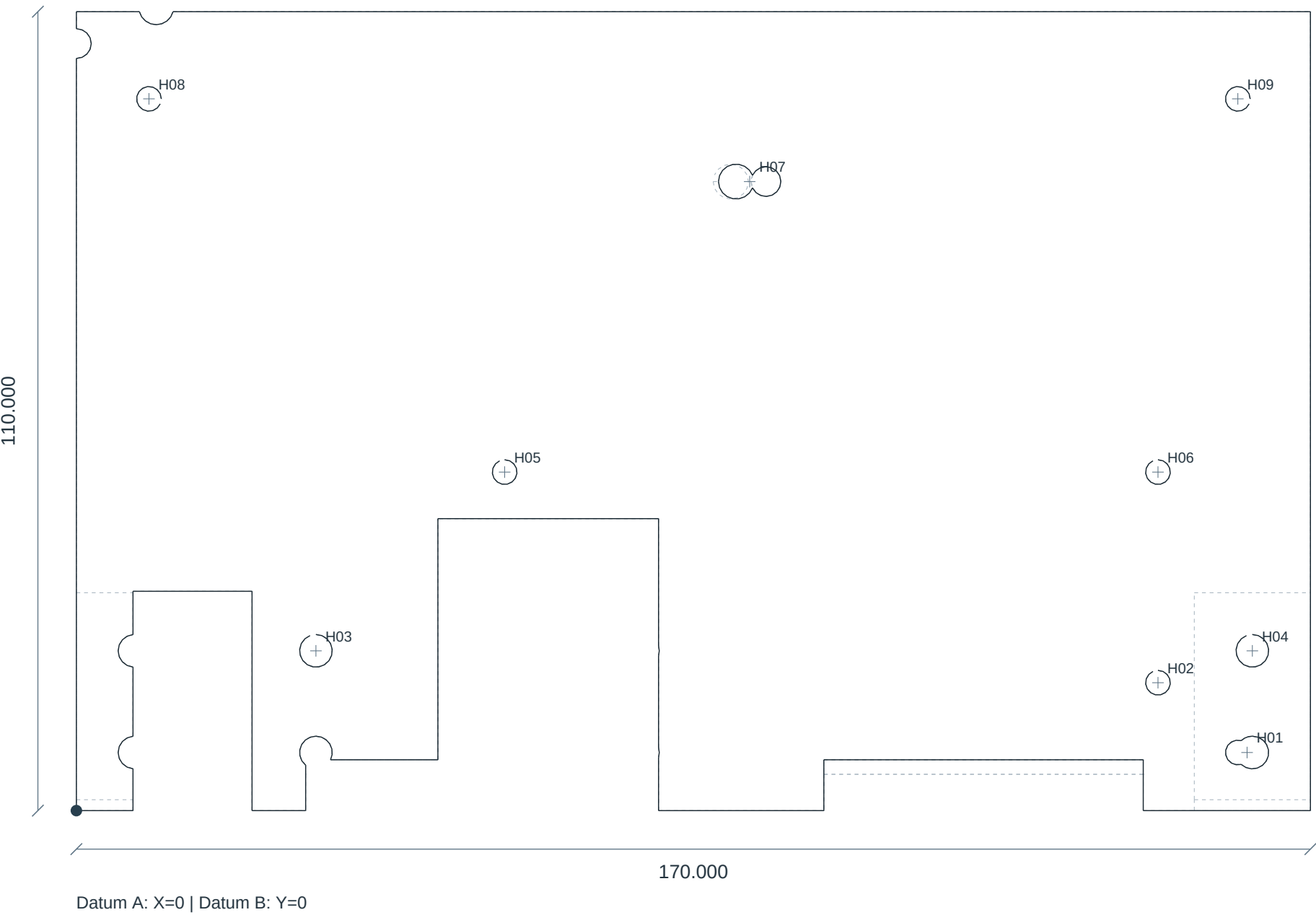
Assembly datum XYZ: 15.000, 142.000, 261.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_POWER_BACKPLANE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	161.275	8.000	5.950 x 4.500
H02	149.000	17.610	DIA 3.400
H03	33.000	22.000	DIA 4.500
H04	162.000	22.000	DIA 4.500
H05	59.000	46.610	DIA 3.400
H06	149.000	46.610	DIA 3.400
H07	92.753	86.610	8.593 x 4.800
H08	10.000	98.000	DIA 3.400
H09	160.000	98.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



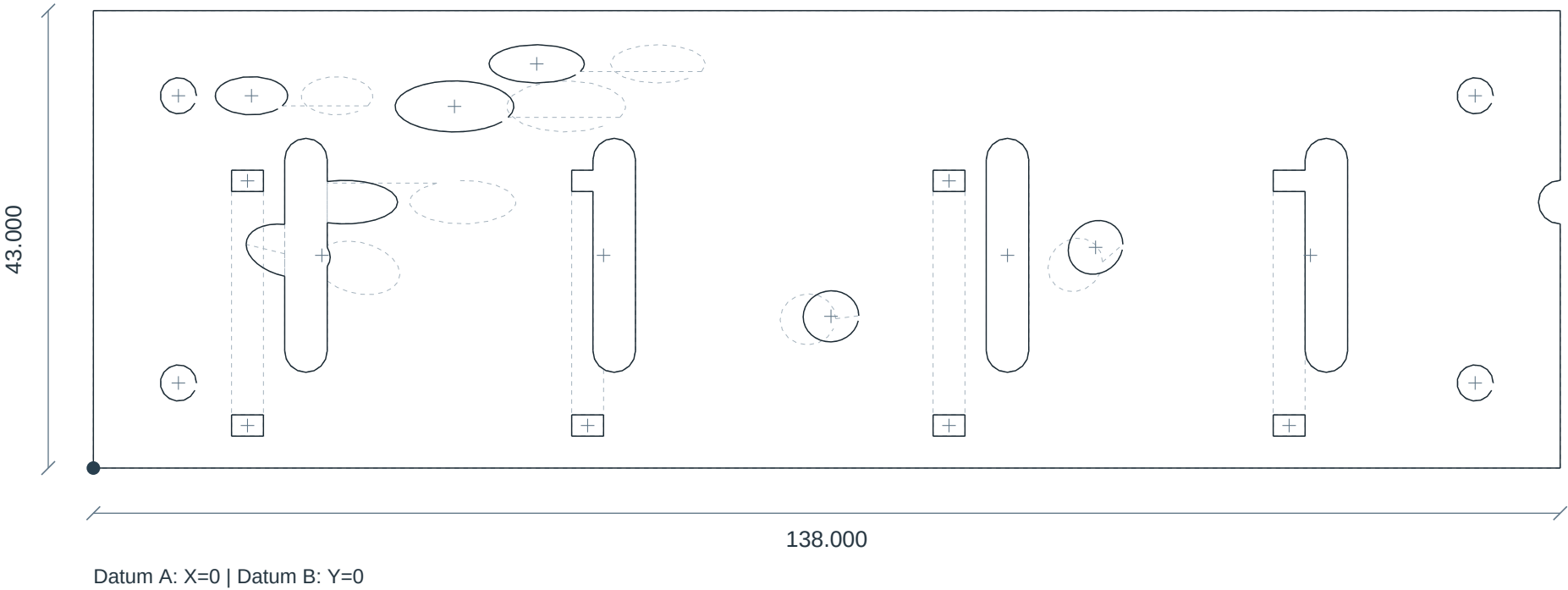
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 15.000, 10.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

CUSTOM_FUSE_SHELF

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	14.500	4.000	3.000 x 2.000
H02	46.500	4.000	3.000 x 2.000
H03	80.500	4.000	3.000 x 2.000
H04	112.500	4.000	3.000 x 2.000
H05	8.000	8.000	DIA 3.400
H06	130.000	8.000	DIA 3.400
H07	69.395	14.265	5.233 x 4.807
H08	21.498	20.000	14.255 x 22.000
H09	48.000	20.000	6.000 x 22.000
H10	86.000	20.000	4.000 x 22.000
H11	114.500	20.000	7.000 x 22.000
H12	94.287	20.748	5.139 x 5.055
H13	14.500	27.000	3.000 x 2.000
H14	80.500	27.000	3.000 x 2.000
H15	33.974	34.000	11.176 x 4.800
H16	8.000	35.000	DIA 3.400
H17	14.877	35.000	6.820 x 3.600
H18	130.000	35.000	DIA 3.400

Full XY schedule follows; cross marks locate centers.

Assembly datum XYZ: 47.000, 10.000, 210.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

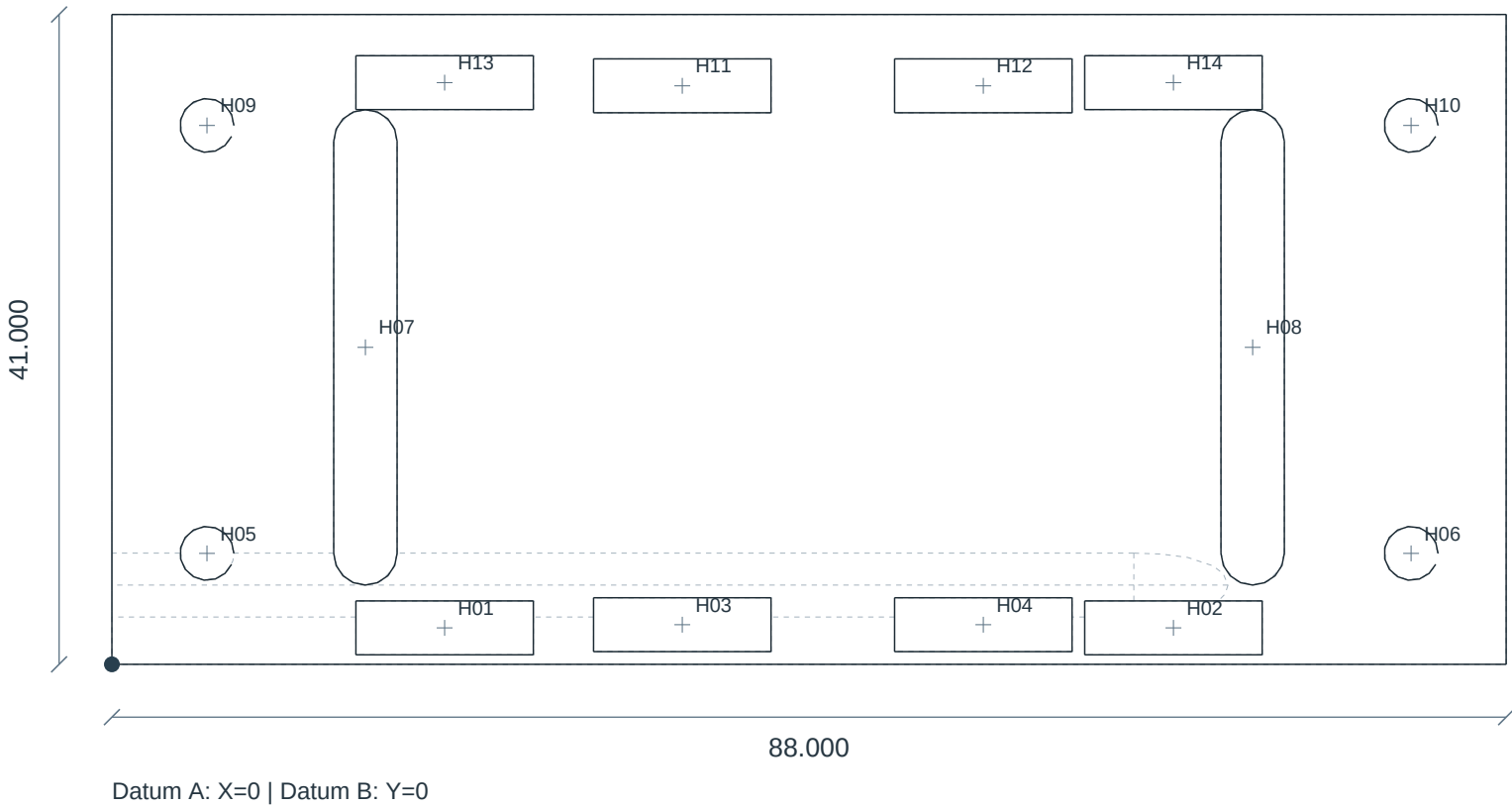
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

ID	FEATURE	X from A	Y from B	DIA	WIDTH	HEIGHT
H01	CUTOUT	14.500	4.000	-	3.000	2.000
H02	CUTOUT	46.500	4.000	-	3.000	2.000
H03	CUTOUT	80.500	4.000	-	3.000	2.000
H04	CUTOUT	112.500	4.000	-	3.000	2.000
H05	HOLE	8.000	8.000	3.400	3.400	3.400
H06	HOLE	130.000	8.000	3.400	3.400	3.400
H07	CUTOUT	69.395	14.265	-	5.233	4.807
H08	CUTOUT	21.498	20.000	-	14.255	22.000
H09	SLOT	48.000	20.000	-	6.000	22.000
H10	SLOT	86.000	20.000	-	4.000	22.000
H11	SLOT	114.500	20.000	-	7.000	22.000
H12	CUTOUT	94.287	20.748	-	5.139	5.055
H13	CUTOUT	14.500	27.000	-	3.000	2.000
H14	CUTOUT	80.500	27.000	-	3.000	2.000
H15	CUTOUT	33.974	34.000	-	11.176	4.800
H16	HOLE	8.000	35.000	3.400	3.400	3.400
H17	CUTOUT	14.877	35.000	-	6.820	3.600
H18	HOLE	130.000	35.000	3.400	3.400	3.400
H19	CUTOUT	41.711	38.000	-	8.966	3.600

Same datum as F-125. Dimensions are CAD nominal. Noncircular width/height are feature bounding dimensions; the STEP / supplied DXF controls the actual contour.

CUSTOM_BOOST_BASE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	21.000	2.300	11.200 x 3.400
H02	67.000	2.300	11.200 x 3.400
H03	36.000	2.500	11.200 x 3.400
H04	55.000	2.500	11.200 x 3.400
H05	6.000	7.000	DIA 3.400
H06	82.000	7.000	DIA 3.400
H07	16.000	20.000	4.000 x 30.000
H08	72.000	20.000	4.000 x 30.000
H09	6.000	34.000	DIA 3.400
H10	82.000	34.000	DIA 3.400
H11	36.000	36.500	11.200 x 3.400
H12	55.000	36.500	11.200 x 3.400
H13	21.000	36.700	11.200 x 3.400
H14	67.000	36.700	11.200 x 3.400

SIDE EDGE PROJECTION - X / DEPTH



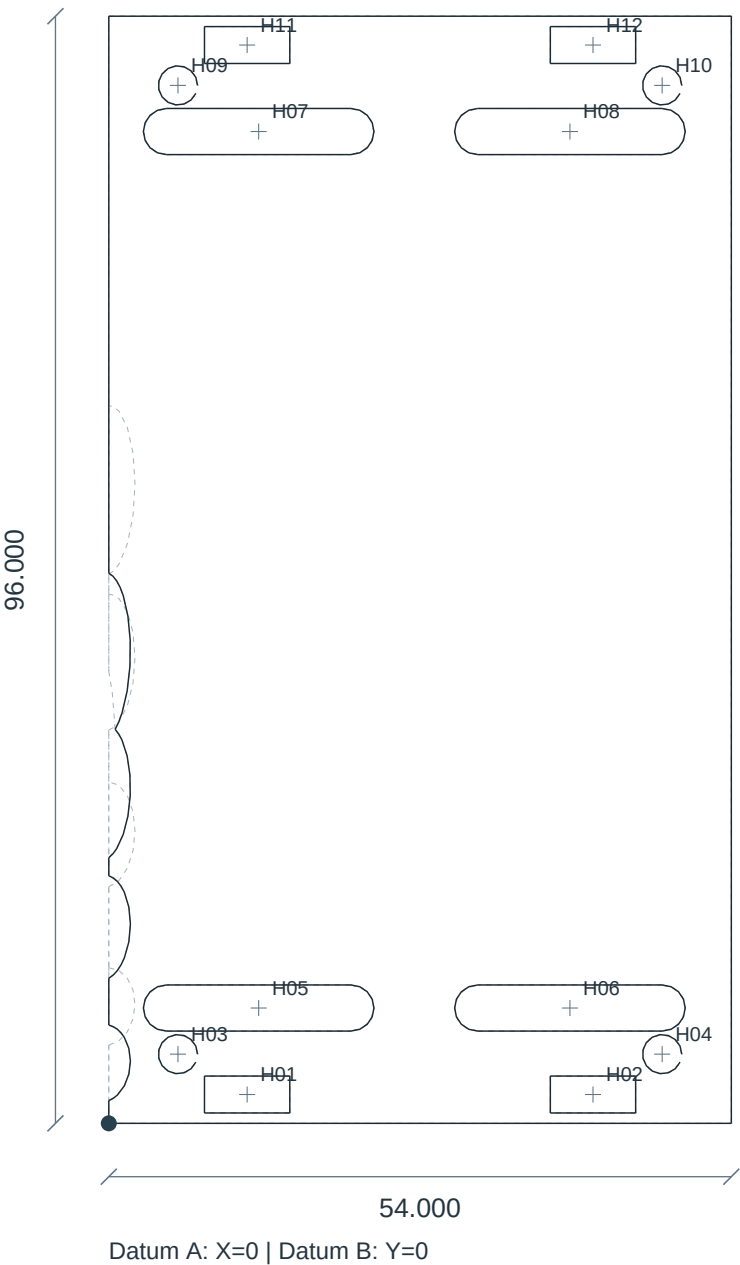
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 130.000, 150.000, 146.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_FUSEBLOCK_TRAY

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

ID	X	Y	SIZE
H01	12.000	2.500	7.400 x 3.200
H02	42.000	2.500	7.400 x 3.200
H03	6.000	6.000	DIA 3.400
H04	48.000	6.000	DIA 3.400
H05	13.000	10.000	20.000 x 4.000
H06	40.000	10.000	20.000 x 4.000
H07	13.000	86.000	20.000 x 4.000
H08	40.000	86.000	20.000 x 4.000
H09	6.000	90.000	DIA 3.400
H10	48.000	90.000	DIA 3.400
H11	12.000	93.500	7.400 x 3.200
H12	42.000	93.500	7.400 x 3.200

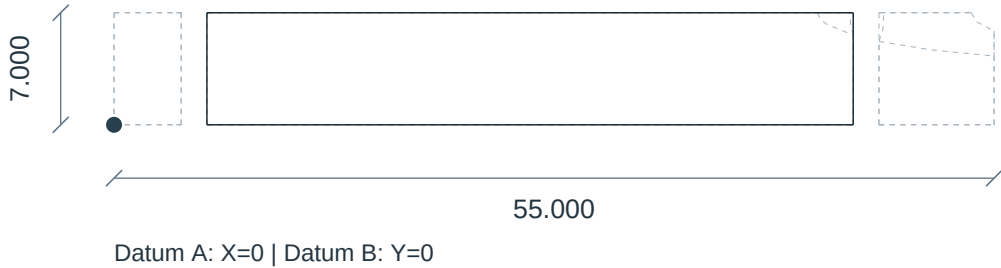
Assembly datum XYZ: 222.000, 48.000, 175.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_JETSON_BASE_RETAINER_LEFT

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 1.500 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

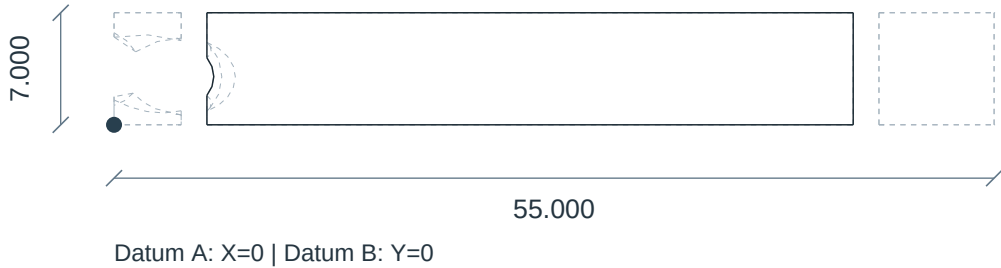
Assembly datum XYZ: 13.500, 55.000, 271.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

CUSTOM_JETSON_BASE_RETAINER_RIGHT

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 1.500 mm

ID	X	Y	SIZE
----	---	---	------

No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



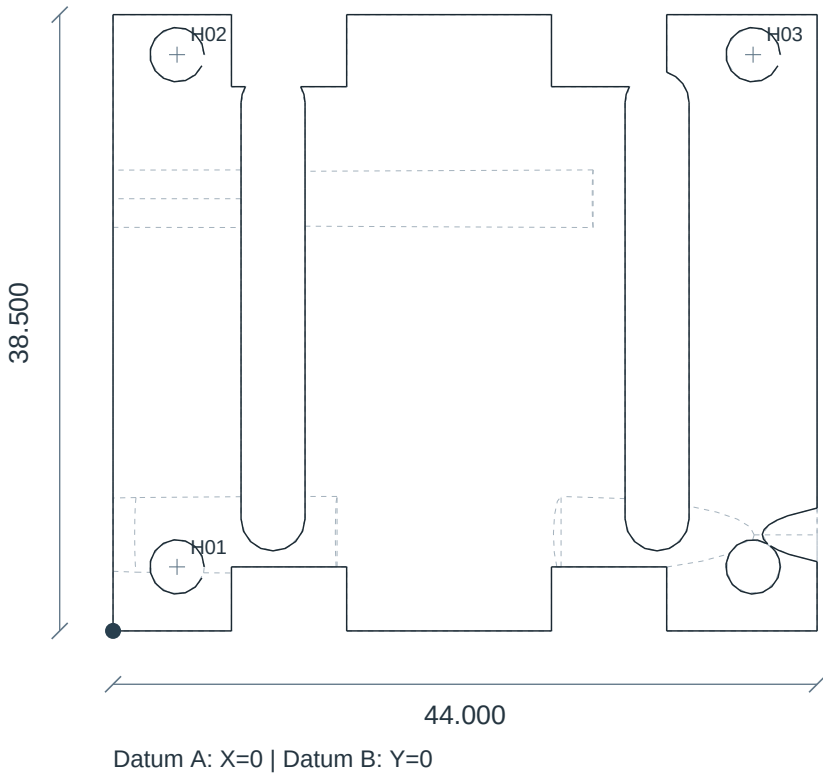
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 119.500, 55.000, 278.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

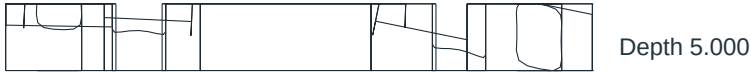
CUSTOM_RELAY_TRAY

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	4.000	4.000	DIA 3.400
H02	4.000	36.000	DIA 3.400
H03	40.000	36.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



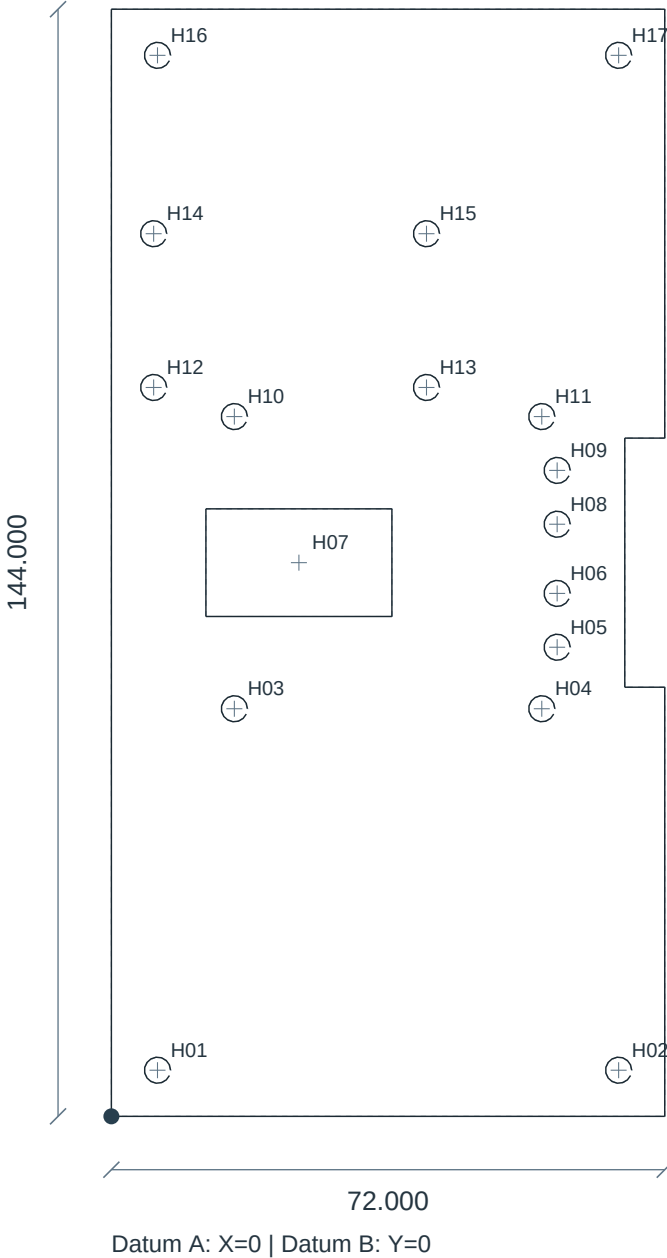
Assembly datum XYZ: 223.000, 9.000, 159.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_CAMERA_BASE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	6.000	6.000	DIA 3.400
H02	66.000	6.000	DIA 3.400
H03	16.000	53.015	DIA 3.400
H04	56.000	53.015	DIA 3.400
H05	58.000	61.015	DIA 3.400
H06	58.000	68.015	DIA 3.400
H07	24.400	72.015	24.200 x 14.000
H08	58.000	77.015	DIA 3.400
H09	58.000	84.015	DIA 3.400
H10	16.000	91.015	DIA 3.400
H11	56.000	91.015	DIA 3.400
H12	5.500	94.800	DIA 3.400
H13	41.000	94.800	DIA 3.400
H14	5.500	114.800	DIA 3.400
H15	41.000	114.800	DIA 3.400
H16	6.000	138.000	DIA 3.400
H17	66.000	138.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



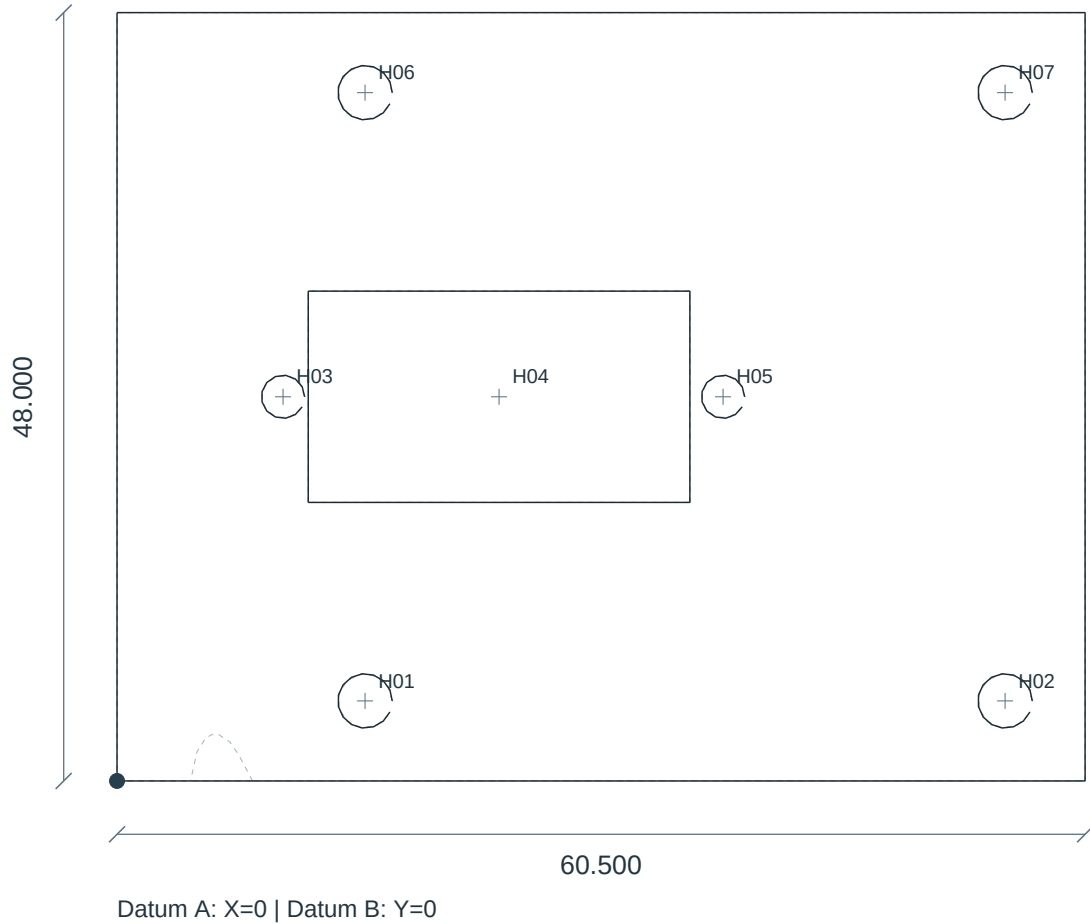
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 218.000, 24.700, 260.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_PAN_SERVO_MOUNT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	15.500	5.000	DIA 3.400
H02	55.500	5.000	DIA 3.400
H03	10.375	24.000	DIA 2.700
H04	23.875	24.000	23.850 x 13.200
H05	37.875	24.000	DIA 2.700
H06	15.500	43.000	DIA 3.400
H07	55.500	43.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



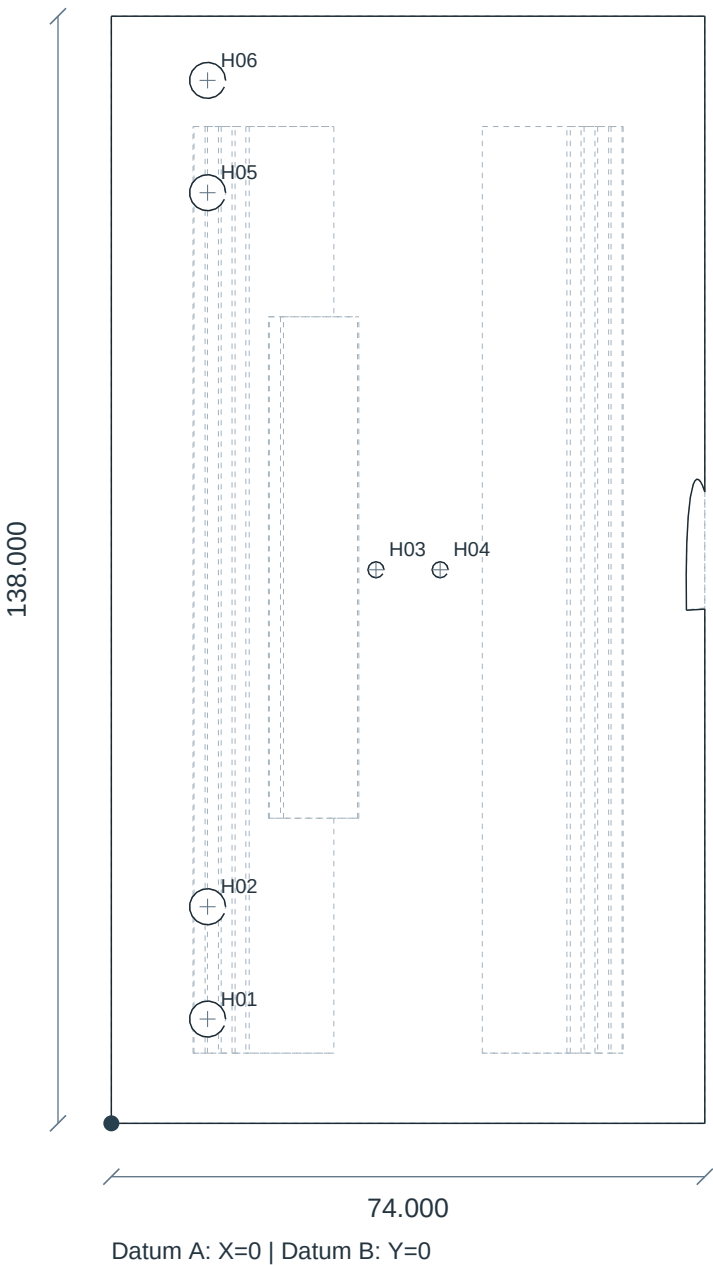
Assembly datum XYZ: 218.500, 72.715, 277.100
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_PAN_YOKE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	12.000	13.000	DIA 4.500
H02	12.000	27.000	DIA 4.500
H03	33.000	69.000	DIA 2.000
H04	41.000	69.000	DIA 2.000
H05	12.000	116.000	DIA 4.500
H06	12.000	130.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



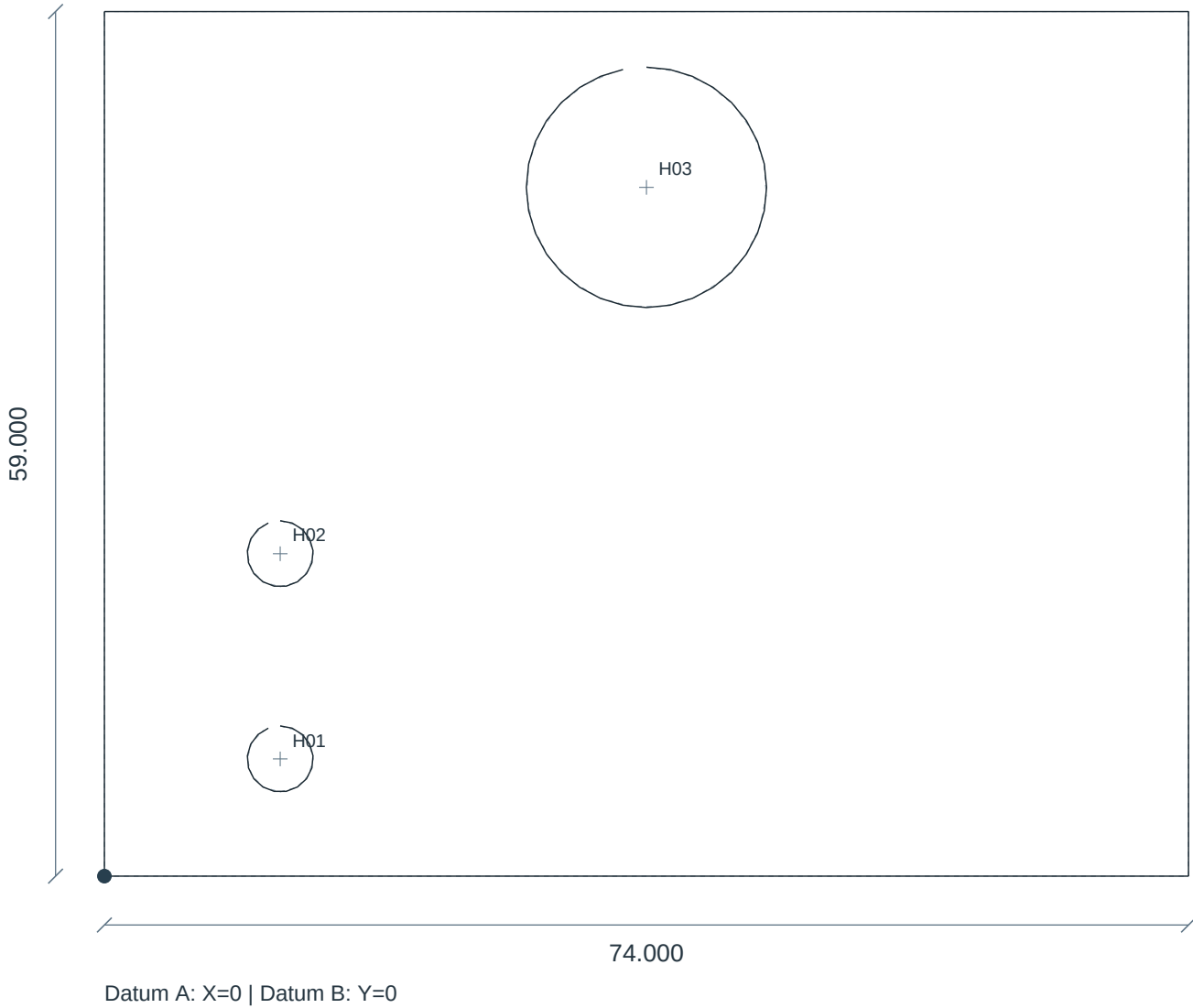
Assembly datum XYZ: 211.000, 27.715, 298.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_PAN_ARM_LEFT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	12.000	8.000	DIA 4.500
H02	12.000	22.000	DIA 4.500
H03	37.000	47.000	DIA 16.400

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 211.000, 32.715, 303.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

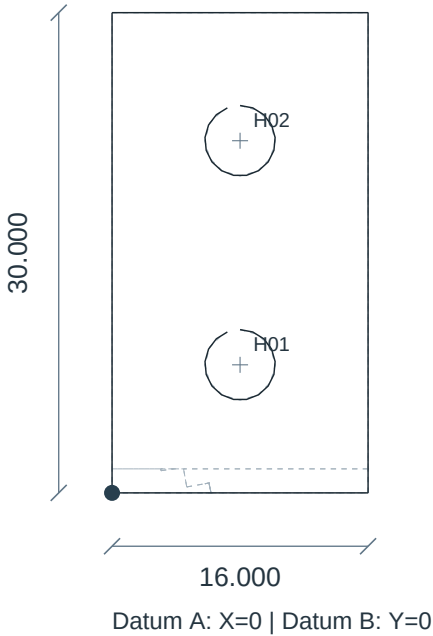
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

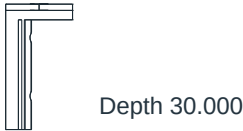
APPROX_CAM_L_LEFT

Qty1 | Modified stock metal L bracket | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	8.000	8.000	DIA 4.400
H02	8.000	22.000	DIA 4.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 215.000, 62.715, 303.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_PAN_ARM_RIGHT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

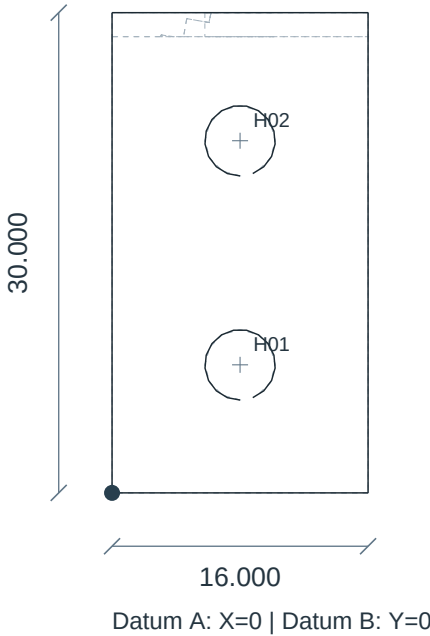
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 211.000, 160.715, 303.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

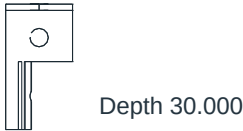
APPROX_CAM_L_RIGHT

Qty1 | Modified stock metal L bracket | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	8.000	8.000	DIA 4.400
H02	8.000	22.000	DIA 4.400



SIDE EDGE PROJECTION - X / DEPTH



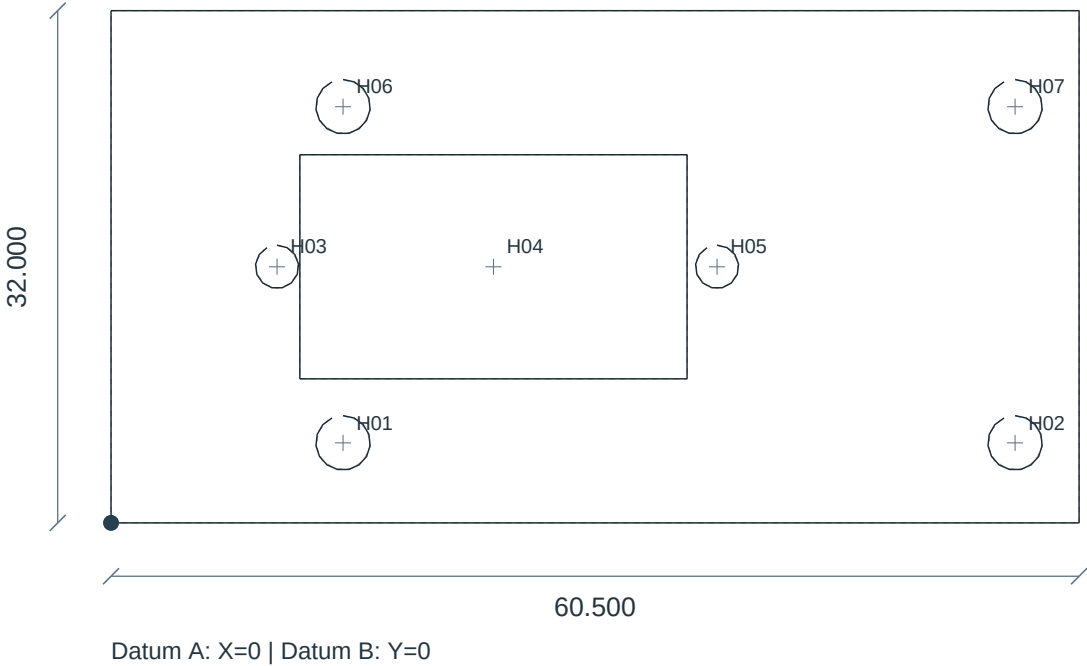
Assembly datum XYZ: 215.000, 135.715, 333.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_TILT_SERVO_MOUNT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	14.500	5.000	DIA 3.400
H02	56.500	5.000	DIA 3.400
H03	10.375	16.000	DIA 2.700
H04	23.900	16.000	24.200 x 14.000
H05	37.875	16.000	DIA 2.700
H06	14.500	26.000	DIA 3.400
H07	56.500	26.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



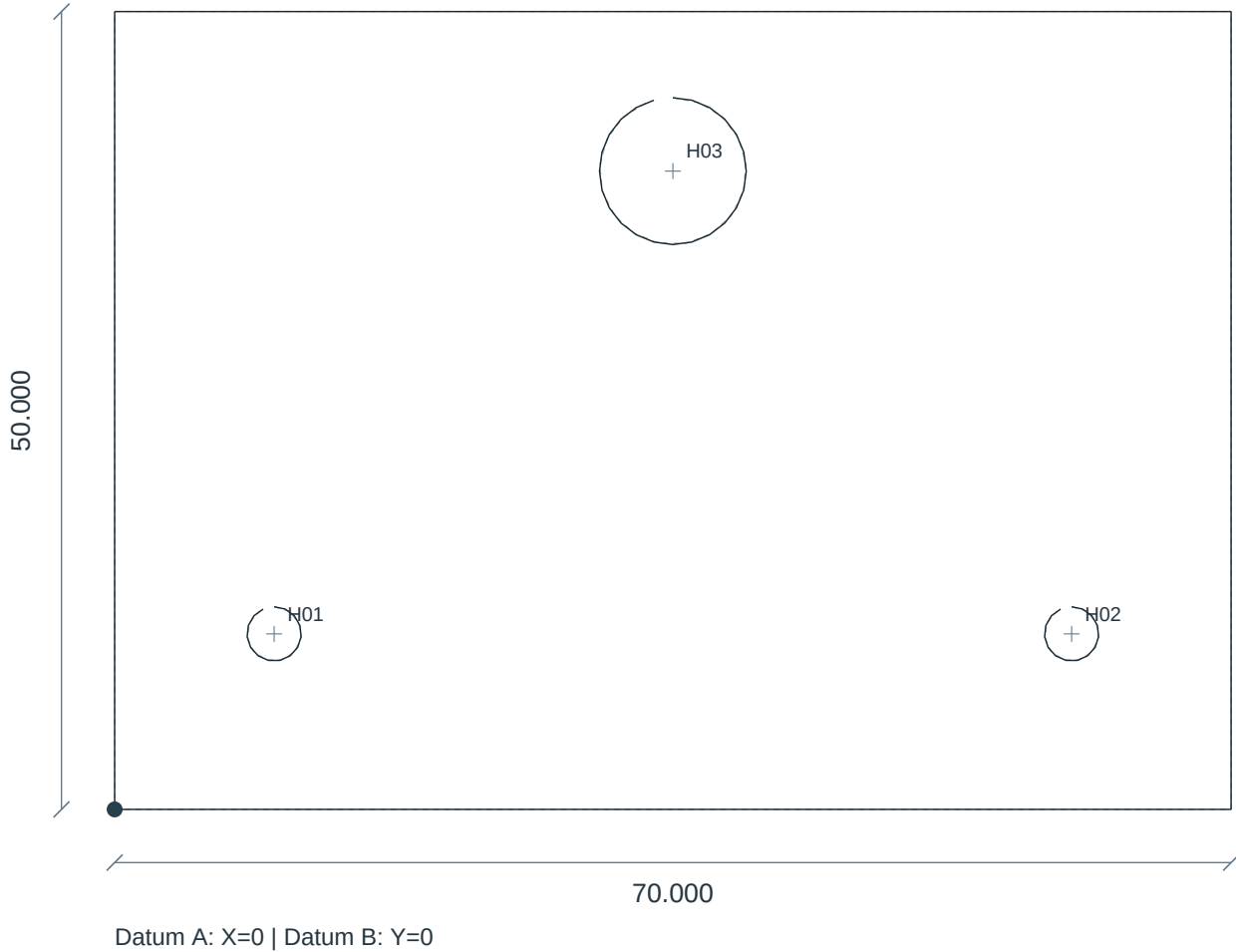
Assembly datum XYZ: 218.500, 179.615, 334.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_TILT_CHEEK_LEFT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	11.000	DIA 3.400
H02	60.000	11.000	DIA 3.400
H03	35.000	40.000	DIA 9.200

SIDE EDGE PROJECTION - X / DEPTH



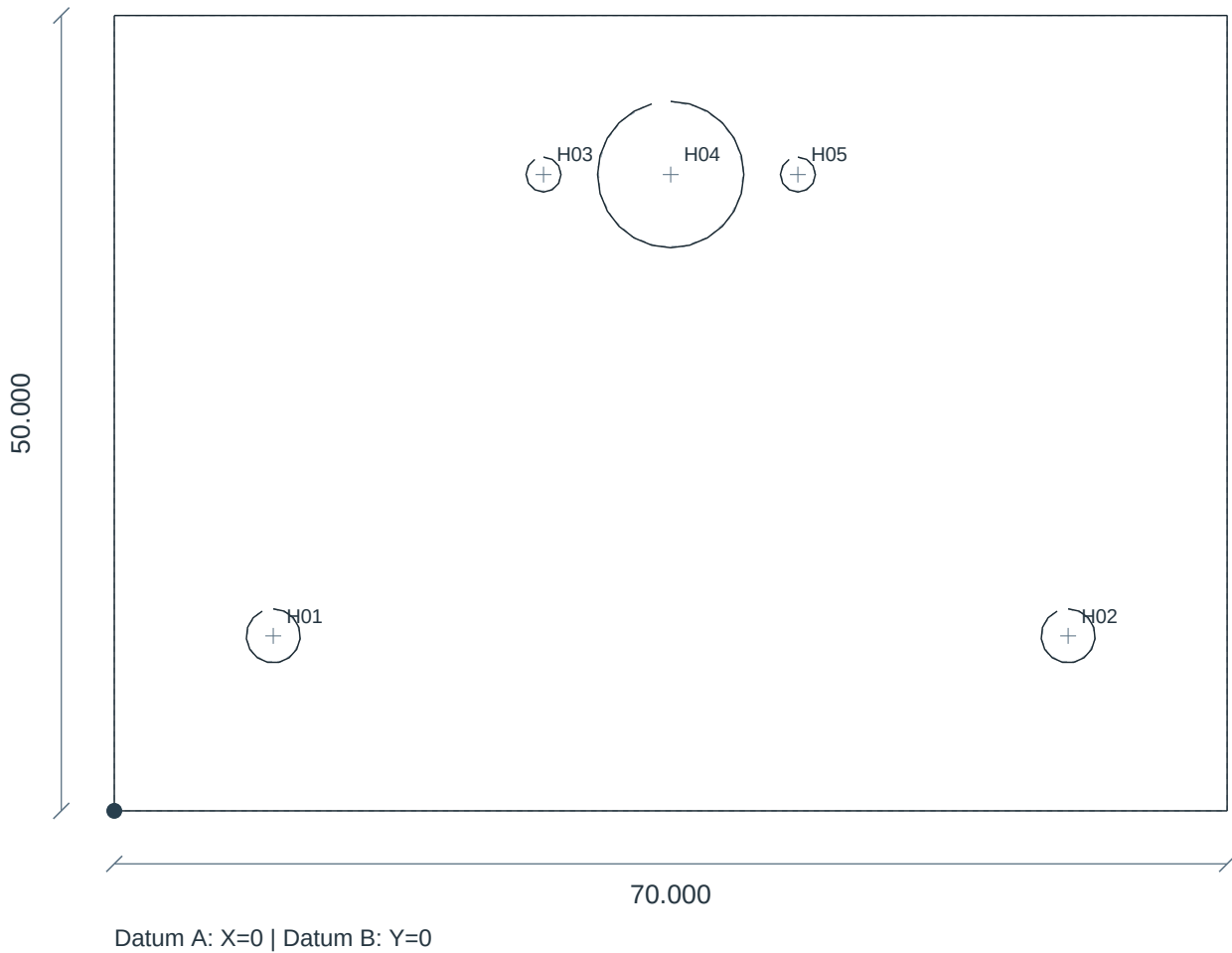
Assembly datum XYZ: 213.000, 41.715, 310.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_TILT_CHEEK_RIGHT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	11.000	DIA 3.400
H02	60.000	11.000	DIA 3.400
H03	27.000	40.000	DIA 2.200
H04	35.000	40.000	DIA 9.200
H05	43.000	40.000	DIA 2.200

SIDE EDGE PROJECTION - X / DEPTH



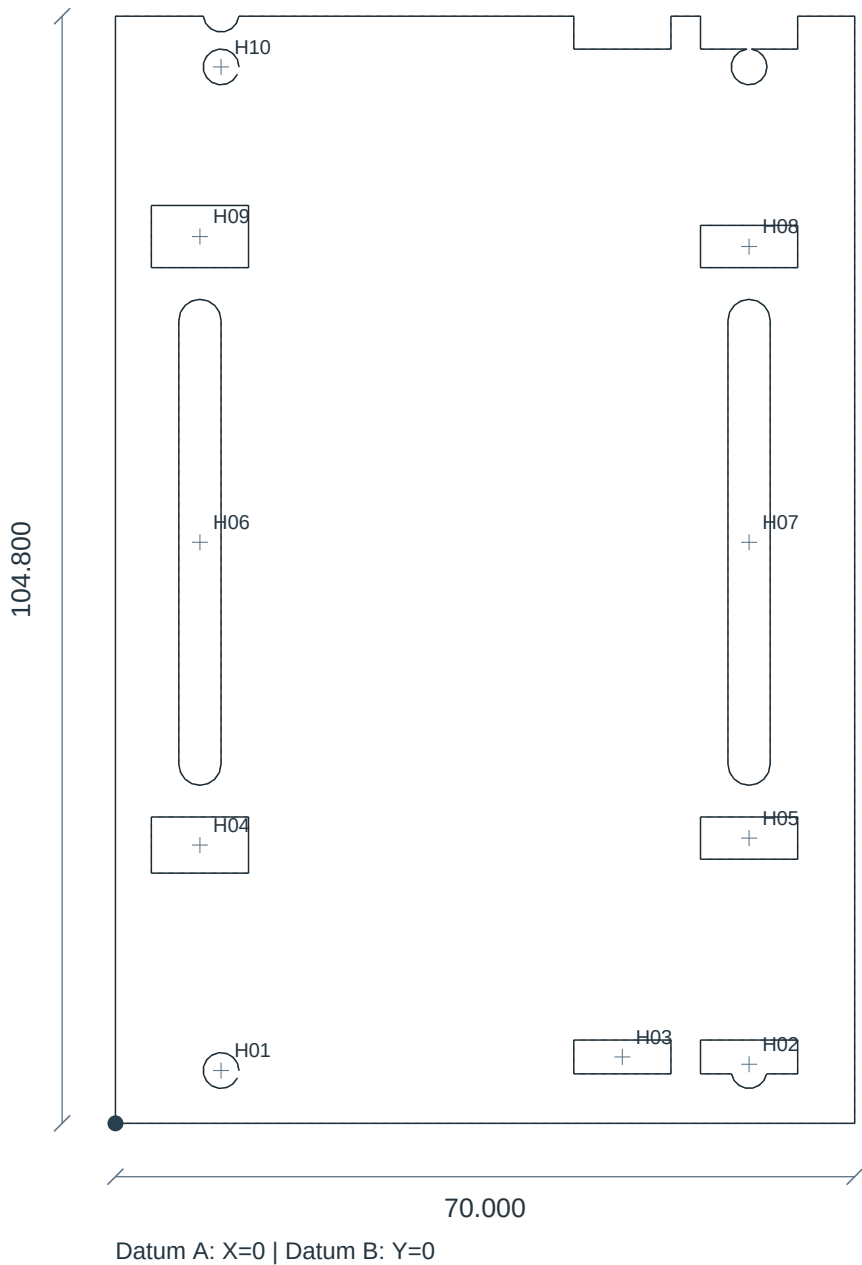
Assembly datum XYZ: 213.000, 151.715, 310.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_C920_SADDLE

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	5.000	DIA 3.400
H02	60.000	5.592	9.200 x 4.585
H03	48.000	6.285	9.200 x 3.200
H04	8.000	26.342	9.200 x 5.315
H05	60.000	27.000	9.200 x 4.000
H06	8.000	55.000	4.000 x 46.000
H07	60.000	55.000	4.000 x 46.000
H08	60.000	83.000	9.200 x 4.000
H09	8.000	83.942	9.200 x 5.885
H10	10.000	100.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

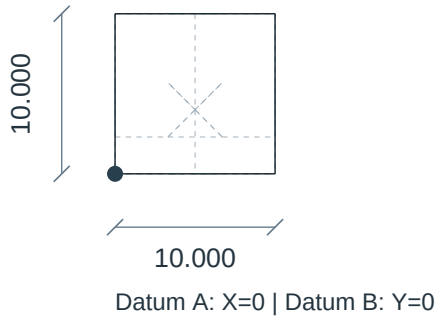
Assembly datum XYZ: 213.000, 41.715, 310.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_TILT_CORNER_LEFT0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 8.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 227.000, 41.715, 325.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

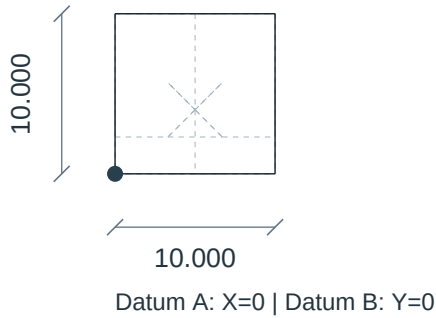
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_TILT_CORNER_LEFT1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 8.000 mm

ID	X	Y	SIZE
----	---	---	------

No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 277.000, 41.715, 325.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

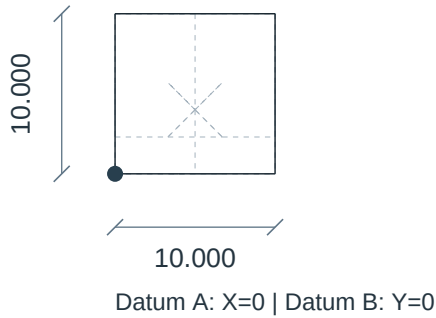
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_TILT_CORNER_RIGHT0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 8.000 mm

ID	X	Y	SIZE
----	---	---	------

No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 227.000, 136.715, 325.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

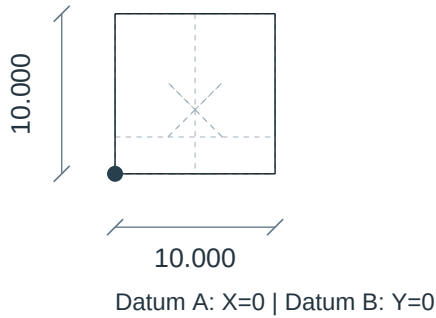
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_TILT_CORNER_RIGHT1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 8.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 277.000, 136.715, 325.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

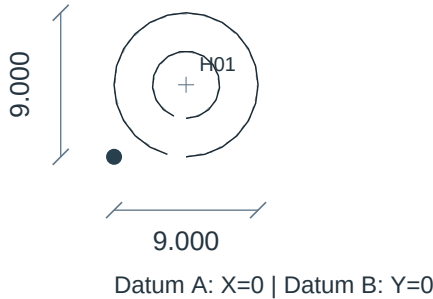
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_TILT_BUSH_LEFT

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	4.500	4.500	DIA 4.200



SIDE EDGE PROJECTION - X / DEPTH



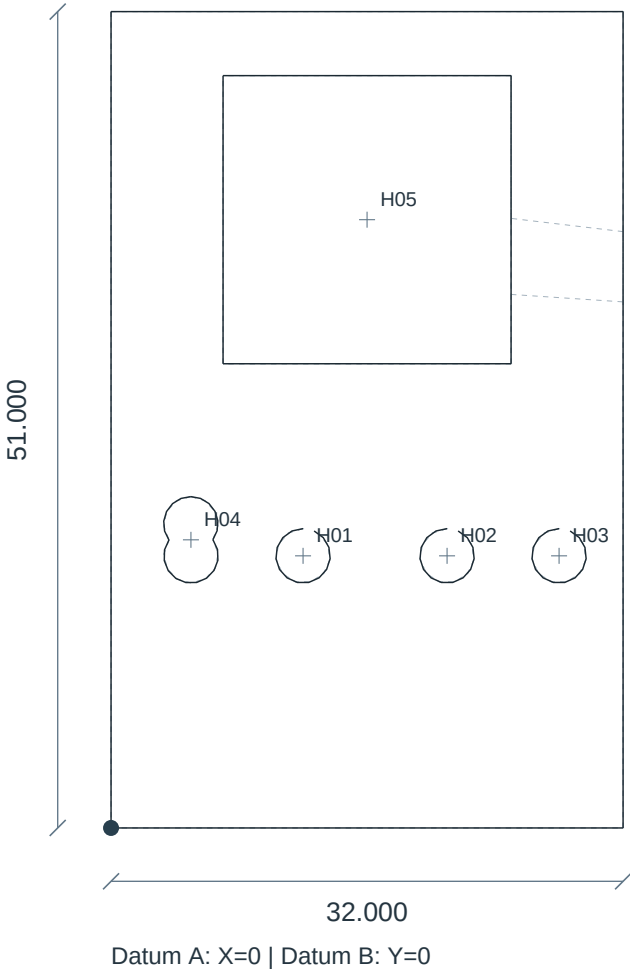
Assembly datum XYZ: 243.500, 32.715, 354.500
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_THERMAL_MOUNT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	12.000	17.000	DIA 3.400
H02	21.000	17.000	DIA 3.400
H03	28.000	17.000	DIA 3.400
H04	4.992	18.000	3.415 x 5.400
H05	16.000	38.000	18.000 x 18.000

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 300.000, 80.715, 215.000
u (0.0, 1.0, 0.0); v (0.0, 0.0, 1.0)

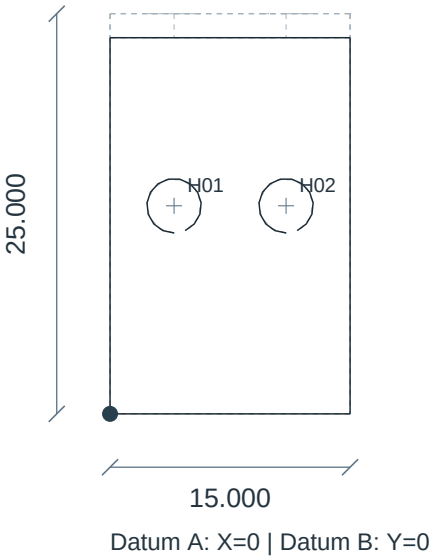
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

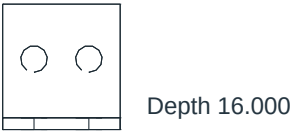
CUSTOM_THERMAL_ANGLE_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 16.000 mm

ID	X	Y	SIZE
H01	4.000	13.000	DIA 3.400
H02	11.000	13.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 300.000, 81.715, 245.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

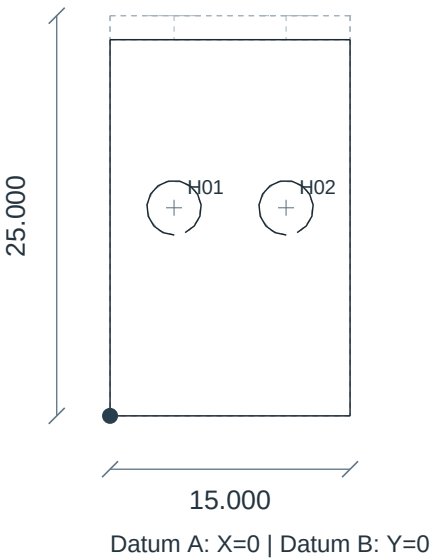
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

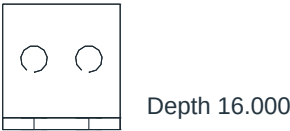
CUSTOM_THERMAL_ANGLE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 16.000 mm

ID	X	Y	SIZE
H01	4.000	13.000	DIA 3.400
H02	11.000	13.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 300.000, 97.715, 245.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

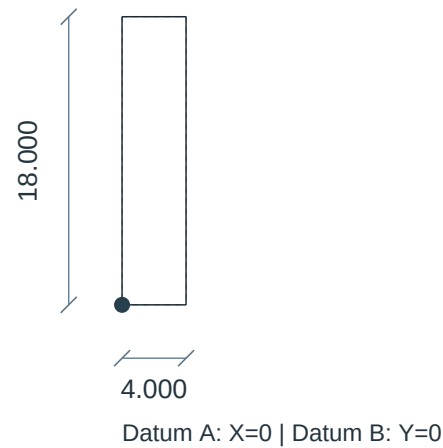
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_THERMAL_EDGE_0

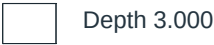
Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 3.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 308.000, 83.715, 263.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

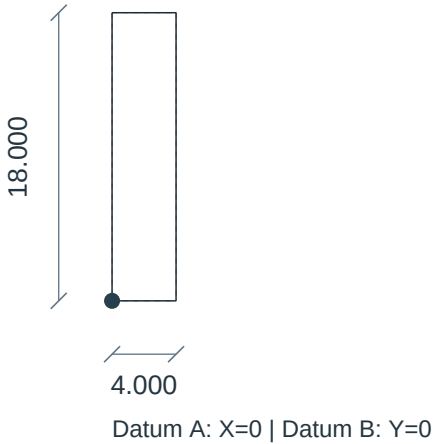
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_THERMAL_EDGE_1

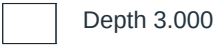
Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 3.000 mm

ID	X	Y	SIZE
----	---	---	------

No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 308.000, 105.715, 263.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

CUSTOM_FRONT BUMPER

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	67.555	41.000	DIA 7.200
H02	175.875	41.000	DIA 7.200



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

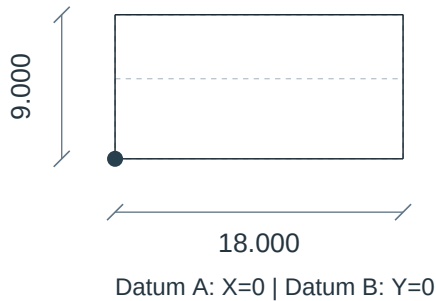
Assembly datum XYZ: 312.939, -25.000, 45.000
u (0.0, 1.0, 0.0); v (0.0, 0.0, 1.0)

CUSTOM_FRONT_STRIKER

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



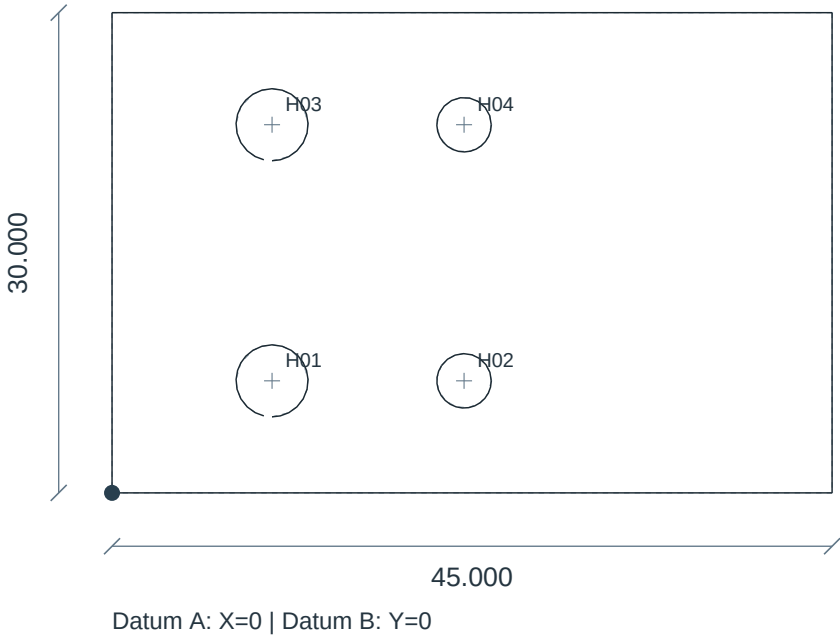
Assembly datum XYZ: 307.939, 87.715, 80.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_GUIDE_FOOT_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	7.000	DIA 4.500
H02	22.000	7.000	DIA 3.400
H03	10.000	23.000	DIA 4.500
H04	22.000	23.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



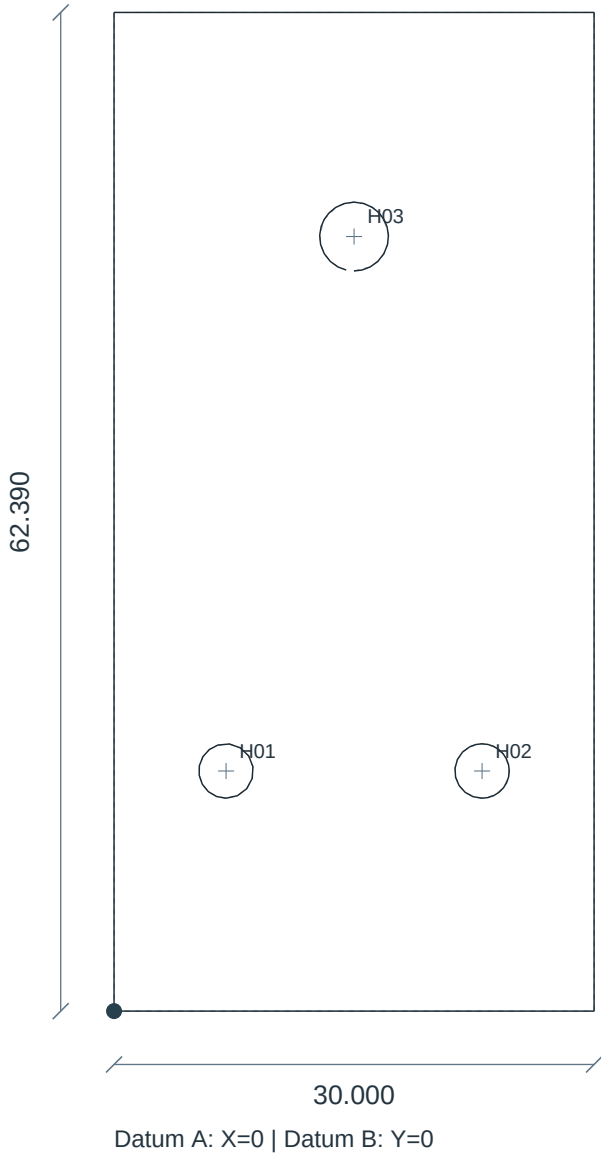
Assembly datum XYZ: 254.939, 57.555, 139.390
u (1.0, 0.0, 0.0); v (0.0, -1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_FRONT_GUIDE_HANGER_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400
H03	15.000	48.390	DIA 4.300

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 299.939, 27.555, 134.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

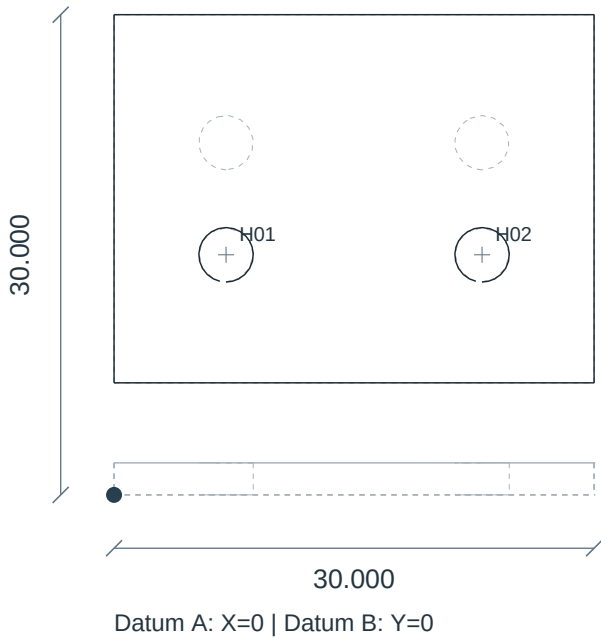
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

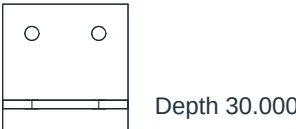
CUSTOM_FRONT_GUIDE_ANGLE_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 299.939, 27.555, 141.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_GUIDE_BUSH_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 307.939, 39.055, 82.500
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

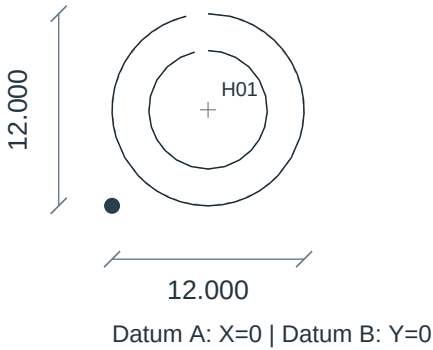
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_STOPPER_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



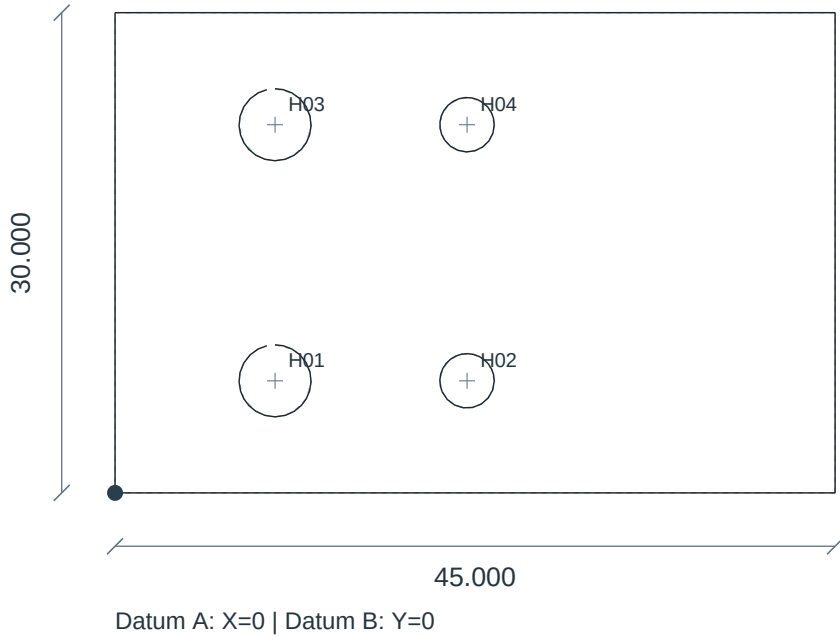
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 299.939, 36.555, 80.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

CUSTOM_FRONT_GUIDE_FOOT_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	10.000	7.000	DIA 4.500
H02	22.000	7.000	DIA 3.400
H03	10.000	23.000	DIA 4.500
H04	22.000	23.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



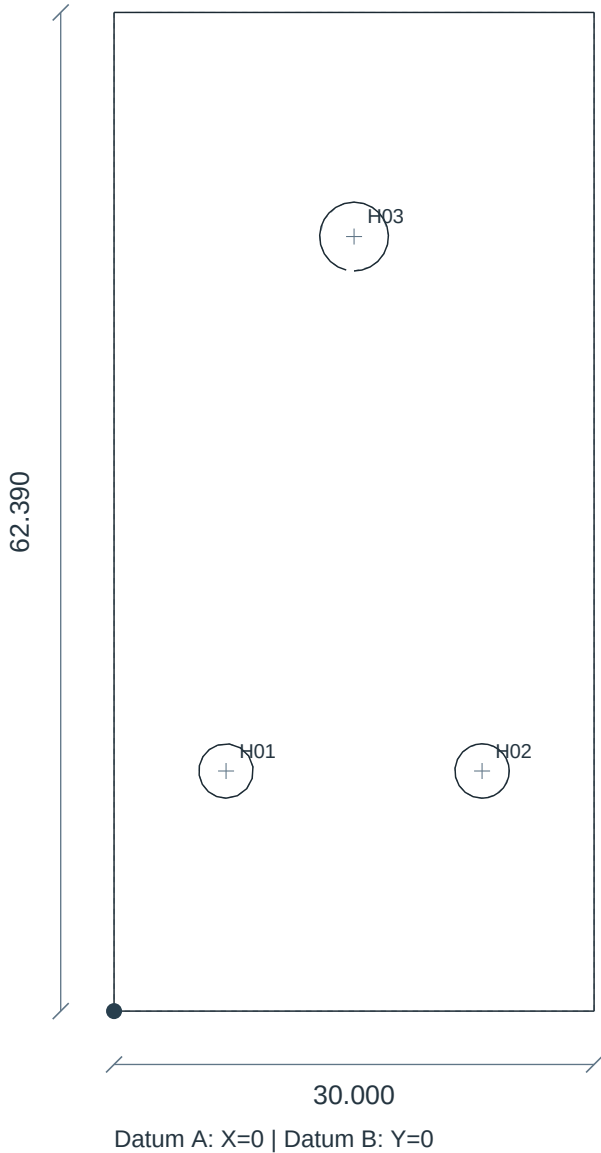
Assembly datum XYZ: 254.939, 135.875, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_FRONT_GUIDE_HANGER_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400
H03	15.000	48.390	DIA 4.300

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 299.939, 135.875, 134.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

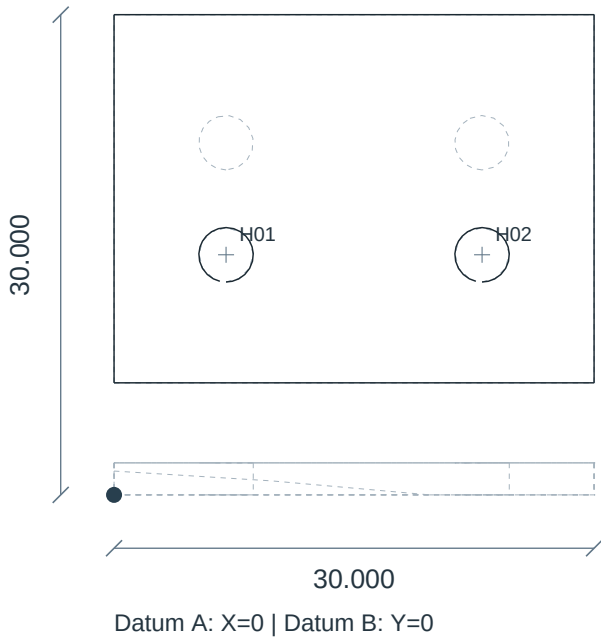
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

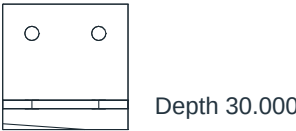
CUSTOM_FRONT_GUIDE_ANGLE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 299.939, 135.875, 141.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_GUIDE_BUSH_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 307.939, 147.375, 82.500
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

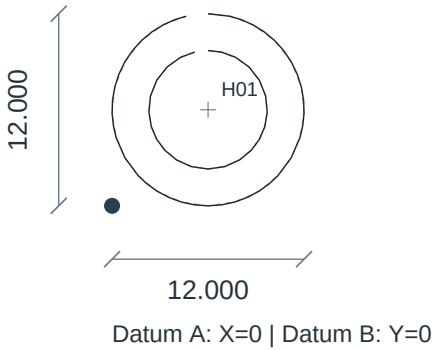
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_STOPPER_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



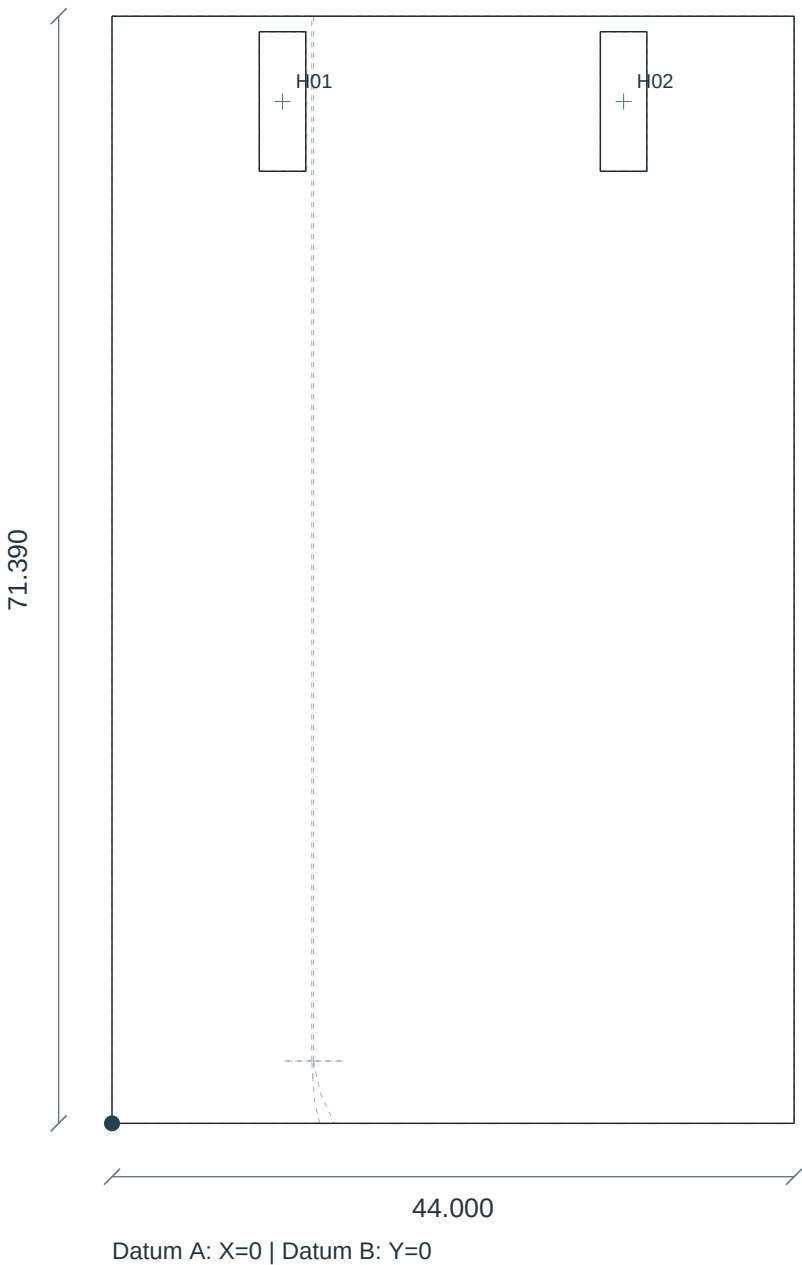
Assembly datum XYZ: 299.939, 144.875, 80.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_SWITCH_BRACKET

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	11.000	65.890	3.000 x 9.000
H02	33.000	65.890	3.000 x 9.000

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 299.939, 74.715, 134.390
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

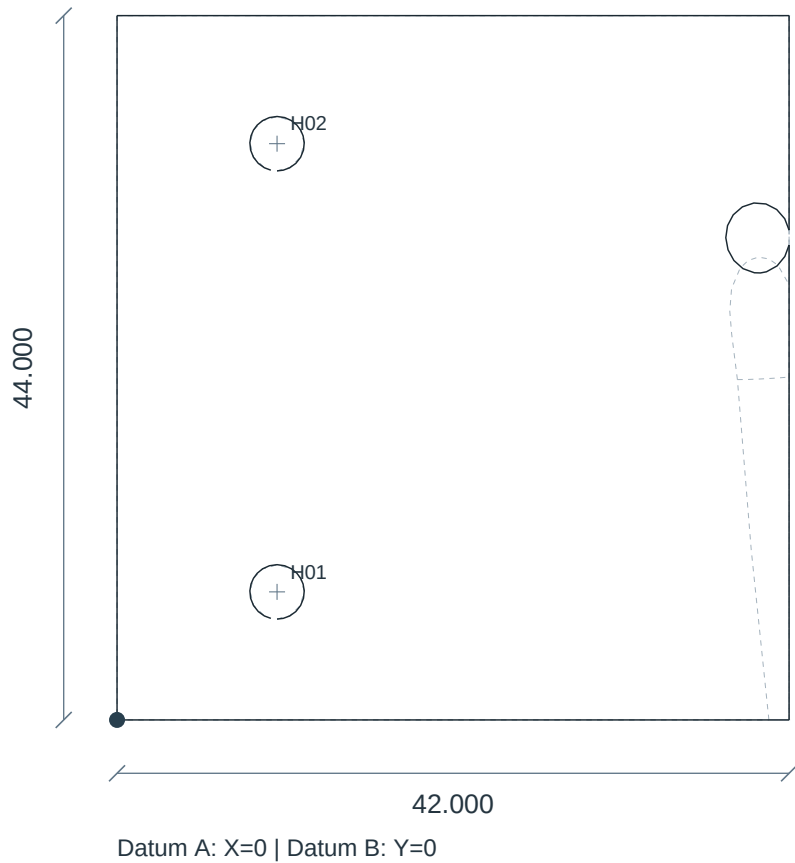
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_SWITCH_FOOT

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	10.000	8.000	DIA 3.400
H02	10.000	36.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 257.939, 118.715, 139.390
u (1.0, 0.0, 0.0); v (0.0, -1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

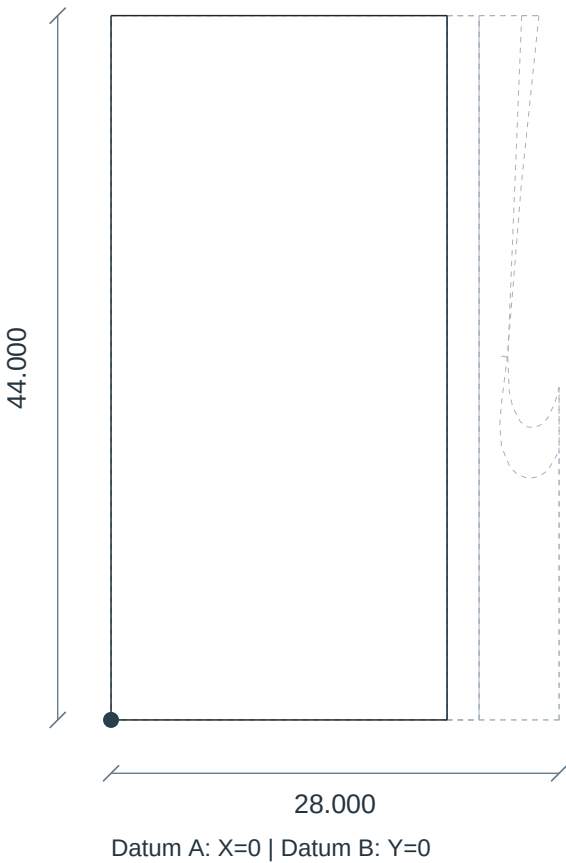
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_SWITCH_ANGLE

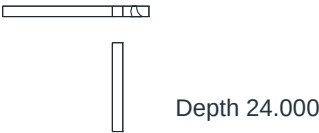
Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 24.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 271.939, 74.715, 117.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

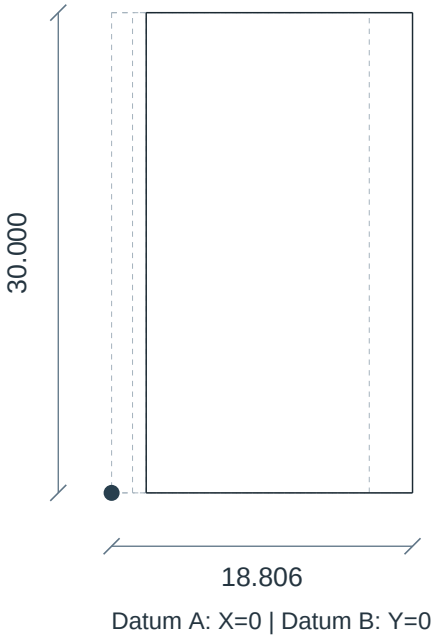
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_CORNER_NOSE_-1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 4.206 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 304.232, -36.545, 55.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

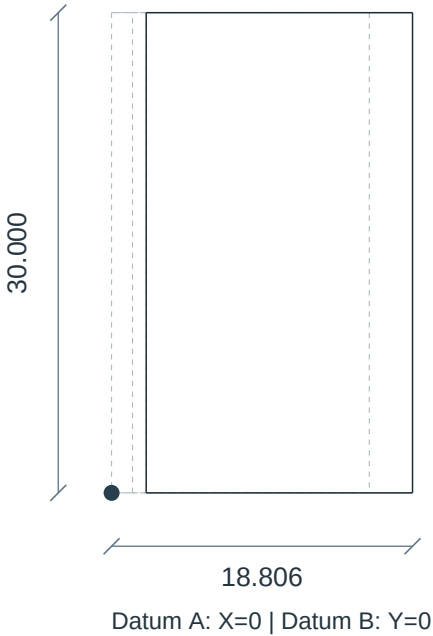
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_FRONT_CORNER_NOSE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 4.206 mm

ID	X	Y	SIZE
----	---	---	------

No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 304.232, 229.975, 85.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR BUMPER

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	67.555	41.000	DIA 7.200
H02	175.875	41.000	DIA 7.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -33.000, 218.430, 45.000
u (0.0, -1.0, 0.0); v (0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

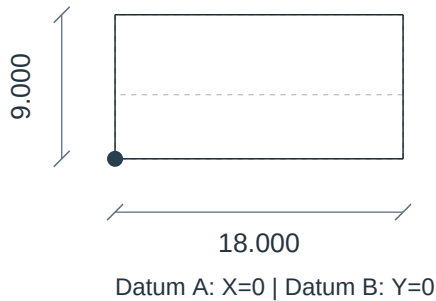
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_REAR_STRIKER

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



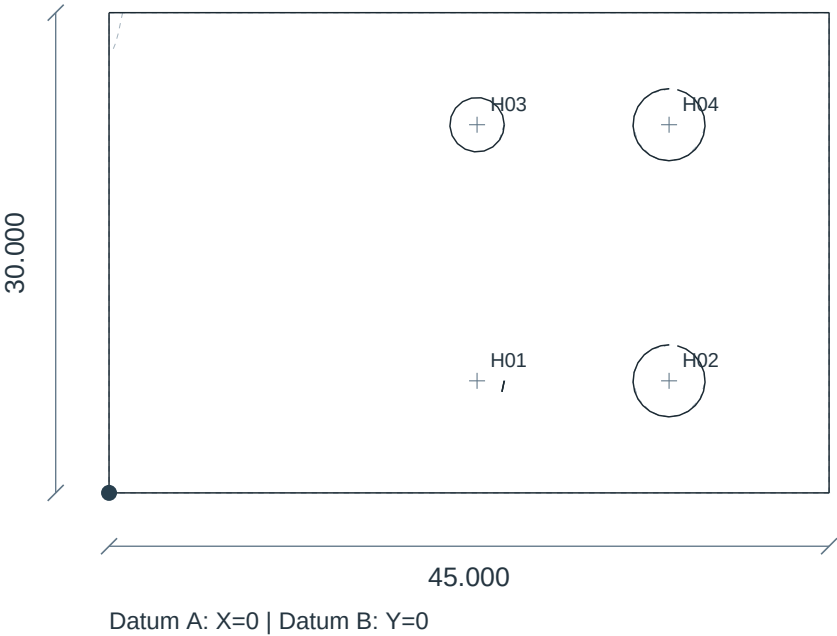
Assembly datum XYZ: -28.000, 87.715, 89.000
u (0.0, 1.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_GUIDE_FOOT_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	23.000	7.000	DIA 3.400
H02	35.000	7.000	DIA 4.500
H03	23.000	23.000	DIA 3.400
H04	35.000	23.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



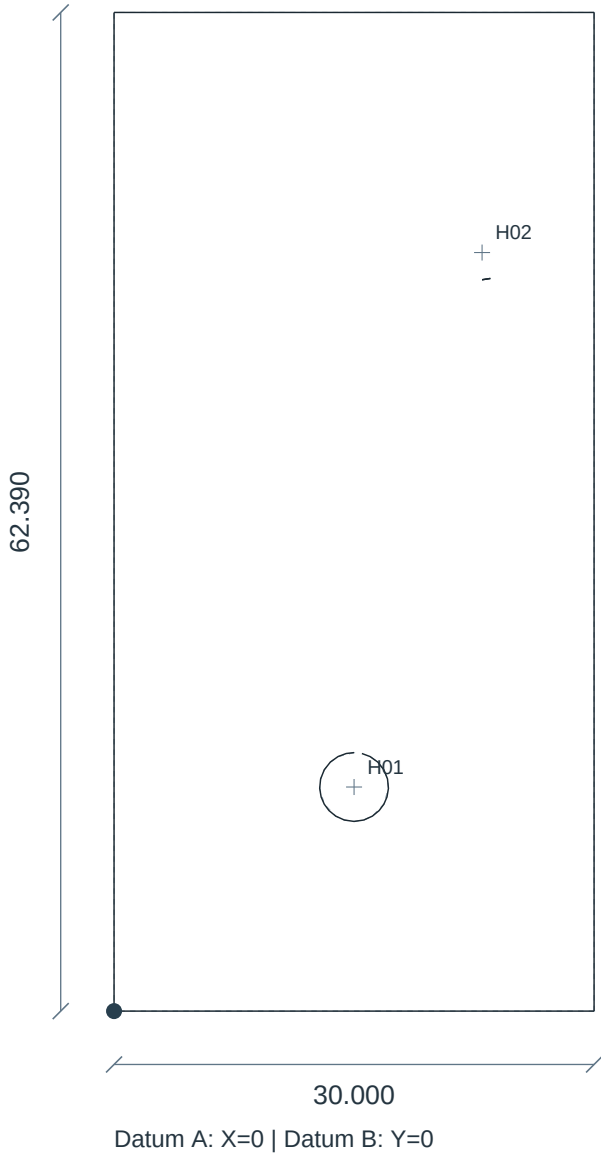
Assembly datum XYZ: -20.000, 165.875, 139.390
u (1.0, 0.0, 0.0); v (0.0, -1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_REAR_GUIDE_HANGER_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	15.000	14.000	DIA 4.300
H02	23.000	47.390	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

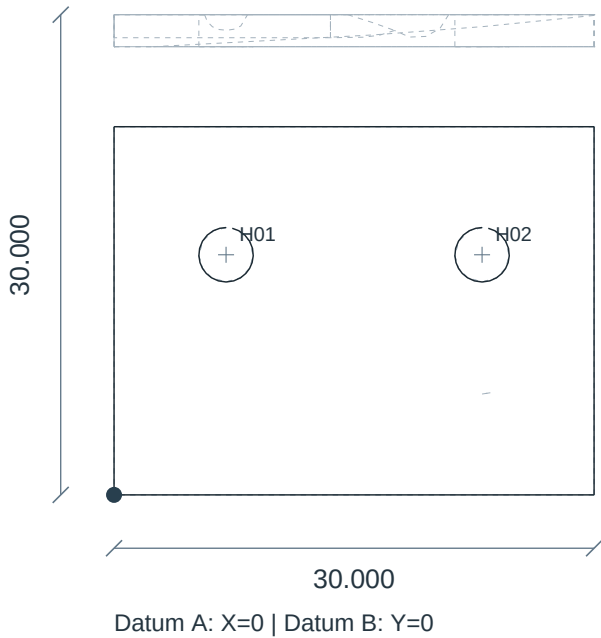
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: -20.000, 135.875, 72.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

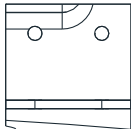
CUSTOM_REAR_GUIDE_ANGLE_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Depth 30.000

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: -20.000, 135.875, 111.390
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

CUSTOM_REAR_GUIDE_BUSH_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -28.000, 147.375, 89.500
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

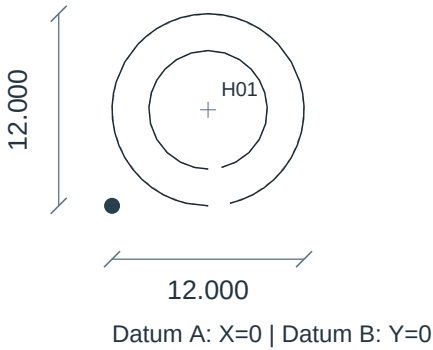
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_STOPPER_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



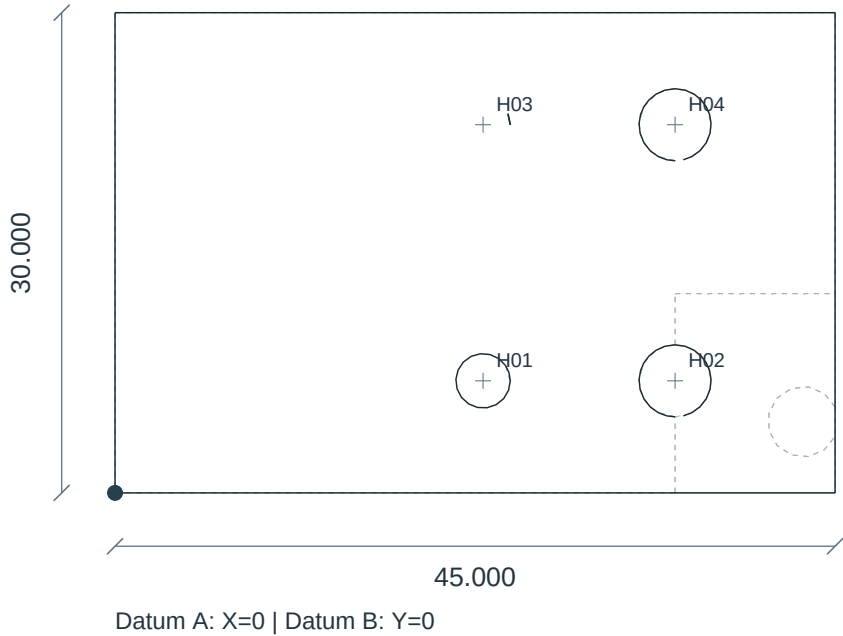
Assembly datum XYZ: -20.000, 144.875, 92.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_GUIDE_FOOT_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	23.000	7.000	DIA 3.400
H02	35.000	7.000	DIA 4.500
H03	23.000	23.000	DIA 3.400
H04	35.000	23.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



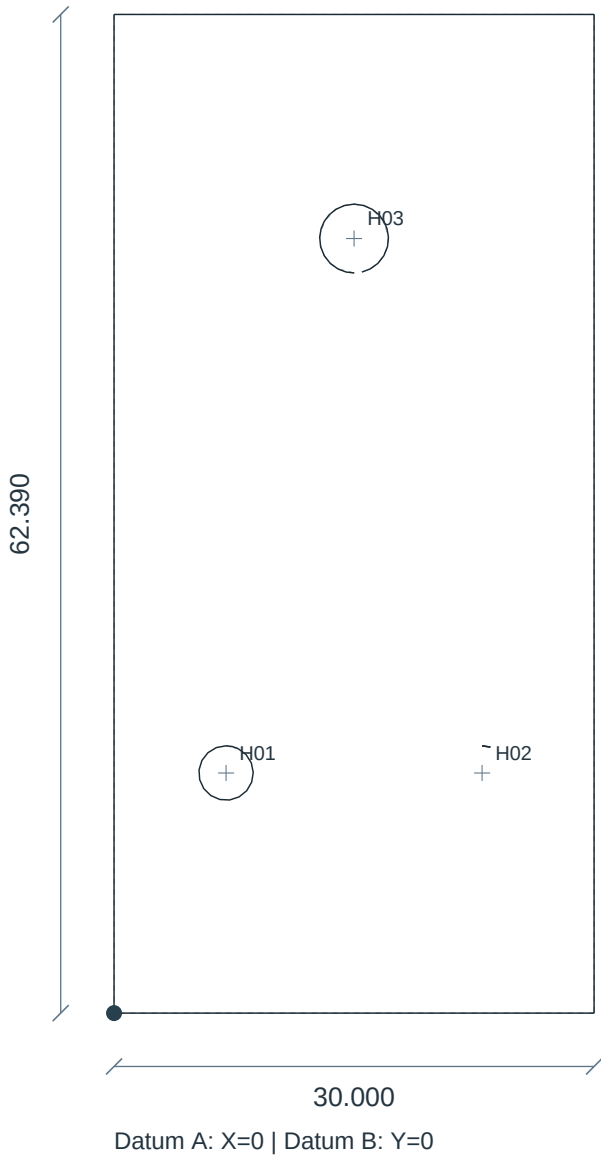
Assembly datum XYZ: -20.000, 27.555, 134.390
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_GUIDE_HANGER_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400
H03	15.000	48.390	DIA 4.300

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

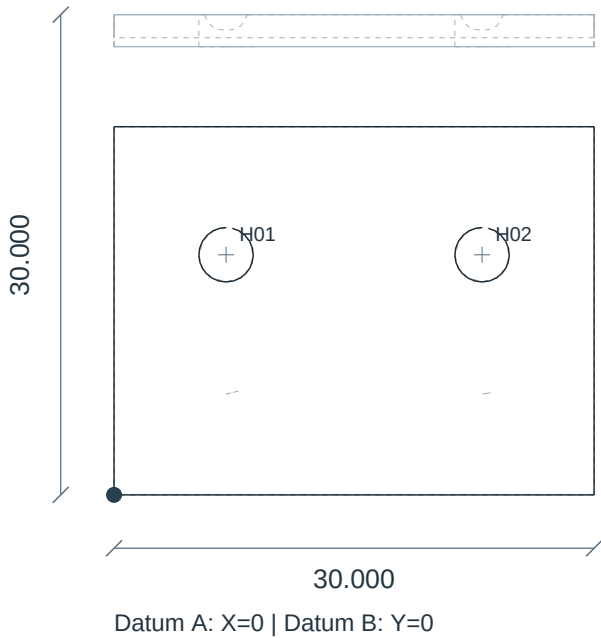
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: -15.000, 27.555, 134.390
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

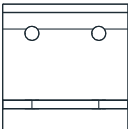
CUSTOM_REAR_GUIDE_ANGLE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Depth 30.000

Assembly datum XYZ: -20.000, 27.555, 111.390
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_GUIDE_BUSH_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

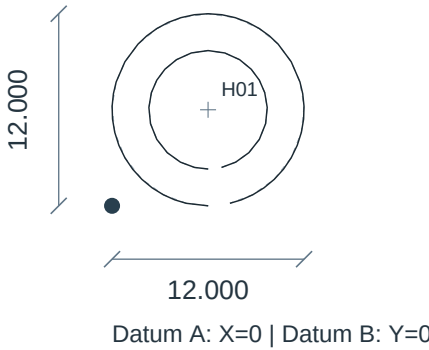
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: -28.000, 39.055, 89.500
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

CUSTOM_REAR_STOPPER_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



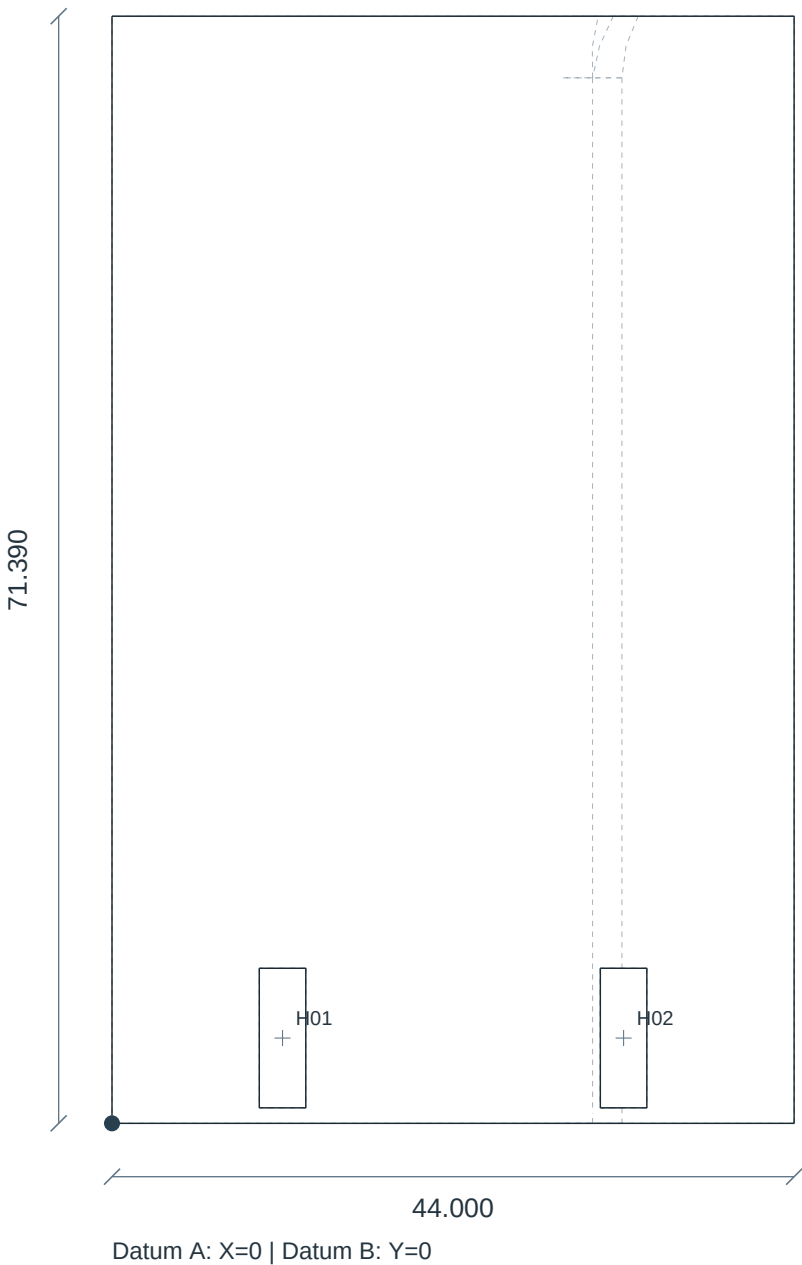
Assembly datum XYZ: -20.000, 36.555, 92.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_SWITCH_BRACKET

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	11.000	5.500	3.000 x 9.000
H02	33.000	5.500	3.000 x 9.000

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -20.000, 74.715, 63.000
u (0.0, 1.0, 0.0); v (0.0, -0.0, 1.0)

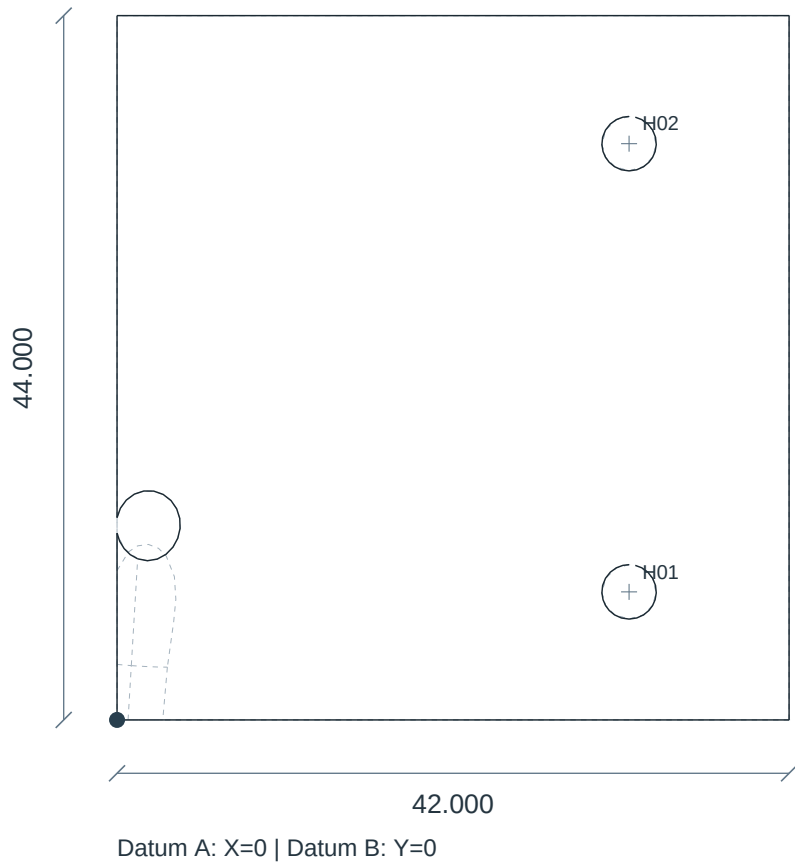
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_SWITCH_FOOT

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	32.000	8.000	DIA 3.400
H02	32.000	36.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -20.000, 118.715, 139.390
u (1.0, 0.0, 0.0); v (0.0, -1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

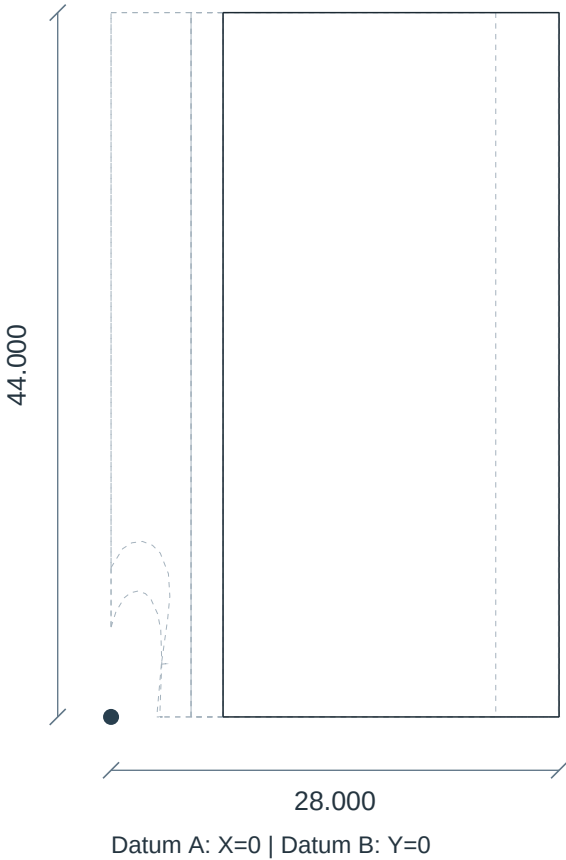
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_SWITCH_ANGLE

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 24.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -20.000, 118.715, 141.390
u (1.0, 0.0, 0.0); v (0.0, -1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

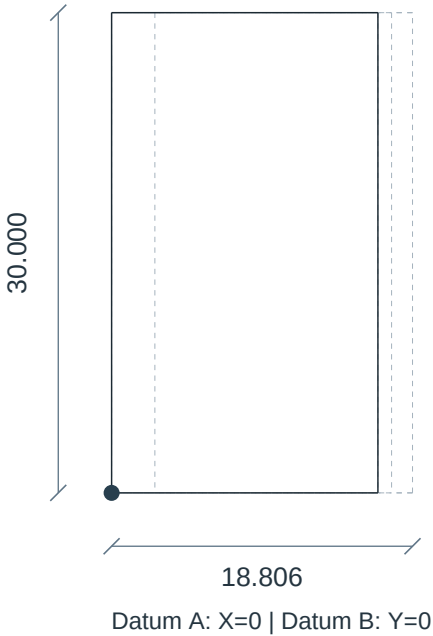
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_CORNER_NOSE_-1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 4.206 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -34.462, 214.156, 85.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

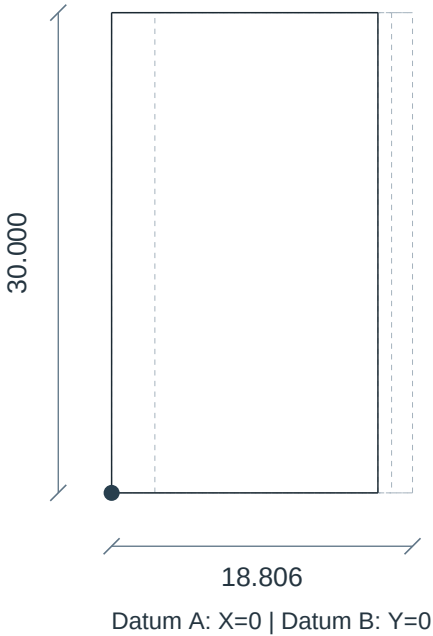
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_REAR_CORNER_NOSE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 4.206 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: -34.462, -20.726, 55.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT BUMPER

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	86.587	41.000	DIA 7.200
H02	243.353	41.000	DIA 7.200



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

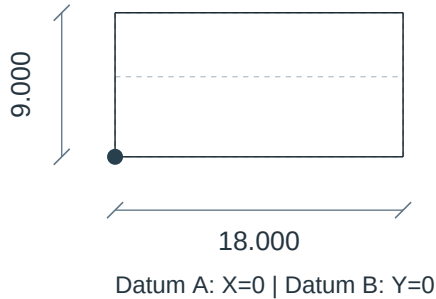
Assembly datum XYZ: -25.000, -33.000, 45.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

CUSTOM_LEFT_STRIKER

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



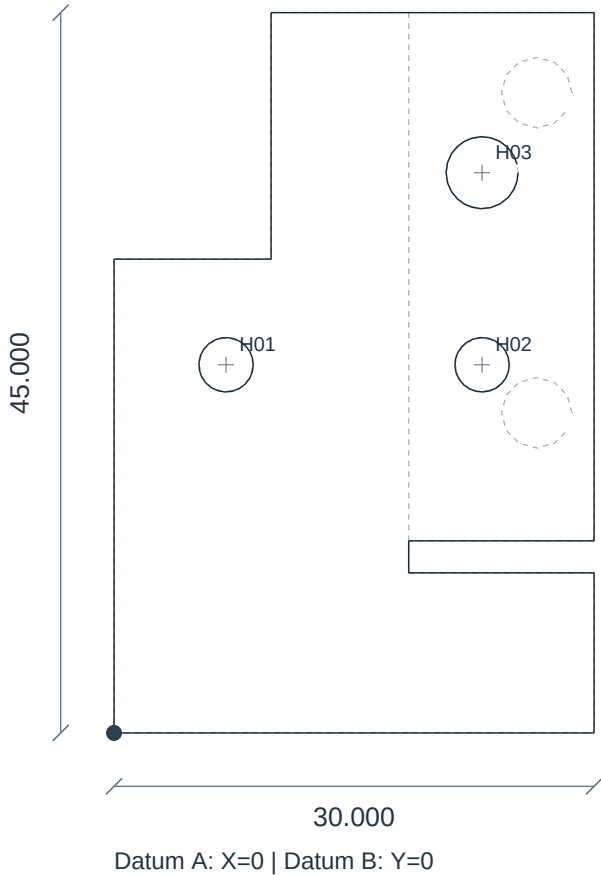
Assembly datum XYZ: 130.970, -28.000, 80.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

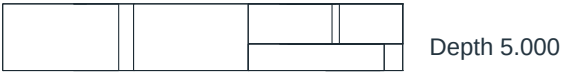
CUSTOM_LEFT_GUIDE_FOOT_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	23.000	DIA 3.400
H02	23.000	23.000	DIA 3.400
H03	23.000	35.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



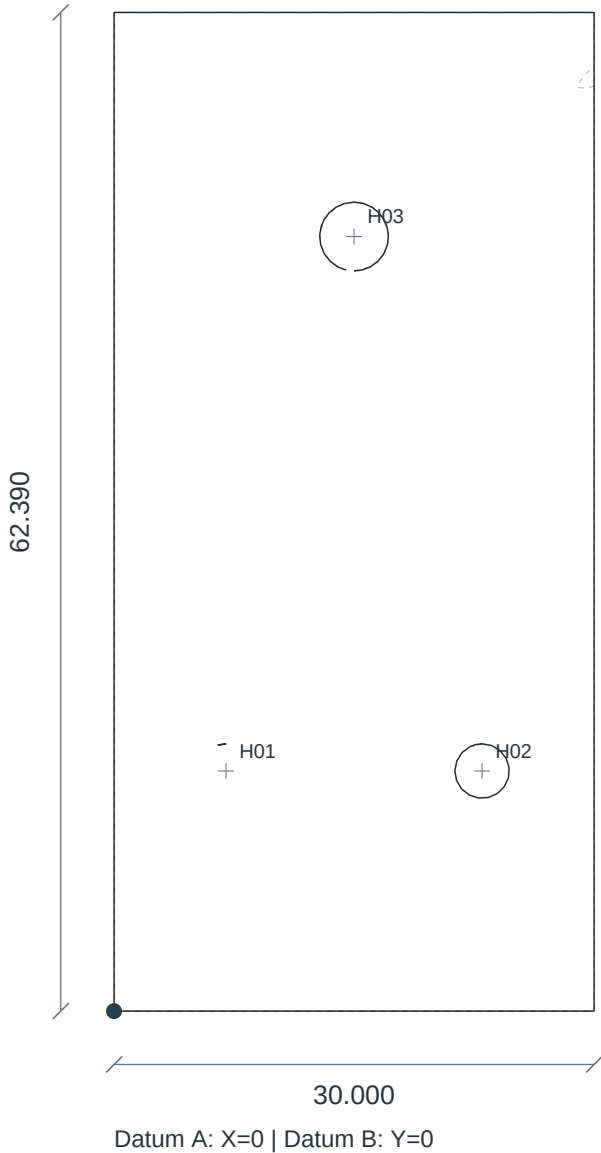
Assembly datum XYZ: 46.587, -20.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_GUIDE_HANGER_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400
H03	15.000	48.390	DIA 4.300

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 46.587, -20.000, 134.390
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

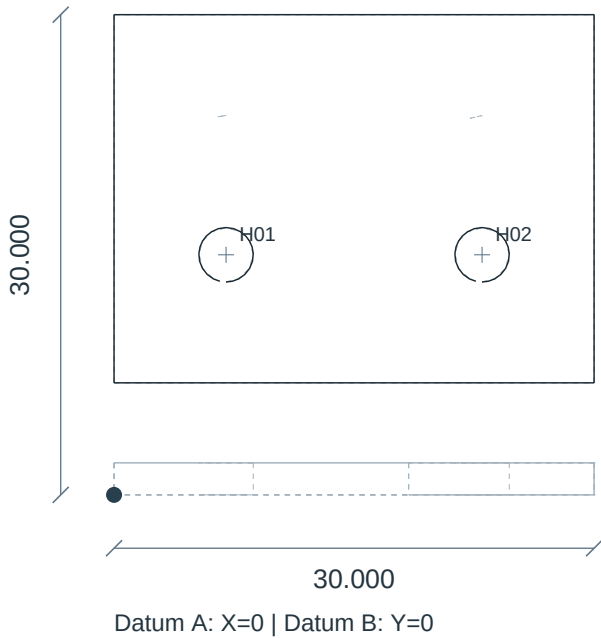
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

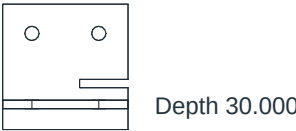
CUSTOM_LEFT_GUIDE_ANGLE_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 46.587, -20.000, 141.390
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_GUIDE_BUSH_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 58.087, -28.000, 82.500
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

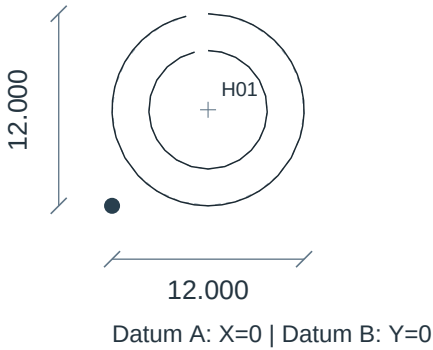
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_STOPPER_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



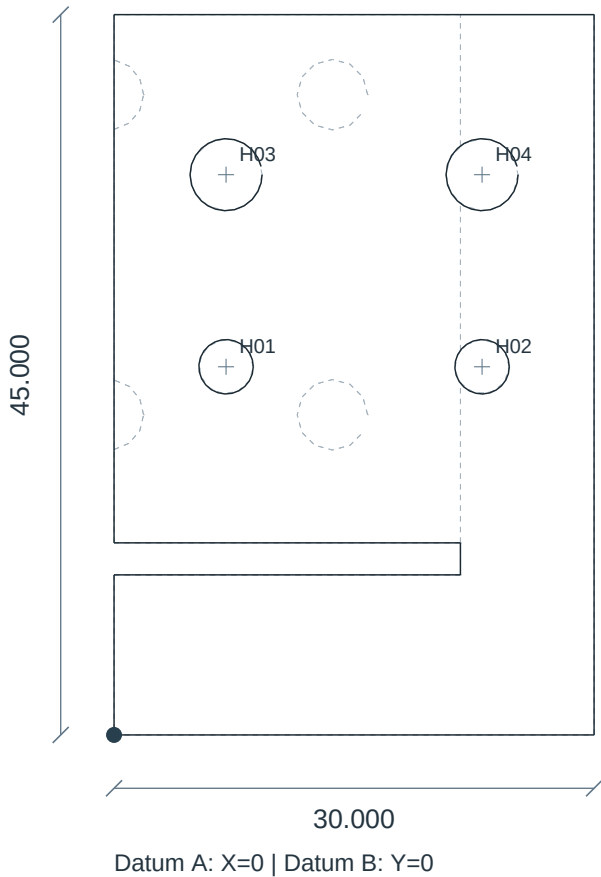
Assembly datum XYZ: 55.587, -20.000, 80.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_GUIDE_FOOT_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	23.000	DIA 3.400
H02	23.000	23.000	DIA 3.400
H03	7.000	35.000	DIA 4.500
H04	23.000	35.000	DIA 4.500

SIDE EDGE PROJECTION - X / DEPTH



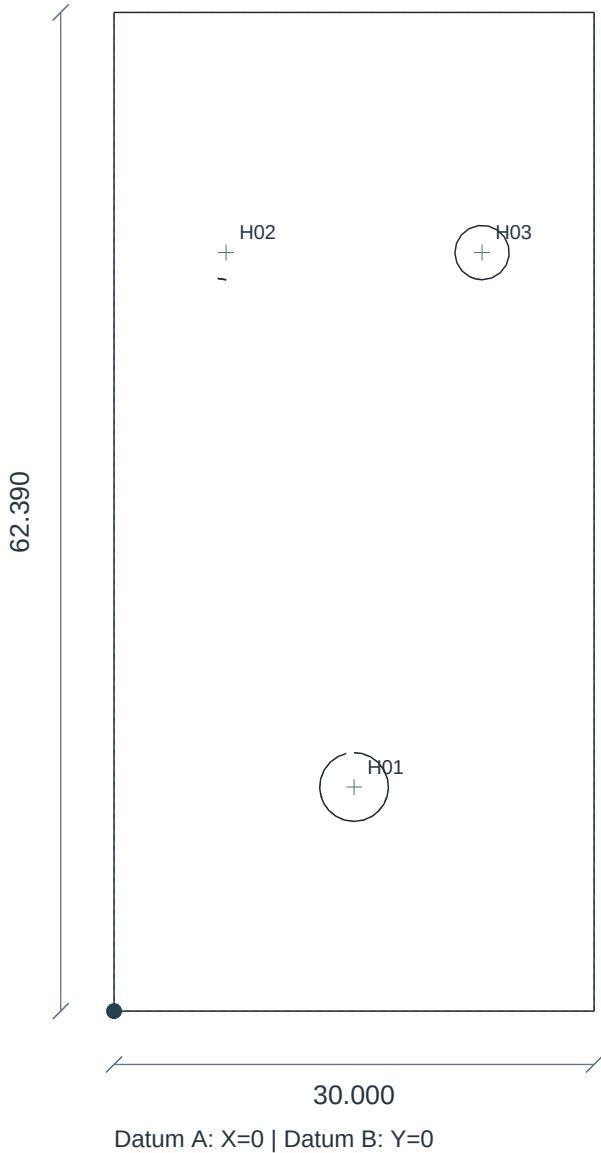
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 203.353, -20.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

CUSTOM_LEFT_GUIDE_HANGER_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

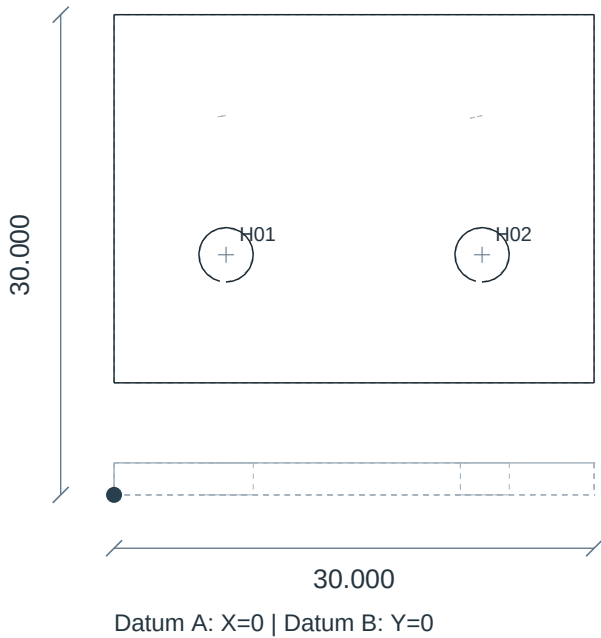
ID	X	Y	SIZE
H01	15.000	14.000	DIA 4.300
H02	7.000	47.390	DIA 3.400
H03	23.000	47.390	DIA 3.400

Assembly datum XYZ: 203.353, -15.000, 72.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

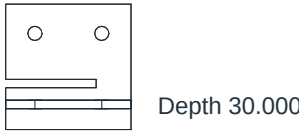
CUSTOM_LEFT_GUIDE_ANGLE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 203.353, -20.000, 141.390
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_GUIDE_BUSH_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 214.853, -28.000, 82.500
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

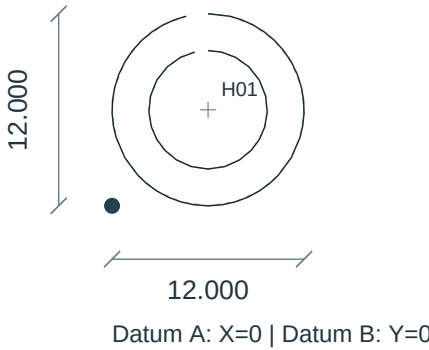
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_STOPPER_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



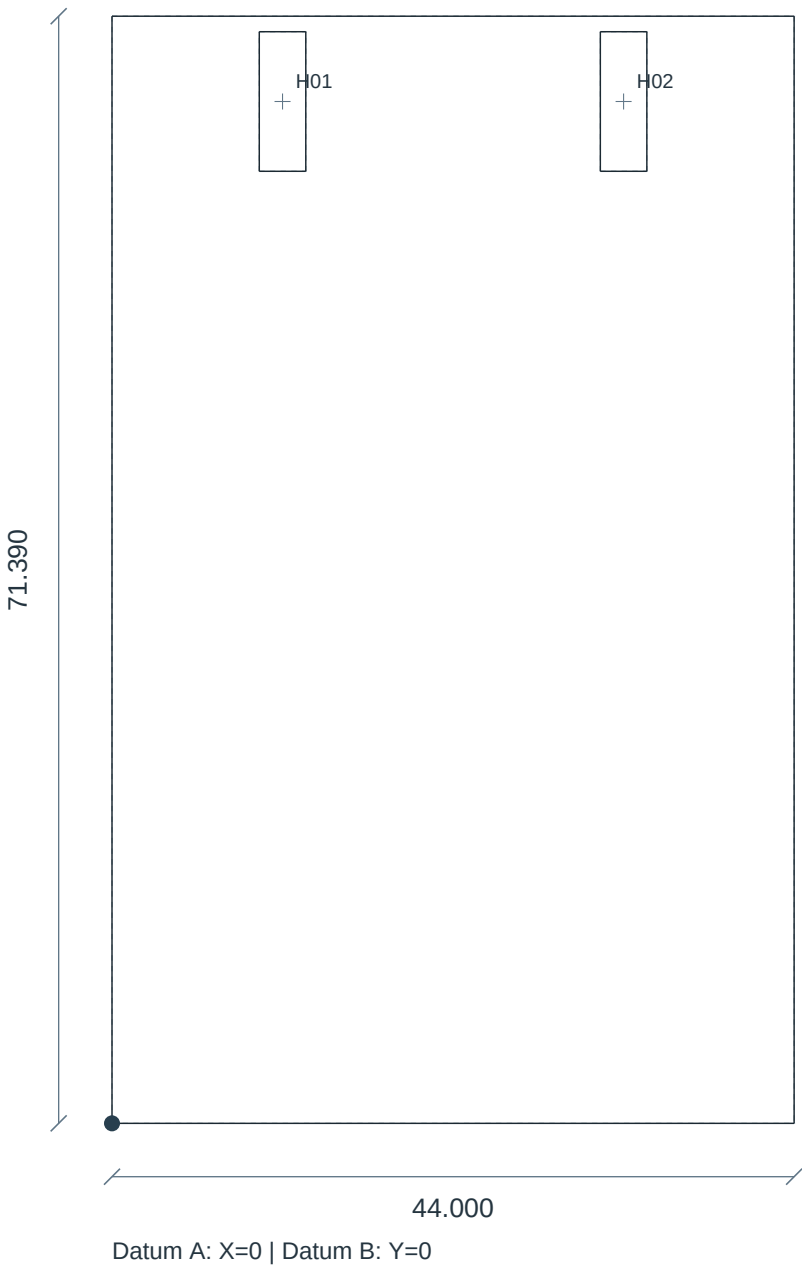
Assembly datum XYZ: 212.353, -20.000, 80.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_LEFT_SWITCH_BRACKET

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	11.000	65.890	3.000 x 9.000
H02	33.000	65.890	3.000 x 9.000

SIDE EDGE PROJECTION - X / DEPTH



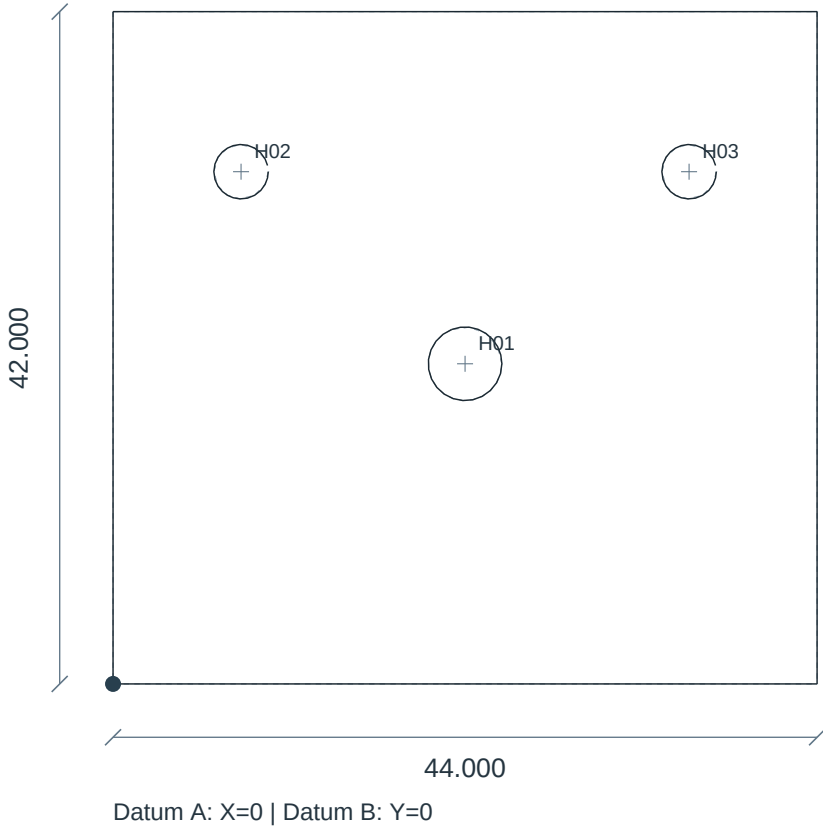
Assembly datum XYZ: 117.970, -20.000, 134.390
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_LEFT_SWITCH_FOOT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	22.000	20.000	DIA 4.600
H02	8.000	32.000	DIA 3.400
H03	36.000	32.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

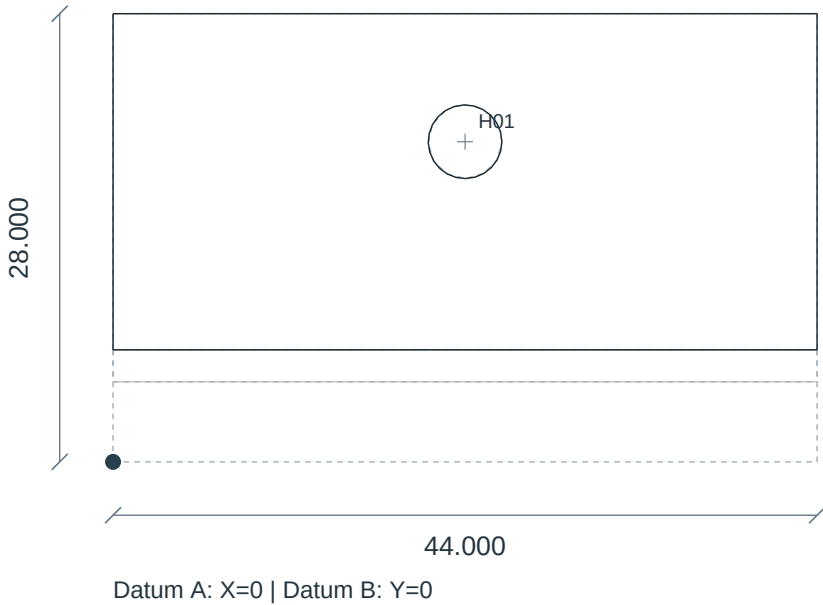
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 117.970, -20.000, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

CUSTOM_LEFT_SWITCH_ANGLE

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 24.000 mm

ID	X	Y	SIZE
H01	22.000	20.000	DIA 4.600



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

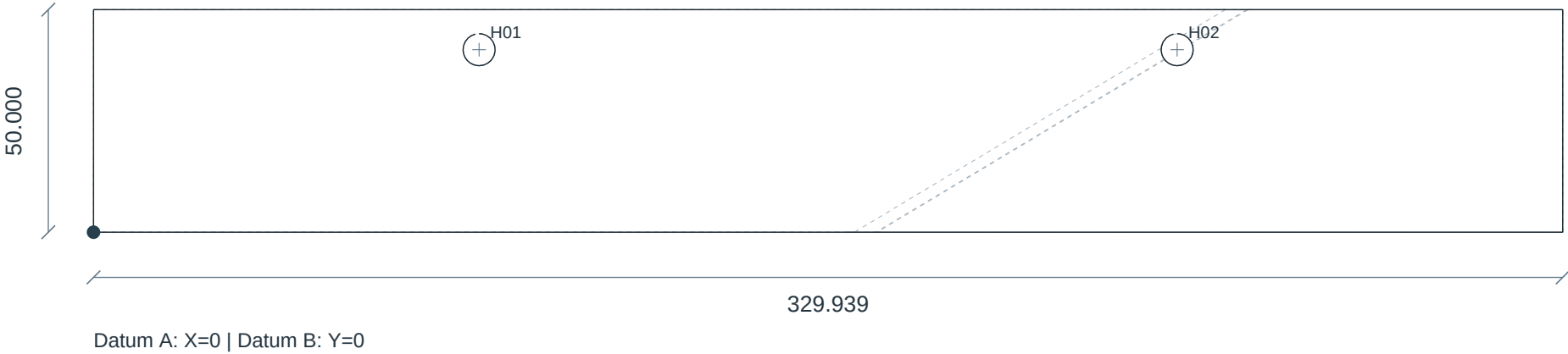
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 117.970, -20.000, 117.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

CUSTOM_RIGHT BUMPER

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm

ID	X	Y	SIZE
H01	86.587	41.000	DIA 7.200
H02	243.353	41.000	DIA 7.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 304.939, 226.430, 45.000
u (-1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

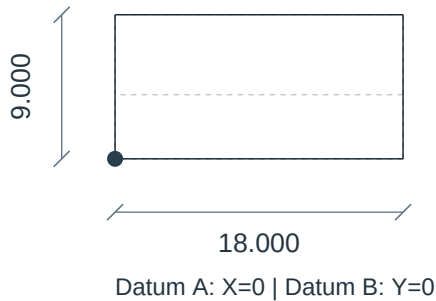
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_STRIKER

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



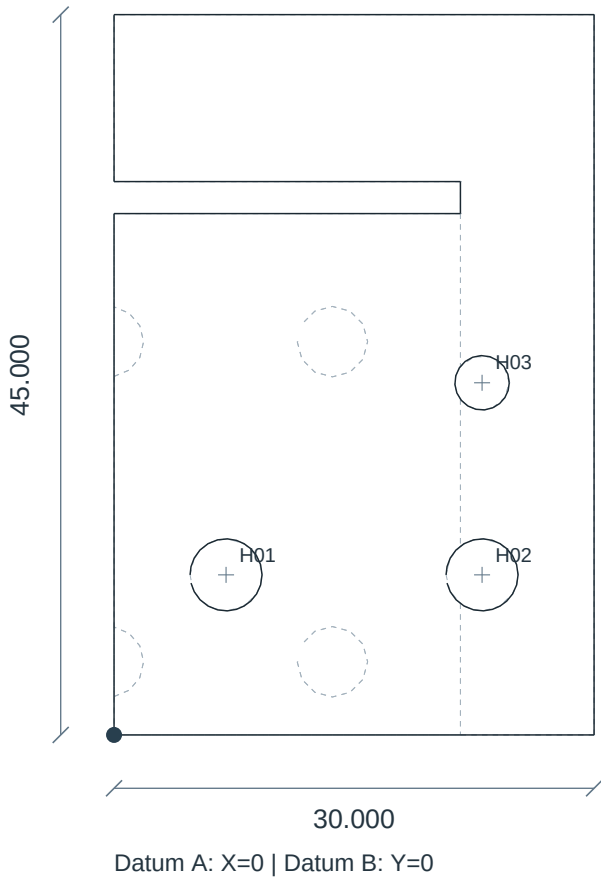
Assembly datum XYZ: 130.970, 221.430, 89.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_GUIDE_FOOT_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	10.000	DIA 4.500
H02	23.000	10.000	DIA 4.500
H03	23.000	22.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



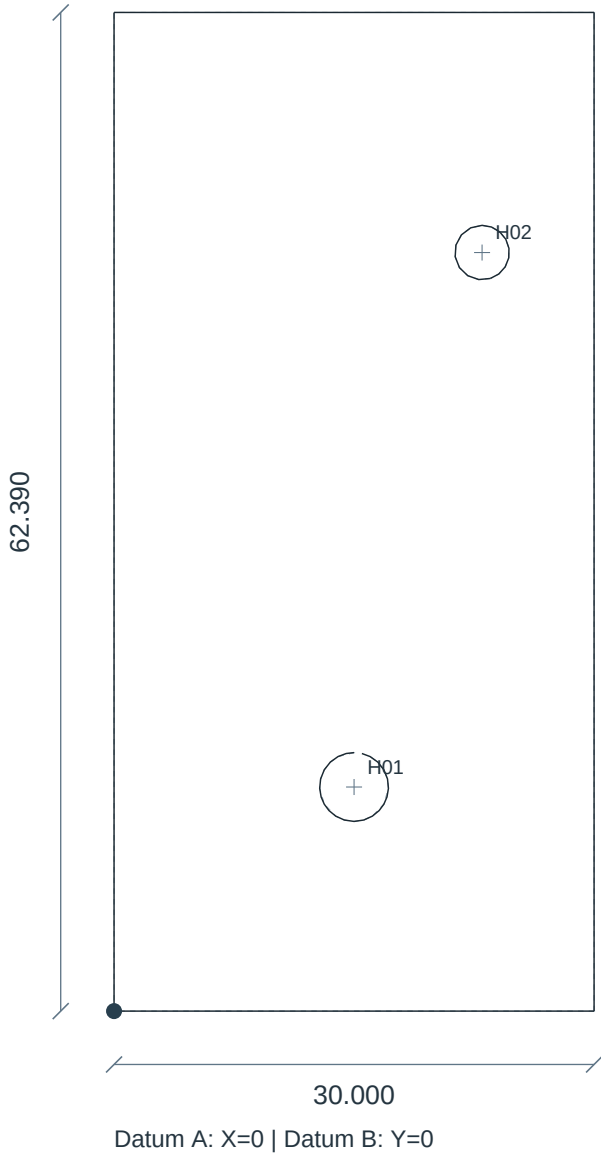
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 203.353, 168.430, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

CUSTOM_RIGHT_GUIDE_HANGER_0

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	15.000	14.000	DIA 4.300
H02	23.000	47.390	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

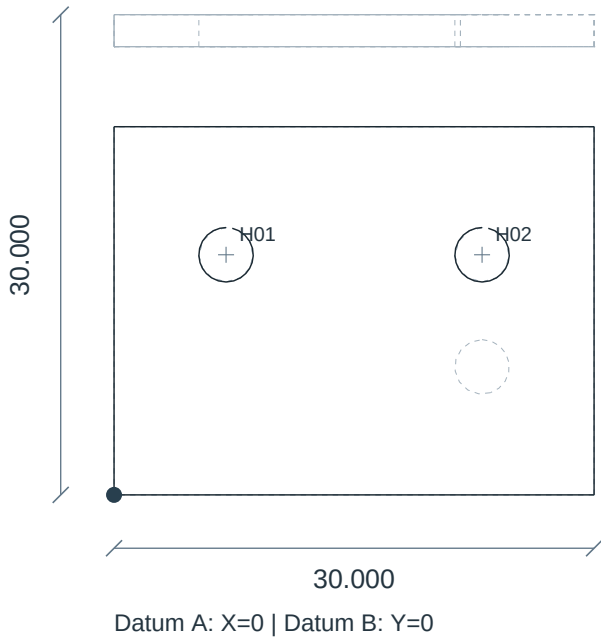
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 203.353, 213.430, 72.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

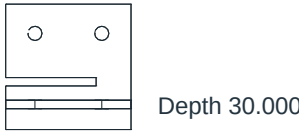
CUSTOM_RIGHT_GUIDE_ANGLE_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 203.353, 213.430, 111.390
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_GUIDE_BUSH_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 214.853, 221.430, 89.500
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

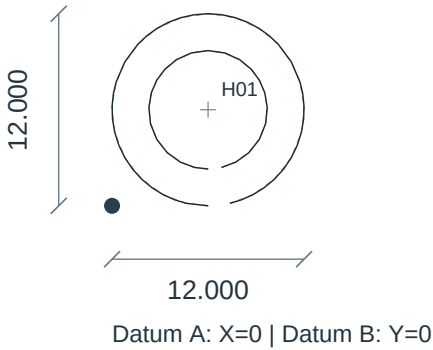
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_STOPPER_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



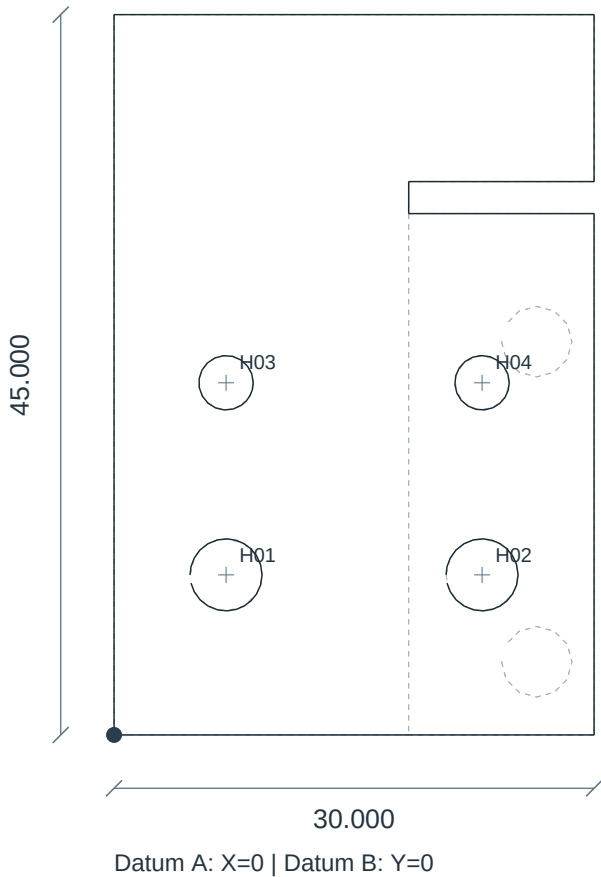
Assembly datum XYZ: 212.353, 213.430, 92.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_GUIDE_FOOT_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	7.000	10.000	DIA 4.500
H02	23.000	10.000	DIA 4.500
H03	7.000	22.000	DIA 3.400
H04	23.000	22.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



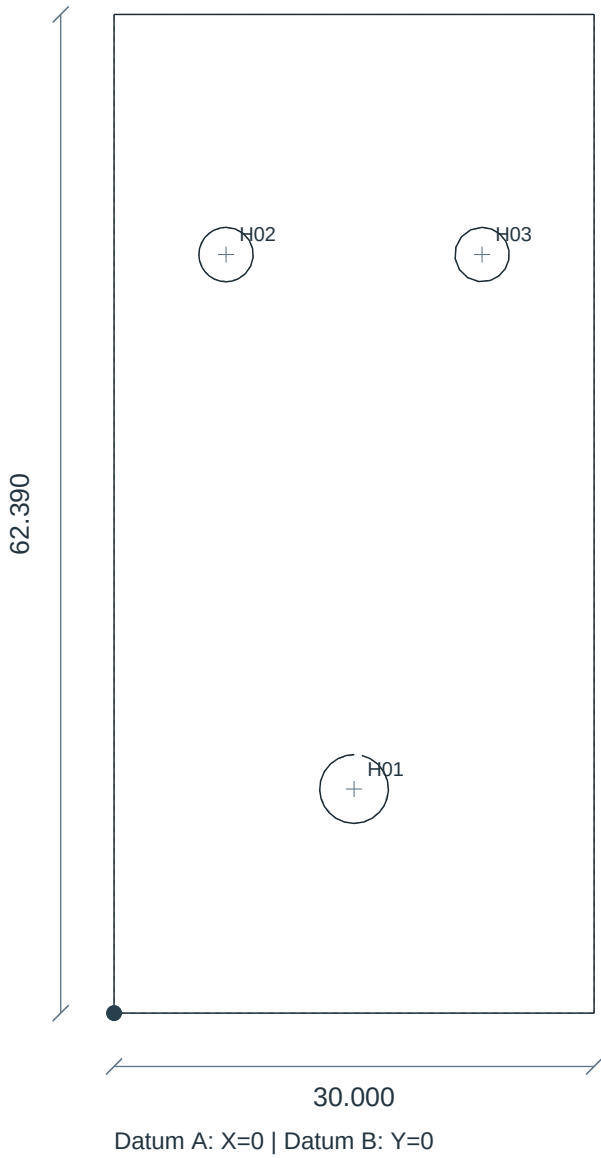
Assembly datum XYZ: 46.587, 168.430, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_GUIDE_HANGER_1

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	15.000	14.000	DIA 4.300
H02	7.000	47.390	DIA 3.400
H03	23.000	47.390	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 46.587, 213.430, 72.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

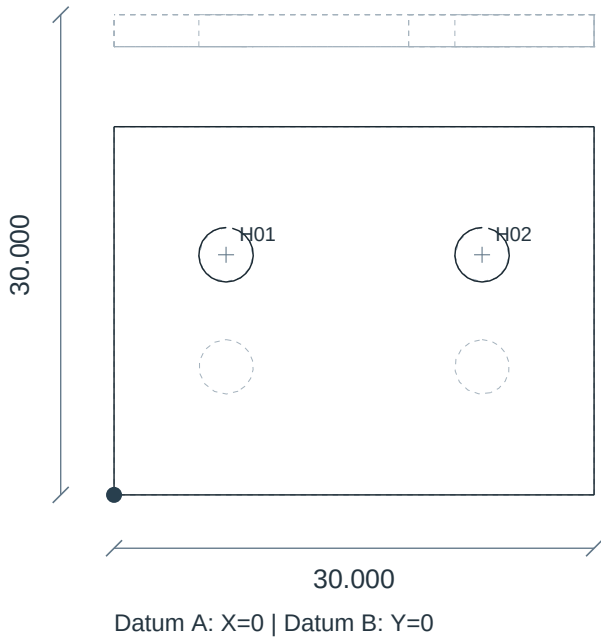
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

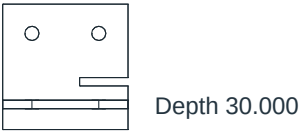
CUSTOM_RIGHT_GUIDE_ANGLE_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 30.000 mm

ID	X	Y	SIZE
H01	7.000	15.000	DIA 3.400
H02	23.000	15.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 46.587, 213.430, 111.390
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_GUIDE_BUSH_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
H01	3.500	3.500	DIA 4.300



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

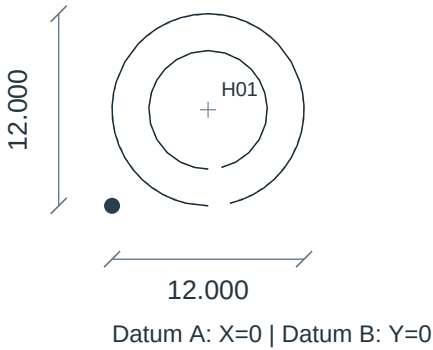
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 58.087, 221.430, 89.500
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

CUSTOM_RIGHT_STOPPER_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 9.000 mm

ID	X	Y	SIZE
H01	6.000	6.000	DIA 7.400



SIDE EDGE PROJECTION - X / DEPTH



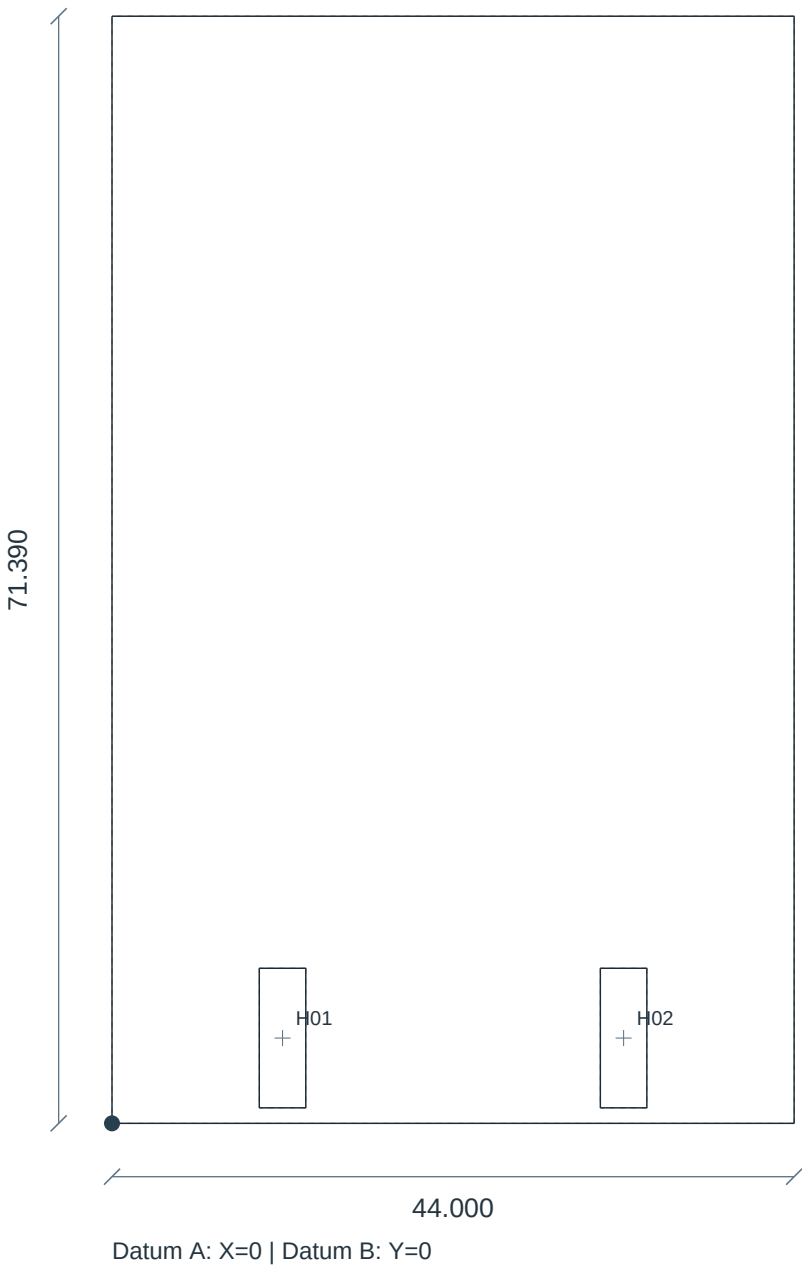
Assembly datum XYZ: 55.587, 213.430, 92.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_RIGHT_SWITCH_BRACKET

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	11.000	5.500	3.000 x 9.000
H02	33.000	5.500	3.000 x 9.000

SIDE EDGE PROJECTION - X / DEPTH



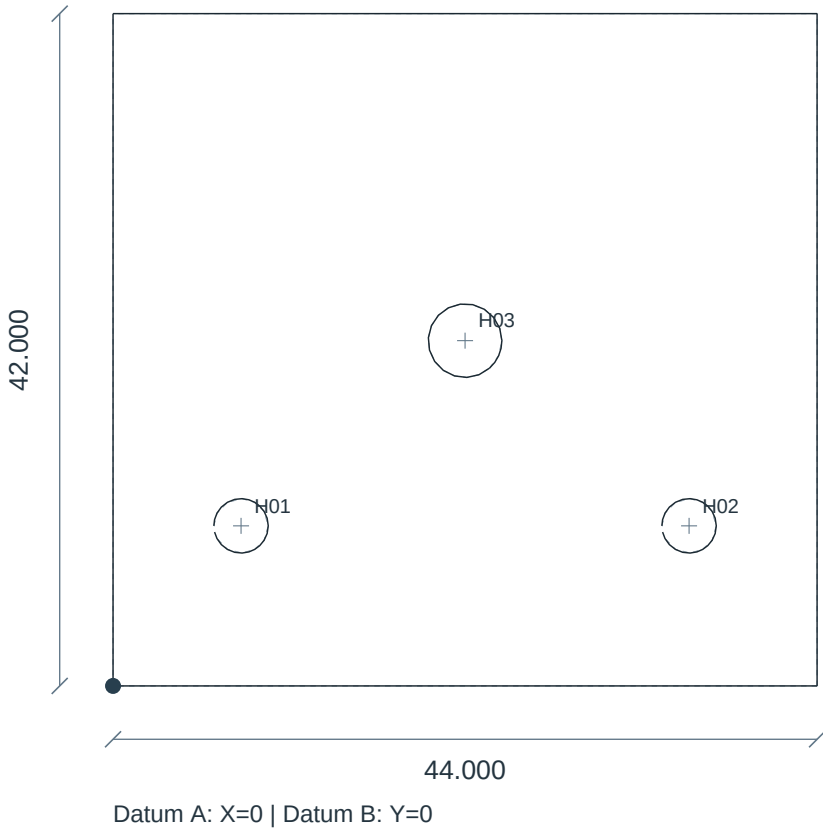
Assembly datum XYZ: 117.970, 213.430, 63.000
u (1.0, 0.0, 0.0); v (-0.0, 0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_RIGHT_SWITCH_FOOT

Qty1 | PVC foam / Fomex | thickness/depth 5.000 mm



ID	X	Y	SIZE
H01	8.000	10.000	DIA 3.400
H02	36.000	10.000	DIA 3.400
H03	22.000	21.570	DIA 4.600

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

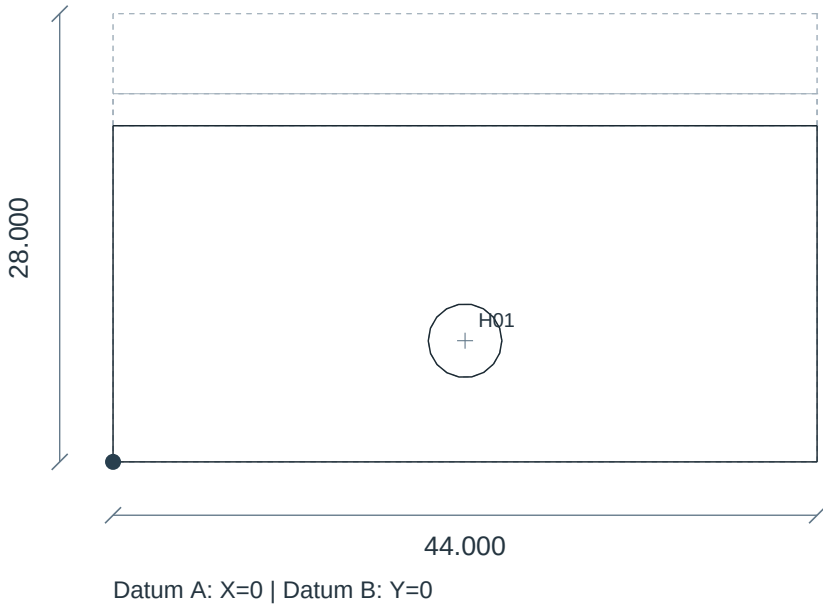
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 117.970, 171.430, 134.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

CUSTOM_RIGHT_SWITCH_ANGLE

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 24.000 mm

ID	X	Y	SIZE
H01	22.000	7.570	DIA 4.600



SIDE EDGE PROJECTION - X / DEPTH



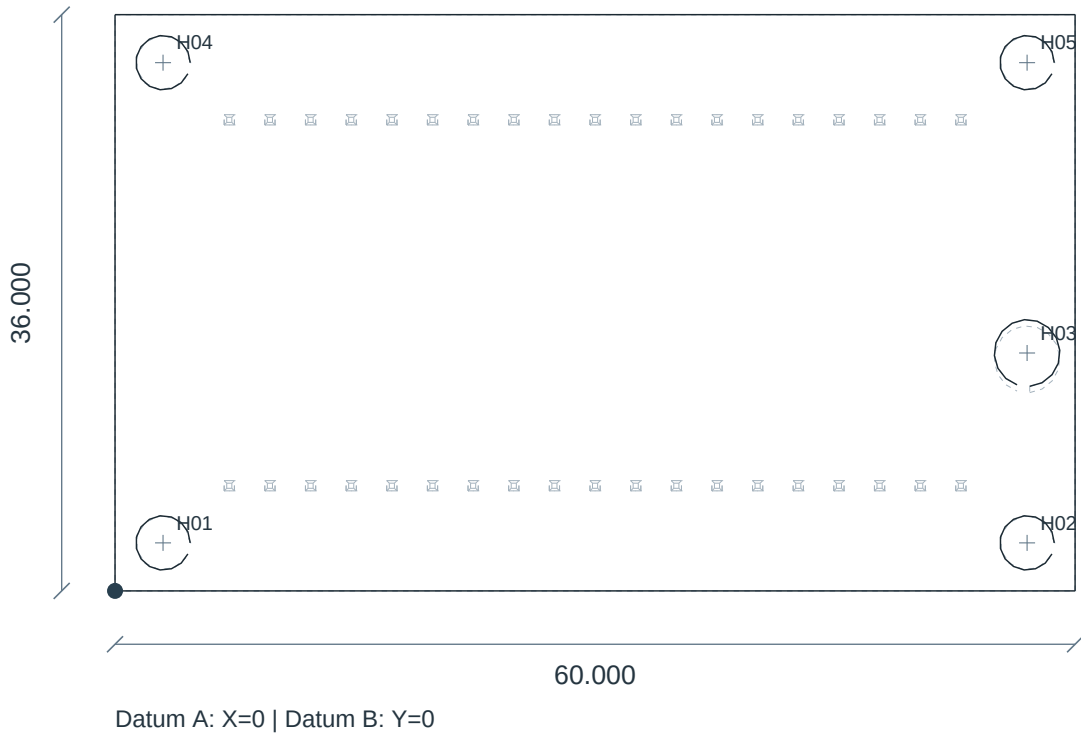
Assembly datum XYZ: 117.970, 185.430, 117.390
u (1.0, 0.0, 0.0); v (-0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_ESP32_CARRIER

Qty1 | PVC foam / Fomex | thickness/depth 2.000 mm



ID	X	Y	SIZE
H01	3.000	3.000	DIA 3.400
H02	57.000	3.000	DIA 3.400
H03	57.000	14.863	4.100 x 4.178
H04	3.000	33.000	DIA 3.400
H05	57.000	33.000	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

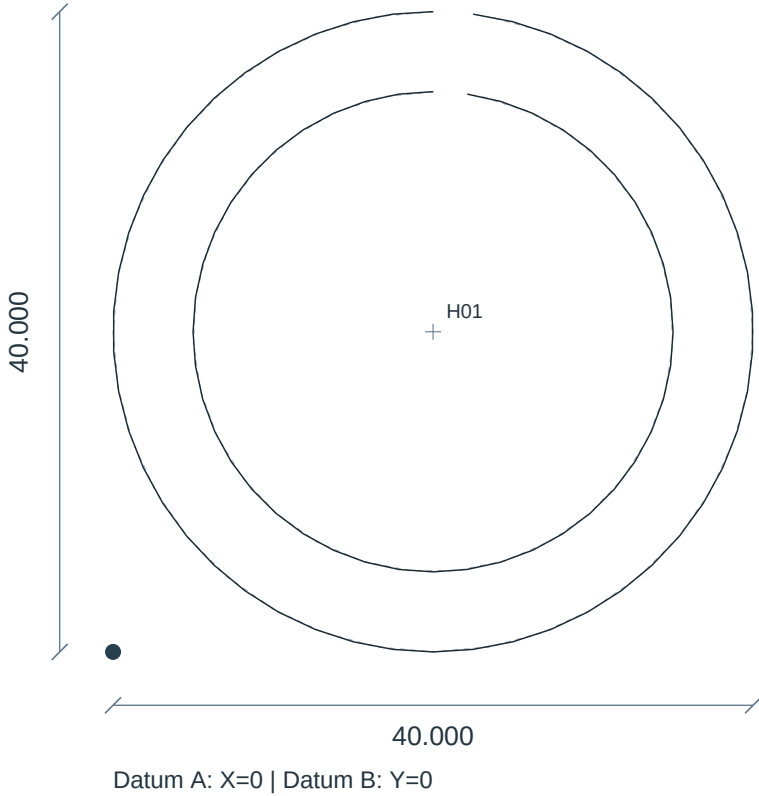
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: 123.000, 118.000, 260.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_ESTOP_ADAPTER_RING

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 2.000 mm

ID	X	Y	SIZE
H01	20.000	20.000	DIA 30.000



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

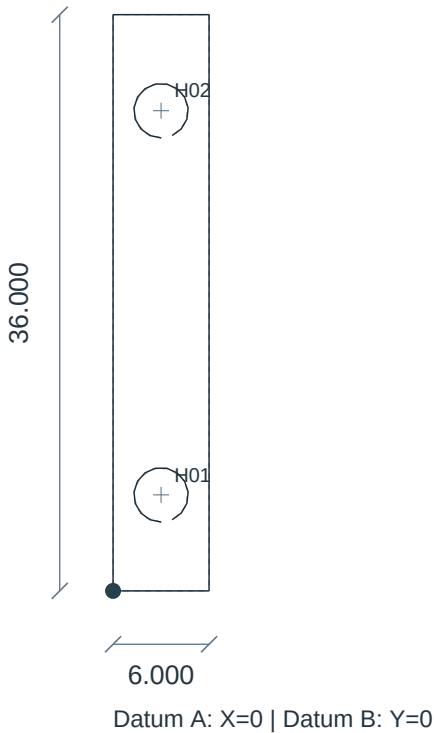
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 295.000, 0.000, 205.000
u (0.0, 1.0, 0.0); v (0.0, 0.0, 1.0)

CUSTOM_SPEAKER_MOUNT_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 3.000 mm

ID	X	Y	SIZE
H01	3.000	6.000	DIA 3.400
H02	3.000	30.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

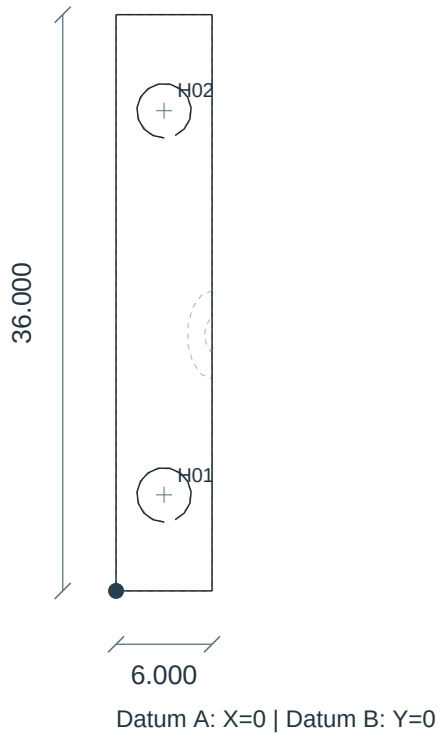
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: -10.000, 118.000, 216.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

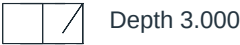
CUSTOM_SPEAKER_MOUNT_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 3.000 mm

ID	X	Y	SIZE
H01	3.000	6.000	DIA 3.400
H02	3.000	30.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

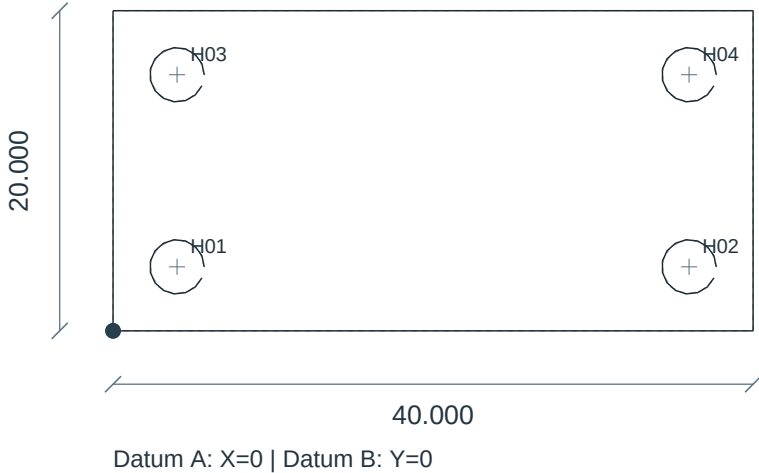
3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

Assembly datum XYZ: -10.000, 198.000, 216.000
u (0.0, 1.0, 0.0); v (-0.0, 0.0, -1.0)

CUSTOM_5V_DISTRIBUTOR

Qty1 | PVC foam / Fomex | thickness/depth 1.600 mm

ID	X	Y	SIZE
H01	4.000	4.000	DIA 3.400
H02	36.000	4.000	DIA 3.400
H03	4.000	16.000	DIA 3.400
H04	36.000	16.000	DIA 3.400



SIDE EDGE PROJECTION - X / DEPTH



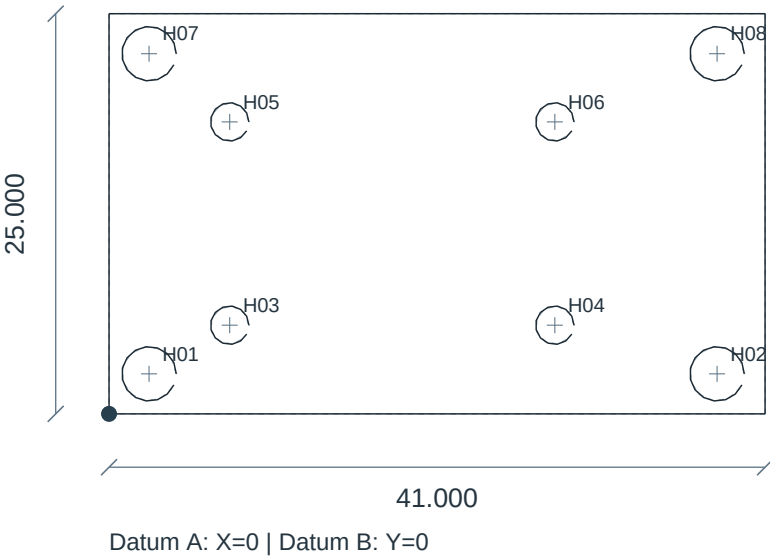
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

Assembly datum XYZ: 228.000, 185.000, 270.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

CUSTOM_ADC_CARRIER

Qty1 | PVC foam / Fomex | thickness/depth 2.000 mm



ID	X	Y	SIZE
H01	2.500	2.500	DIA 3.400
H02	38.000	2.500	DIA 3.400
H03	7.540	5.540	DIA 2.400
H04	27.860	5.540	DIA 2.400
H05	7.540	18.240	DIA 2.400
H06	27.860	18.240	DIA 2.400
H07	2.500	22.500	DIA 3.400
H08	38.000	22.500	DIA 3.400

SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 221.000, 117.000, 265.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

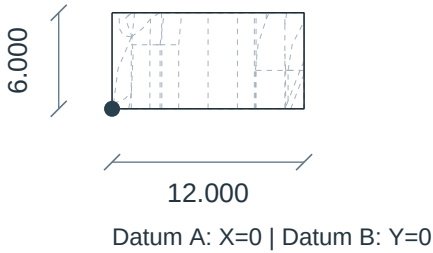
DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_HARNESS_CLAMP_0

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 10.000 mm

ID	X	Y	SIZE
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No inner loops on the datum face.



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 9.000, 208.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, 1.0, 0.0)

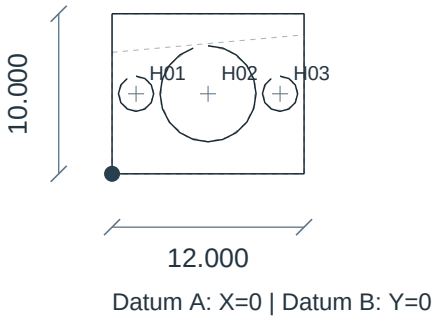
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_HARNESS_CLAMP_1

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 74.000, 214.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

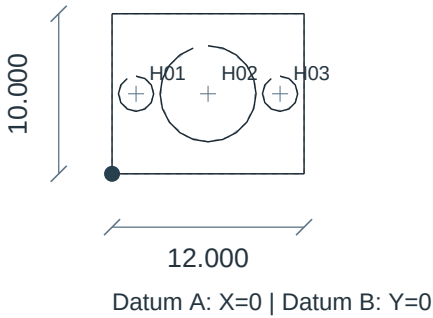
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_HARNESS_CLAMP_2

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 144.000, 214.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

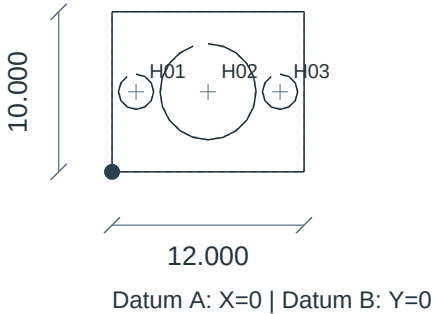
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_HARNESS_CLAMP_3

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 229.000, 214.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

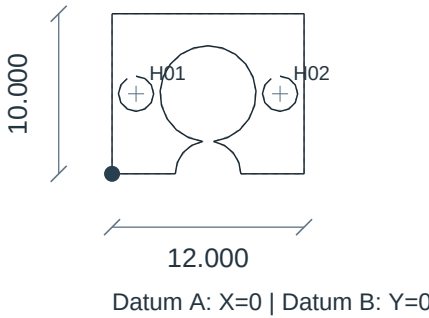
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_HARNESS_CLAMP_4

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 9.000, -15.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

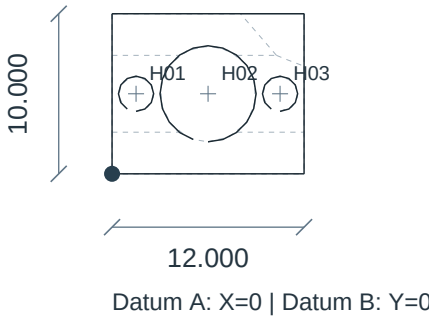
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_HARNESS_CLAMP_5

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 84.000, -21.000, 250.000
u (1.0, 0.0, 0.0); v (0.0, 0.0, -1.0)

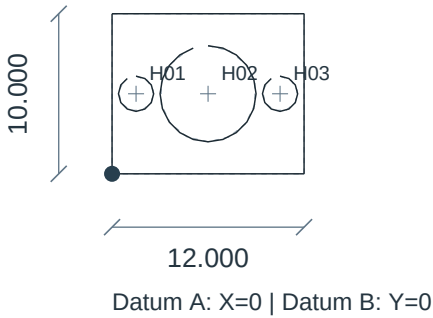
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

3D relief / multiple depths: individual STEP and side projection control. Dashed projected edges are not through-cut instructions.

CUSTOM_HARNESS_CLAMP_6

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 174.000, -15.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

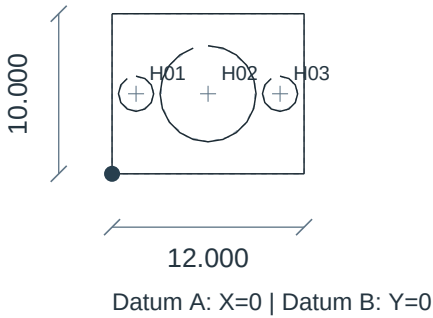
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_HARNESS_CLAMP_7

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 6.000 mm

ID	X	Y	SIZE
H01	1.500	5.000	DIA 2.200
H02	6.000	5.000	DIA 6.000
H03	10.500	5.000	DIA 2.200



SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 274.000, -15.000, 240.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

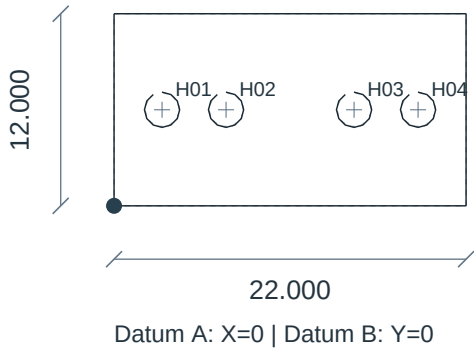
VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

CUSTOM_TILT_HORN_COUPLER

Qty1 | Stock material - VERIFY ON HARDWARE | thickness/depth 3.000 mm

ID	X	Y	SIZE
H01	3.000	6.000	DIA 2.200
H02	7.000	6.000	DIA 2.200
H03	15.000	6.000	DIA 2.200
H04	19.000	6.000	DIA 2.200



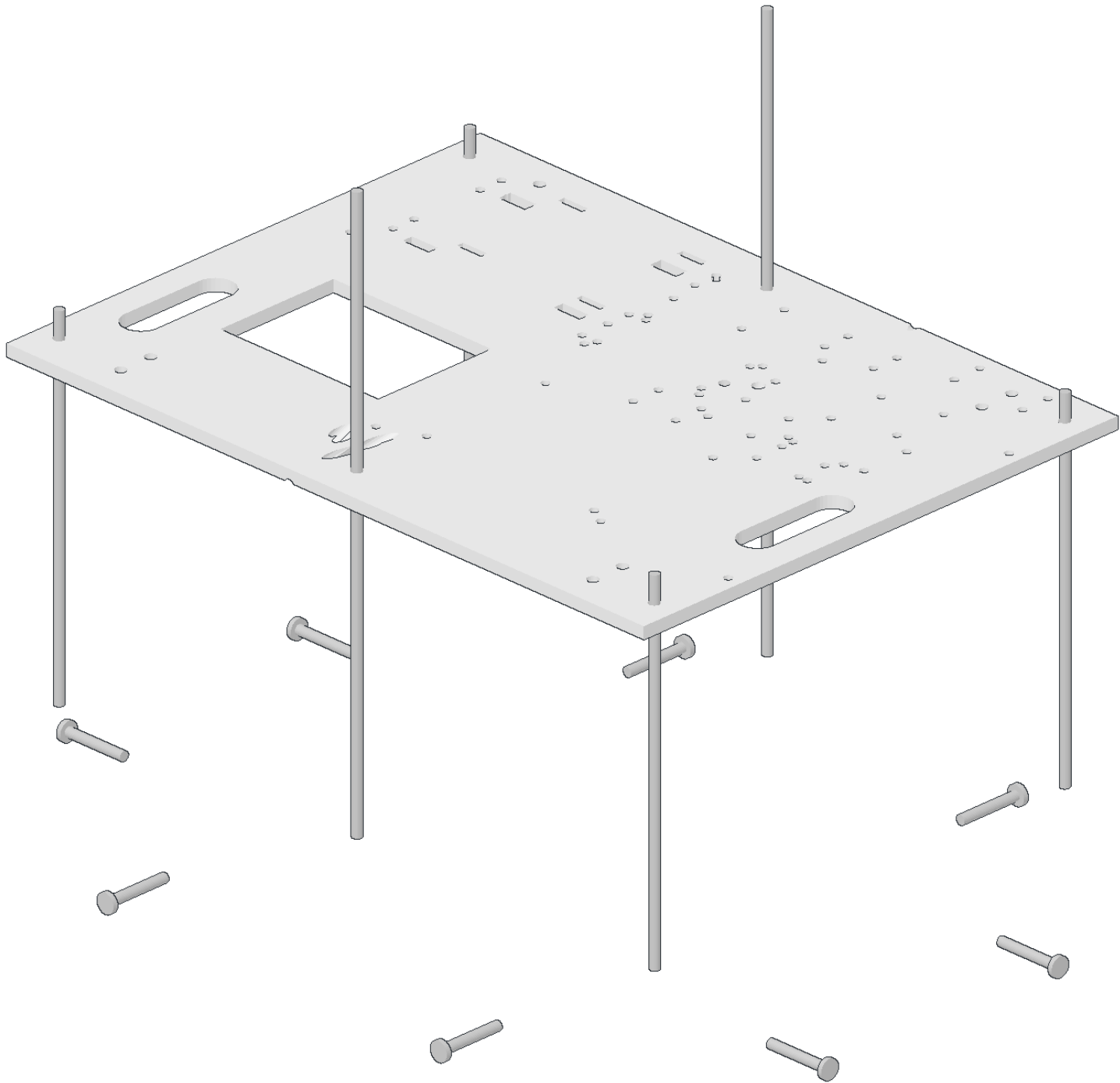
SIDE EDGE PROJECTION - X / DEPTH



Assembly datum XYZ: 237.000, 154.715, 344.000
u (1.0, 0.0, 0.0); v (0.0, -0.0, 1.0)

VERIFIED CAD NOMINAL: geometry measured from the final model; hardware interfaces may require ASSEMBLY ADJUST.

DXF through-cut profile supplied; units mm, arc tolerance0.01; no kerf allowance.

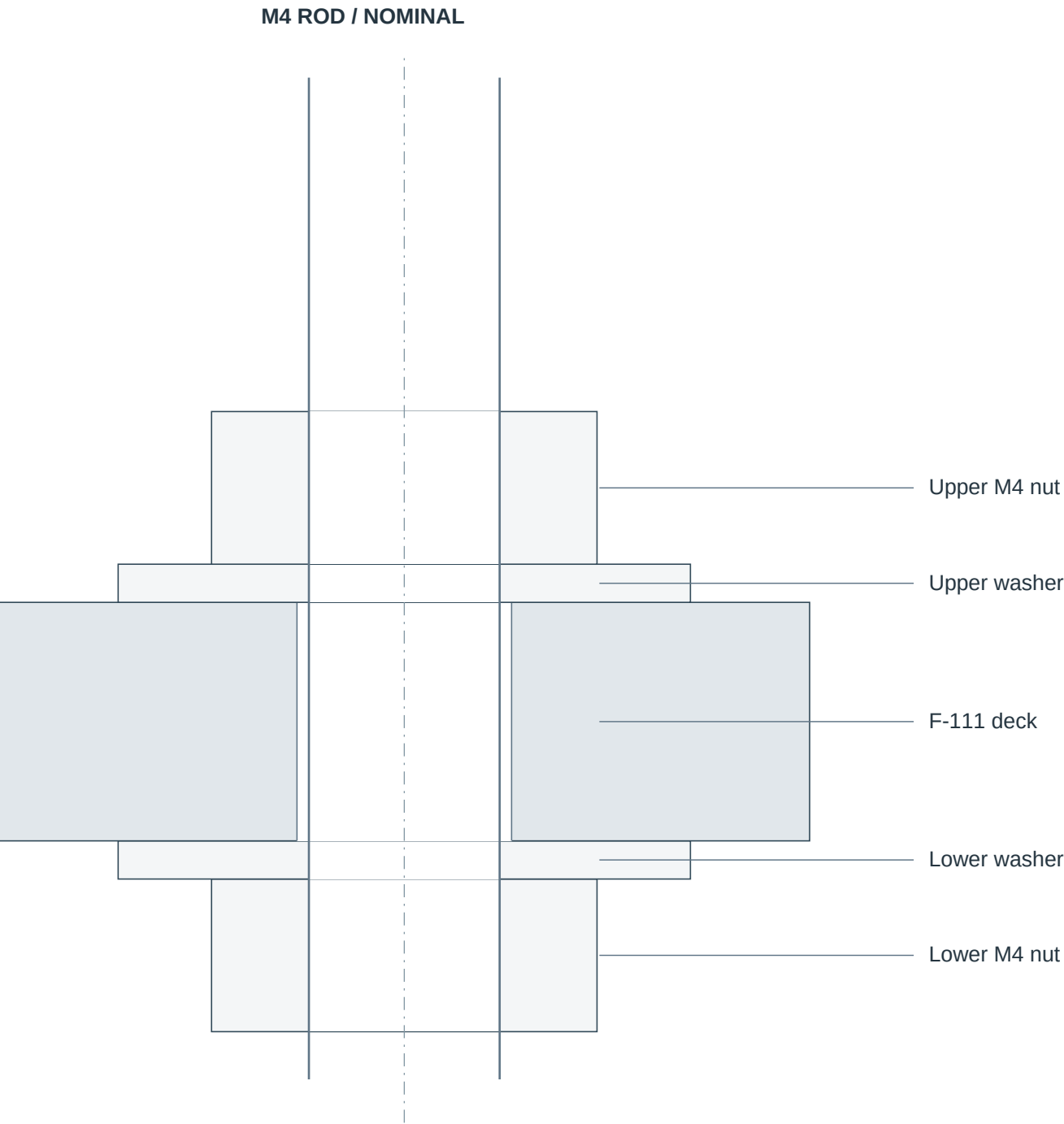


POST STACK

- 01. M4 rod supports deck at Z255..260mm in the model.
- 02. Use washer bearing faces on each foam-sheet clamp location and lock with modeled nuts.
- 03. Preserve clearances to board stand-offs and cable passages.
- 04. Rod length, nut engagement, tightening force and actual material strength: VERIFY ON HARDWARE.

Use final STEP part relationships and the matching F drawing. Standard hardware heads / holes are nominal; verify purchased fasteners, clamp load and tool access on hardware.

LOCAL STACK AT ROD 0 / DECK - CAD Z LEVELS

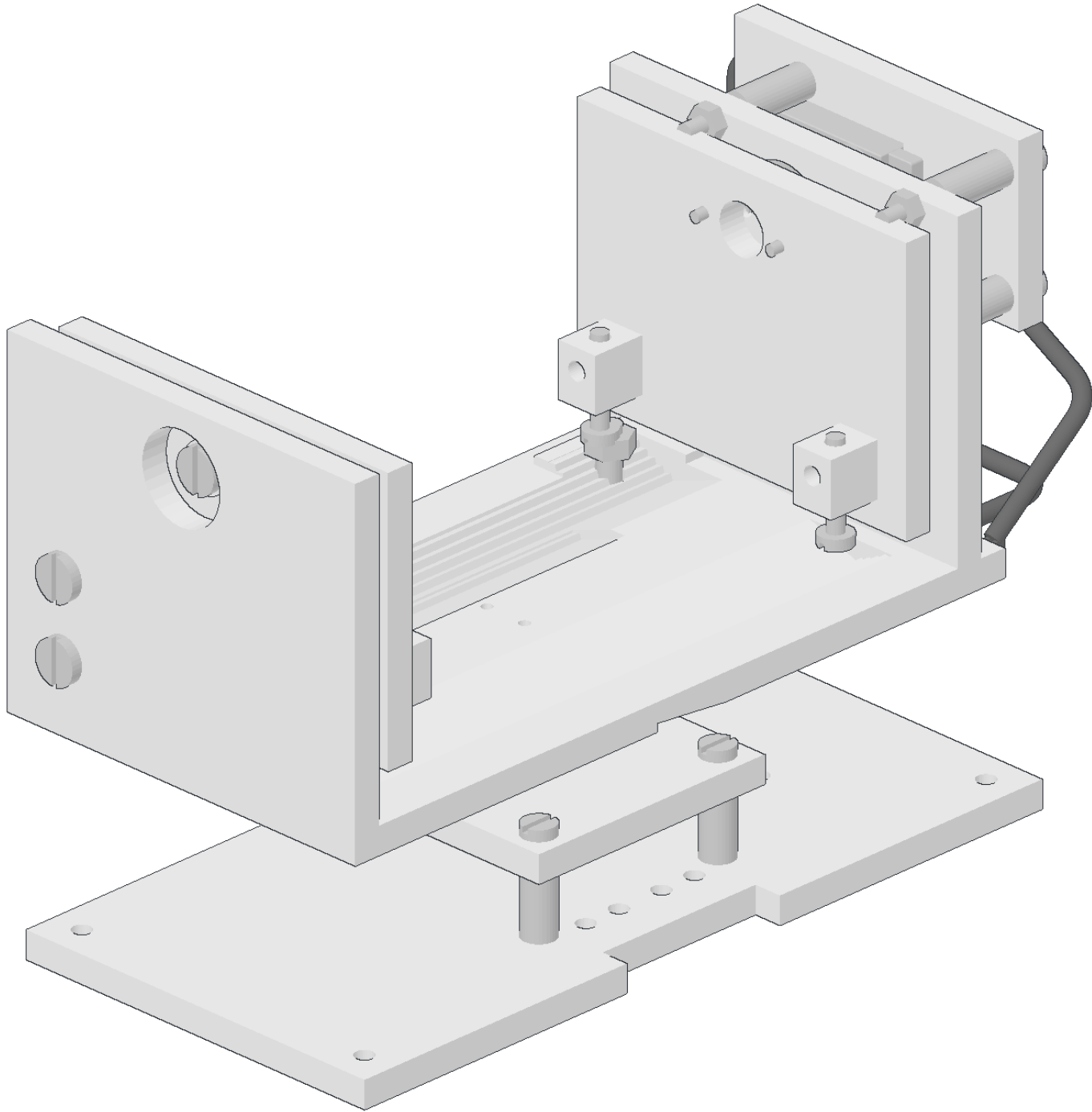


LAYER	Z MIN	Z MAX	T mm
Lower M4 nut	251.000	254.200	3.200
Lower washer	254.200	255.000	0.800
F-111 deck	255.000	260.000	5.000
Upper washer	260.000	260.800	0.800
Upper M4 nut	260.800	264.000	3.200

The deck is clamped between two washers and two nuts. This local stack follows the existing CAD heights. Plate width is cropped for visibility; threads are nominal cylinders, not thread profiles.

ROD IDs	QTY	CAD CUT LENGTH mm
0 / 1 / 4 / 5	4	146.610
2 / 3	2	239.610

Local X-Z stack representation, not a new section cut or altered CAD. Select actual washer/nut stock to match nominal interfaces. Confirm engagement, material bearing and locking method on the physical assembly.



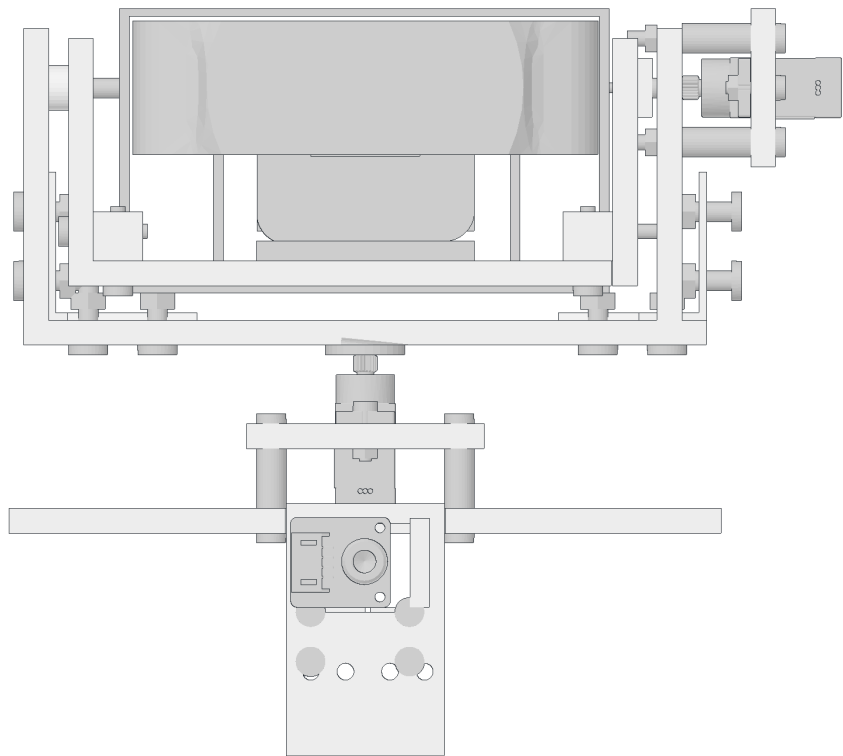
CAMERA SUPPORT

- 01. Pan servo fixes to its mounting plate; horn drives yoke.
- 02. Two arms support the tilt axis and servo mount.
- 03. Tilt cheek / bushing and horn coupler locate the rotating platform.
- 04. Trimmed L brackets and yoke relief match the final STEP; keep mounting/support faces.

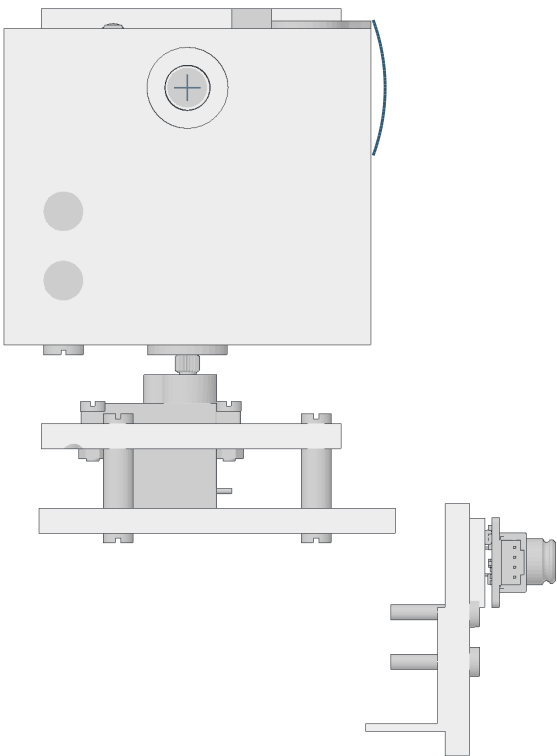
Use final STEP part relationships and the matching F drawing. Standard hardware heads / holes are nominal; verify purchased fasteners, clamp load and tool access on hardware.

TILT +/-20 deg | range reference

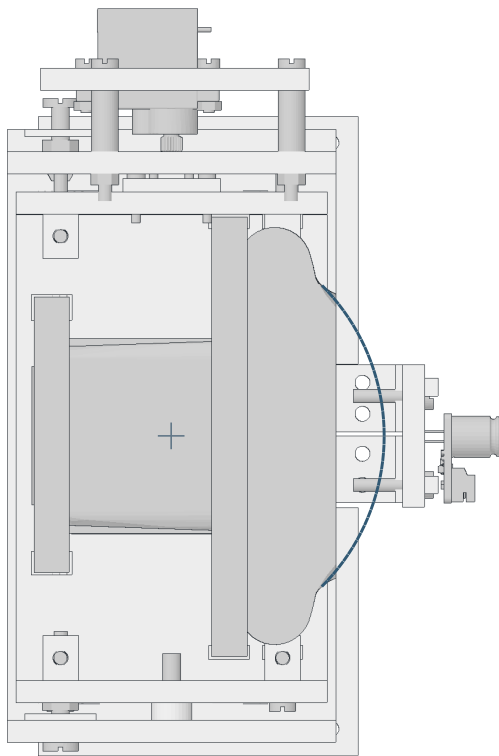
PAN +/-45 deg | range reference



FRONT +X



LEFT -Y

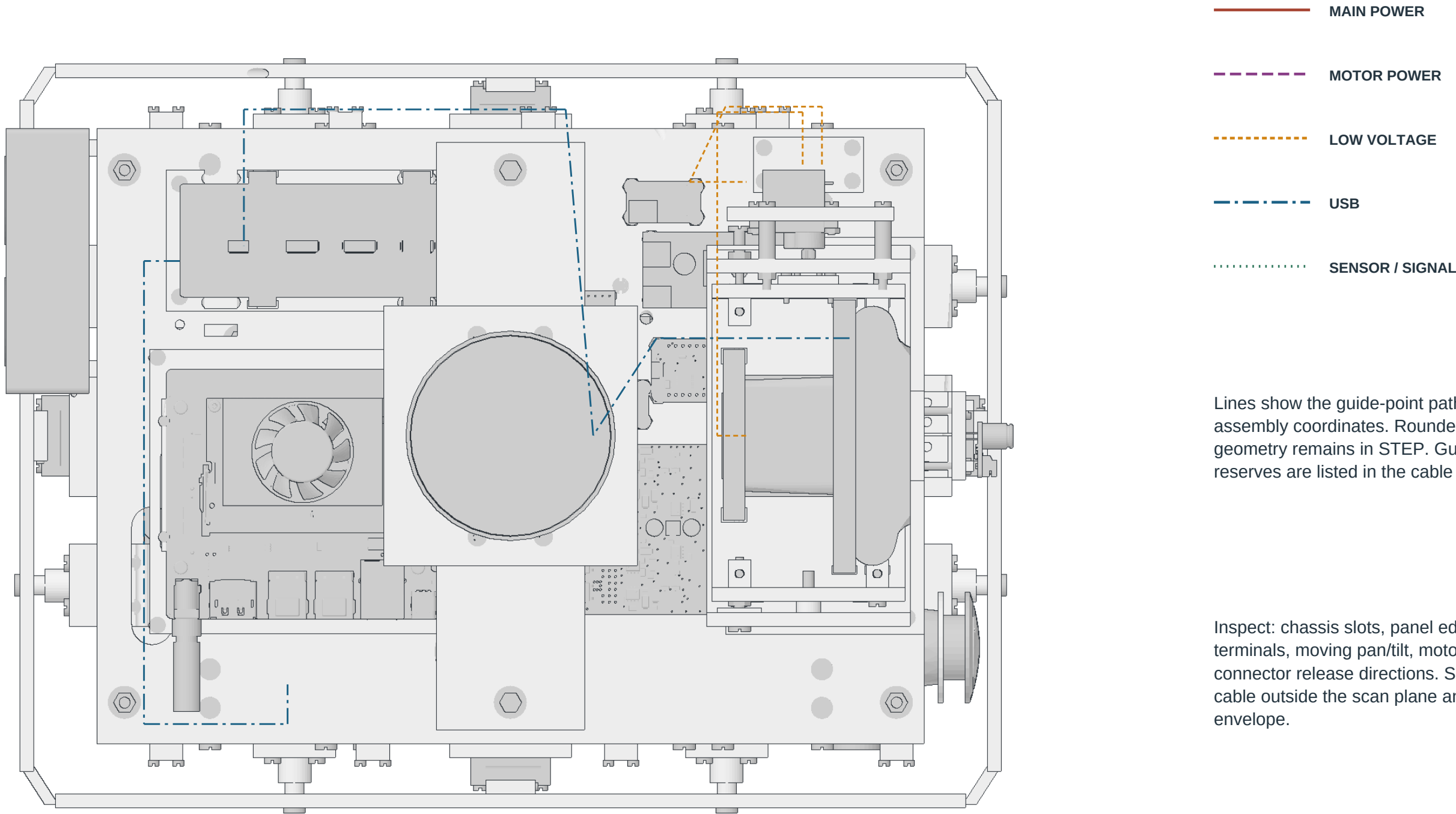


TOP +Z

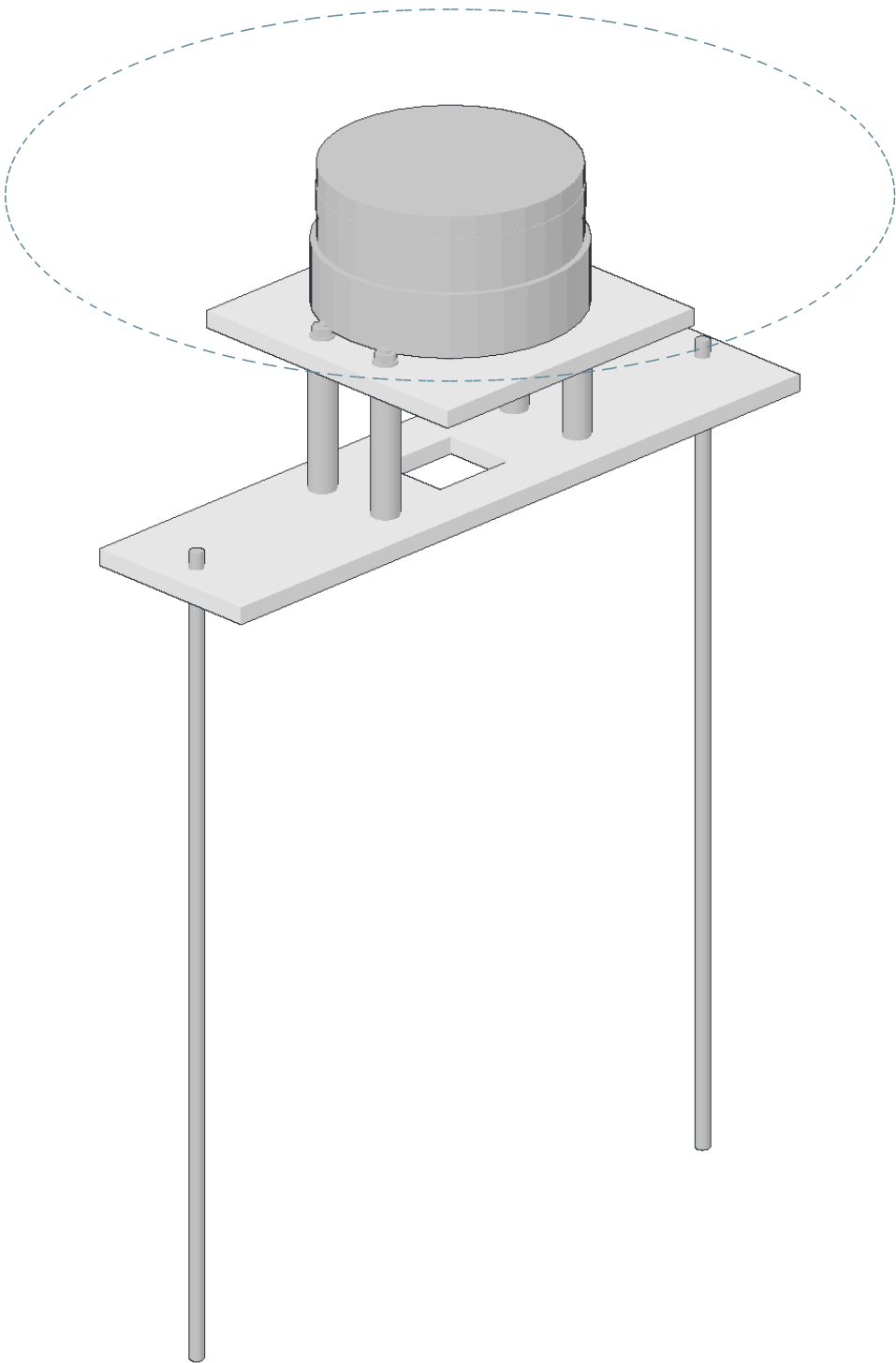
Pan axis: (248,96.715,298), +Z. Tilt axis: (248,96.715,350), +Y in the neutral pan frame. Prior baseline clearance check: Pan -45..+45 deg and Tilt -20..+20 deg. The coaxial bushing is an intentional mating interface.

The yoke and two L brackets include continuous tilt clearance relief. C920 is retained by saddle / pad / straps. Its exact1/4-20 center is not a fabricated datum. Thermal sensor remains fixed to the front wall below the moving camera.

C920 USB: camera guide -> moving pan service guide -> rear service loop -> hub -> Jetson. Prior baseline: routes checked at 35 poses; not rerun for this reissue. Physical flex, strain relief and servo load require hardware adjustment.



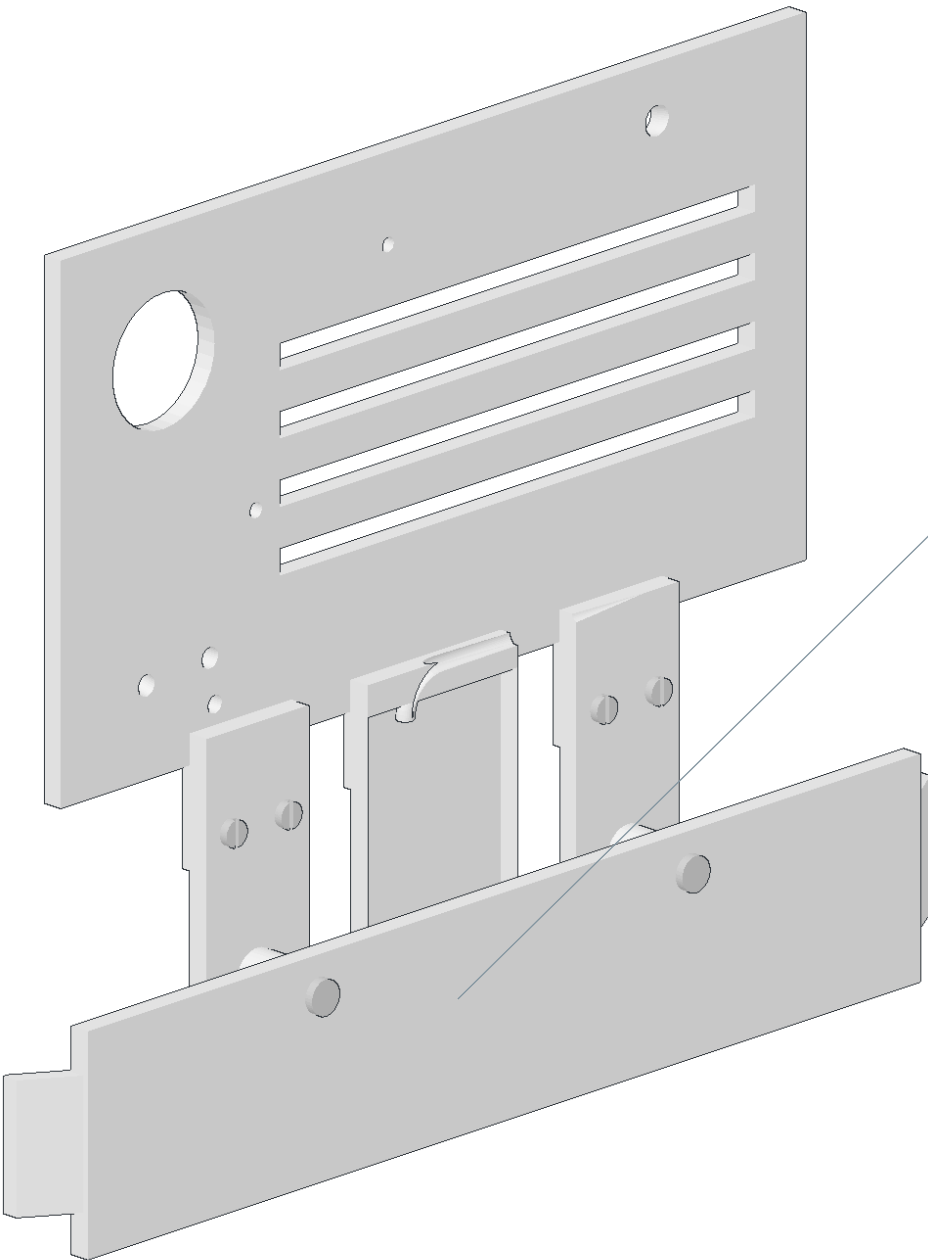
Physical cable routes are separate from functional circuits. Drawing line styles remain distinct in monochrome. Conductor breakout / cavity mapping and exact cable bend radii are VERIFY ON HARDWARE.



SCAN PLANE REFERENCE
The dashed ring is a visualization, not a physical part.
Z=438mm is the raised optical-window envelope
reference; verify the exact beam height on the purchased
sensor.

Mount plate above the bridge on four40mm risers. Keep
cable slack below the scan window. Maintain a level plate
and inspect all360 degrees for obstructions after wiring.

The current scan reference lies above the camera / brackets and support rods. Optical obscuration, beam height and LiDAR performance are hardware checks. Plate / bridge / riser geometry is controlled by the final STEP.

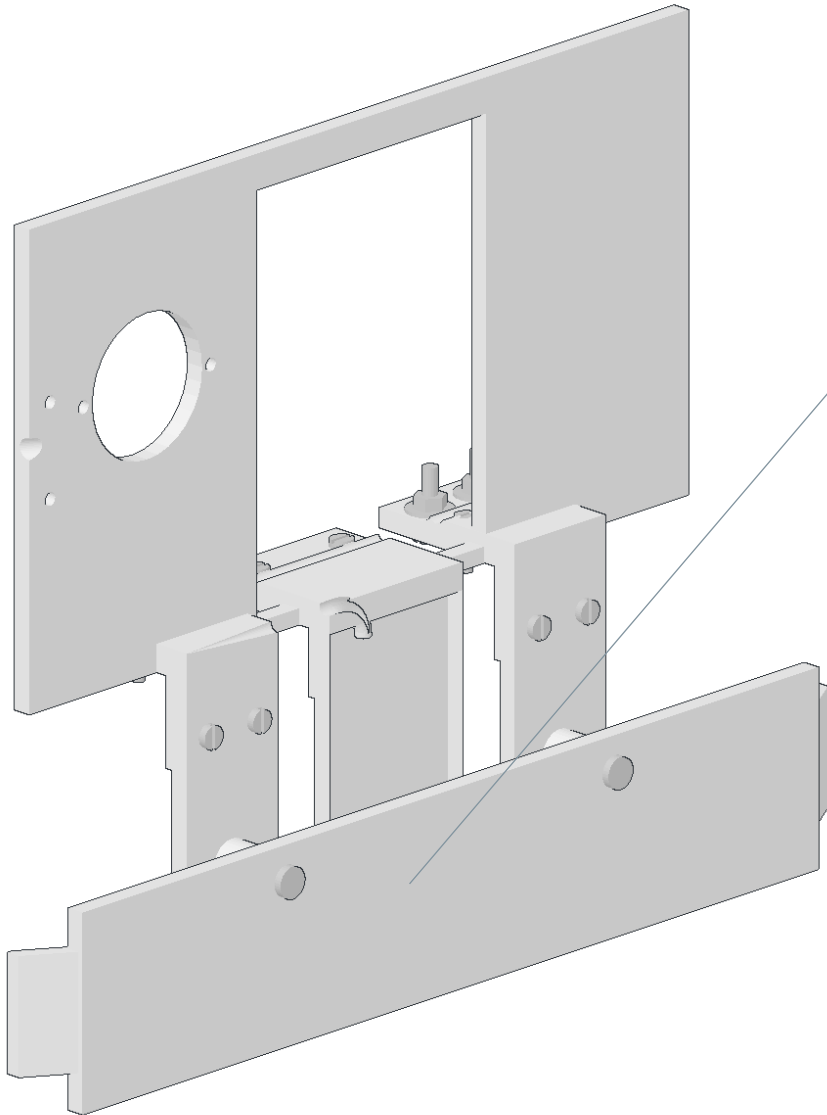


FRONT BUMPER
Named switch LS-F closes the modeled mounting relationship.

Impact pushes the bumper inward, toward the chassis. Guide rods constrain motion; the striker depresses the switch lever. Springs return the bumper. Verify switch actuation before the mechanical stop and release after unloading.

← **Inward direction -X**

Wire COM/NC according to the project signal table; identify actual terminal markings on the switch. Stroke and preload are ASSEMBLY ADJUST, not a newly specified numerical travel.



028

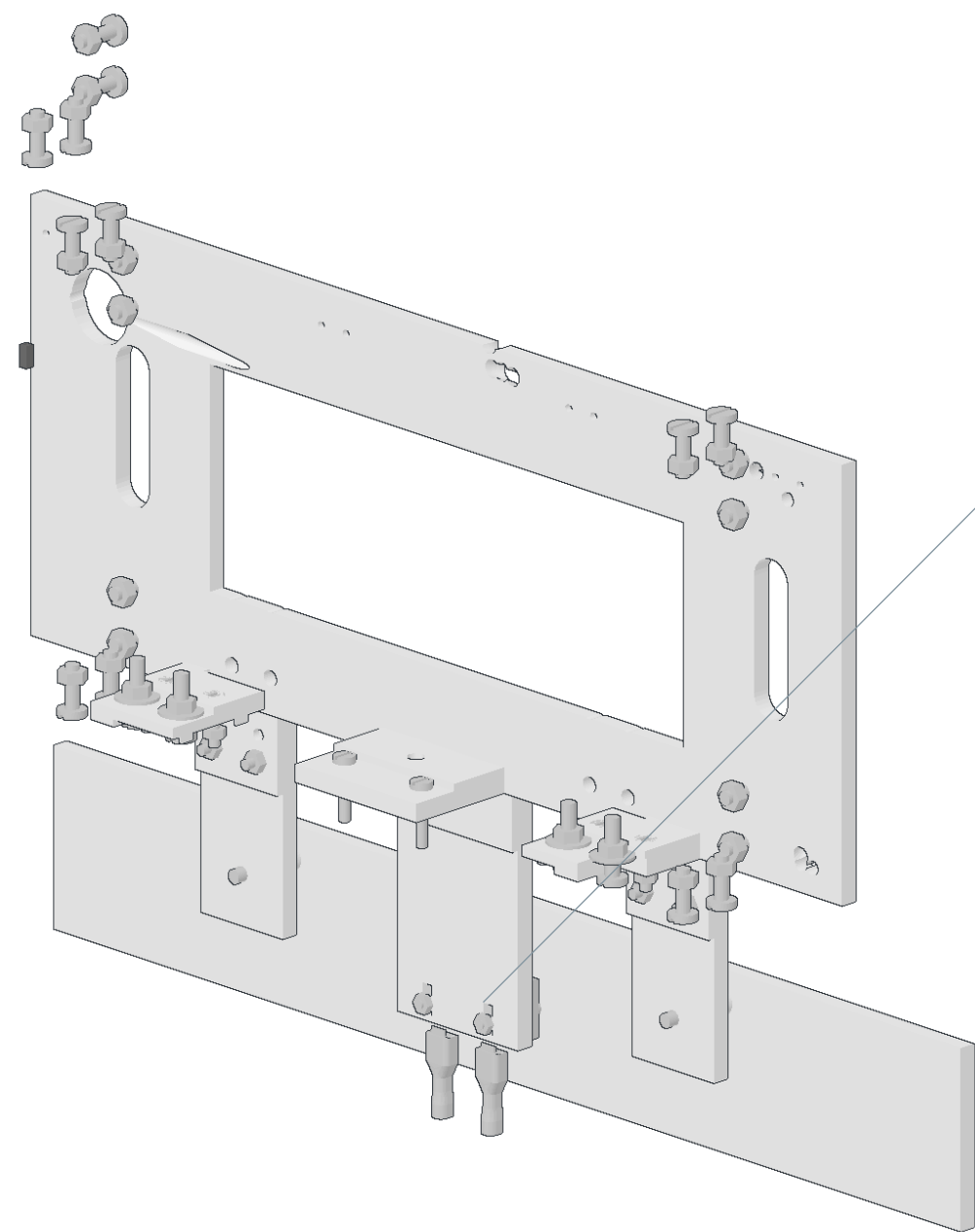
REAR BUMPER
Named switch LS-R closes the modeled mounting relationship.

Impact pushes the bumper inward, toward the chassis. Guide rods constrain motion; the striker depresses the switch lever. Springs return the bumper. Verify switch actuation before the mechanical stop and release after unloading.



Inward direction +X

Wire COM/NC according to the project signal table; identify actual terminal markings on the switch. Stroke and preload are ASSEMBLY ADJUST, not a newly specified numerical travel.



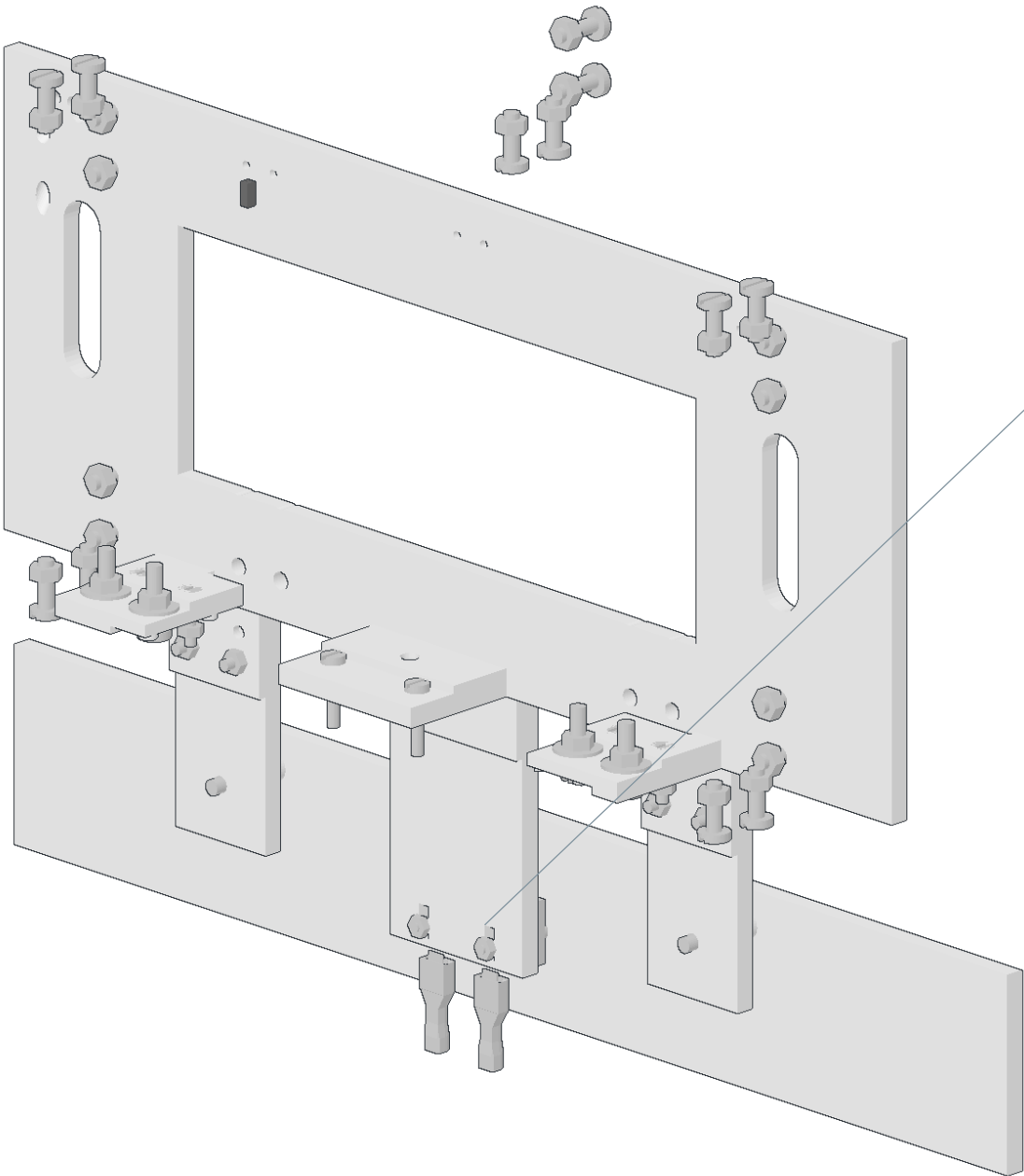
029

LEFT BUMPER
Named switch LS-L closes the modeled mounting relationship.

Impact pushes the bumper inward, toward the chassis. Guide rods constrain motion; the striker depresses the switch lever. Springs return the bumper. Verify switch actuation before the mechanical stop and release after unloading.

← Inward direction +Y

Wire COM/NC according to the project signal table; identify actual terminal markings on the switch. Stroke and preload are ASSEMBLY ADJUST, not a newly specified numerical travel.

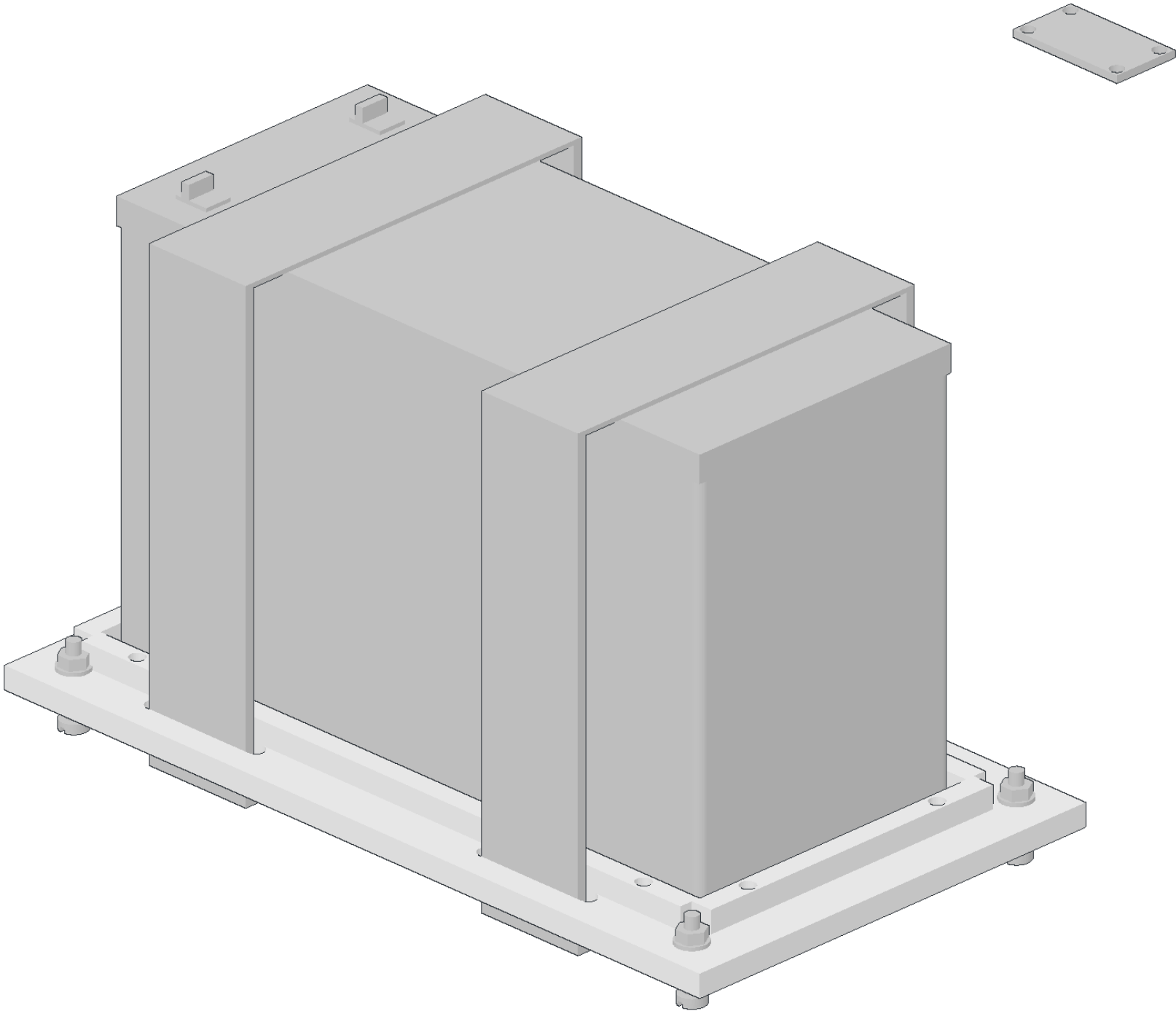


RIGHT BUMPER
Named switch LS-Rt closes the modeled mounting relationship.

Impact pushes the bumper inward, toward the chassis. Guide rods constrain motion; the striker depresses the switch lever. Springs return the bumper. Verify switch actuation before the mechanical stop and release after unloading.

← Inward direction -Y

Wire COM/NC according to the project signal table; identify actual terminal markings on the switch. Stroke and preload are ASSEMBLY ADJUST, not a newly specified numerical travel.

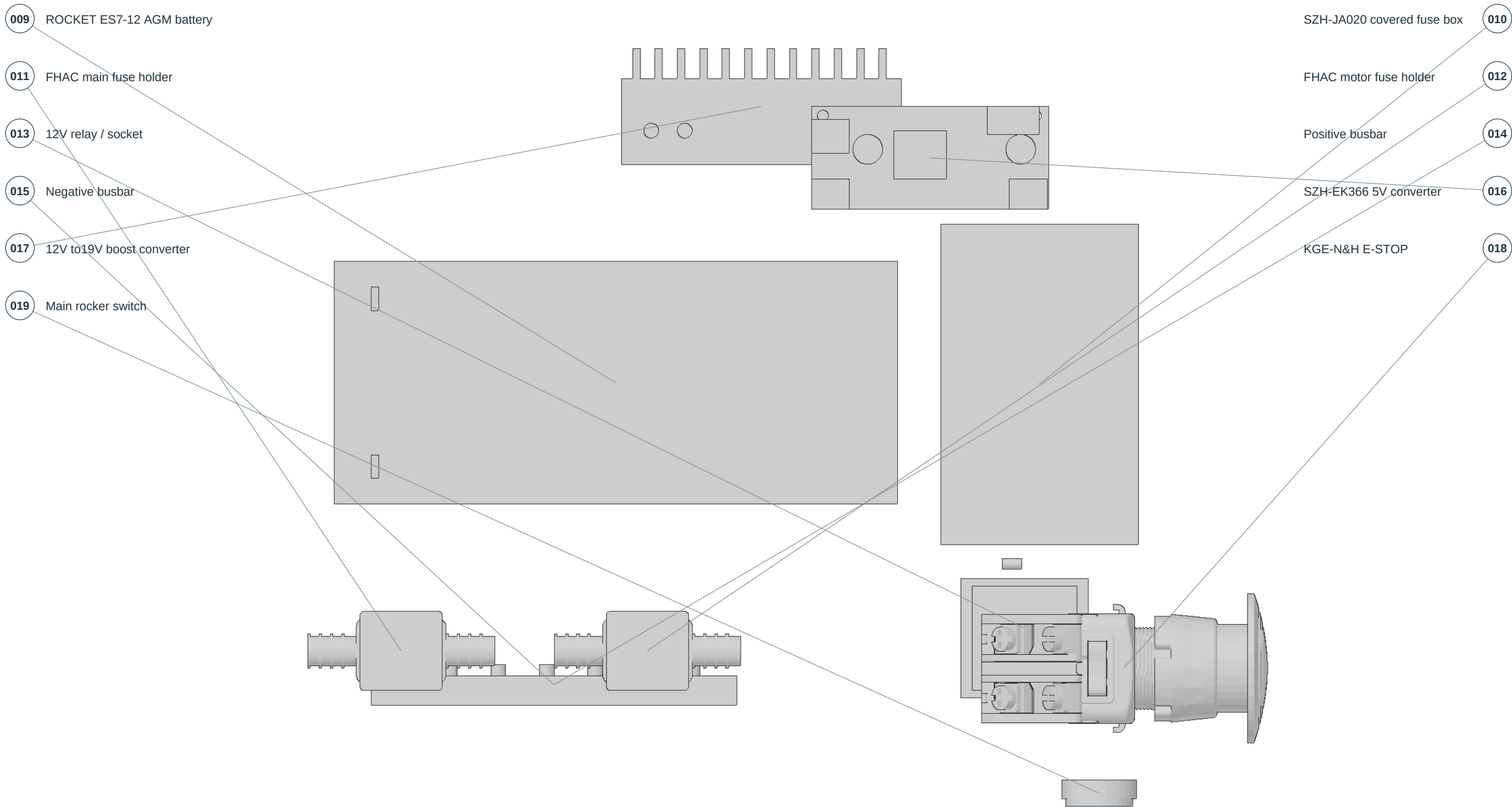


BATTERY RETENTION

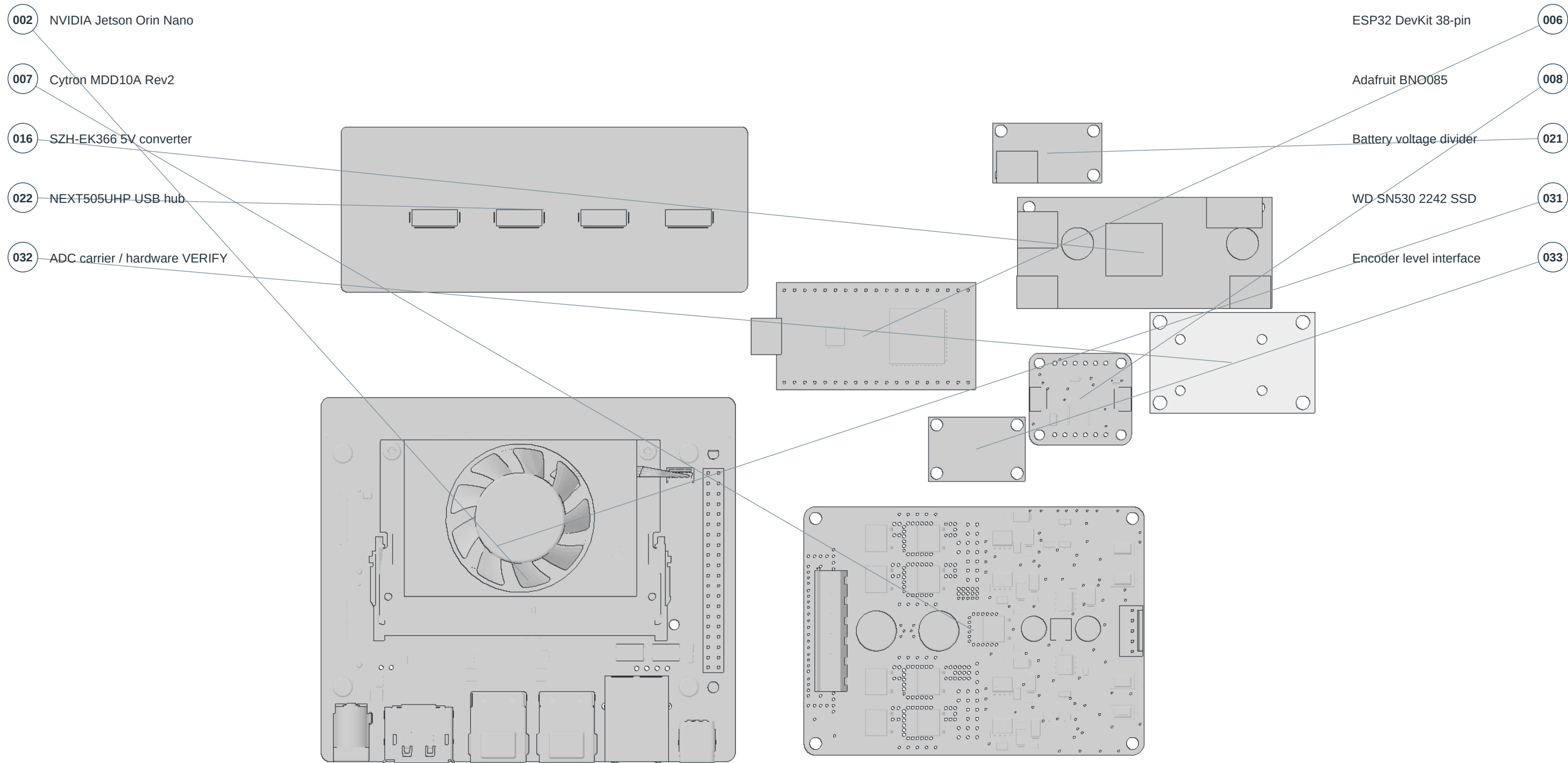
- 01. Set battery onto its support plate between four stops.
- 02. Pass both retaining straps and secure the modeled buckles without covering terminals.
- 03. Route positive cable to the main fuse and return to negative distribution.
- 04. For removal: disconnect supply/harness, remove the rear strip and rear stop, release straps, then withdraw -X per the existing panel notes. Confirm physical passage before final wiring.

Items: 009, 011

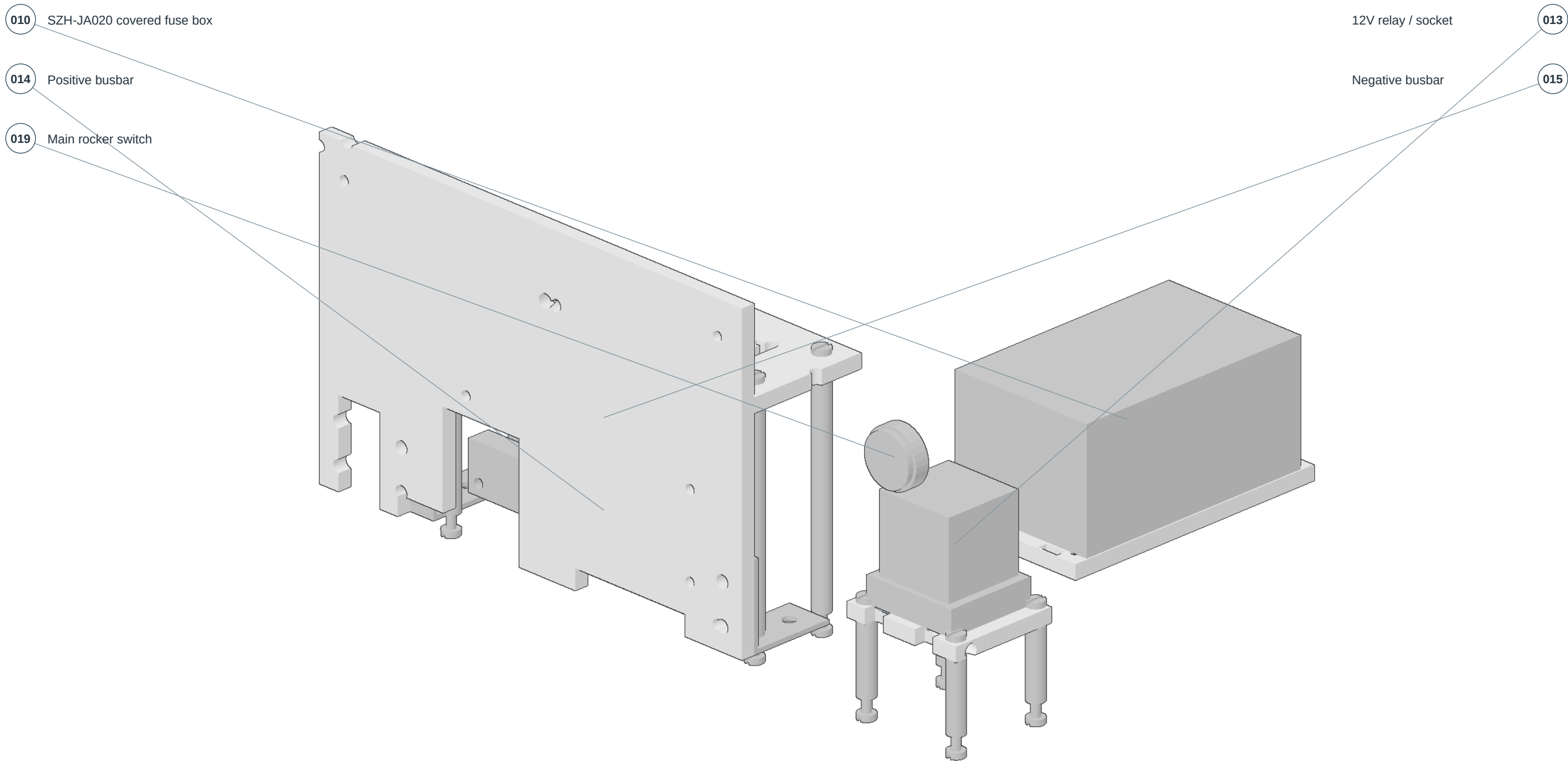
Use final STEP part relationships and the matching F drawing. Standard hardware heads / holes are nominal; verify purchased fasteners, clamp load and tool access on hardware.



TOP view - power components only; other layers hidden for visibility. Coordinates remain final STEP coordinates.

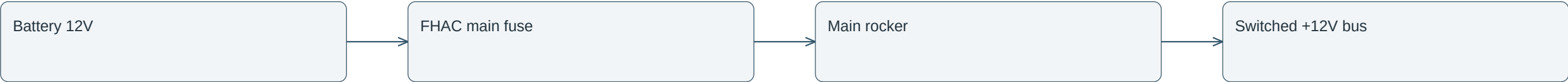


TOP view - control layer isolated. Existing ESP32 38-pin and EK366 geometry retained. Carrier/provision does not certify the unmodeled ADC module.



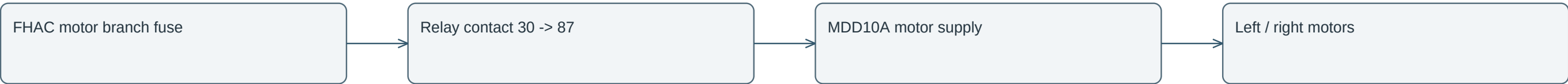
Fuse box: covered envelope and removable cover access. Busbars: lug / terminal approach and cable space. Rocker: outward actuator and rear terminal clearance. Panel cutout, stud thread and fixing holes are ASSEMBLY ADJUST / VERIFY ON HARDWARE.

12V DISTRIBUTION - FUNCTIONAL CONNECTIONS

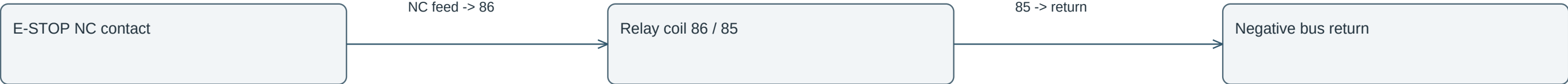


NEGATIVE BUS: battery return, motor return, converter inputs, relay coil return and sense return.

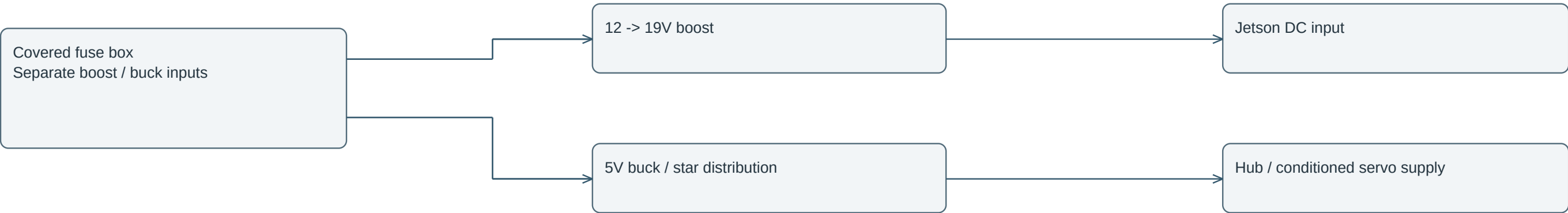
MOTOR CONTACT PATH / FROM SWITCHED +12V BUS



RELAY COIL PATH / FROM SWITCHED +12V BUS

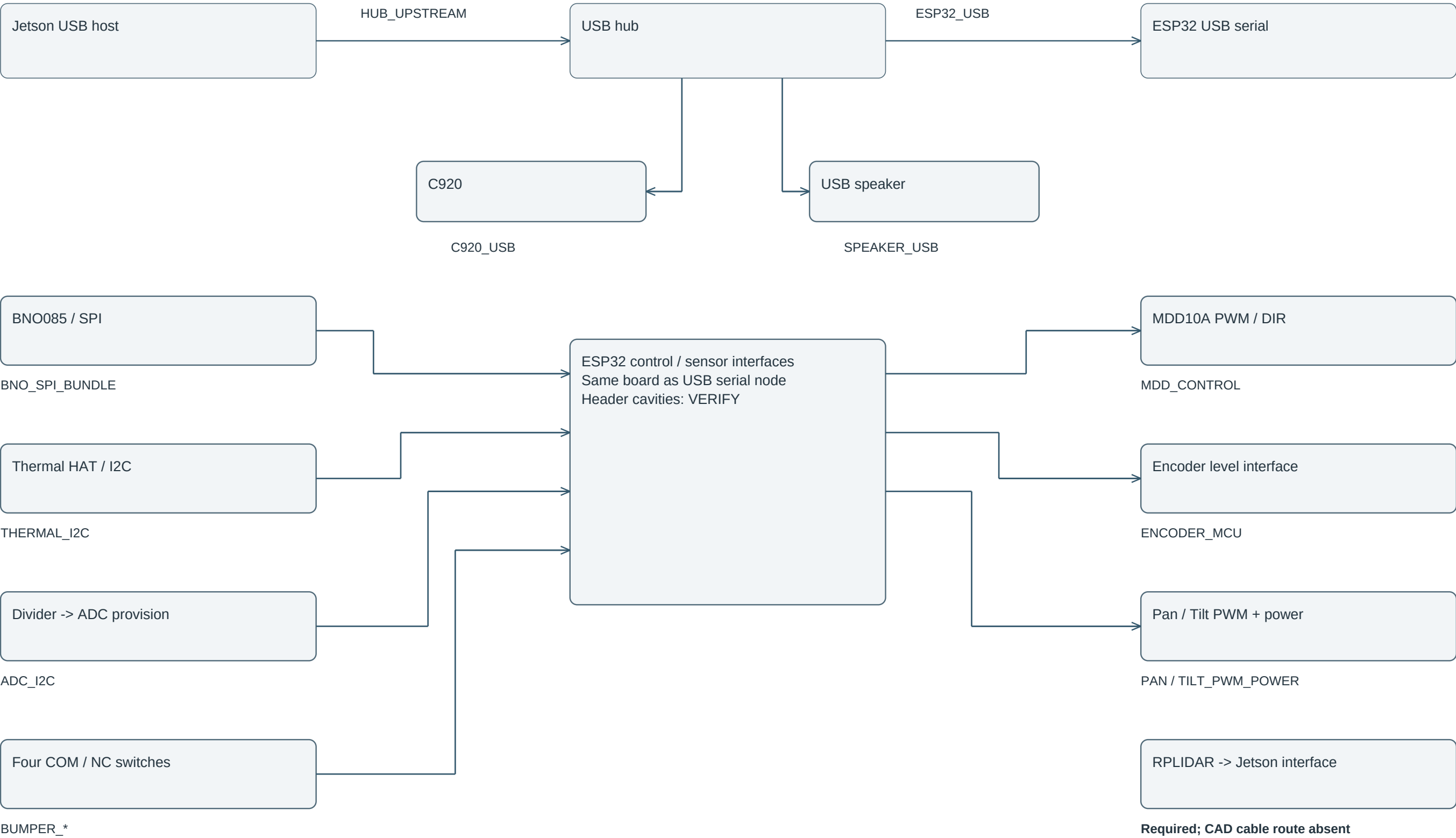


FUSED ELECTRONICS / FROM SWITCHED +12V BUS

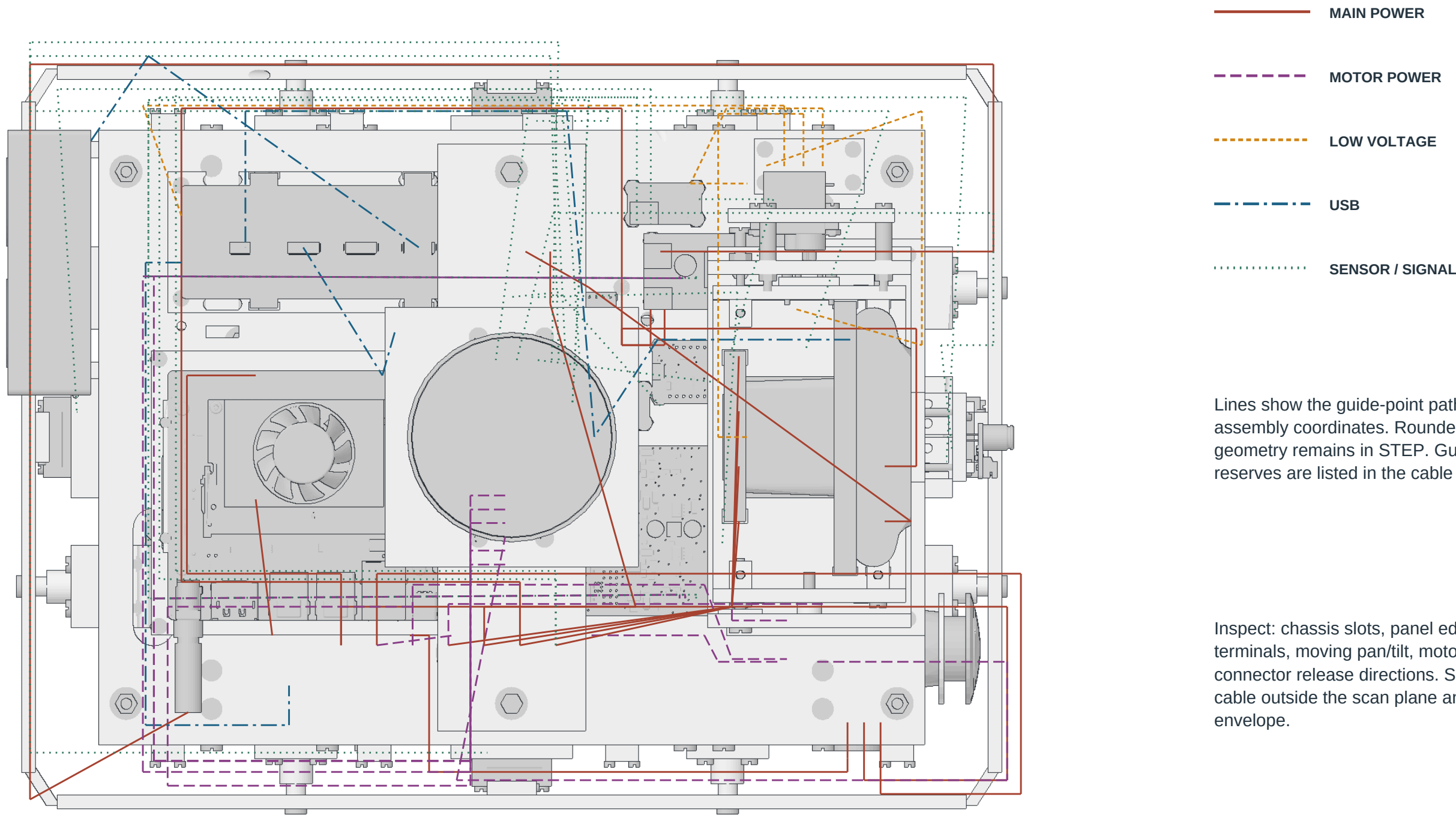


E-STOP interrupts the motor relay coil; other electronics can remain powered. Input / output returns follow the actual converter terminals. Ratings, relay markings, suppression polarity and the servo conditioning provision require hardware verification; see E-211.

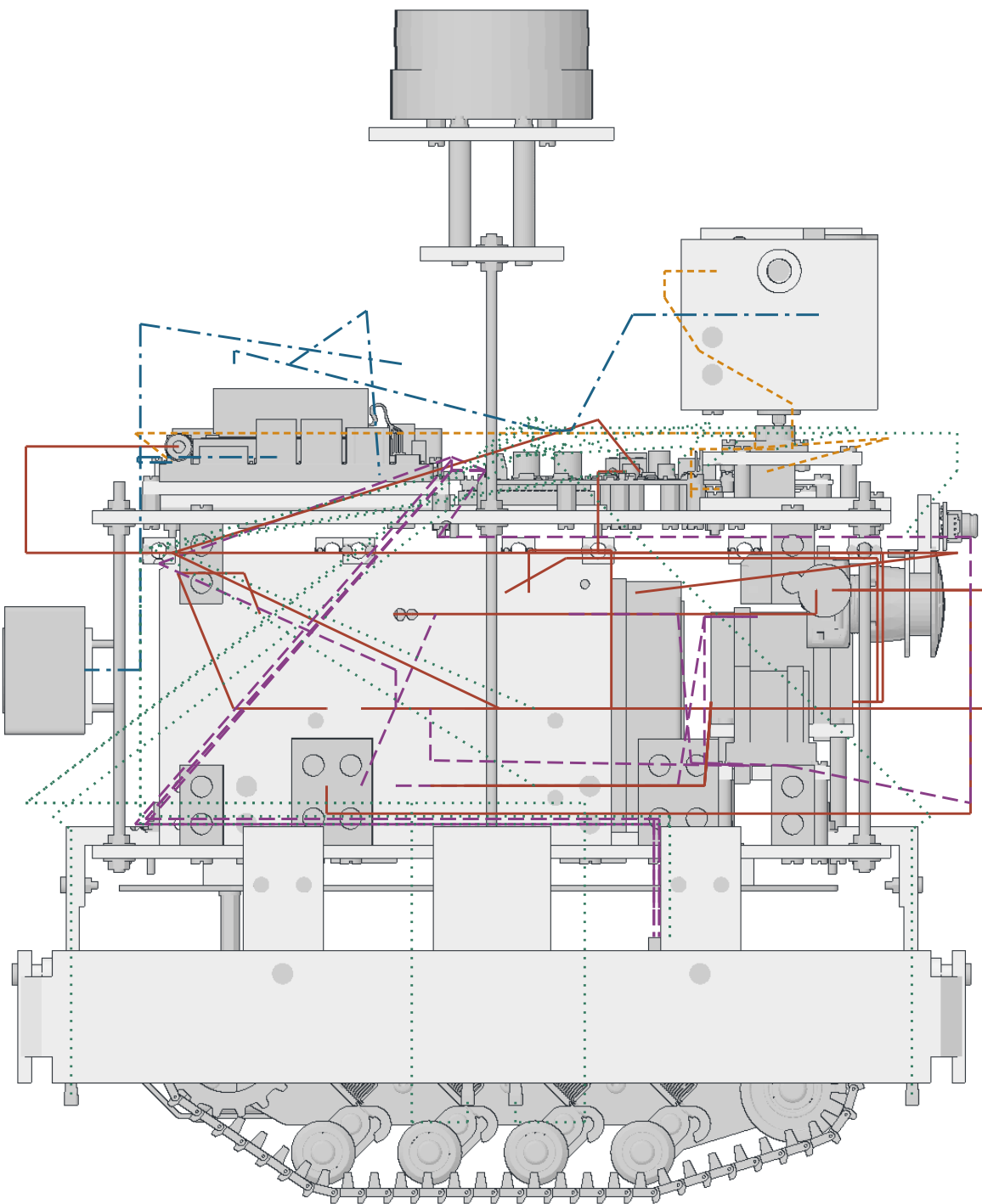
USB / CONTROL INTERFACES - ROUTE NAMES MATCH THE CABLE SCHEDULE



These lines show interface relationships, not physical cable position or pin cavities. Thermal data uses the ESP32 link. ADC board, encoder logic levels, RPLIDAR cable and electrical breakouts require closure in E-211 before final wiring.



Physical cable routes are separate from functional circuits. Drawing line styles remain distinct in monochrome. Conductor breakout / cavity mapping and exact cable bend radii are VERIFY ON HARDWARE.

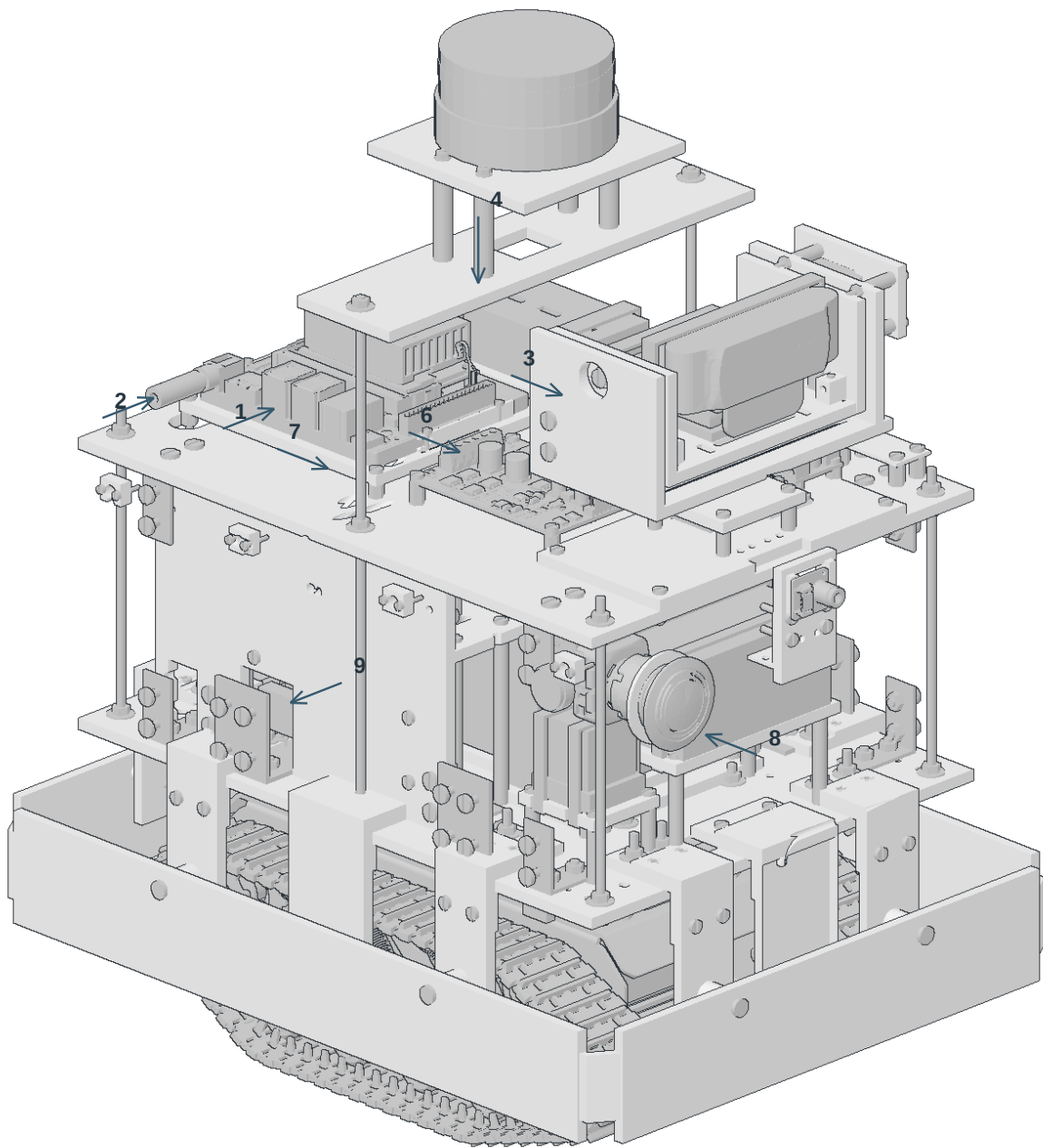


- MAIN POWER
- MOTOR POWER
- LOW VOLTAGE
- USB
- SENSOR / SIGNAL

Lines show the guide-point path in final assembly coordinates. Rounded sweep geometry remains in STEP. Guide segments / reserves are listed in the cable schedule.

Inspect: chassis slots, panel edges, battery terminals, moving pan/tilt, motor tracks and connector release directions. Secure excess cable outside the scan plane and moving envelope.

Physical cable routes are separate from functional circuits. Drawing line styles remain distinct in monochrome. Conductor breakout / cavity mapping and exact cable bend radii are VERIFY ON HARDWARE.



PORT	FORM	APPROACH AXIS
1 JETSON_USB0	USB-A	[0, -1, 0]
2 JETSON_DC	DC5.5/2.5	[0, -1, 0]
3 ESP32_USB	USB-C	[-1, 0, 0]
4 HUB_USB0	USB-A	[0, 0, 1]
6 MDD_POS	Screw terminal	[-1, 0, 0]
7 BAT_POS	FDFD female FASTON	[-1, 0, 0]
8 FUSE_OUT0	FASTON6.3 assumed	[1, 0, 0]
9 BUS_POS0	Screw lug	[0, 1, 0]

Arrows show nominal connector insertion / removal axes. They are not certified hand envelopes. Battery/fuse/busbar terminal work needs panel access; Jetson USB/DC require the exposed port side. Confirm plug body, latch and hand access on hardware.

ITEM	ACCESS CLASS	SERVICE SEQUENCE
Battery	Rear strip / stop removal; harness access first	Isolate source, disconnect terminals, release straps, remove the rear strip / rear stop and withdraw -X; confirm clearance.
Fuse box cover	Front/side panel access may be required	Remove cover along its actual latch axis; leave wire slack.
E-STOP / rocker actuation	Direct exterior access	Keep head/actuator clear; terminal service requires panel removal.
Jetson USB / DC	Port side exposed	Unplug along connector axis; release harness ties if needed.
ESP32	Open upper control area	Unplug USB and remove carrier/retainer as required.
Camera adjustment	Direct at top assembly	Support camera while releasing straps; protect service loops.
LiDAR removal	Top fasteners, then +Z	Disconnect cable and remove through-bolt retainers.
Relay / busbar	Panel removed	Label terminals before disconnecting; do not force cable lugs.

Service classification is based on visible CAD structure. Tool / finger envelopes, latch travel and connector SKU details require hardware checks. Do not assume a closed panel permits terminal access.

INTERFACE	MODEL / DATA STATE	REQUIRED HARDWARE RESOLUTION
RPLIDAR data / power	No RPLIDAR route is present in the final wires table.	Select the actual sensor interface cable; route it below the scan window to the Jetson interface without loading the tower.
ADC module	CUSTOM_ADC_CARRIER is modeled; the net table names ADS1115.	Confirm the actual ADC module, supply and pinout. Do not treat the carrier as a purchased circuit board.
Voltage divider feed	VOLTAGE_POS originates at BUS_POS6; its note requires an auxiliary fuse.	Add the required auxiliary fuse to the actual wiring. The modeled route does not document that fuse inline.
Servo supply	Net notes call for conditioned 4.8V; the buck/star is labeled 5V.	Confirm the selected servo-compatible regulated supply / conditioning and common signal return.
Power negative cores	Return tails meet pair guide geometry only.	Terminate positive and negative on separate insulated conductors. A shared guide point is not an electrical junction.
Relay / E-STOP / fuses	Functional topology follows the current net table.	Verify actual terminal markings, coil suppression direction, contact behavior, conductor capacity and selected fuse values.
Connector cavities	Ports provide body positions and nominal approach directions.	Match the purchased connector markings, keys and motor/encoder cavity mapping before making harnesses.

This sheet records known limits of the existing mechanical CAD and wiring references. It does not change the approved assembly geometry. These items remain physical wiring / integration actions and must not be read as already tested.

ROUTE	FROM -> TO	L mm	STOCK	RESERVE	SIGNAL / NOTE
BAT_MAIN	BAT_POS -> INLINE0_IN	52.1	-	-	12AWG battery positive main fuse close to terminal
BAT_RETURN	BAT_NEG -> BUS_NEG0	195.4	-	-	14AWG ground
MAIN_SWITCH	INLINE0_OUT -> ROCKER_IN	225.7	-	-	14AWG fused12V
MAIN_BUS	ROCKER_OUT -> BUS_POS0	427.9	-	-	14AWG switched12V
MAIN_LED_RETURN	ROCKER_GND -> BUS_NEG1	414.8	-	-	22AWG LED return
MOTOR_BRANCH_FUSE	BUS_POS1 -> INLINE1_IN	70.4	-	-	14AWG motor branch fuse
MOTOR_RELAY30	INLINE1_OUT -> RELAY_30	111.8	-	-	14AWG contact30 feed
MOTOR_RELAY87	RELAY_87 -> MDD_POS	476.2	-	-	14AWG switched motor supply
MOTOR_RETURN	BUS_NEG2 -> MDD_NEG	398.0	-	-	14AWG motor return
COIL_FEED	BUS_POS2 -> ESTOP_IN	209.4	-	-	22AWG coil feed through NC E-stop
COIL_NC	ESTOP_OUT -> RELAY_86	101.7	-	-	22AWG E-stop NC to coil86
COIL_RETURN	RELAY_85 -> BUS_NEG3	139.3	-	-	22AWG coil85 return
FUSE_INPUT_BOOST	BUS_POS3 -> FUSE_IN0	144.0	-	-	14AWG fused distribution input
FUSE_INPUT_BUCK	BUS_POS4 -> FUSE_IN1	147.1	-	-	14AWG fused distribution input
FUSE_INPUT_AUX	BUS_POS5 -> FUSE_IN2	152.2	-	-	14AWG fused distribution input
FUSE_INPUT_SPARE	BUS_POS6 -> FUSE_IN3	158.4	-	-	14AWG fused distribution input
BOOST_INPUT	FUSE_OUT0 -> BOOST_IN_POS	233.4	-	-	14AWG boost input
BOOST_RETURN	BUS_NEG4 -> BOOST_IN_NEG	260.2	-	-	14AWG boost return
JETSON_19V_PAIR	BOOST_OUT_POS -> JETSON_DC	822.9	-	-	18AWG pair: BOOST OUT+/OUT- to760K center+/sleeve-; endpoint2 joined pair, no exposed branch
BUCK_INPUT	FUSE_OUT1 -> BUCK_IN_POS	237.4	-	-	14AWG5V buck input
BUCK_RETURN	BUS_NEG5 -> BUCK_IN_NEG	608.3	-	-	14AWG buck return
5V_STAR	BUCK_OUT_POS -> D5_IN	167.3	-	-	5V paired supply; negative fromBUCK_OUT_NEG in same paired jacket

Guide-route lengths are nominal CAD path lengths. Moving pose maxima are in FINAL_MOTION.json. Unmodeled negative-core breakouts require actual insulated termination; they are not electrical shorts.

ROUTE	FROM -> TO	L mm	STOCK	RESERVE	SIGNAL / NOTE
5V_HUB	D5_HUB -> HUB_DC	283.4	-	-	5V paired supply; hub jack diameter/polarity to verify
HUB_UPSTREAM	HUB_UPSTREAM -> JETSON_USB0	239.0	270	31.0	USB3 captive270mm
C920_USB	C920_USB -> HUB_USB0	396.9	1500	1103.1	USB2 captive1500mm; residual service loop requires routing release
ESP32_USB	HUB_USB1 -> ESP32_USB	102.6	1000	897.4	ComsABFC100 A-C1000mm; MCU powered only here
SPEAKER_USB	HUB_USB3 -> SPEAKER_USB	271.8	-	-	USB speaker captive length unspecified
BNO_SPI_BUNDLE	MCU_SPI_SCK -> BNO_SPI_SCK	64.2	-	-	7coreSPI:3V3/GND/SCK18/MISO19/MOSI23/CS5/INT13. PS0/PS1 straps and reset per Adafruit; exact pin cavities pending
THERMAL_I2C	MCU_SDA -> THERMAL_I2C	325.5	-	-	4core3V3/GND/I2C GPIO21/22
PAN_PWM_POWER	D5_PAN -> MG90S_PAN_CABLE	190.2	250	59.8	Servo JR:power4.8V conditioned,ground,GPIO32; signal joins atstar
TILT_PWM_POWER	D5_TILT -> MG90S_TILT_CABLE	130.1	250	119.9	Servo JR:power4.8V conditioned,ground,GPIO33; flexible movement loop
MDD_CONTROL	MCU_PWM_L -> MDD_PWM_L	157.5	-	-	5corePWM25/DIR27/PWM26/DIR14/commonGND; Cytron order verify
BUMPER_FRONT	MCU_BUMP_F -> BUMPER_FRONT_COM	521.4	-	-	22AWG NC/COM paired cable; signal GPIO4 with3.3V pullup,NC closed toGND; open=contact or wire break
BUMPER_REAR	MCU_BUMP_REAR -> BUMPER_REAR_COM	535.6	-	-	22AWG NC/COM paired cable; signal GPIO2 with3.3V pullup,NC closed toGND; open=contact or wire break
BUMPER_LEFT	MCU_BUMP_LEFT -> BUMPER_LEFT_COM	888.1	-	-	22AWG NC/COM paired cable; signal GPIO16 with3.3V pullup,NC closed toGND; open=contact or wire break
BUMPER_RIGHT	MCU_BUMP_RIGHT -> BUMPER_RIGHT_COM	737.4	-	-	22AWG NC/COM paired cable; signal GPIO17 with3.3V pullup,NC closed toGND; open=contact or wire break
MOTOR_L_M1	MDD_L1 -> MOTOR_L_M1	540.7	-	-	Motor PH6 motor pair; current through PH contacts must fit stall rating
MOTOR_L_M2	MDD_L2 -> MOTOR_L_M2	547.8	-	-	Motor PH6 motor pair; current through PH contacts must fit stall rating
ENCODER_L	MOTOR_L_ENC -> ENCODER_IN	719.0	-	-	4coreencoder VCC,GND,A,B through levelinterface; physical motor cavity mapping pending
MOTOR_R_M1	MDD_R1 -> MOTOR_R_M1	677.1	-	-	Motor PH6 motor pair; current through PH contacts must fit stall rating
MOTOR_R_M2	MDD_R2 -> MOTOR_R_M2	701.8	-	-	Motor PH6 motor pair; current through PH contacts must fit stall rating
ENCODER_R	MOTOR_R_ENC -> ENCODER_IN	630.8	-	-	4coreencoder VCC,GND,A,B through levelinterface; physical motor cavity mapping pending
ENCODER_MCU	ENCODER_OUT -> MCU_ENC_L	77.3	-	-	4signals GPIO34/35/36/39; additional external input resistors
VOLTAGE_SENSE	DIVIDER_SIG -> ADC_ANALOG	90.0	-	-	Analog <=3.3V after divider; ADS1115 added

Guide-route lengths are nominal CAD path lengths. Moving pose maxima are in FINAL_MOTION.json. Unmodeled negative-core breakouts require actual insulated termination; they are not electrical shorts.

ROUTE	FROM -> TO	L mm	STOCK	RESERVE	SIGNAL / NOTE
ADC_I2C	ADC_I2C -> MCU_SCL	299.0	-	-	3V3,GND,SDA21,SCL22
VOLTAGE_POS	BUS_POS6 -> DIVIDER_POS	576.6	-	-	Voltage measurement positive through auxiliary fuse required
VOLTAGE_NEG	BUS_NEG6 -> DIVIDER_NEG	562.8	-	-	Sense ground
BOOST_19V_RETURN_TAIL	BOOST_OUT_NEG -> BOOST_OUT_POS	-	-	-	Negative conductor joins insulated19V pair; not an electrical short
BUCK_5V_RETURN_TAIL	BUCK_OUT_NEG -> BUCK_OUT_POS	-	-	-	Negative conductor enters separate-core5V jacket; not an electrical junction

Guide-route lengths are nominal CAD path lengths. Moving pose maxima are in FINAL_MOTION.json. Unmodeled negative-core breakouts require actual insulated termination; they are not electrical shorts.

ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
001	Hiwonder tracked chassis	1	PURCHASED	MANUFACTURER_CAD / MANUFACTURER_DRAWING / REPAIRED_MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
002	NVIDIA Jetson Orin Nano	1	PURCHASED	MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
003	Logitech C920	1	PURCHASED	COMMUNITY_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
004	MLX90640 Thermal Camera HAT	1	PURCHASED	MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
005	RPLIDAR A2M12	1	PURCHASED	MANUFACTURER_DRAWING / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
006	ESP32 DevKit 38-pin	1	PURCHASED	ADAPTED_COMMUNITY_CAD / STANDARD_RECONSTRUCTION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
007	Cytron MDD10A Rev2	1	PURCHASED	MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
008	Adafruit BNO085	1	PURCHASED	MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
009	ROCKET ES7-12 AGM battery	1	PURCHASED	DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
010	SZH-JA020 covered fuse box	1	PURCHASED	DATASHEET / PHOTO_ESTIMATE / SELLER_DIMENSION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
011	FHAC main fuse holder	1	PURCHASED	MANUFACTURER_DRAWING / PHOTO_ESTIMATE / STANDARD_RECONSTRUCTION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
012	FHAC motor fuse holder	1	PURCHASED	MANUFACTURER_DRAWING / PHOTO_ESTIMATE / STANDARD_RECONSTRUCTION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
013	12V relay / socket	1	PURCHASED	DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
014	Positive busbar	1	PURCHASED	ASSUMED	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
015	Negative busbar	1	PURCHASED	ASSUMED	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
016	SZH-EK366 5V converter	1	PURCHASED	ASSUMED / PHOTO_ESTIMATE / SELLER_DIMENSION / STANDARD_RECONSTRUCTION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
017	12V to19V boost converter	1	PURCHASED	PHOTO_ESTIMATE / SELLER_DIMENSION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
018	KGE-N&H E-STOP	1	PURCHASED	MANUFACTURER_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
019	Main rocker switch	1	PURCHASED	PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
020	USB speaker SP001 envelope	1	PURCHASED	ASSUMED / PHOTO_ESTIMATE / STANDARD_RECONSTRUCTION	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
021	Battery voltage divider	1	PURCHASED	ASSUMED	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
022	NEXT505UHP USB hub	1	PURCHASED	DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
023	MG90S pan servo	1	PURCHASED	COMMUNITY_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
024	MG90S tilt servo	1	PURCHASED	COMMUNITY_CAD	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
025	Pan horn	1	PURCHASED	PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
026	Tilt horn	1	PURCHASED	PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
027	LS-F FRONT V-J166	1	PURCHASED	ASSUMED / DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
028	LS-R REAR V-J166	1	PURCHASED	ASSUMED / DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
029	LS-L LEFT V-J166	1	PURCHASED	ASSUMED / DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
030	LS-Rt RIGHT V-J166	1	PURCHASED	ASSUMED / DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	CAD placement / envelope; actual mating features: VERIFY ON HARDWARE
031	WD SN530 2242 SSD	1	PURCHASED	DATASHEET / PHOTO_ESTIMATE	A-001 / A-005	Existing provision. Verify hardware compatibility / procurement / electrical interface.
032	ADC carrier / hardware VERIFY	1	FABRICATED	CUSTOM	F-221	Existing provision. Verify hardware compatibility / procurement / electrical interface.
033	Encoder level interface	1	PURCHASED	ASSUMED / DATASHEET	A-001 / A-005	Existing provision. Verify hardware compatibility / procurement / electrical interface.
101	CUSTOM_BOTTOM_PLATE	1	FABRICATED	CUSTOM	F-101	PVC foam / Fomex; through profile
102	CUSTOM_BATTERY_PLATE	1	FABRICATED	CUSTOM	F-102	PVC foam / Fomex; through profile
103	CUSTOM_CHASSIS_PAD_0_0	1	FABRICATED	CUSTOM	F-103	PVC foam / Fomex; 3D relief - STEP controls
104	CUSTOM_CHASSIS_PAD_0_1	1	FABRICATED	CUSTOM	F-104	PVC foam / Fomex; through profile
105	CUSTOM_CHASSIS_PAD_1_0	1	FABRICATED	CUSTOM	F-105	PVC foam / Fomex; 3D relief - STEP controls
106	CUSTOM_CHASSIS_PAD_1_1	1	FABRICATED	CUSTOM	F-106	PVC foam / Fomex; through profile

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
107	CUSTOM_CHASSIS_PAD_2_0	1	FABRICATED	CUSTOM	F-107	PVC foam / Fomex; 3D relief - STEP controls
108	CUSTOM_CHASSIS_PAD_2_1	1	FABRICATED	CUSTOM	F-108	PVC foam / Fomex; through profile
109	CUSTOM_CHASSIS_PAD_3_0	1	FABRICATED	CUSTOM	F-109	PVC foam / Fomex; 3D relief - STEP controls
110	CUSTOM_CHASSIS_PAD_3_1	1	FABRICATED	CUSTOM	F-110	PVC foam / Fomex; through profile
111	CUSTOM_ELECTRONICS_PLATE	1	FABRICATED	CUSTOM	F-111	PVC foam / Fomex; through profile
112	CUSTOM_LEFT_PANEL	1	FABRICATED	CUSTOM	F-112	PVC foam / Fomex; through profile
113	CUSTOM_RIGHT_PANEL	1	FABRICATED	CUSTOM	F-113	PVC foam / Fomex; through profile
114	CUSTOM_REAR_PANEL	1	FABRICATED	CUSTOM	F-114	PVC foam / Fomex; 3D relief - STEP controls
115	CUSTOM_FRONT_PANEL	1	FABRICATED	CUSTOM	F-115	PVC foam / Fomex; through profile
116	CUSTOM_LIDAR_BRIDGE	1	FABRICATED	CUSTOM	F-116	PVC foam / Fomex; through profile
117	CUSTOM_LIDAR_PLATE	1	FABRICATED	CUSTOM	F-117	PVC foam / Fomex; through profile
118	CUSTOM_JETSON_CRADLE	1	FABRICATED	CUSTOM	F-118	PVC foam / Fomex; 3D relief - STEP controls
119	CUSTOM_BATTERY_STOP_FRONT	1	FABRICATED	CUSTOM	F-119	PVC foam / Fomex; through profile
120	CUSTOM_BATTERY_STOP_REAR	1	FABRICATED	CUSTOM	F-120	PVC foam / Fomex; through profile
121	CUSTOM_BATTERY_STOP_LEFT	1	FABRICATED	CUSTOM	F-121	PVC foam / Fomex; through profile
122	CUSTOM_BATTERY_STOP_RIGHT	1	FABRICATED	CUSTOM	F-122	PVC foam / Fomex; through profile
123	CUSTOM_HUB_TRAY	1	FABRICATED	CUSTOM	F-123	PVC foam / Fomex; 3D relief - STEP controls
124	CUSTOM_POWER_BACKPLANE	1	FABRICATED	CUSTOM	F-124	PVC foam / Fomex; 3D relief - STEP controls
125	CUSTOM_FUSE_SHELF	1	FABRICATED	CUSTOM	F-125	PVC foam / Fomex; 3D relief - STEP controls
126	CUSTOM_BOOST_BASE	1	FABRICATED	CUSTOM	F-126	PVC foam / Fomex; 3D relief - STEP controls
127	CUSTOM_FUSEBLOCK_TRAY	1	FABRICATED	CUSTOM	F-127	PVC foam / Fomex; 3D relief - STEP controls
128	CUSTOM_JETSON_BASE_RETAINER_LEFT	1	FABRICATED	CUSTOM	F-128	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
129	CUSTOM_JETSON_BASE_RETAINER_RIGHT	1	FABRICATED	CUSTOM	F-129	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
130	CUSTOM_RELAY_TRAY	1	FABRICATED	CUSTOM	F-130	PVC foam / Fomex; 3D relief - STEP controls
131	CUSTOM_CAMERA_BASE	1	FABRICATED	CUSTOM	F-131	PVC foam / Fomex; through profile
132	CUSTOM_PAN_SERVO_MOUNT	1	FABRICATED	CUSTOM	F-132	PVC foam / Fomex; through profile
133	CUSTOM_PAN_YOKE	1	FABRICATED	CUSTOM	F-133	PVC foam / Fomex; 3D relief - STEP controls
134	CUSTOM_PAN_ARM_LEFT	1	FABRICATED	CUSTOM	F-134	PVC foam / Fomex; through profile
135	APPROX_CAM_L_LEFT	1	FABRICATED	MODIFIED_STOCK_BRACKET	F-135	Modified stock metal L bracket; 3D relief - STEP controls
136	CUSTOM_PAN_ARM_RIGHT	1	FABRICATED	CUSTOM	F-136	PVC foam / Fomex; through profile
137	APPROX_CAM_L_RIGHT	1	FABRICATED	MODIFIED_STOCK_BRACKET	F-137	Modified stock metal L bracket; 3D relief - STEP controls
138	CUSTOM_TILT_SERVO_MOUNT	1	FABRICATED	CUSTOM	F-138	PVC foam / Fomex; through profile
139	CUSTOM_TILT_CHEEK_LEFT	1	FABRICATED	CUSTOM	F-139	PVC foam / Fomex; through profile
140	CUSTOM_TILT_CHEEK_RIGHT	1	FABRICATED	CUSTOM	F-140	PVC foam / Fomex; through profile
141	CUSTOM_C920_SADDLE	1	FABRICATED	CUSTOM	F-141	PVC foam / Fomex; through profile
142	CUSTOM_TILT_CORNER_LEFT0	1	FABRICATED	CUSTOM	F-142	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
143	CUSTOM_TILT_CORNER_LEFT1	1	FABRICATED	CUSTOM	F-143	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
144	CUSTOM_TILT_CORNER_RIGHT0	1	FABRICATED	CUSTOM	F-144	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
145	CUSTOM_TILT_CORNER_RIGHT1	1	FABRICATED	CUSTOM	F-145	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
146	CUSTOM_TILT_BUSH_LEFT	1	FABRICATED	CUSTOM	F-146	Stock material - VERIFY ON HARDWARE; through profile
147	CUSTOM_THERMAL_MOUNT	1	FABRICATED	CUSTOM	F-147	PVC foam / Fomex; 3D relief - STEP controls
148	CUSTOM_THERMAL_ANGLE_0	1	FABRICATED	CUSTOM	F-148	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
149	CUSTOM_THERMAL_ANGLE_1	1	FABRICATED	CUSTOM	F-149	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
150	CUSTOM_THERMAL_EDGE_0	1	FABRICATED	CUSTOM	F-150	Stock material - VERIFY ON HARDWARE; through profile

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
151	CUSTOM_THERMAL_EDGE_1	1	FABRICATED	CUSTOM	F-151	Stock material - VERIFY ON HARDWARE; through profile
152	CUSTOM_FRONT BUMPER	1	FABRICATED	CUSTOM	F-152	PVC foam / Fomex; through profile
153	CUSTOM_FRONT_STRIKER	1	FABRICATED	CUSTOM	F-153	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
154	CUSTOM_FRONT_GUIDE_FOOT_0	1	FABRICATED	CUSTOM	F-154	PVC foam / Fomex; through profile
155	CUSTOM_FRONT_GUIDE_HANGER_0	1	FABRICATED	CUSTOM	F-155	PVC foam / Fomex; through profile
156	CUSTOM_FRONT_GUIDE_ANGLE_0	1	FABRICATED	CUSTOM	F-156	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
157	CUSTOM_FRONT_GUIDE_BUSH_0	1	FABRICATED	CUSTOM	F-157	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
158	CUSTOM_FRONT_STOPPER_0	1	FABRICATED	CUSTOM	F-158	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
159	CUSTOM_FRONT_GUIDE_FOOT_1	1	FABRICATED	CUSTOM	F-159	PVC foam / Fomex; through profile
160	CUSTOM_FRONT_GUIDE_HANGER_1	1	FABRICATED	CUSTOM	F-160	PVC foam / Fomex; through profile
161	CUSTOM_FRONT_GUIDE_ANGLE_1	1	FABRICATED	CUSTOM	F-161	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
162	CUSTOM_FRONT_GUIDE_BUSH_1	1	FABRICATED	CUSTOM	F-162	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
163	CUSTOM_FRONT_STOPPER_1	1	FABRICATED	CUSTOM	F-163	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
164	CUSTOM_FRONT_SWITCH_BRACKET	1	FABRICATED	CUSTOM	F-164	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
165	CUSTOM_FRONT_SWITCH_FOOT	1	FABRICATED	CUSTOM	F-165	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
166	CUSTOM_FRONT_SWITCH_ANGLE	1	FABRICATED	CUSTOM	F-166	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
167	CUSTOM_FRONT_CORNER_NOSE_-1	1	FABRICATED	CUSTOM	F-167	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
168	CUSTOM_FRONT_CORNER_NOSE_1	1	FABRICATED	CUSTOM	F-168	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
169	CUSTOM_REAR BUMPER	1	FABRICATED	CUSTOM	F-169	PVC foam / Fomex; through profile
170	CUSTOM_REAR_STRIKER	1	FABRICATED	CUSTOM	F-170	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
171	CUSTOM_REAR_GUIDE_FOOT_0	1	FABRICATED	CUSTOM	F-171	PVC foam / Fomex; through profile
172	CUSTOM_REAR_GUIDE_HANGER_0	1	FABRICATED	CUSTOM	F-172	PVC foam / Fomex; through profile

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
173	CUSTOM_REAR_GUIDE_ANGLE_0	1	FABRICATED	CUSTOM	F-173	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
174	CUSTOM_REAR_GUIDE_BUSH_0	1	FABRICATED	CUSTOM	F-174	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
175	CUSTOM_REAR_STOPPER_0	1	FABRICATED	CUSTOM	F-175	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
176	CUSTOM_REAR_GUIDE_FOOT_1	1	FABRICATED	CUSTOM	F-176	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
177	CUSTOM_REAR_GUIDE_HANGER_1	1	FABRICATED	CUSTOM	F-177	PVC foam / Fomex; through profile
178	CUSTOM_REAR_GUIDE_ANGLE_1	1	FABRICATED	CUSTOM	F-178	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
179	CUSTOM_REAR_GUIDE_BUSH_1	1	FABRICATED	CUSTOM	F-179	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
180	CUSTOM_REAR_STOPPER_1	1	FABRICATED	CUSTOM	F-180	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
181	CUSTOM_REAR_SWITCH_BRACKET	1	FABRICATED	CUSTOM	F-181	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
182	CUSTOM_REAR_SWITCH_FOOT	1	FABRICATED	CUSTOM	F-182	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
183	CUSTOM_REAR_SWITCH_ANGLE	1	FABRICATED	CUSTOM	F-183	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
184	CUSTOM_REAR_CORNER_NOSE_-1	1	FABRICATED	CUSTOM	F-184	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
185	CUSTOM_REAR_CORNER_NOSE_1	1	FABRICATED	CUSTOM	F-185	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
186	CUSTOM_LEFT BUMPER	1	FABRICATED	CUSTOM	F-186	PVC foam / Fomex; through profile
187	CUSTOM_LEFT_STRIKER	1	FABRICATED	CUSTOM	F-187	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
188	CUSTOM_LEFT_GUIDE_FOOT_0	1	FABRICATED	CUSTOM	F-188	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
189	CUSTOM_LEFT_GUIDE_HANGER_0	1	FABRICATED	CUSTOM	F-189	PVC foam / Fomex; through profile
190	CUSTOM_LEFT_GUIDE_ANGLE_0	1	FABRICATED	CUSTOM	F-190	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
191	CUSTOM_LEFT_GUIDE_BUSH_0	1	FABRICATED	CUSTOM	F-191	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
192	CUSTOM_LEFT_STOPPER_0	1	FABRICATED	CUSTOM	F-192	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
193	CUSTOM_LEFT_GUIDE_FOOT_1	1	FABRICATED	CUSTOM	F-193	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
194	CUSTOM_LEFT_GUIDE_HANGER_1	1	FABRICATED	CUSTOM	F-194	PVC foam / Fomex; through profile

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
195	CUSTOM_LEFT_GUIDE_ANGLE_1	1	FABRICATED	CUSTOM	F-195	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
196	CUSTOM_LEFT_GUIDE_BUSH_1	1	FABRICATED	CUSTOM	F-196	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
197	CUSTOM_LEFT_STOPPER_1	1	FABRICATED	CUSTOM	F-197	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
198	CUSTOM_LEFT_SWITCH_BRACKET	1	FABRICATED	CUSTOM	F-198	PVC foam / Fomex; through profile
199	CUSTOM_LEFT_SWITCH_FOOT	1	FABRICATED	CUSTOM	F-199	PVC foam / Fomex; through profile
200	CUSTOM_LEFT_SWITCH_ANGLE	1	FABRICATED	CUSTOM	F-200	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
201	CUSTOM_RIGHT BUMPER	1	FABRICATED	CUSTOM	F-201	PVC foam / Fomex; 3D relief - STEP controls
202	CUSTOM_RIGHT_STRIKER	1	FABRICATED	CUSTOM	F-202	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
203	CUSTOM_RIGHT_GUIDE_FOOT_0	1	FABRICATED	CUSTOM	F-203	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
204	CUSTOM_RIGHT_GUIDE_HANGER_0	1	FABRICATED	CUSTOM	F-204	PVC foam / Fomex; through profile
205	CUSTOM_RIGHT_GUIDE_ANGLE_0	1	FABRICATED	CUSTOM	F-205	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
206	CUSTOM_RIGHT_GUIDE_BUSH_0	1	FABRICATED	CUSTOM	F-206	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
207	CUSTOM_RIGHT_STOPPER_0	1	FABRICATED	CUSTOM	F-207	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
208	CUSTOM_RIGHT_GUIDE_FOOT_1	1	FABRICATED	CUSTOM	F-208	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
209	CUSTOM_RIGHT_GUIDE_HANGER_1	1	FABRICATED	CUSTOM	F-209	PVC foam / Fomex; through profile
210	CUSTOM_RIGHT_GUIDE_ANGLE_1	1	FABRICATED	CUSTOM	F-210	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
211	CUSTOM_RIGHT_GUIDE_BUSH_1	1	FABRICATED	CUSTOM	F-211	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
212	CUSTOM_RIGHT_STOPPER_1	1	FABRICATED	CUSTOM	F-212	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
213	CUSTOM_RIGHT_SWITCH_BRACKET	1	FABRICATED	CUSTOM	F-213	PVC foam / Fomex; through profile
214	CUSTOM_RIGHT_SWITCH_FOOT	1	FABRICATED	CUSTOM	F-214	PVC foam / Fomex; through profile
215	CUSTOM_RIGHT_SWITCH_ANGLE	1	FABRICATED	CUSTOM	F-215	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
216	CUSTOM_ESP32_CARRIER	1	FABRICATED	CUSTOM	F-216	PVC foam / Fomex; 3D relief - STEP controls

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ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
217	CUSTOM_ESTOP_ADAPTER_RING	1	FABRICATED	CUSTOM	F-217	Stock material - VERIFY ON HARDWARE; through profile
218	CUSTOM_SPEAKER_MOUNT_0	1	FABRICATED	CUSTOM	F-218	Stock material - VERIFY ON HARDWARE; through profile
219	CUSTOM_SPEAKER_MOUNT_1	1	FABRICATED	CUSTOM	F-219	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
220	CUSTOM_5V_DISTRIBUTOR	1	FABRICATED	CUSTOM	F-220	PVC foam / Fomex; through profile
222	CUSTOM_HARNESS_CLAMP_0	1	FABRICATED	CUSTOM	F-222	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
223	CUSTOM_HARNESS_CLAMP_1	1	FABRICATED	CUSTOM	F-223	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
224	CUSTOM_HARNESS_CLAMP_2	1	FABRICATED	CUSTOM	F-224	Stock material - VERIFY ON HARDWARE; through profile
225	CUSTOM_HARNESS_CLAMP_3	1	FABRICATED	CUSTOM	F-225	Stock material - VERIFY ON HARDWARE; through profile
226	CUSTOM_HARNESS_CLAMP_4	1	FABRICATED	CUSTOM	F-226	Stock material - VERIFY ON HARDWARE; through profile
227	CUSTOM_HARNESS_CLAMP_5	1	FABRICATED	CUSTOM	F-227	Stock material - VERIFY ON HARDWARE; 3D relief - STEP controls
228	CUSTOM_HARNESS_CLAMP_6	1	FABRICATED	CUSTOM	F-228	Stock material - VERIFY ON HARDWARE; through profile
229	CUSTOM_HARNESS_CLAMP_7	1	FABRICATED	CUSTOM	F-229	Stock material - VERIFY ON HARDWARE; through profile
230	CUSTOM_TILT_HORN_COUPLER	1	FABRICATED	CUSTOM	F-230	Stock material - VERIFY ON HARDWARE; through profile
501	Custom-length insulating spacer 11.610; add-required if kit cannot stack to length	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
502	Custom-length insulating spacer 40.610; add-required if kit cannot stack to length	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
503	M2 nominal thread, no helical detail	4	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
504	M2.5 nominal thread, no helical detail	12	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
505	M2.5 spacer L10; verify stock kit length	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
506	M2.5x10; symbolic head, verify actual tool/head	8	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
507	M2.5x18; symbolic head, verify actual tool/head	8	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
508	M2.5x8; symbolic head, verify actual tool/head	4	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification

Quantities are model-derived instances. Source CAD internals are grouped as purchased assemblies. Reference sweep/keep-out/service shapes are excluded; harness routes are in the separate cable schedule.

ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
509	M2x10; symbolic head, verify actual tool/head	6	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
510	M2x12; symbolic head, verify actual tool/head	20	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
511	M2x6; symbolic head, verify actual tool/head	2	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
512	M3 nominal thread, no helical detail	68	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
513	M3 spacer L10; verify stock kit length	20	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
514	M3 spacer L3; verify stock kit length	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
515	M3 spacer L6; verify stock kit length	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
516	M3x10; symbolic head, verify actual tool/head	72	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
517	M3x12; symbolic head, verify actual tool/head	48	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
518	M3x14; symbolic head, verify actual tool/head	8	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
519	M3x16; symbolic head, verify actual tool/head	36	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
520	M3x30; symbolic head, verify actual tool/head	4	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
521	M3x8; symbolic head, verify actual tool/head	20	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
522	M4 nominal thread, no helical detail	128	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
523	M4x12; symbolic head, verify actual tool/head	80	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
524	M4x15; symbolic head, verify actual tool/head	1	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
525	M4x16; symbolic head, verify actual tool/head	16	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
526	M4x25; symbolic head, verify actual tool/head	4	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
527	Modeled hardware [0.8, 12.0, 12.0] mm envelope	48	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
528	Modeled hardware [0.8, 7.0, 7.0] mm envelope	8	FASTENER	DATASHEET	B-001 / D series	Quantity = modeled instances; verify stock specification
529	Modeled hardware [6.0, 6.0, 13.9] mm envelope	4	FASTENER	CUSTOM	B-001 / D series	Quantity = modeled instances; verify stock specification
530	Official2.5mm hole, additionalM2 fastening	4	FASTENER	CUSTOM	B-001 / D series	Quantity = modeled instances; verify stock specification

Quantities are model-derived instances. Source CAD internals are grouped as purchased assemblies. Reference sweep/keep-out/service shapes are excluded; harness routes are in the separate cable schedule.

ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
531	Sleeve or spacer stack L16.000, additional when absent from kit	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
532	Sleeve or spacer stack L39.610, additional when absent from kit	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
533	Sleeve or spacer stack L75.610, additional when absent from kit	4	FASTENER	ASSUMED	B-001 / D series	Quantity = modeled instances; verify stock specification
534	APPROX_BACKPLANE_L	2	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
535	APPROX_FDFD	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
536	APPROX_FDFD_5p5	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
537	APPROX_FRONT_RETURN_SPRING	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
538	APPROX_LEFT_RETURN_SPRING	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
539	APPROX_MEDIUM_L	4	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
540	APPROX_REAR_RETURN_SPRING	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
541	APPROX_RIGHT_RETURN_SPRING	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
542	APPROX_SMALL_L_0_23_LEFT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
543	APPROX_SMALL_L_0_23_RIGHT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
544	APPROX_SMALL_L_0_245_LEFT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
545	APPROX_SMALL_L_0_245_RIGHT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
546	APPROX_SMALL_L_1_23_LEFT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
547	APPROX_SMALL_L_1_23_RIGHT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
548	APPROX_SMALL_L_1_245_LEFT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
549	APPROX_SMALL_L_1_245_RIGHT	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
550	CUSTOM_BATTERY_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
551	CUSTOM_BOOST_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
552	CUSTOM_C920_PAD	1	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate

Quantities are model-derived instances. Source CAD internals are grouped as purchased assemblies. Reference sweep/keep-out/service shapes are excluded; harness routes are in the separate cable schedule.

ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
553	CUSTOM_C920_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
554	CUSTOM_ESP32_SOCKET_RAIL	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
555	CUSTOM_FUSEBLOCK_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
556	CUSTOM_FUSE_TIE	4	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
557	CUSTOM_HUB_PAD	1	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
558	CUSTOM_HUB_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
559	CUSTOM_LIDAR_RISER	4	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
560	CUSTOM_RELAY_STRAP	2	PURCHASED	CUSTOM	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
561	DRAWING_BASED_760K_BARREL	1	PURCHASED	DATASHEET	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
562	DRAWING_BASED_760K_HANDLE	1	PURCHASED	DATASHEET	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
563	DRAWING_BASED_760K_NECK	1	PURCHASED	MANUFACTURER_DRAWING	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
564	DRAWING_BASED_FRONT_GUIDE_ROD	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
565	DRAWING_BASED_LEFT_GUIDE_ROD	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
566	DRAWING_BASED_M4_ROD	6	PURCHASED	DATASHEET	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
567	DRAWING_BASED_REAR_GUIDE_ROD	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
568	DRAWING_BASED_RIGHT_GUIDE_ROD	2	PURCHASED	ASSUMED	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
569	OFFICIAL_ADS1115_ADDED	27	PURCHASED	MANUFACTURER_CAD	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
570	OFFICIAL_TE187_FRONT	2	PURCHASED	MANUFACTURER_CAD	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
571	OFFICIAL_TE187_LEFT	2	PURCHASED	MANUFACTURER_CAD	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
572	OFFICIAL_TE187_REAR	2	PURCHASED	MANUFACTURER_CAD	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
573	OFFICIAL_TE187_RIGHT	2	PURCHASED	MANUFACTURER_CAD	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate
574	PHOTO_BASED_760K_RING	1	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate

Quantities are model-derived instances. Source CAD internals are grouped as purchased assemblies. Reference sweep/keep-out/service shapes are excluded; harness routes are in the separate cable schedule.

ITEM	PART NAME	QTY	TYPE	SOURCE	DRAWING No.	NOTE
575	PHOTO_BASED_STRAP_BUCKLE	2	PURCHASED	PHOTO_ESTIMATE	B-001 / D series	Model-derived item/group; purchased contour and mounting details may be approximate

Quantities are model-derived instances. Source CAD internals are grouped as purchased assemblies. Reference sweep/keep-out/service shapes are excluded; harness routes are in the separate cable schedule.

Fastener list / model counts

B-002-1

ITEM	HARDWARE / SOURCE NOTE	QTY	SOURCE	NOTE
501	Custom-length insulating spacer 11.610; add-required if kit cannot stack to length	4	ASSUMED	Quantity = modeled instances; verify stock specification
502	Custom-length insulating spacer 40.610; add-required if kit cannot stack to length	4	ASSUMED	Quantity = modeled instances; verify stock specification
503	M2 nominal thread, no helical detail	4	DATASHEET	Quantity = modeled instances; verify stock specification
504	M2.5 nominal thread, no helical detail	12	DATASHEET	Quantity = modeled instances; verify stock specification
505	M2.5 spacer L10; verify stock kit length	4	ASSUMED	Quantity = modeled instances; verify stock specification
506	M2.5x10; symbolic head, verify actual tool/head	8	DATASHEET	Quantity = modeled instances; verify stock specification
507	M2.5x18; symbolic head, verify actual tool/head	8	DATASHEET	Quantity = modeled instances; verify stock specification
508	M2.5x8; symbolic head, verify actual tool/head	4	DATASHEET	Quantity = modeled instances; verify stock specification
509	M2x10; symbolic head, verify actual tool/head	6	DATASHEET	Quantity = modeled instances; verify stock specification
510	M2x12; symbolic head, verify actual tool/head	20	DATASHEET	Quantity = modeled instances; verify stock specification
511	M2x6; symbolic head, verify actual tool/head	2	DATASHEET	Quantity = modeled instances; verify stock specification
512	M3 nominal thread, no helical detail	68	DATASHEET	Quantity = modeled instances; verify stock specification
513	M3 spacer L10; verify stock kit length	20	ASSUMED	Quantity = modeled instances; verify stock specification
514	M3 spacer L3; verify stock kit length	4	ASSUMED	Quantity = modeled instances; verify stock specification
515	M3 spacer L6; verify stock kit length	4	ASSUMED	Quantity = modeled instances; verify stock specification
516	M3x10; symbolic head, verify actual tool/head	72	DATASHEET	Quantity = modeled instances; verify stock specification
517	M3x12; symbolic head, verify actual tool/head	48	DATASHEET	Quantity = modeled instances; verify stock specification
518	M3x14; symbolic head, verify actual tool/head	8	DATASHEET	Quantity = modeled instances; verify stock specification
519	M3x16; symbolic head, verify actual tool/head	36	DATASHEET	Quantity = modeled instances; verify stock specification
520	M3x30; symbolic head, verify actual tool/head	4	DATASHEET	Quantity = modeled instances; verify stock specification
521	M3x8; symbolic head, verify actual tool/head	20	DATASHEET	Quantity = modeled instances; verify stock specification
522	M4 nominal thread, no helical detail	128	DATASHEET	Quantity = modeled instances; verify stock specification

M4 rods / L brackets / straps also appear in the main BOM. Manufacturer-native subassembly internal screws are not a separate purchase count.

ITEM	HARDWARE / SOURCE NOTE	QTY	SOURCE	NOTE
523	M4x12; symbolic head, verify actual tool/head	80	DATASHEET	Quantity = modeled instances; verify stock specification
524	M4x15; symbolic head, verify actual tool/head	1	DATASHEET	Quantity = modeled instances; verify stock specification
525	M4x16; symbolic head, verify actual tool/head	16	DATASHEET	Quantity = modeled instances; verify stock specification
526	M4x25; symbolic head, verify actual tool/head	4	DATASHEET	Quantity = modeled instances; verify stock specification
527	Modeled hardware [0.8, 12.0, 12.0] mm envelope	48	DATASHEET	Quantity = modeled instances; verify stock specification
528	Modeled hardware [0.8, 7.0, 7.0] mm envelope	8	DATASHEET	Quantity = modeled instances; verify stock specification
529	Modeled hardware [6.0, 6.0, 13.9] mm envelope	4	CUSTOM	Quantity = modeled instances; verify stock specification
530	Official2.5mm hole, additionalM2 fastening	4	CUSTOM	Quantity = modeled instances; verify stock specification
531	Sleeve or spacer stack L16.000, additional when absent from kit	4	ASSUMED	Quantity = modeled instances; verify stock specification
532	Sleeve or spacer stack L39.610, additional when absent from kit	4	ASSUMED	Quantity = modeled instances; verify stock specification
533	Sleeve or spacer stack L75.610, additional when absent from kit	4	ASSUMED	Quantity = modeled instances; verify stock specification

M4 rods / L brackets / straps also appear in the main BOM. Manufacturer-native subassembly internal screws are not a separate purchase count.

STEP	TASK	PART / ASSEMBLY RELATION
01	Cut / prepare fabricated parts	Use F profiles, thickness and datum tables; retain part labels. Relief parts follow STEP, not through-cut DXF.
02	Finish edges and dry-fit	Remove sharp edges; verify stock hole / bracket interfaces before final tightening.
03	Lower frame on chassis	Fit chassis pads, lower plate, rods and lower L brackets to existing chassis interfaces.
04	Battery and power backplane	Install support/stops/straps and distribution components before enclosing the frame.
05	Upper deck / panels	Fit deck on M4 rods with washers/nuts. Keep service-side panels removable until wiring checks finish.
06	Control electronics	Install cradles/carriers, M3 standoffs, Jetson, ESP32, MDD10A, sensors and interface provisions.
07	LiDAR tower	Bridge, risers, sensor plate and LiDAR; route cable below scan reference.
08	Camera / thermal assembly	Pan base, yoke, tilt frame, saddle/pad/straps, fixed thermal support; apply verified clearance relief.
09	Bumpers / controls	Fit four guide/return/lever mechanisms; mount rocker and E-STOP with exterior access.
10	Harness and connector checks	Follow E pages and cable schedule. Secure service loops and excess length outside moving/scan regions.
11	Physical acceptance	Confirm polarity/rails/pinout, relay/E-STOP response, manual motion, retention and final interference.

Keep power disconnected while mounting and routing. The listed order follows the modeled closed-deck structure. Purchased subassemblies remain intact unless their manufacturer instructions require otherwise.

GROUP	CHECK	ACCEPTANCE / RECORD
MECHANICAL	Part identity / outline / thickness	Match F drawing, material and individual STEP.
MECHANICAL	Bracket / fastener direction	Confirm rod washers, nut seating, spacer length and supported surfaces.
MECHANICAL	Battery retention	Case seated against stops; straps secure; removal route understood.
MECHANICAL	Pan/Tilt and bumpers	Full slow manual travel; no drag; switches actuate and release before stops.
MECHANICAL	LiDAR plane	Level sensor; no bracket or cable in the scan window.
ELECTRICAL	Battery /19V /5V polarity	Identify actual terminals; verify rail voltage and polarity before connection.
ELECTRICAL	Fuse / relay / E-STOP	Confirm selected ratings, actual pin markings, contact operation and motor disable behavior.
ELECTRICAL	Controller interfaces	Verify GPIO/header cavities, common returns, ADC and encoder levels.
CABLE	USB / connectors	All plugs latch; insertion/removal and excess-length storage possible.
CABLE	Bending / moving parts	No sharp edge contact, tension, track contact or scan obstruction; strain relief secured.

Record physical results before powered integration. VERIFIED CAD does not replace electrical commissioning, servo load/torque checks or actual material/fastener strength confirmation.